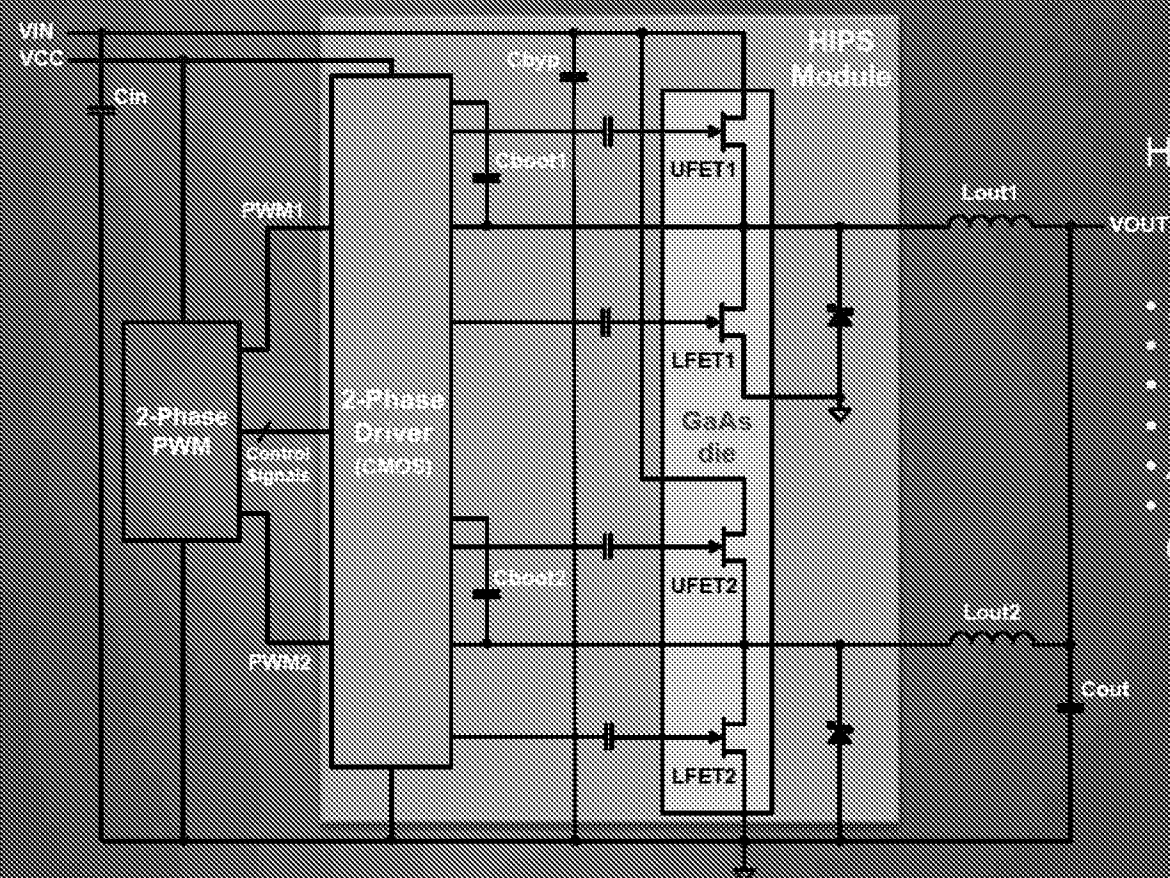


APPENDIX 1

Part 3 of 5

Integrated Power Stage Approach: HIPS



HIPS: Heterogeneously Integrated Power Stage

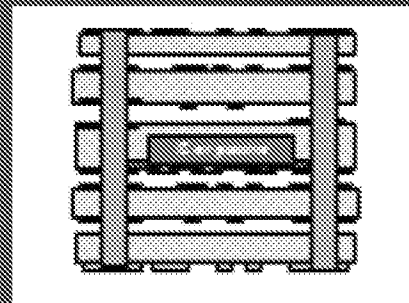
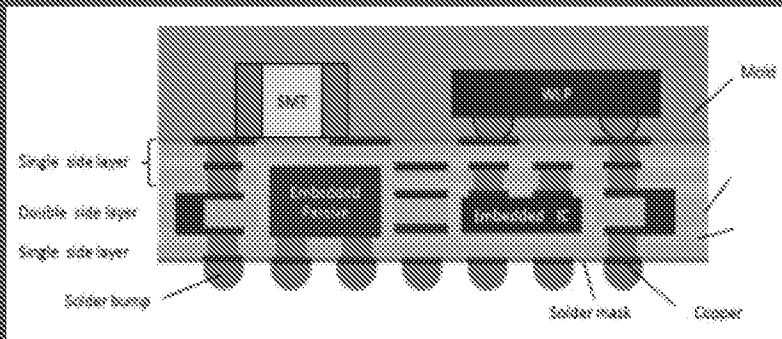
- Monolithic Quad GaAs die
- 2-phase CMOS driver
- Cby
- Boot caps
- AC-coupled gate caps
- Schottky diodes (for diode emulation)

Integrates performance-critical components

HIPS Module

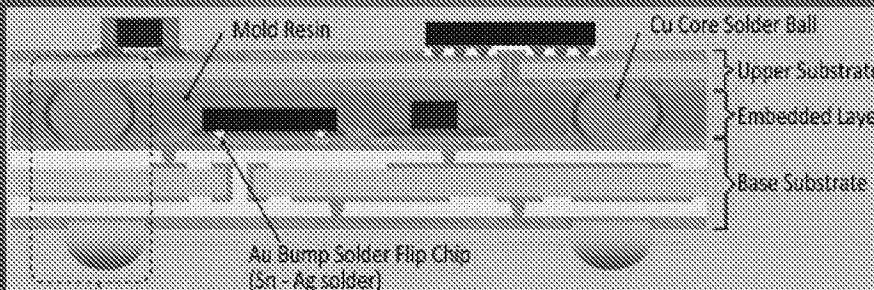
Available technology to enable compact multi-chip HIPS module:
Embedded Die in Substrate

Two motivations: (1) Performance (*minimize parasitics for multi-MHz operation*)
 (2) PowerStage package size

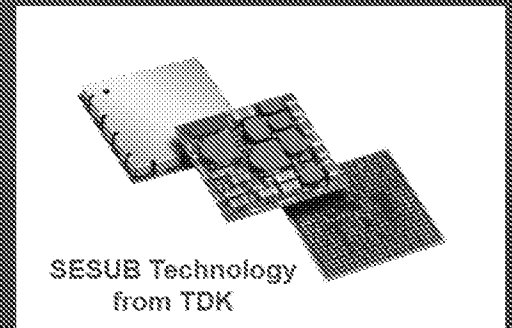


ECP®
 Technology
 from AT&S

Chipset™ Technology from FCI and Fujikura



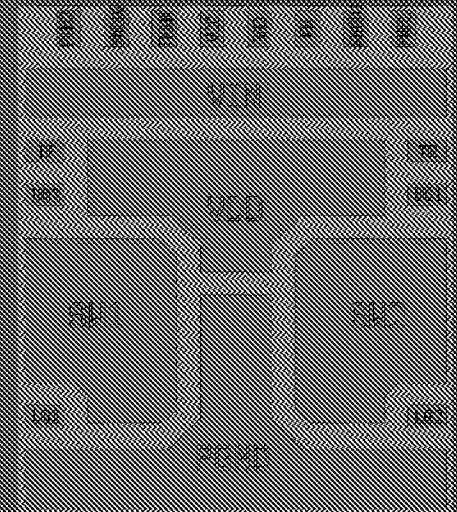
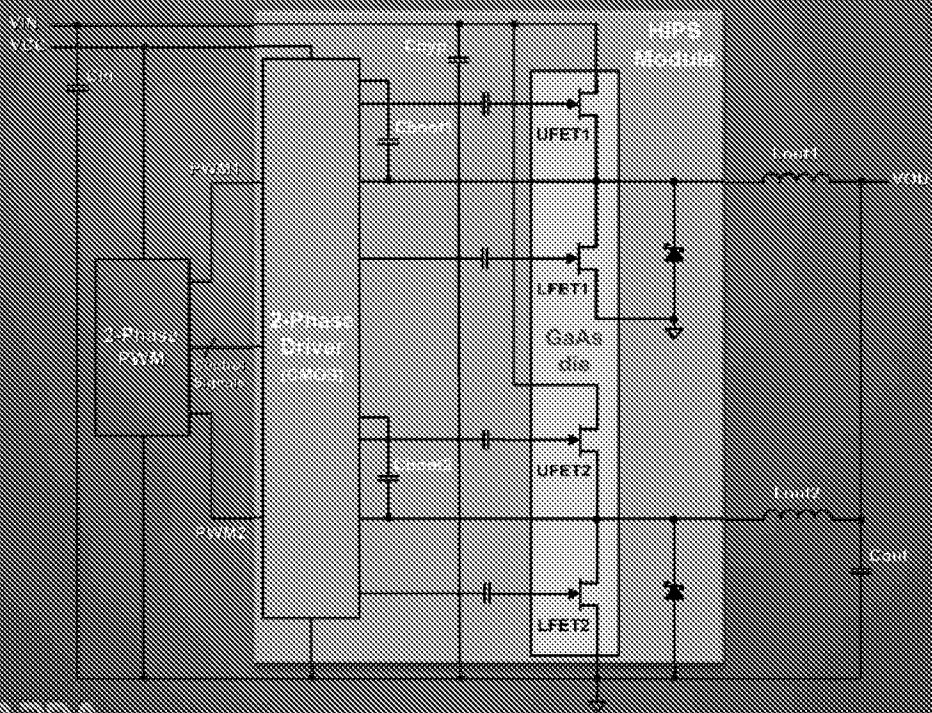
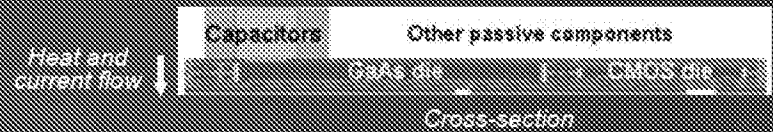
MCeP® Technology from Shinko Electric



SESUB Technology
 from TDK

HIPS Approach

- **Embed Active components**
 - Quad GaAs FET die
 - CMOS 2-phase driver
 - **Integrate performance-critical components**
 - Driver & FET
 - Cbyp, Boot caps, gate caps, Schottky's
- 
- The diagram shows a top-down view of a power MOSFET driver IC. It features a central square die labeled 'Quad GaAs FET die' and a surrounding rectangular region labeled 'CMOS 2-phase driver'. A red arrow points from the top right towards the center, labeled 'Heat and current flow'. The IC is mounted on a substrate with various pads and connections.

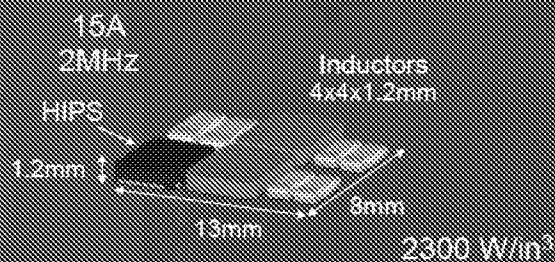
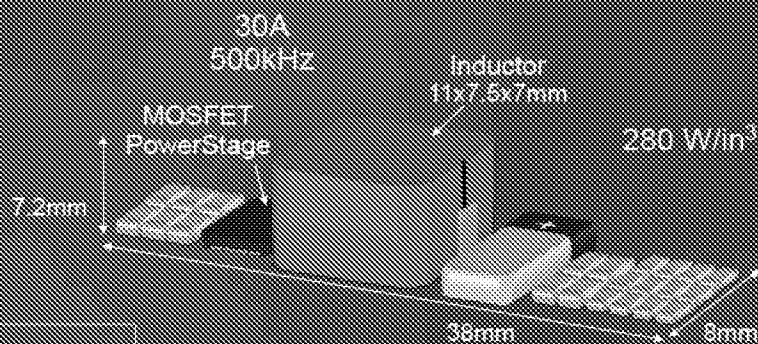
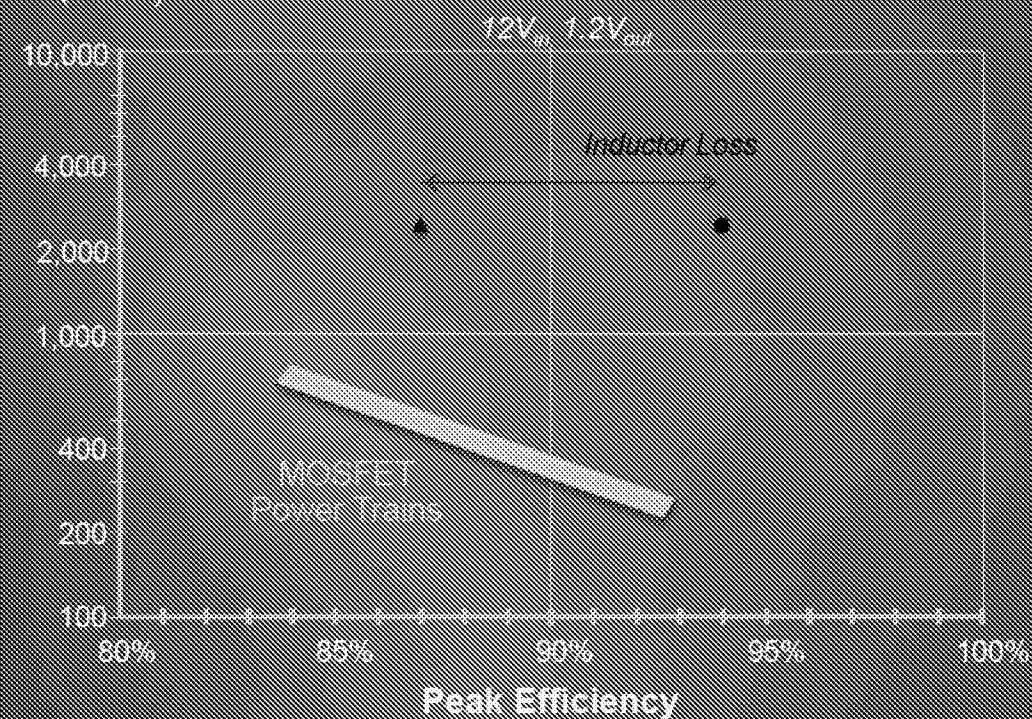


Bottom View
(preliminary)

5x4x1mm QFN package
15A capability
1-5MHz

HIPS Performance

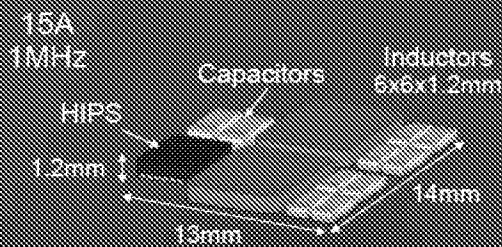
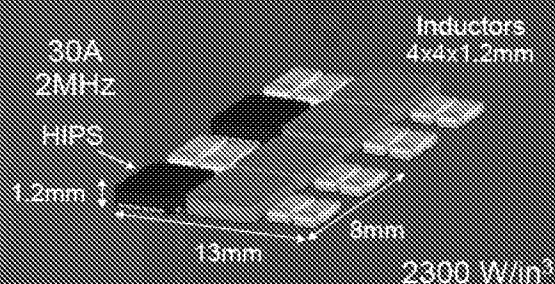
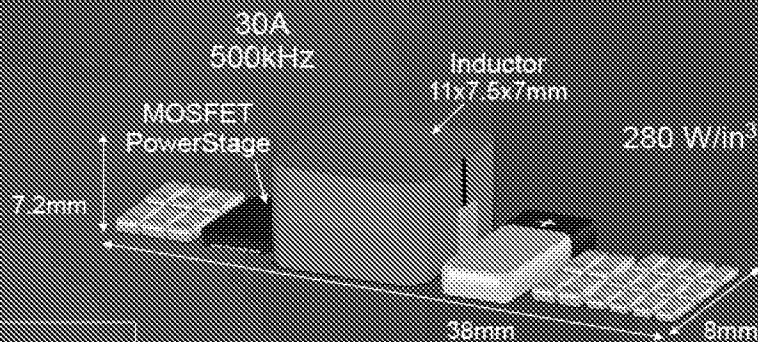
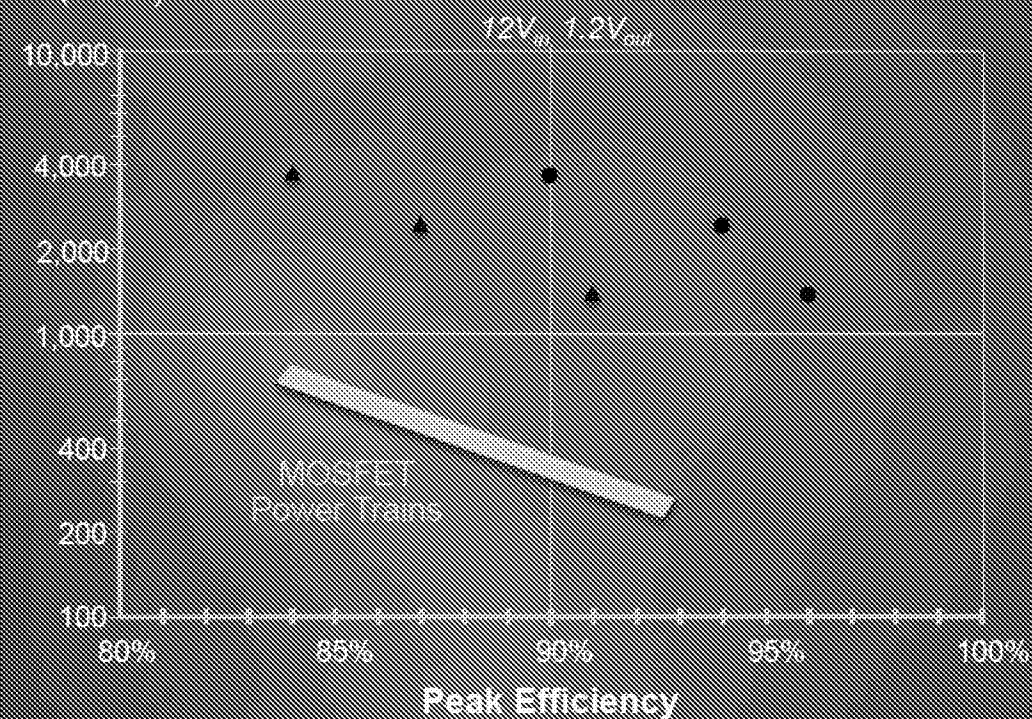
Power Train
Density
(W/in³)



- Power Stage efficiency
- Power Train efficiency

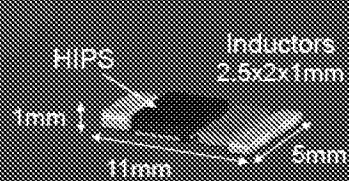
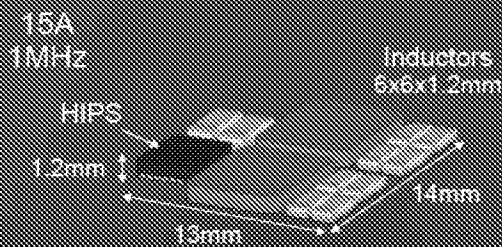
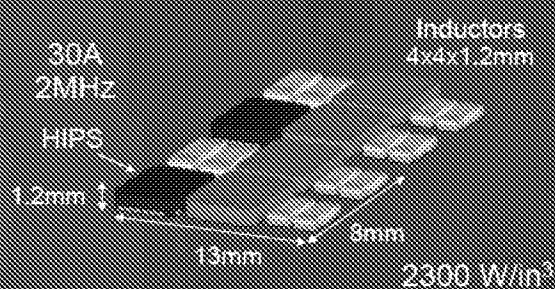
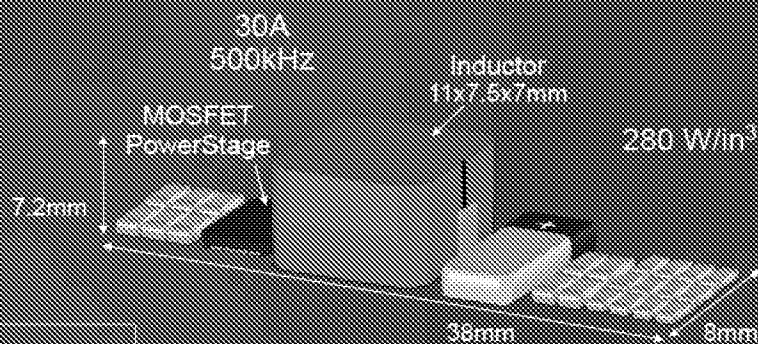
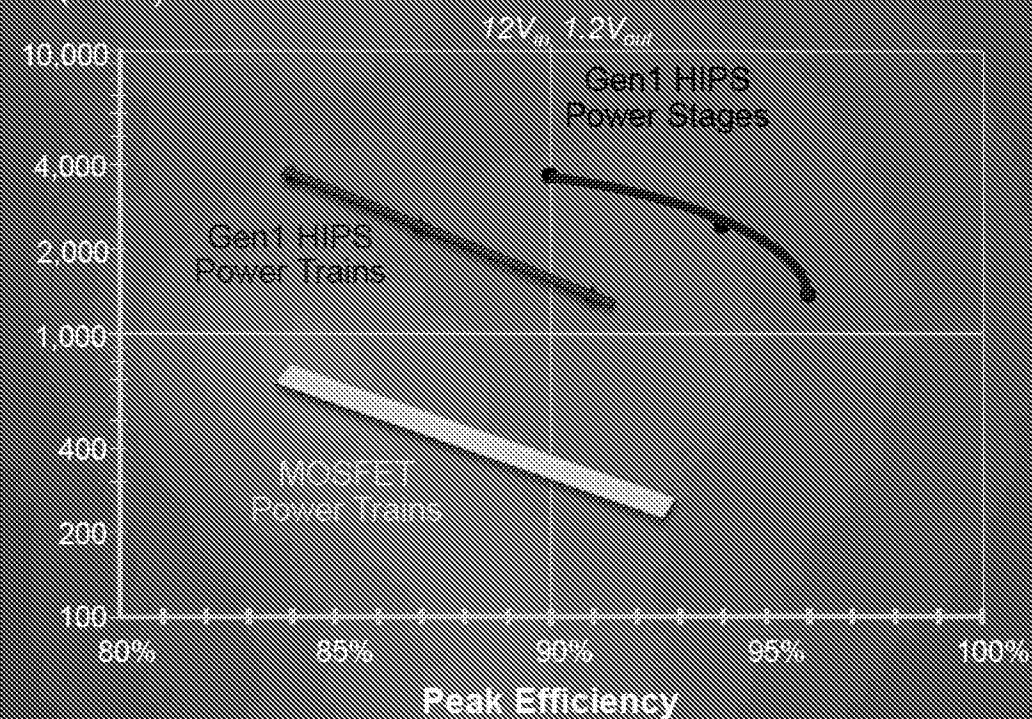
HIPS Performance

Power Train
Density
(W/in³)

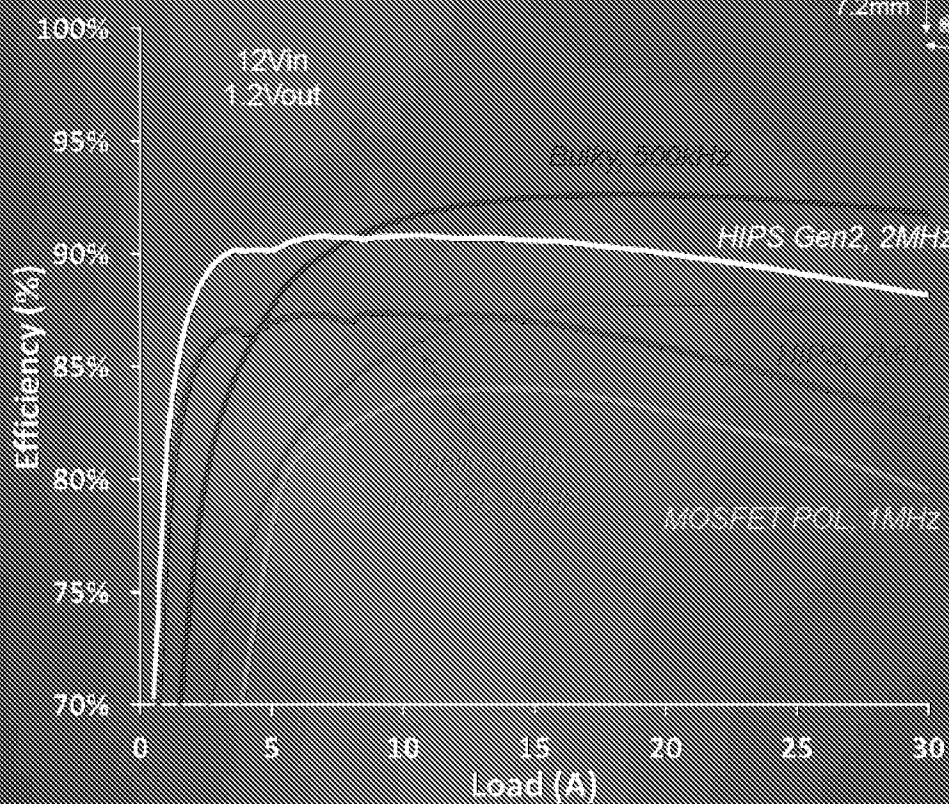


HIPS Performance

Power Train
Density
(W/in³)

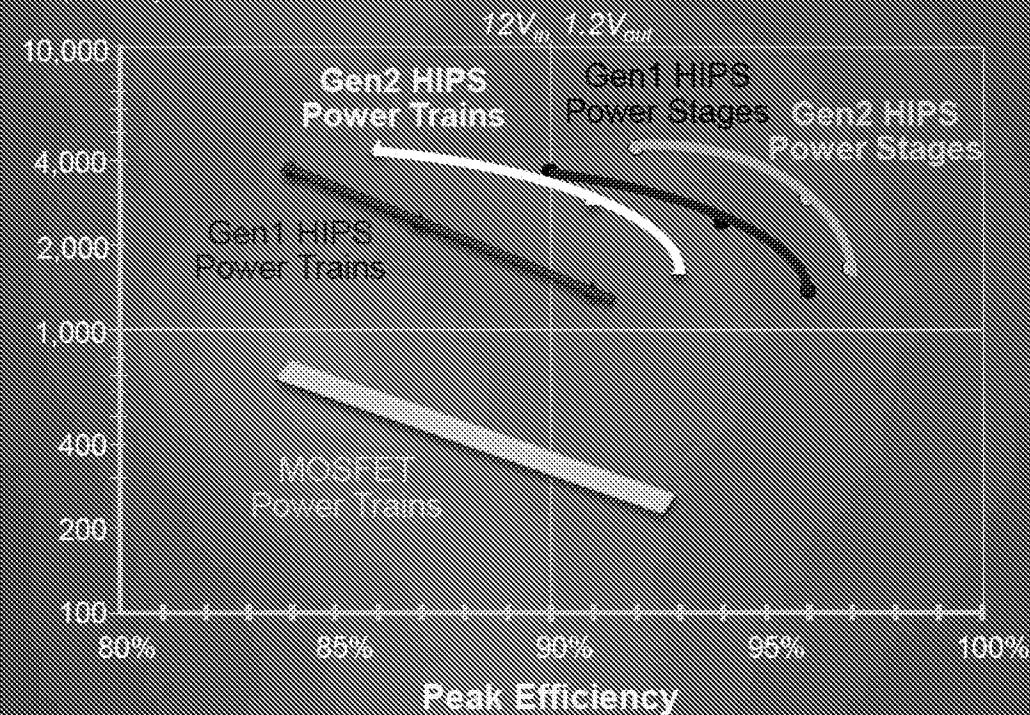


Efficiency Examination

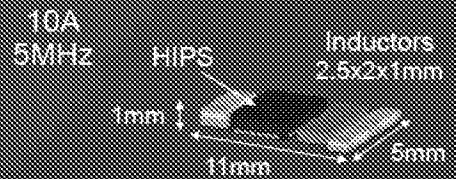
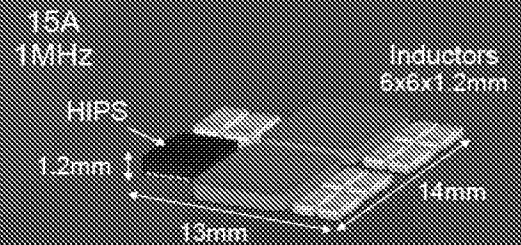
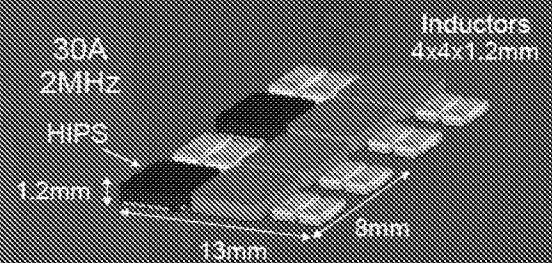
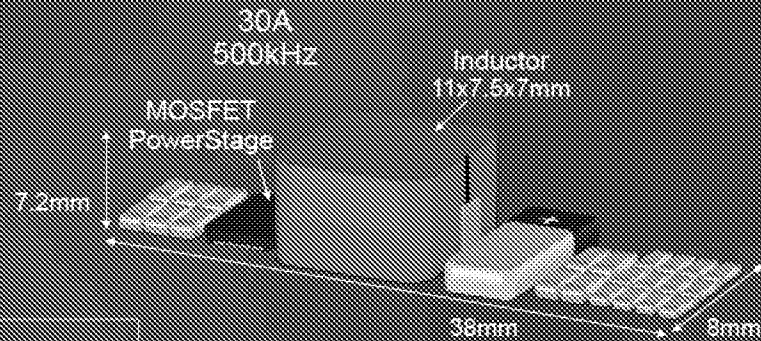


Gen2 HIPS Performance

Power Train
Density
(W/in³)



Examines Gen2 performance in 1-5MHz range
Can push to higher F_{sw} also



Summary

- **HIPS Module**

- Integrate performance-critical components
- Dramatically increase F_{sw} to shrink the output filter

- **Embedded Die in Substrate (3D Packaging) can enable new levels of power density in 12Vin POLs**

- **Compound semiconductors required to fully realize benefits**

- GaAs is ideal for 12V input

- **Enable higher switching frequencies**

- Up to 5MHz is only a starting point → HIPS roadmap to 10's of MHz



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 02/20/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT

PAPER NUMBER

2838

DATE MAILED: 02/20/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/044,063

03/09/2011

Patrizio Vinciarelli

00614-0166001

1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	05/20/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 02/20/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	05/20/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

26214 7590 02/20/2015 FISH & RICHARDSON P.C. P.O. BOX 1022 MINNEAPOLIS, MN 55440-1022	EXAMINER LEE, JYE-JUNE
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ART UNIT 2838	PAPER NUMBER
------------------	--------------

DATE MAILED: 02/20/2015

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 10/23/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-6,11,16-23,28,44-47,52,57,58,89 and 90. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the amendment filed on 10/23/2014.

EXAMINER'S AMENDMENT

3. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given by Applicants representative Rex Huang on 02/10/2015.

The application has been amended as follows:

Regarding the **Abstract**, the abstract is amended to read as follows.

Low-voltage outputs are provided by full-bridge rectification using controlled switches with fault detection monitoring of circuit conditions and disabling switches upon detection of a fault to decouple the converter from the system. Common-source dual MOSFET devices include elements arranged in alternating patterns on the die. Common-source dual synchronous rectifiers include control circuitry powered from the voltage across the complementary switch. A DC-to-DC transformer converts power using a fixed voltage transformation ratio. A clamp phase may be used to reduce power

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losses, control the output resistance, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage.

A new point of load converter includes input driver circuitry removed from and output circuitry located at the point of load, with a transformer located near the output circuit and an AC bus between the driver circuit and the primary winding of the transformer.

Regarding the Claims, **claims 7-10, 12-15, 24-27, 29-43, 48-51, 53-56, 59-88** are cancelled.

Claims 1, 18, 42, 44, and 89 are amended so the claims would read as follows.

1. A method of converting power via a transformer for supplying a low voltage output comprising:

providing a secondary winding having a first terminal and a second terminal;

providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction;

connecting the first series circuit across an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first

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series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

18. A method of providing fault tolerance in a power converter comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present,

disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the

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second switch connected to a second terminal of the output of the full-bridge rectifier circuit; and

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

42. Cancelled.

44. A fault-tolerant converter comprising:

a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;

a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and

a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.

89. A method of delivering power to an output, comprising:

providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;

providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;

connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open to block current flow through the

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first series circuit in a direction that would short the output, wherein a short across one of the first or second switch does not short the output;

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switch does not short the output;

converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

Reasons For Allowance

4. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, and 90** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second

switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...”.

The aforementioned limitations in combination with all remaining limitations of the

Art Unit: 2838

respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

5. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

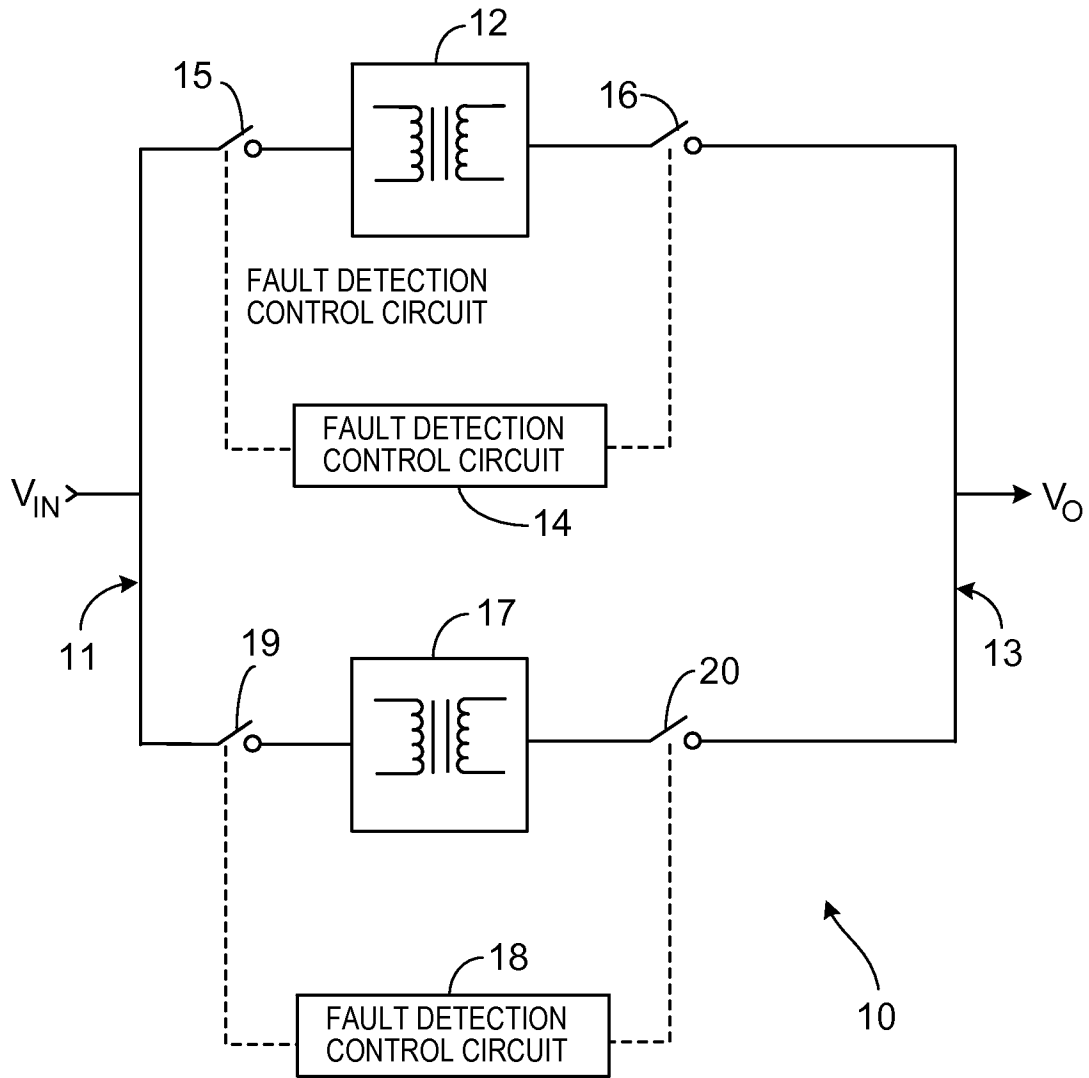
Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

OK TO ENTER: /J.L./

02/08/2015



(PRIOR ART)

FIG. 1



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 06/05/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT

PAPER NUMBER

2838

DATE MAILED: 06/05/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/044,063

03/09/2011

Patrizio Vinciarelli

00614-0166001

1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	09/08/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 06/05/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	09/08/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

26214	7590	06/05/2015
FISH & RICHARDSON P.C. P.O. BOX 1022 MINNEAPOLIS, MN 55440-1022		

EXAMINER	
LEE, JYE-JUNE	

ART UNIT	PAPER NUMBER
2838	

DATE MAILED: 06/05/2015

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the RCE filed on 05/18/2015.

Continued Examination Under 37 CFR 1.114

3. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after allowance or after an Office action under *Ex Parte Quayle*, 25 USPQ 74, 453 O.G. 213 (Comm'r Pat. 1935). Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, prosecution in this application has been reopened pursuant to 37 CFR 1.114. Applicant's submission filed on 05/18/2015 has been entered.

Reasons For Allowance

4. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, 90-114** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...".

The aforementioned limitations in combination with all remaining limitations of the respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

Art Unit: 2838

5. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

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/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

Notice of Allowability	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the RCE filed on 05/18/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-6, 11, 16-23, 28, 44-47, 52, 57, 58 and 89-114. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>05/18/2015</u> 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/ADOLF BERHANE/ Primary Examiner, Art Unit 2838	
--	--

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	13/044,063	Examiner :	Jye-June Lee
Filed :	March 9, 2011	Conf. No. :	1081
Title :	FAULT TOLERANT POWER CONVERTER		

MAIL STOP RCE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the references listed on the enclosed PTO-SB-08 or Disclosure Form.

Cited U.S. patents will be provided on request.

This filing is being made with the filing of a Request for Continued Examination. No fee is required.

Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: May 18, 2015_____

/Rex I. Huang/_____
Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0166001	Application No. 13/044,063
	Applicant Patrizio Vinciarelli	
	Filing Date March 9, 2011	Group Art Unit 2838

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
/JL/	1	6,031,743	02/29/2000	Carpenter et al.			
/JL/	2	6,035,261	03/07/2000	Carpenter et al.			
/JL/	3	6,275,958	08/14/2001	Carpenter et al.			
/JL/	4	6,466,458	10/15/2002	Zhang et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	5							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	6	

Examiner Signature /JYE-JUNE LEE/	Date Considered 05/28/2015
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 01/17/2017
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT

PAPER NUMBER

2838

DATE MAILED: 01/17/2017

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/682,187

04/09/2015

Patrizio Vinciarelli

00614-0166004

3198

TITLE OF INVENTION: A Fault Tolerant Power Converter having Converter Operating Cycles

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	04/17/2017

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 01/17/2017
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/682,187	04/09/2015	Patrizio Vinciarelli	00614-0166004	3198

TITLE OF INVENTION: A Fault Tolerant Power Converter having Converter Operating Cycles

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	04/17/2017

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (**Please first reapply any previously paid issue fee shown above**)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/682,187	04/09/2015	Patrizio Vinciarelli	00614-0166004	3198

26214 7590 01/17/2017
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

DATE MAILED: 01/17/2017

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/682,187	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 11/10/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 61-98. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

Examiner's Amendment

1. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given via telephone with Applicants representative Rex Huang on 9/26/2016.

2. The application has been amended as follows:

Regarding **claims 72 and 81**, the claims are amended so the claims would read as follows.

72. A method of converting power received from an input source at an input voltage via a power transformer for supplying a load at a DC output voltage, comprising

providing a driver circuit having an input for receiving power from the source and a driver output,

providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC output voltage to the load;

providing the power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit;

forming a resonant circuit including a winding of the power transformer, the resonant circuit being characterized by a characteristic resonant period;

providing a power bus between the driver circuit and the power transformer;
operating the driver circuit to drive the power transformer in a series of converter
operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration each having a
duration less than the characteristic resonant period, during which one or more switches
in the driver circuit are ON and power is transferred from the input to the output via the
power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped,
the clamp phase being characterized by (i) essentially zero voltage across the clamped
winding and (ii) an average value of current flowing in the clamped winding, the average
value of current having a polarity, wherein the average is taken over the duration of the
clamp phase, and the polarity is the same as a polarity of an average value of current
flowing in the clamped winding during the power transfer phase immediately preceding
the clamp phase, and (iii) essentially zero current is conducted between the secondary
winding and the output; and

deploying the driver circuit as a self-contained unit at a first location separated
from the output circuit.

81. An apparatus comprising:

a first package comprising a driver circuit having an input for receiving power
from an input source and a driver output;

an output circuit having an AC input and an output for supplying a DC output

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voltage to a load, the output circuit being disposed outside of the first package;

a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit;

a resonant circuit formed in part by one of the windings of the power transformer, the resonant circuit having a characteristic resonant period; and

a power bus for electrical connection to the driver circuit and the power transformer;

wherein the driver circuit is configured to drive the power transformer in a series of converter operating cycles, each converter operating cycle comprising

(a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

Reasons For Allowance

3. **Claims 61-98** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 61**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...operating the primary switches in a series of converter operating cycles, each converter operating cycle characterized by (a) two power transfer phases of essentially equal duration during which (i) one or more of the primary switches are ON, (ii) a primary winding of the transformer is driven with power received from the input, (iii) a magnetizing current flows through the primary winding and increases in magnitude, and (iv) power is transferred between the input and the output via the transformer; and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output."

Regarding **claim 72**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC

Art Unit: 2838

output voltage to the load; providing the power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit; forming a resonant circuit including a winding of the power transformer, the resonant circuit being characterized by a characteristic resonant period; providing a power bus between the driver circuit and the power transformer; operating the driver circuit to drive the power transformer in a series of converter operating cycles, each converter operating cycle characterized by (a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the power transformer and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output; and deploying the driver circuit as a self-contained unit at a first location separated from the output circuit.”.

Regarding **claim 81**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...a first package comprising a driver circuit having an input for receiving power from an input source and a driver output; an output

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circuit having an AC input and an output for supplying a DC output voltage to a load, the output circuit being disposed outside of the first package; a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit; a resonant circuit formed in part by one of the windings of the power transformer, the resonant circuit having a characteristic resonant period; and a power bus for electrical connection to the driver circuit and the power transformer; wherein the driver circuit is configured to drive the power transformer in a series of converter operating cycles, each converter operating cycle comprising (a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the power transformer and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.”

The aforementioned limitations in combination with all remaining limitations of the

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respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

4. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office		Attorney Docket No. 00614-0166004	Application No. 14/682,187
	Applicant VLT, Inc.			
	Filing Date April 9, 2015		Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	6,031,743	02/29/2000	Carpenter et al.			
	2	6,035,261	03/07/2000	Carpenter et al.			
	3	6,275,958	08/14/2001	Carpenter et al.			
	4	6,466,458	10/15/2002	Zhang et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
	5						Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	6	

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0166004	Application No. 14/682,187
	Applicant VLT, Inc.	
	Filing Date April 9, 2015	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.						

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	2.							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	3.	Office Action; U.S. Appln. No. 13/044,063; Examiner Nguyen; May 23, 2014
	4.	Reply to Action of May 23, 2014; U.S. 13/044,063; Examiner Nguyen; Oct. 24, 2014
	5.	Notice of Allowance; U.S. Appln. No. 13/044,063; Examiner Nguyen; Feb. 20, 2015
	6.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312; U.S. Appln. No. 13/044,063; Examiner Nguyen; Mar. 3, 2015
	7.	Supplemental Notice of Allowability; U.S. Appln. No. 13/044,063; Examiner Nguyen; Apr. 20, 2015
	8.	Notice of Allowance; U.S. Appln. No. 13/044,063; Examiner Nguyen; Jun. 5, 2015

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0166004	Application No.
	Applicant VLT, Inc.		
	Filing Date	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	6,788,033	09/07/2004	Vinciarelli			
	2	6,911,848	06/28/2005	Vinciarelli			
	3	6,930,893	08/16/2005	Vinciarelli			
	4	6,934,166	08/23/2005	Vinciarelli			
	5	6,969,909	11/29/2005	Briere			
	6	6,975,098	12/13/2005	Vinciarelli			
	7	6,984,965	01/10/2006	Vinciarelli			
	8	6,985,341	01/10/2006	Vinciarelli, et al.			
	9	7,145,786	12/05/2005	Vinciarelli			
	10	RE40,072	02/19/2008	Prager, et al.			
	11	7,561,446	07/14/2009	Vinciarelli			
	12	7,768,807	08/2010	Chen, et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	13							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	14	AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009)
	15	ISL6210 (Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008)

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
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Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0166004	Application No. 14/682,187
		Applicant VLT, Inc.	
		Filing Date April 9, 2015	Group Art Unit 2838

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	2013/0069608	03-2013	Gakhar et al.			
	2.	6,300,749	10-2001	Castelli et al.			
	3.	6,603,292	08-2003	Schouten et al.			
	4.	6,700,361	03-2004	Gregorius, Peter			
	5.	7,521,909	04-2009	Dow et al.			
	6.	8,232,783	07-2012	Yanagawa, Kenji			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

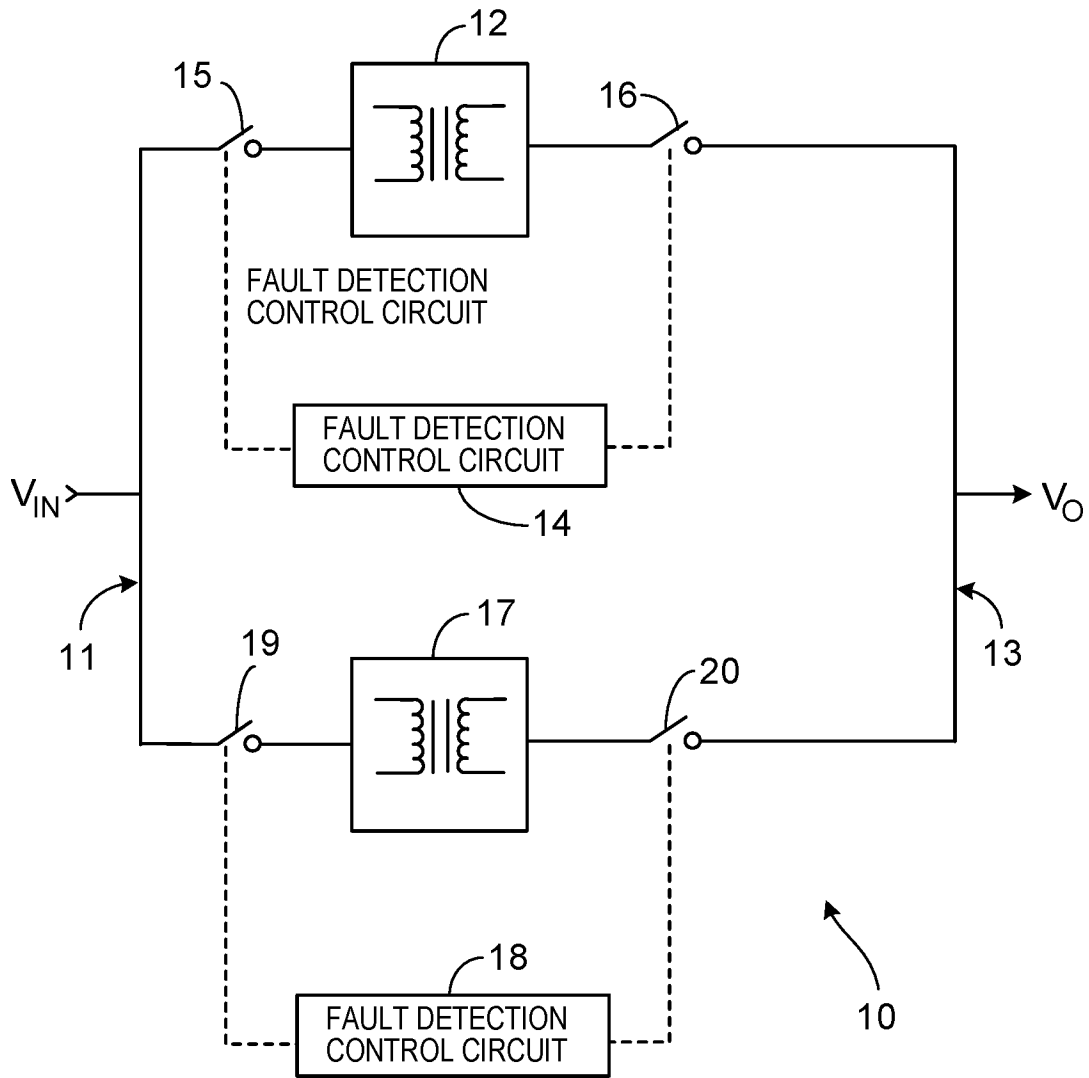
Examiner Signature /JYE-JUNE LEE/	Date Considered 01/09/2017
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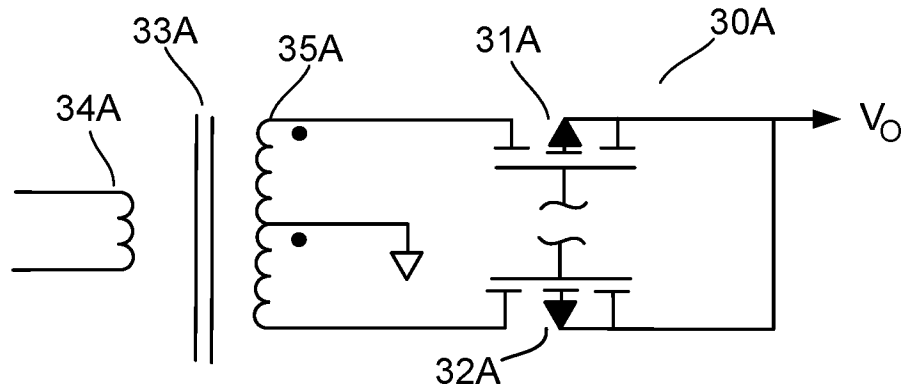
OK TO ENTER: /J.L./

12/27/2016

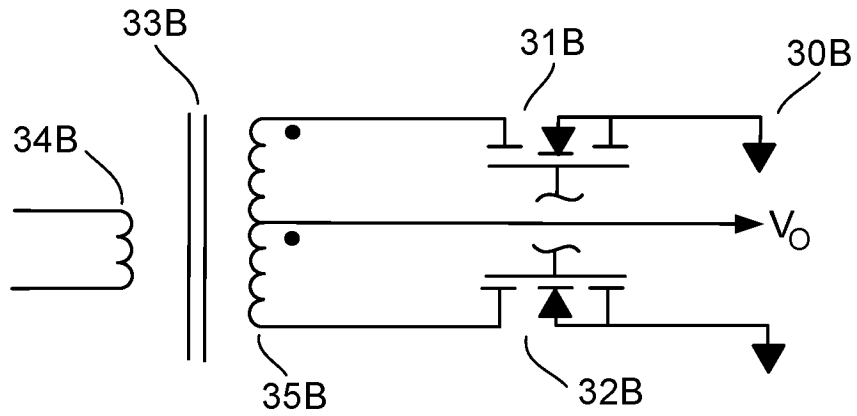


(PRIOR ART)

FIG. 1



(PRIOR ART)

FIG. 2A

(PRIOR ART)

FIG. 2B

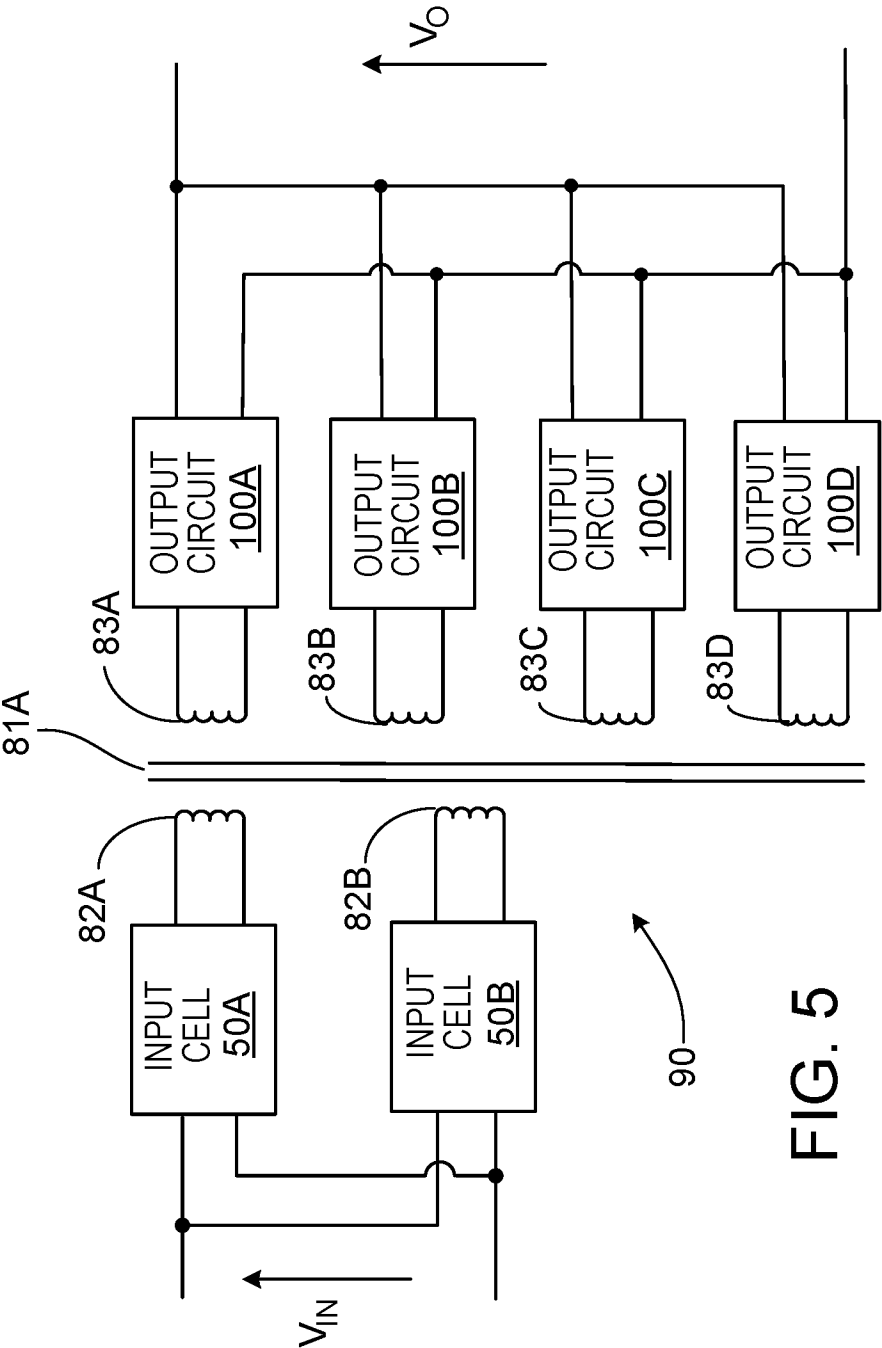


FIG. 5

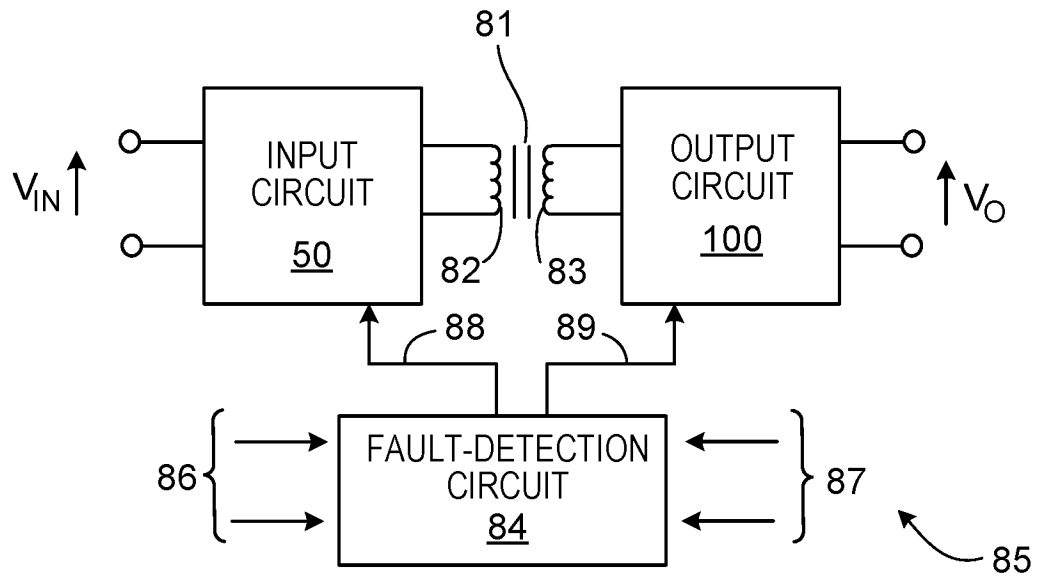
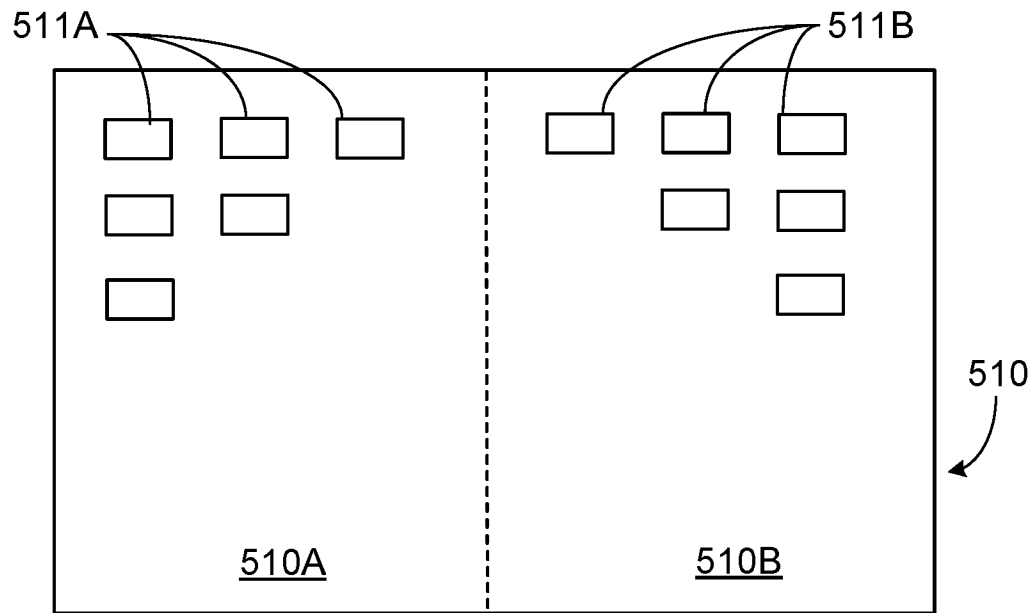


FIG. 6



(PRIOR ART)
FIG. 9

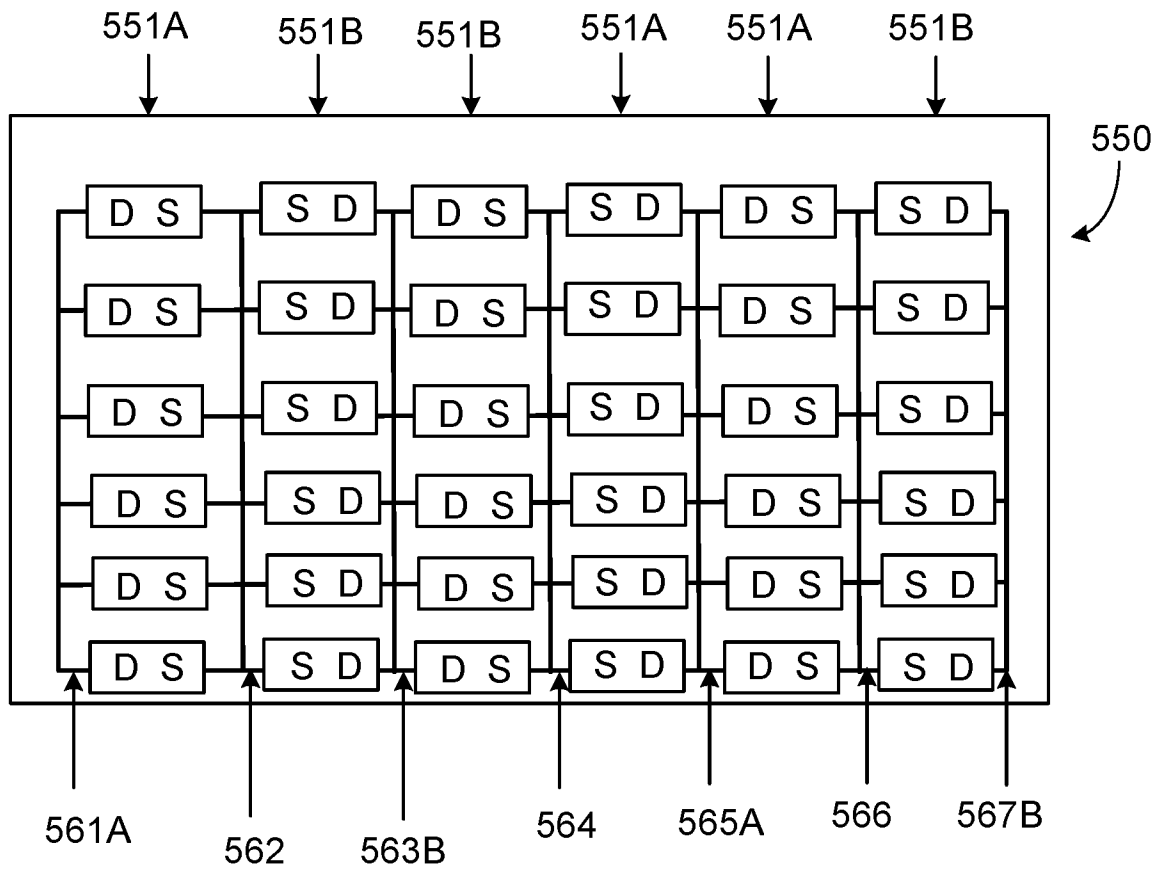


FIG. 10

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	14/682,187	Examiner :	Jye-June Lee
Filed :	April 9, 2015		
Title :	FAULT TOLERANT POWER CONVERTER		

ALLOWED CLAIMS

61. A method of converting power received from an input source at an input voltage via a transformer for delivery via an output at an output voltage to a load, where the load may vary over a normal operating range, comprising:

using an input circuit having a plurality of primary switches to drive a primary winding of the transformer with power received from the input source;

using output circuitry having a plurality of secondary switches for rectifying power received from a secondary winding of the transformer to conduct current from the secondary winding to the output;

operating the primary switches in a series of converter operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration during which (i) one or more of the primary switches are ON, (ii) a primary winding of the transformer is driven with power received from the input, (iii) a magnetizing current flows through the primary winding and increases in magnitude, and (iv) power is transferred between the input and the output via the transformer; and

(b) a clamp phase during which a winding of the transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

62. The method of claim 61 further comprising:

forming a resonant circuit including the transformer and having a characteristic resonant frequency and period; wherein a resonant current flows in the primary winding during the power transfer phases; and

turning one or more primary switches OFF essentially at times when the resonant current returns to zero.

63. The method of claim 61 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

64. The method of claim 61 further comprising using an inductance to limit the rate of change of current in the transformer; and wherein

the primary and secondary switches are operated in the series of converter operating cycles and each converter operating cycle further comprises:

a freewheel phase during which a circuit including a winding of the transformer is clamped, the freewheel phase being characterized by (i) essentially zero voltage across the clamped circuit and (ii) an average value of current flowing in the clamped circuit, the average value of current having a polarity, wherein the average is taken over the duration of the initial clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer interval immediately preceding the freewheel phase, (iii) a current flowing in the inductance declines in magnitude to essentially zero current, and (iv) the secondary switches conduct current between the secondary winding and the output.

65. The method of claim 64 wherein each clamp phase occurs after a respective freewheel phase; and the freewheel phase ends, and the secondary switches are turned OFF, essentially when the current between the secondary winding and the output returns to zero.

66. The method of claim 65 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

67. The method of claim 61 further comprising:
varying the duration of the clamp phase to control an effective output resistance of the power conversion.

68. The method of claim 61 further comprising:
varying the duration of the clamp phase to regulate conversion factor defined as the ratio of input voltage to output voltage.

69. The method of claim 61 further comprising:
varying the duration of the clamp phase to reduce power losses at light loads.

70. The method of claim 61 further comprising:
varying the duration of the clamp phase to control the rate of change of the output voltage.

71. The method of claim 61 further comprising:
varying the duration of the clamp phase to regulate the output voltage.

72. A method of converting power received from an input source at an input voltage via a power transformer for supplying a load at a DC output voltage, comprising

providing a driver circuit having an input for receiving power from the source and a driver output,

providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC output voltage to the load;

providing the power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit;

forming a resonant circuit including a winding of the power transformer, the resonant circuit being characterized by a characteristic resonant period;

providing a power bus between the driver circuit and the power transformer; operating the driver circuit to drive the power transformer in a series of converter operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output; and

deploying the driver circuit as a self-contained unit at a first location separated from the output circuit.

73. The method of claim 72, comprising deploying the output circuit as a self-contained unit that is separate from the self-contained unit of the driver circuit.

74. The method of claim 72 wherein providing an output circuit comprises providing an output circuit that comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

75. The method of claim 74 wherein providing an output circuit comprises providing an output circuit that comprises the first series circuit and a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

76. The method of claim 74, comprising deploying the first and second switches of the output circuit on a die having the load, wherein the driver circuit is located outside of the die.

77. The method of claim 76 wherein the load comprises a processor core.

78. The method of claim 76, comprising providing control circuitry to operate the first and second switches of the output circuit, and deploying the control circuitry on the die having the load.

79. The method of claim 72 wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and

the method comprises turning one or more of the primary switches OFF essentially at times when the resonant current returns to zero.

80. The method of claim 79 wherein a magnetizing current flows through the primary winding during the power transfer phases, and each converter operating cycle comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF, and

the method comprises using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

81. An apparatus comprising:

a first package comprising a driver circuit having an input for receiving power from an input source and a driver output;

an output circuit having an AC input and an output for supplying a DC output voltage to a load, the output circuit being disposed outside of the first package;

a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit;

a resonant circuit formed in part by one of the windings of the power transformer, the resonant circuit having a characteristic resonant period; and

a power bus for electrical connection to the driver circuit and the power transformer;

wherein the driver circuit is configured to drive the power transformer in a series of converter operating cycles, each converter operating cycle comprising

(a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

82. The apparatus of claim 81 wherein the output circuit and the power transformer are provided in a second package that is separate from the first package.

83. The apparatus of claim 81 wherein the output circuit comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output

of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

84. The apparatus of claim 83 wherein the output circuit comprises a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

85. The apparatus of claim 83 wherein the first and second switches of the output circuit are provided on a die having the load.

86. The apparatus of claim 85 wherein the load comprises a processor core.

87. The apparatus of claim 85, comprising control circuitry to operate the first and second switches of the output circuit, wherein the control circuitry is provided on the die having the load.

88. The apparatus of claim 81, wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and wherein the apparatus comprises a control circuit to operate the primary switches, and the control circuit is configured to turn OFF one or more of the primary switches essentially at times when the resonant current returns to zero.

89. The method of claim 62 further comprising:
varying the duration of the clamp phase to control an effective output resistance of the power conversion.

90. The method of claim 62 further comprising:

varying the duration of the clamp phase to regulate conversion factor defined as the ratio of input voltage to output voltage.

91. The method of claim 62 further comprising:

varying the duration of the clamp phase to reduce power losses at light loads.

92. The method of claim 62 further comprising:

varying the duration of the clamp phase to control the rate of change of the output voltage.

93. The method of claim 62 further comprising:

varying the duration of the clamp phase to regulate the output voltage.

94. The method of claim 65 further comprising:

varying the duration of the clamp phase to control an effective output resistance of the power conversion.

95. The method of claim 65 further comprising:

varying the duration of the clamp phase to regulate conversion factor defined as the ratio of input voltage to output voltage.

96. The method of claim 65 further comprising:

varying the duration of the clamp phase to reduce power losses at light loads.

97. The method of claim 65 further comprising:

varying the duration of the clamp phase to control the rate of change of the output voltage.

98. The method of claim 65 further comprising:

varying the duration of the clamp phase to regulate the output voltage.

First Named Inventor : Patrizio Vinciarelli
Serial No. : 14/682,187
Filed : April 9, 2015
Page : 9 of 9

Attorney's Docket No.: 00614-0166004 / VA0166-4
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P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 08/13/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/091,346

04/05/2016

Patrizio Vinciarelli

00614-0178001

8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	11/13/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

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I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

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If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

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(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	11/13/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 _____

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

26214 7590 08/13/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT	PAPER NUMBER
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2842

DATE MAILED: 08/13/2018

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability

Application No.

15/091,346

Applicant(s)

VINCIARELLI ET AL.

Examiner

CASSANDRA COX

Art Unit

2842

**AtA (First Inventor to File)
Status**

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 05/30/18.
☐ A declaration(s)/affidavit(s) under 37 CFR 1.130(b) was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-56, 121-127 and 131. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____
3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material
4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____
5. ☐ Examiner's Amendment/Comment
6. ☒ Examiner's Statement of Reasons for Allowance
7. ☐ Other _____

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 1-56, 121-127- and 131 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 1-29, 121-127- and 131 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source, comprising :providing one or more output circuits for deployment in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the

Art Unit: 2842

rest of the limitations of the base claims and any intervening claims. Claims 30-56 are allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the rest of the limitations of the base claims and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 05/21/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/091,346

04/05/2016

Patrizio Vinciarelli

00614-0178001

8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/21/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/21/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-110000

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---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

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☐ Applicant asserting small entity status. See 37 CFR 1.27
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NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

26214	7590	05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)		
P.O. BOX 1022		
MINNEAPOLIS, MN 55440-1022		

EXAMINER
COX, CASSANDRA F

ART UNIT	PAPER NUMBER
2842	

DATE MAILED: 05/21/2018

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<i>Notice Requiring Inventor's Oath or Declaration</i>	Application No. 15/091,346	Applicant(s) Patrizio Vinciarelli	
	Examiner COX, CASSANDRA F	Art Unit 2842	

This notice is an attachment to the Notice of Allowability (PTOL-37), or the Notice of Allowability For A Design Application (PTOL-37D).

An inventor's oath or declaration in compliance with 37 CFR 1.63 or 1.64 executed by or with respect to each inventor has not yet been submitted.

An oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each inventor (for any inventor for which a compliant oath, declaration, or substitute statement has not yet been submitted) **MUST** be filed no later than the date on which the issue fee is paid. See 35 U.S.C. 115(f). Failure to timely comply will result in ABANDONMENT of this application.

A properly executed inventor's oath to declaration has not been received for the following inventor(s):

If applicant previously filed one or more oaths, declarations, or substitute statements, applicant may have received an informational notice regarding deficiencies therein.

The following deficiencies are noted:

INFORMAL ACTION PROBLEMS

- A properly executed inventor's oath or declaration has not been received for the following inventor(s): **Patrizio Vinciarelli**.

Applicant may submit the inventor's oath or declaration at any time before the Notice of Allowance and Fee(s) Due, PTOL-85, is mailed.

Questions relating to this Notice should be directed to the Application Assistance Unit at 571-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/091,346	Applicant(s) VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 02/08/18.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-56,121-127 and 131. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

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1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 1-56, 121-127- and 131 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 1-29, 121-127- and 131 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source, comprising :providing one or more output circuits for deployment in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the

Art Unit: 2842

rest of the limitations of the base claims and any intervening claims. Claims 30-56 are allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the rest of the limitations of the base claims and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 12/11/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 12/11/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/616,288

06/07/2017

Patrizio Vinciarelli

00614-0180001

7438

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	03/11/2019

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

**By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450**

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 12/11/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date indicated below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	03/11/2019

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438
26214	7590	12/11/2018	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2842	
DATE MAILED: 12/11/2018				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/616,288	Applicant(s) VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

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1. ☒ This communication is responsive to 09/25/18.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 132-164 and 193. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

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Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

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☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other ____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. | |

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 132-164 and 193 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 132-150 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising: providing one or more rail selection circuits on the one or more semiconductor chips, each having inputs connected to the output rails and an output connected to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches connected to provide a selectable path for current flow from a selected one of the rails to the output; wherein each rail selection circuit is operative to enable a respective rail switch to conduct current from selected output rail to its respective output in combination with the rest of the limitations of the base claims and any intervening claims. Claims 151-164 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising providing one or more driver circuits; providing one or more output

Art Unit: 2842

circuits; providing a power bus; and providing a first capacitor; wherein the driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts; wherein the one or more output circuits are adapted to receive power from the driver circuit at a peak-to-peak bus voltage and supply power to the one or more semiconductor chips at a load voltage, wherein the peak-to-peak bus voltage is a multiple of the load voltage and that multiple is at least six in combination with the rest of the limitations of the base claims and any intervening claims. Claim 164 is allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more rail selection circuits disposed on the one or more semiconductor chips, each having inputs electrically coupled to the output rails and an output electrically coupled to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches electrically coupled to provide a selectable path for current to flow from a selected one of the output rails to the respective output in combination with the rest of the limitations of the base claim and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

4. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2842

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842

Notice of References Cited	Application/Control No. 15/616,288		Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner CASSANDRA COX		Art Unit 2842	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-6,118,186 A	09-2000	Scott; Harold C.	H02P9/04	290/1A
*	B	US-2006/0291265 A1	12-2006	Schrom; Gerhard	G11C17/18	365/94
*	C	US-7,245,509 B1	07-2007	Chan; Chun-Kong	H02M7/538	363/17
*	D	US-2015/0188407 A1	07-2015	Golder; Barnaby	H02J1/00	307/31
*	E	US-9,765,750 B2	09-2017	Ferguson; Tod	F02P5/15	1/1
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	1.	Cao et al., "A Family of Zero Current Switching Switched-Capacitor DC-DC converters," Applied Power Electronics Conference and Exposition (APEC), 2010 Twenty-Fifth Annual IEEE, Mar. 18, 2010
	2.	Cao et al., "Zero-Current-Switching Multilevel Modular Switched-Capacitor DC-DC Converter" IEEE Transactions on Industry Applications, 46(6):2536-2544, Sep. 2010
	3.	Oraw et al., "Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter," Telecommunications Energy Conference, IEEE 2007, pp. 628-635 Sept. 30 – Oct. 4, 2007
	4.	Seeman et al., "Analysis and optimization of switched-capacitor dc-dc converters," Power Electronics, IEEE Transactions on, vol. 23, pp. 841–851, March 2008
	5.	Yeung et al., "Generalised analysis of switched-capacitor step-down quasi-resonant converter," Electronics Letters, vol. 38, pp. 263-264, 2002
	6.	Yeung et al., "Zerocurrent switching switched-capacitor quasiresonant step-down converter," Electric Power Applications, IEE Proceedings -, vol. 149, pp. 111-121, 2002

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	5,438,294	8/1/1995	Smith			
	2.	9,087,656	7/21/2015	Vinciarelli			
	3.	2013/0030534	1/31/2013	DeLurio et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	4.	Notice of Allowance in U.S. Application No. 15/091,346, dated May 21, 2018, 9 pages
	5.	Notice of Allowance in U.S. Application No. 15/091,346, dated August 13, 2018, 8 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Sheet 1 of 2

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	6,031,743	02/29/2000	Carpenter et al.			
	2.	6,035,261	03/07/2000	Carpenter et al.			
	3.	6,275,958	08/14/2001	Carpenter et al.			
	4.	6,300,749	10/01/2001	Castelli et al.			
	5.	6,421,262	07/16/2002	Saxelby et al.			
	6.	6,466,458	10/15/2002	Zhang et al.			
	7.	6,603,292	08/01/2003	Schouten et al.			
	8.	6,700,361	03/01/2004	Gregorius, Peter			
	9.	6,788,033	09/07/2004	Vinciarelli			
	10.	6,911,848	06/28/2005	Vinciarelli			
	11.	6,934,166	08/23/2005	Vinciarelli			
	12.	6,969,909	11/29/2005	Briere			
	13.	6,985,341	01/10/2006	Vinciarelli, et al.			
	14.	7,145,786	12/05/2006	Vinciarelli			
	15.	7,521,909	04/01/2009	Dow et al.			
	16.	7,561,446	07/14/2009	Vinciarelli			
	17.	7,768,807	08/01/2010	Chen, et al.			
	18.	8,232,783	07/01/2012	Yanagawa, Kenji			
	19.	RE40,072	02/19/2008	Prager, et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation
							Yes No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	20.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312 in U.S. Application No. 13/044,063, dated March 3, 2015, 17 pages

Examiner Signature	Date Considered
/CASSANDRA F COX/	12/04/2018

EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Substitute Disclosure Form

Sheet 2 of 2

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
Information Disclosure Statement by Applicant (Use several sheets if necessary)	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit

(37 CFR §1.98(b))

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	21.	AN2738, Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009
	22.	Burton et al., "FIVR - Fully Integrated Voltage Regulators on 4th Generation Intel ® Core™ SoCs," Applied Power Electronics Conference and Exposition (APEC), 2014 Twenty-Ninth Annual IEEE, March 16, 2014, 432-439
	23.	ISL6210, Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008
	24.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated February 20, 2015, 15 pages
	25.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated June 5, 2015, 11 pages
	26.	Notice of Allowance in U.S. Application No. 14/682,187, dated January 17, 2017, 22 pages
	27.	Office Action in U.S. Application No. 13/044,063, Examiner Nguyen, dated May 23, 2014, 24 pages
	28.	Reply to Action of May 23, 2014 in U.S. Application No. 13/044,063, dated October 24, 2014, 39 pages
	29.	Supplemental Notice of Allowability in U.S. Application No. 13/044,063, Examiner Nguyen, dated April 20, 2015, 5 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

 ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/
 Substitute Disclosure Form

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No.
	Applicant VL T, Inc.	
	Filing Date	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	14/731,287		Vinciarelli et al.			06/04/2015
	2.	14/822,561		Vinciarelli			08/10/2015
	3.	14/874,054		Vinciarelli			10/02/2015
	4.	6,930,893	08/16/2005	Vinciarelli			
	5.	6,975,098	12/13/2005	Vinciarelli			
	6.	6,984,965	01/10/2006	Vinciarelli			
	7.	2013/0069608	03-2013	Gakhar et al.			
	8.	9,112,422	08/18/2015	Vinciarelli			
	9.	9,166,481	10/15/2015	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	10.	JPH02280666	11/16/1990	JP			English Abstract	
	11.	EP2863531	4/22/2015	EP				

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	12.	DiBene II et al., "A 400 Amp Fully Integrated Silicon Voltage Regulator with in-die Magnetically Coupled Embedded Inductors," retrieved from http://www.psma.com/sites/default/files/uploads/tech-forums-nanotechnology/resources/400a-fully-integrated-silicon-voltage-regulator.pdf , retrieved on August 3, 2016, 25 pages.
	13.	European Office Communication in European Application No. 16173089.0, dated April 10, 2017, 2 pages
	14.	Extended Search Report in European Application No. 16173089.0, dated February 14, 2017, 8 pages
	15.	Miller, "12V PowerStage in Embedded Die System-in-Package," APEC 2015, 33 pages
	16.	Okudaira and Matsuse, "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor," Power Conversion Conference 2002, 1496-1501
	17.	Watson, "New Techniques in the Design of Distributed Power Systems," Dissertation, Virginia Polytechnic Institute and State University, 1998, 32 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Substitute Disclosure Form

Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office		Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.			
	Filing Date June 7, 2017		Group Art Unit 2842	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	8,582,333	11/12/2013	Oraw et al.			
	2.	9,660,537	05/23/2017	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Substitute Disclosure Form



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

26214 7590 05/23/2014
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
----------	--------------

2838

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

05/23/2014

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
 - If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
 - Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).
- Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 05/02/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-90 is/are pending in the application.
5a) Of the above claim(s) 7-10, 12-15, 24-27, 29-43, 48-51, 53-56 and 59-88 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1, 3, 4, 16, 18-23, 28, 44, 46, 57, 58, 89 and 90 is/are rejected.
- 8) ☒ Claim(s) 2, 5, 6, 11, 17, 45, 47, and 52 is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 05/20/2011 is/are: a) ☐ accepted or b) ☒ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>03/09/2011</u> . | 4) ☐ Other: _____ |

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the application filed on 03/09/2011, the preliminary amendment filed on 07/06/2012, and the response to restriction requirement filed on 05/02/2014.

Election/Restrictions

3. Applicant's election without traverse of claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 78, 89, and 90 in the reply filed on 05/02/2014 is acknowledged.

Drawings

4. Figures 1, 2A, 2B and 9 should be designated by a legend such as --Prior Art-- because only that which is old is illustrated. See MPEP § 608.02(g). Corrected drawings in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. The replacement sheet(s) should be labeled "Replacement Sheet" in the page header (as per 37 CFR 1.84(c)) so as not to obstruct any portion of the drawing figures. If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

5. The drawings are objected to because of the following informalities.

Regarding Fig. 1, 5, and 6, the unlabeled rectangular box(es) shown in the drawings should be provided with descriptive text labels.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

6. The lengthy specification has not been checked to the extent necessary to determine the presence of all possible minor errors. Applicant's cooperation is

Art Unit: 2838

requested in correcting any errors of which applicant may become aware in the specification.

7. The abstract of the disclosure is objected to due to its undue length.

Applicant is reminded of the proper language and format for an abstract of the disclosure

The abstract should be in narrative form and generally limited to a single paragraph on a separate sheet within the range of 50 to 150 words. The form and legal phraseology often used in patent claims, such as "means" and "said," should be avoided. The abstract should describe the disclosure sufficiently to assist readers in deciding whether there is a need for consulting the full patent text for details.

The language should be clear and concise and should not repeat information given in the title. It should avoid using phrases which can be implied, such as, "The disclosure concerns," "The disclosure defined by this invention," "The disclosure describes," etc.

Correction is required. See MPEP § 608.01(b).

Claim Objections

8. **Claims 1, 2 and 5** are objected to because of the following informalities:

regarding **claim 1**, in line 9, "the output" should read as "an output";

regarding **claim 2**, in line 5, "the second polarity is opposite the ..." should read as "the second polarity is opposite of the ...";

regarding **claim 5**, in line 5, "the second polarity is opposite the ..." should read as "the second polarity is opposite of the ...";

Appropriate correction is required.

Claim Rejections - 35 USC § 112

9. The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

10. **Claims 18, 20-23, 28, and 58** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

Regarding **claim 18**, in lines 13 and 14, “the output” renders the claims to be indefinite as it is unclear whether the term “the output” refers to “the output of the full-bridge rectifier circuit” or “the output of power converter”. For examining purposes, the office interprets “the output” as “the output of the full-bridge rectifier circuit”.

Regarding **claim 20**, in lines 7, 8, 10, and 11, “the output” renders the claims to be in definite as it is unclear whether the term “the output” refers to “the output of the full-bridge rectifier circuit” or “the output of power converter”. For examining purposes, the office interprets “the output” as “the output of the full-bridge rectifier circuit”.

Regarding **claim 58**, in lines 2 and 3, the bold underlined part in “**each of the** first, second, **third and fourth switches**” lacks antecedent basis.

Claims 19, 21-23, and 28 depend directly or indirectly from a rejected claim above and are therefore also rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-

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AIA), second paragraph for the reasons set above.

Claim Rejections - 35 USC § 103

11. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

12. **Claims 1, 3, 4, 16, 44, 46, and 57** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen et al. (US Patent 7,768,807 B2, hereafter referred to as “Chen”) in view of ISL6210 (Datasheet ISL6210, “Dual Synchronous Rectified MOSFET Drivers,” Intersil Americas Inc., December 2008).

Regarding **claim 1**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) a method of converting power via a transformer (transformer of Fig. 2) for supplying a low voltage output (V_{hi}) comprising: providing a secondary winding (right hand side winding of transformer) having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); providing a first switch (S1) and a second switch (S3) each adapted to block current flow in an OFF state in at least one direction; forming a first series circuit with the first switch connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); connecting the first terminal of the secondary winding to the first junction; connecting the first series circuit across *an* output (top and bottom nodes of Co) with

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each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Chen does not disclose preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a first switch from turning ON if a second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the first switch from turning ON if the second switch fails to open and preventing the second switch from turning ON if the first switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

Regarding **claim 3**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) further comprising: providing a third switch (S2) and a fourth switch (S4) each adapted to block current in an OFF state in at least one direction; forming a second series circuit

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with the third switch connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Regarding **claim 4**, as best understood, Chen does not disclose preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a third switch from turning ON if a fourth switch fails to open; and preventing the fourth switch from turning ON if the third switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two

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switches.

Regarding **claim 16**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) forming a second series circuit with a first capacitor (C1) connected to a second capacitor (C2) at a second junction (common node of C1 and C2), the second series circuit having two end terminals (top node of C1 and bottom node of C2); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection).

Regarding **claim 44**, Chen discloses (see Fig. 2 and Fig. 3) a fault-tolerant converter comprising: a transformer (transformer of Fig. 2) having a primary winding (left hand side winding of transformer) and a secondary winding (right hand side winding of transformer), the secondary winding having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); a first series circuit having a first switch (S1) and a second switch (S3) each adapted to block current flow in an OFF state in at least one direction; the first switch being connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); the first junction being connected to the first terminal of the secondary winding; the first series circuit being connected across an output (top and bottom nodes of Co) of the converter with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series

circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Chen does not disclose a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open; and to prevent the second switch from turning ON if the first switch fails to open.

However, ISL6210 discloses (see p. 6) a switch control circuit (ISL6210) configured to prevent a first switch from turning ON if a second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the converter of Chen to include the switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open; and to prevent the second switch from turning ON if the first switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

Regarding **claim 46**, Chen discloses (see Fig. 2 and Fig. 3) comprising a second series circuit that comprises a third switch (S2) and a fourth switch (S4) each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4), the

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second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Regarding **claim 57**, Chen discloses (see Fig. 2 and Fig. 3) comprising a second series circuit having a first capacitor (C1) connected to a second capacitor (C2) at a second junction (common node of C1 and C2) that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals (top node of C1 and bottom node of C2) connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter (see connections).

13. **Claims 18, and 20** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009).

Regarding **claim 18**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) a method of providing fault tolerance in a power converter (converter of Fig. 2)

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comprising: supplying an alternating current (AC in transformer) from a secondary winding (right hand side winding of transformer) to an input of a full-bridge rectifier circuit (circuit comprising S1, S2, S3, S4, C1, C2) having switches (S1, S2, S3, S4), operating the switches as rectifiers to provide full-wave rectification of the alternating current (see Fig. 3) at an output of the full-bridge rectifier circuit (top node of S2 and bottom node of S4), connecting the output of the full-bridge rectifier circuit to an output of the power converter (top and bottom nodes of Co), and providing a first one of the switches connected in series with a second one of the switches across the output with the first switch connected to a first terminal of the output and the second switch connected to a second terminal of the output.

Chen does not disclose monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present.

However, AN2738 discloses (see p. 7) monitoring selected circuit conditions (VCC); determining from the selected circuit conditions whether a predetermined fault is present (when VCC goes below a threshold), disabling the switches in response to a determination that the predetermined fault is present (turns off IC outputs).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present as disclosed in AN2738, because

it can help prevent faults in the circuit which can help protect the circuit elements from excessive electrical stress.

Regarding **claim 20**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) further comprising: providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction; forming a second series circuit with the third switch connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

14. **Claim 21** is rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738, and further in view of ISL6210.

Regarding **claim 21**, as best understood, Chen does not disclose preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a third switch from turning ON if a fourth switch fails to open; and preventing the fourth switch from turning ON if the third switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

15. **Claims 89 and 90** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen.

Regarding **claim 89**, Chen discloses (see Fig. 2 and Fig. 3) a method of delivering power to an output (V_{hl}), comprising: providing a secondary winding (right hand side winding of transformer) having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); providing a first switch (S1), a second switch (S3), a third switch (S2) and a fourth switch (S4) each adapted to block current flow in an OFF state in at least one direction; forming a first series circuit with the first switch connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); forming a second series circuit with the third switch connected to the fourth switch at a second junction

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(common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the first series circuit across an output (top and bottom nodes of Co) with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)); connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)); converting power via the transformer to deliver a unipolar voltage (V_{hl}) to the output.

Chen does not disclose wherein the unipolar voltage is five Volts or less.

However, it would have been obvious to one of ordinary skill in the art at the time the invention was made to have the unipolar voltage be five Volts or less, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or working ranges involves only routine skill in the art. In re

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Aller, 105 USPQ 233.

Regarding **claim 90**, Chen does not disclose wherein the unipolar output is approximately 1 Volt.

However, it would have been obvious to one of ordinary skill in the art at the time the invention was made to have the unipolar voltage be approximately 1 Volt, since it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. In re Boesch, 617 F.2d 272, 205 USPQ 215 (CCPA 1980).

Allowable Subject Matter

16. **Claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58** are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Regarding **claims 2, 17, 19, 45, and 58**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “sensing across each of the first and second switches, a respective voltage polarity; and turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Regarding **claims 5 and 22**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “sensing across each of the third and fourth switches, a respective voltage polarity; and turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Regarding **claims 23 and 47**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “providing a first control circuit to operate the first switch; providing power to operate the first control circuit and first switch from across the third switch; providing a third control circuit to operate the third switch; and providing power to operate the third control circuit and third switch from across the first switch.”.

Regarding **claims 28 and 52**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “providing a second control circuit to operate the second switch; providing power to operate the second control circuit and second switch from across the fourth switch; providing a fourth control circuit to operate the fourth switch; and providing power to operate the fourth control circuit and fourth switch from across the second switch.”.

17. Regarding **claims 6, 11, and 52**, claims 6 and 11 are objected to due to their dependency on claim 5, and claim 52 is objected to due to its dependency on claim 47.

18. **Claim 58** would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), 2nd paragraph, set forth in this Office action

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and to include all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Regarding **claim 58**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “wherein the switch control circuit is configured to sense across each of the first, second, third, and fourth switches, a respective voltage polarity, and to turn each of the first, second, third, and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/MONICA LEWIS/
Supervisory Patent Examiner, Art Unit 2838

/JYE-JUNE LEE/

Examiner, Art Unit 2838

Notice of References Cited	Application/Control No. 13/044,063	Applicant(s)/Patent Under Reexamination VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	Page 1 of 1

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	E	US-			
	F	US-			
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009)
	V	ISL6210 (Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008)
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 03/09/2011

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

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	Attorney Docket Number	00614-0166001	

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	Art Unit			
	Examiner Name			
	Attorney Docket Number	00614-0166001		

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A New Quasi-Resonant Inverter

With Two-way Short-circuit Switch Across A Resonant Capacitor

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Abstract

Two frequencies are often used in induction heating, that is, a lower frequency f_L is for preheating and a higher frequency f_H is for main heating in order to decrease thermal stress. In this case, each of f_H and f_L is generated by the different inverter, and f_H approximately equals several times to ten times of f_L . Then we propose a new quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has features that its operating frequency is adjustable and the frequency range is so wide that the higher frequency can be ten times as much as the lower frequency. It is considered that the output current has only a fundamental wave by properly manipulating the shorting time of the resonant capacitor even though the output current is not a sinusoidal wave.

Key words : quasi-resonant inverter, adjustable frequency, capacitor shorting, dual frequency, induction heating

1 Introduction

The purpose of this study is to propose a novel quasi-resonant inverter whose operating frequency can be adjustable over a wide range for induction heating and to present a frequency control method such that higher harmonics of the output current may be suppressed.

Dual frequencies which are a higher frequency f_H and a lower frequency f_L , are often used for induction heating [1]-[5]. According to [1], it is to be desired that the ratio of f_H to f_L should be 3.3 in case of gear hardening. It is impossible for a conventional resonant inverter to generate two frequencies because its operating frequency is kept constant i.e. the resonant frequency. Then we reported a quasi-resonant inverter that the operating frequency was adjustable [6]-[9], however, the higher operating frequency f_H was about three times as much as the lower operating frequency f_L and the operating frequency range was not so wide. Practically it is also proposed that the ratio f_H/f_L should be about ten according to [2]. Further, in contour induction hardening [3] which consists of preheating and final heating, the lower frequency f_L is used in the preheating and the higher frequency f_H is used in the final heating for a result of high residual compressive stresses and a precise case of heated depth. In both cases, the ratio f_H/f_L needs to be about ten or more and the reported inverter [6]-[9] could not be used. It is desired that the operating frequency of the inverter can be widely changed for various purposes.

Thus we propose a new adjustable frequency quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has a feature that its operating frequency range is wide and the higher frequency can be ten times as much as the

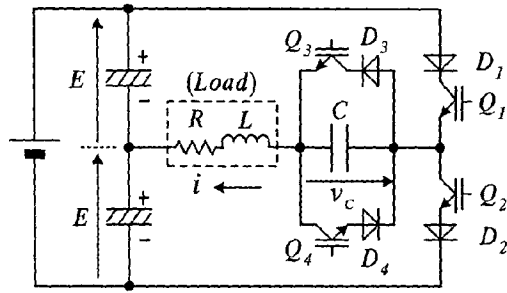


Fig.1 Circuit configuration

lower frequency. However the output current i is not a sinusoidal wave and includes many harmonics. The harmonic current causes the surface of a work to be overheated, and it is needed that the harmonics should be suppressed. It is considered that the output current has only a fundamental wave by properly manipulating the shorting time of the resonant capacitor even though the output current is not a sinusoidal wave.

2 Circuit Configuration and Operation Principle

Fig.1 shows a proposed inverter circuit using IGBTs as switching devices. In Fig.1, R means an equivalent resistance and L means an equivalent inductance of an induction heater with an iron-core matching transformer. Q_3 and Q_4 play the role of shorting the resonant capacitor C , whereas Q_1 and Q_2 work as switches for changing the polarity of the power supply voltage E .

Fig.2 shows typical operation waveforms of the output current i and the capacitor voltage v_c and mode classifications. It is supposed that the resonant capacitor has been charged to $-V_m$ before the mode 1 begins. The operation in each mode is disclosed as follows on condition that the IGBTs $Q_1 - Q_4$ and the diodes $D_1 - D_4$ are ideal switching devices.

1) Mode 1 : On-signal is supplied to Q_1 and Q_2 . But Q_3 cannot turn on because the voltage across D_3 is negative. Then the discharge current i flows under LC resonant state.

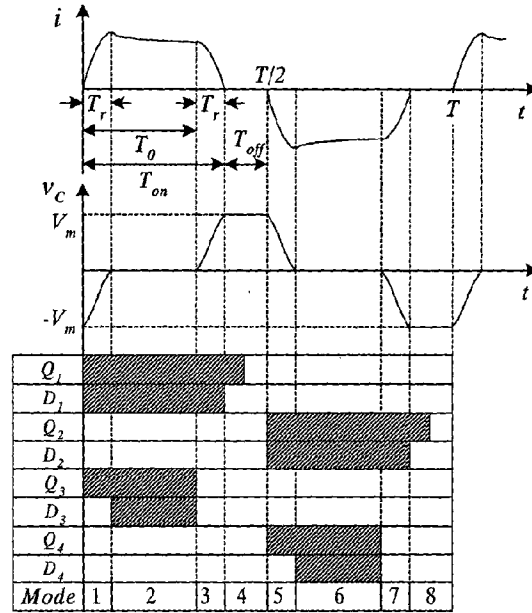


Fig.2 Typical operation waveforms and mode classifications

2) Mode 2 : When the capacitor voltage v_c returns to zero, Q_3 and D_3 turn on, and the non-oscillatory current i is supplied to the load.

3) Mode 3 : After desired time T_θ passes, Q_3 is made off. Then the charge current i decreases under LC resonant state and the capacitor voltage v_c rises. When the current i returns to zero, the diode D_1 turns off.

4) Mode 4 : The circuit becomes in a standby state until the beginning of the next mode. After desired period $T/2$ passes, the antisymmetric operation starts.

In Mode 5 - 8, the circuit operates in the same manner as the operation in Mode 1 - 4. Thus the typical output current i and the capacitor voltage v_c are obtained as in Fig.2. In Fig.2, the operating frequency f is adjustable by manipulating T and the absolute value of the current i is adjustable by manipulating the capacitor shorting time T_θ . At the same time, harmonics of the current i are variable by manipulating the capacitor shorting time T_θ .

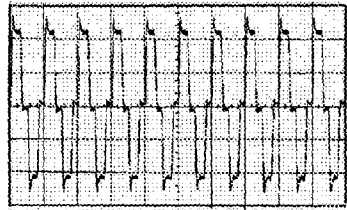
It is basically possible that the operating frequency f

Table 1 Parameters

Equivalent inductance L	82 μH
Resonant capacitance C	0.5 μF
Equivalent resistance R	about 4 Ω



(a) $f = 1 \text{ kHz}$



(b) $f = 10 \text{ kHz}$

(Abscissa : 100 $\mu\text{s}/\text{div}$, Ordinate axis : 5A/div)

Fig.3 Examples of waveforms output current i

becomes infinitely low by increasing the cycle time T then the operating frequency range can be made very wide. Besides, the harmonics can be suppressed by properly manipulating the capacitor shorting time T_o .

3 Experimental Waveforms

A prototype inverter system is manufactured and the parameters are shown in Table 1. The value of R is rough and depends on the frequency f of the supplied current. When the power supplied into the work is controlled, the value of R must be known by some means *i.e.* detecting during the operation [9].

Examples of the waveforms of the output current i are illustrated in Fig.3 when d.c. power voltage $E = 50\text{V}$. As shown in Fig.3, it is clear that the prototype inverter

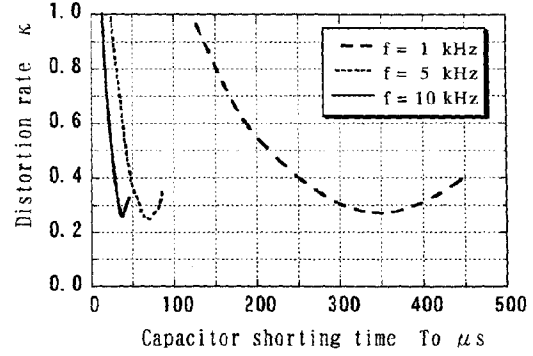


Fig.4 Relations between the capacitor shorting time T_o and the distortion rate κ of the output current (Analytical results)

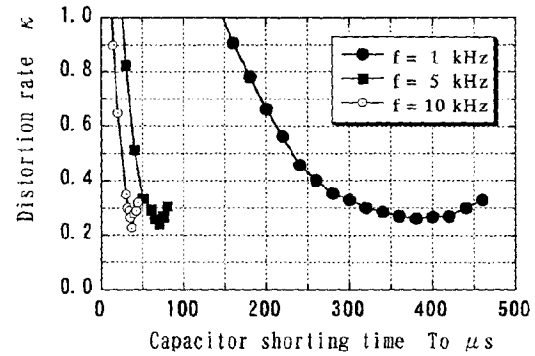


Fig.5 Relations between the capacitor shorting time T_o and the distortion rate κ of the output current (Experimental results)

operates in accordance with the theory in Fig.2 and the higher operating frequency can be ten times as much as the lower operating frequency. But the waveforms are not sinusoidal and it is assumed that many harmonics are included in the current.

4 Harmonics Characteristics and Harmonics Control

4.1 Harmonics Characteristic

Fig.4 and Fig.5 show relations between the capacitor shorting time T_o and the distortion rate κ of the output

current, where Fig.4 indicates analytical results and Fig.5 indicates experimental results. The distortion rate κ is defined as (1).

$$\kappa = \frac{\sqrt{I^2 - I_1^2}}{I_1} = \sqrt{\left(\frac{I}{I_1}\right)^2 - 1} \quad (1)$$

where I_1 is the effective value of the fundamental wave of the output current i and I is the effective value of the output current i .

The analytical characteristics are very similar to the experimental characteristics. It is clear that the distortion rate κ depends on T_0 and has a minimum value. Therefore higher harmonics can be suppressed by manipulating T_0 properly and then the surface of a work is prevented from overheating.

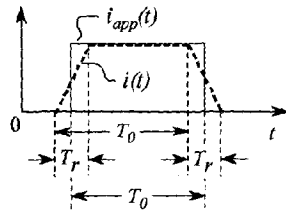
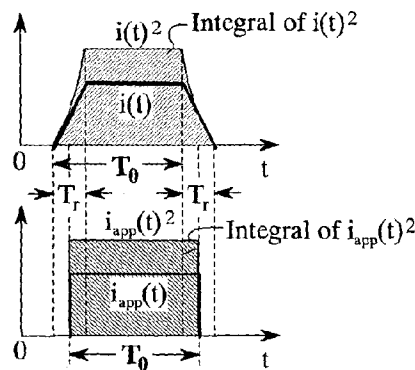
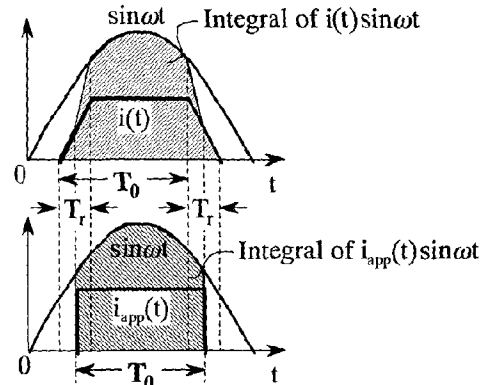


Fig.6 Approximation of i by square wave



(a) Calculation of the effective value of the current



(b) Calculation of the fundamental current

Fig.7 Validity of the approximation of i for calculating (2) and (3)

4.2 Harmonics Control

The fundamental wave content λ_1 of the output current should be the maximum in order to minimize the distortion rate κ . Thus the capacitor shorting time T_0 is such manipulated that λ_1 may be the maximum. The fundamental wave content λ_1 is defined as (2).

$$\lambda_1 = \frac{I_1}{I} \quad (2)$$

Each of I_1 and I is calculated by (3) and (4) as well known.

$$I_1 = \frac{4}{\sqrt{2}T} \int_0^{T/2} i(t) \cdot \sin \omega t dt \quad (3),$$

$$I = \sqrt{\frac{2}{T} \int_0^{T/2} i(t)^2 dt} \quad (4).$$

Strictly speaking, the equation of i should be obtained by solving the circuit equation in each mode 1 to 4 in Fig.2 but it is too complicated to solve the circuit equations during the operation. Then the waveform of the output current i is approximated by a square wave. Though the shape of the output current i is like a trapezoid wave shown in Fig.2 or Fig.3, for convenience, let

the waveform of the output current i be a square wave i_{app} which has dead time as shown in Fig.6. Let us assume that the approximated wave i_{app} has the maximum value that is equal to the trapezoid wave i and has on-time T_0 that is equal to the capacitor shorting time.

Fig.7 shows the validity of the approximation of i in order to simply calculate (3) and (4). In Fig.7, the upper sketches in (a) and (b) illustrate the calculation images of the effective value and the fundamental wave of the trapezoid wave i , and the lower sketches illustrate the images of the approximated wave i_{app} . In Fig.7(a), each shadowed portion expresses the integral of i^2 and the integral of i_{app}^2 . It is considered that both are almost equal and we can use i_{app} instead of i for calculating the effective value in (4). Likewise, In Fig.7(b), each shadowed portion expresses the integral of $i \sin \alpha$ and the integral of $i_{app} \sin \alpha$. It is deemed that both are almost equal and we can use i_{app} instead of i for calculating the fundamental wave in (3).

Fig.8 illustrates an equivalent square wave of the output current i in order to calculate the fundamental wave content λ_1 in (2) and λ_1 is given by (5).

$$\lambda_1 = \frac{2\sqrt{T}}{\pi} \cdot \frac{\sin \frac{\omega T_0}{2}}{\sqrt{T_0}} \quad (5)$$

where $\omega = 2\pi/T = 2\pi f$ and f means the operating frequency i.e. the fundamental output frequency.

The equation $d\lambda_1/dT_0 = 0$ is solved in order to obtain the capacitor shorting time T_{OM} which gives the maximum value of the fundamental wave content λ_1 . T_{OM} is obtained as (6).

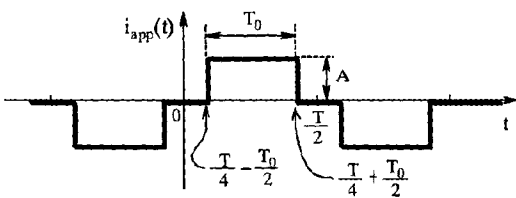


Fig.8 Equivalent square wave of the output current i

$$T_{OM} = 0.371/f \quad (6)$$

We can minimize the higher harmonics of the output current by the manipulated variable T_{OM} in (6).

4.3 Experimental Results of Harmonics Control

Fig.9 shows the relation between the operating frequency f and the capacitor shorting time T_{OM} for minimizing the distortion rate. The calculated values by (6) agree well with the experimental results. Thus (6) is appropriate for controlling the operating frequency so that the harmonics including the output current may be suppressed.

Fig.10 discloses an example of experimental results of harmonics control by using (6) in case of $f = 1$ kHz and $f = 10$ kHz. It is apparent that higher harmonics of the current are suppressed and the fundamental content becomes about 1.0 even though the operating frequency f is adjusted high or low. There is not a serious problem although a little higher harmonics of the output current are included, because the power is proportion to the square of the current and the higher harmonics of the power are very little.

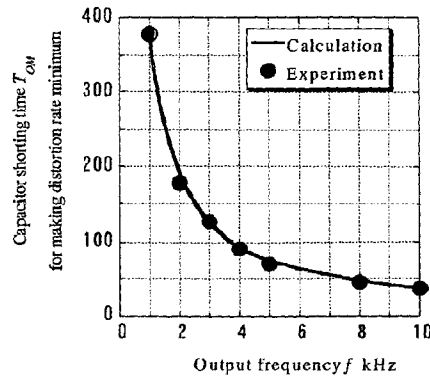


Fig.9 Relation between the operating frequency f and the capacitor shorting time T_{OM} for minimizing the distortion rate

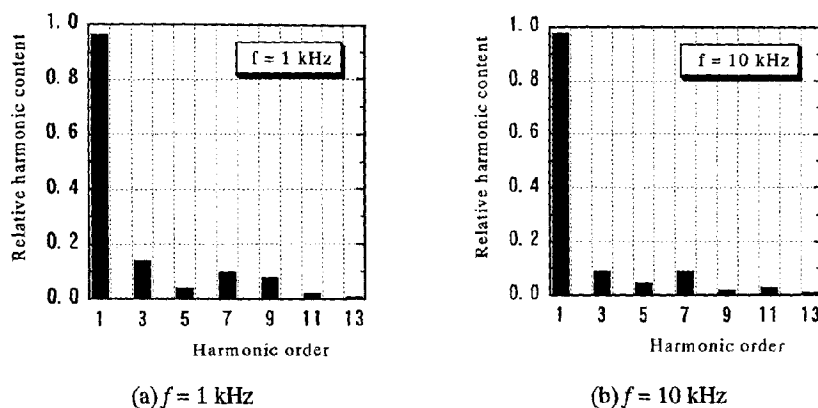


Fig.10 Example of experimental results of harmonics control

5 Conclusions

A new quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has a feature that its operating frequency range is wide and the higher frequency can be ten times as much as the lower frequency. Although the output current is not a sinusoidal wave and includes many harmonics, the harmonics can be suppressed by manipulating the capacitor shorting time properly and then the surface of a work is prevented from overheating. According to the present inverter, dual frequency induction heating which consists of the preheating and the final heating can be possible with one inverter system.

In the future, we will discuss how to control the output power supplied into the work, where the equivalent output resistance and the equivalent loss resistance are unknown and depend on the operating frequency [9].

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Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter

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Abstract – A new non-isolated Intermediate Bus Converter is proposed using a divide-by-4 switched capacitor network. A load adaptive scheme is designed that maximizes the efficiency for all load currents. Detailed analysis using steady-state charge transfer is provided with a static charge constraint concept for degenerate switched capacitor networks. Simple average models for output impedance, capacitor and switch RMS currents, and efficiency are defined. A bus stability criterion is established, and a novel current splitting technique is introduced that reduces the capacitors' RMS currents. A 300W 48V to 12V prototype is designed using a provided efficiency-specified design procedure that achieves 96%-97% over the full range of load current, 1A-25A.

I. INTRODUCTION

This paper introduces a new approach to 48V to 12V Intermediate Bus Architectures. Commonly, unregulated, isolated half bridge topologies are used as bus converters. The proposed innovative approach targets applications where isolation is not required between a 48V backplane and a 12V bus. Removing the isolation transformer saves considerable board area and reduces converter cost. Reliability analyses can be conducted for a particular application to assess the risk of current faults on the 48V side propagating to the 12V side of the system. In many applications, this risk should be acceptable; therefore a non-isolated Intermediate Bus Architecture has merit in many practical systems.

Some unregulated 300W isolated bus converters that are commercially available claim peak efficiencies up to 95-96% [1-3]. This paper suggests using a high efficiency, switched capacitor voltage divider that adapts to the load current, as shown in Fig. 1. The proposed non-isolated topology achieves up to 96%-97% efficiency over the entire load range. A particular improvement in efficiency is achieved at light load conditions, which is advantageous for reducing energy consumption in systems where usage is light. Conventional, magnetic-based, switching power converters have poor light load efficiency, which is a significant disadvantage since switching losses dominate when systems are lightly used. The time integral of light load power losses can amount to a tremendous energy waste.

Characteristic high efficiency and high power density are key advantages of switched cap networks over magnetic-based power converters. A 98% peak efficiency, 60W 12V to 6V switched capacitor divider was demonstrated in [4] with

very high power density. An automotive application in [5] showed a 98% peak efficiency with a 600W 42V to 14V switched capacitor converter. The converter proposed in Fig. 1 divides the input by 4; where as these other application only divide by two or three. It requires more switches; hence the losses are greater. 98% efficiency is possible with this topology if the parasitic resistance and capacitances of the MOSFETs are small enough and if larger capacitance is used in the network. A tradeoff between converter cost and efficiency must be made. A 97% peak efficiency is a reasonable balance with cost and can compete with commercially available isolated converters [1].

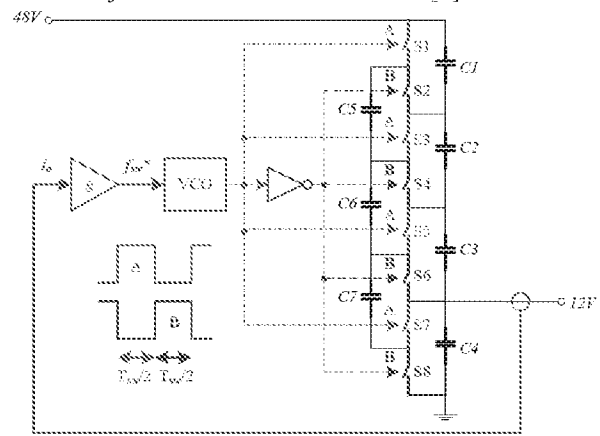


Figure 1 Proposed load adaptive non-isolated bus converter.

The proposed converter in Fig. 1 is a two phase switched capacitor network. The phases are switched with a fixed 50% duty cycle. The proposed load adaptation scheme modulates the switching frequency such that the efficiency is maximized as the load current varies. Efficiency is determined by losses in the network. Switched-capacitor energy losses dominate when the load current is high, and MOSFET capacitor charging and discharging losses dominate at light load. The loss incurred from switching the capacitors is proportional to the load current but inversely related to the switching frequency. Specifically, the converter has a DC output impedance that is inversely proportional to the switching frequency. The adaptation scheme reduces the output resistance by increasing the switching frequency at heavy loads. Conversely, MOSFET capacitor losses are proportional to switching frequency. Thus a low switching frequency reduces MOSFET capacitor losses. Hence a balance must be found between switched cap and MOSFET cap losses. A

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simple formula for the optimal switching frequency is provided in this paper that minimizes the sum of these losses, hence maximizing efficiency. The optimal switching frequency is found to be proportional to the load current.

A simple charge transfer analysis is described and used to formulate useful models for output resistance, switch currents, and capacitor currents. These models are used as design equations. The output resistance determines the conversion efficiency and the deviation in the output voltage from the ideal voltage. The models for the currents are used to select appropriate devices. The capacitor RMS currents are critical design parameters, which primarily determine the capacitor selections.

II. CHARGE TRANSFER ANALYSIS

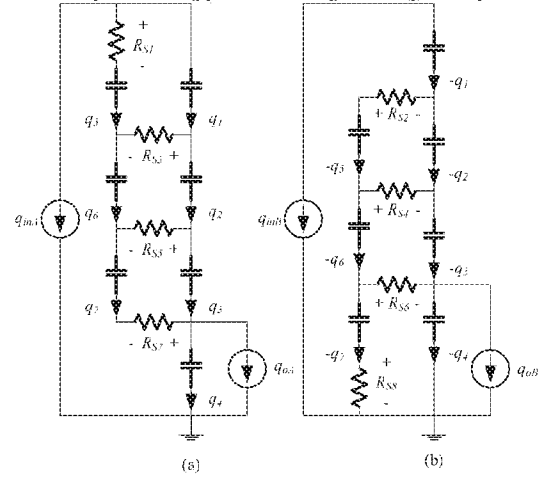
This section develops an analysis framework that simplifies modeling steady state behavior. The framework is based on a simple KCL analysis. The technique is preferred over other modeling techniques, which are possibly more complicated and do not scale well for higher order networks. The switched capacitor network in Fig. 1 is 7th order. Periodic steady state analysis of such a high order system via solving state equations is very cumbersome numerically and impractical, if at all possible, symbolically. State space averaging techniques have been established. [6] proposes a state space averaging approach, but illustrates the method with only a 2nd order system. Unfortunately the approach does not scale well to a 7th order network. [5] provides a power loss analysis based on a detailed capacitor voltage study, which would be difficult to generalize. [7] presents a simple time-constant analysis, which is intuitive for 2nd or 3rd order systems.

A generalized analysis framework is needed that can address the scalability of switched capacitor networks. An analysis technique that is based on steady state charge balance is most appropriate. The charge transfer analysis technique described in [8] and elaborated in [9] provides a complete steady-state framework for modeling switched capacitor networks. From this simple approach, models for steady state output resistance, switch currents, and capacitor currents can be obtained. These models are useful for calculating conversion efficiency and selecting suitable devices. Moreover, the relationships are quite simple as compared with the complexity and ambiguity of state space models. The derived models are intuitive and are very practical design equations.

A. KCL System with a Static Charge Constraint

The charge transfer analysis begins by identifying the circuits of the converter in each phase. The phases are named 'A' for the circuit (Fig. 2a) when the odd numbered switches are on, and 'B' for the circuit (Fig. 2b) during the on-state of the even numbered switches. The charge balance mechanism requires that the net charge entering each capacitor is zero. In other words, all cycle-by-cycle average capacitor currents must equal zero, i.e. amp-second balance. The charge entering each branch (e.g. q_{inA} , q_1 , q_2 , etc.) is labeled in Fig. 2. Note

that for charge balance, the phase charges of each capacitor, C_i , must be equal and opposite, i.e., $q_{C1A} = -q_{C1B} = q_1$.



transferred to the load, absorbed by each capacitor, and drawn from the input source. These vectors are arranged as follows

$$\begin{aligned} a_A &= \begin{bmatrix} a_{aA} & (a_c)^T & a_{mA} \end{bmatrix}^T \\ a_B &= \begin{bmatrix} a_{aB} & (-a_c)^T & a_{mB} \end{bmatrix}^T \end{aligned} \quad (3)$$

The terms a_{aA} , a_{aB} , a_{mA} , and a_{mB} are scalar quantities. The term $a_c = [a_{c1} \ a_{c2} \ a_{c3} \ a_{c4} \ a_{c5} \ a_{c6} \ a_{c7}]^T$ is a vector with components for each capacitor. The capacitor vector component values are characteristic of a particular network topology. The capacitor vector is also the unique factor in determining the output resistance, the switch currents, and the capacitor currents. Moreover, these relationships are generalized in terms of this unique charge transfer vector. The relationships are the same for all switch capacitor networks, only the length and values of the charge transfer vector is different. Hence, charge transfer analysis is widely applicable to other switched capacitor networks. Though, as in the case of the switched capacitor divider presented in this paper, the extension of the methodology to degenerate networks requires identifying necessary static charge constraints. The vector components for the proposed topology in Fig. 1 are

$$\begin{aligned} a_{aA} &= a_{aB} = \frac{1}{2}, & a_{mA} &= -\frac{1}{2}, & a_{mB} &= \frac{1}{4} \\ a_c &= \begin{bmatrix} \frac{1}{4} & 0 & -\frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{bmatrix}^T \end{aligned} \quad (4)$$

These components uniquely characterize the charge transfer between the capacitors, the input source, and the load. They are useful for deriving the steady state output resistance of the switched capacitor network.

C. Output Resistance Model

A model for the network is an ideal transformer with a turns ratio equal to the number of stationary capacitors, N [8,9]. All switched capacitor energy losses, hard-switching losses, and parasitic resistive losses are modeled as a lumped output resistance, R_o , as depicted in Fig. 3.

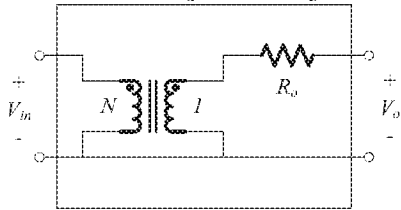


Figure 3 Ideal transformer model of switched capacitor network with losses represented by the output resistance R_o [8,9].

The output resistance varies with switching frequency and has two distinct limits. A slow switching limit, R_{SSL} , quantifies the resistance at low switching frequencies. The output resistance is inversely proportional to slow switching frequencies, where the proportionality constant is a function of the network capacitances. Conversely for high switching frequency, the output resistance approaches a minimum, R_{FSL} . This minimum is determined primarily by the on-state resistances, R_{dson} , of the switches and secondarily by the equivalent series resistances, R_c , of the capacitors. Typically the R_{dson} is much greater than the R_c , since each capacitor is

composed of many parallel MLCCs in order to carry large RMS currents. The output resistance and the two limits are illustrated in Fig. 4.

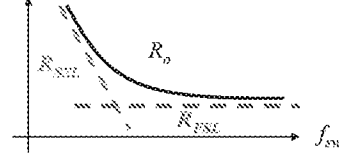


Figure 4 Variation of output resistance with the switching frequency, f_{sw} .

The output resistance approaches the fast switching limit (R_{FSL}) at high frequency and the slow switching limit (R_{SSL}) at low frequency.

The effective output resistance, R_o , is equal to the Euclidian distance between the fast switching limit and the slow switching limit. Specifically,

$$R_o = \sqrt{R_{SSL}^2 + R_{FSL}^2} \quad (5)$$

D. Slow Switching Limit

The slow switching limit describes the circuit behavior when the switching frequency is low enough such that each pair of capacitors in parallel will have equal voltages at the end of a half cycle. The output resistance for the slow switching limit is determined by considering the capacitor voltage deviation as it varies with the amount of charge transferred. By Tellegen's Theorem, a vector of branch voltages that satisfies KVL is orthogonal to a vector of branch currents that satisfies KCL [9]. This theorem is applicable to the switch capacitor network, since the average currents in each phase are the charge transferred per half cycle. Hence, the charge transfer vectors are orthogonal to the respective voltages.

$$\begin{aligned} a_A \cdot v_A &= 0 \\ a_B \cdot v_B &= 0 \end{aligned} \quad (6)$$

where v_A and v_B are the respective voltage vectors with components for the output voltage and each capacitor voltage. For analyzing the output impedance, zero input voltage can be assumed, $v_{in}=0$. The inner products can be rewritten as sums

$$(a_{aA} + a_{aB})v_o + \sum_i (a_{cA}v_{cA} + a_{cB}v_{cB}) = 0 \quad (7)$$

Or equally

$$v_o + \sum_i a_{ci} \Delta v_{ci} = 0 \quad (8)$$

$$\Delta v_{ci} = (v_{cA} - v_{cB}) = \frac{q_{ci}}{C_i} = \frac{q_o a_{ci}}{C_i}$$

The total charge delivered to the load is constant, hence $(a_{aA} + a_{aB}) = 1$. Dividing (8) by the total output charge, q_o , it follows that

$$\frac{v_o}{q_o} + \sum_i \frac{(a_{ci})^2}{C_i} = 0 \quad (9)$$

Further (9) describes the output resistance for the slow switching limit, noting that the output current is $i_o = q_o f_{sw}$. Making this substitution and rearranging (9), the slow switching limit is found to be

$$R_{SSL} = -\frac{v_o}{i_o} = \sum_i \frac{(a_{ci})^2}{C_i f_{sw}} = \frac{1}{f_{sw} C_{eq}} \quad (10)$$

where C_{eq} describes the equivalent capacitance of the network

$$C_{eq} = \left(\sum_i \frac{(a_{ci})^2}{C_i} \right)^{-1} \quad (11)$$

The equivalent capacitance is a useful concept. Particularly it creates a compact representation for the output resistance and also simplifies the notation when considering optimal switching frequency, which is described in a later section. The equivalent capacitance also can be a design parameter. The output resistance of the network represents the switched-cap losses in the circuit. Lower output resistance results in lower losses, and hence produces a higher efficiency. Therefore it is desirable that the equivalent capacitance be as large as possible. Of course the amount of capacitance will be constrained in a practical design based on cost and board area. The equivalent capacitance for the network in Fig. 1 is

$$C_{eq} = \frac{16(C_1 C_3 C_5 C_6 C_7)}{\left(C_3 C_5 C_6 C_7 + C_1 C_5 C_6 C_7 + C_1 C_3 C_6 C_7 \right.} \quad (12)$$

$$\left. + 4 C_1 C_3 C_5 C_7 + 9 C_1 C_3 C_5 C_6 \right)$$

E. Fast Switching Limit

The fast switching limit models the circuit behavior, where the parallel capacitors do not have equal voltages at the end of each half cycle. The circuit switches between phases much faster than RC time constants of the paralleled capacitors. It is expected from the slow switching limit that, R_{SSL} , goes to zero as the switching frequency goes to infinity. In other words the effect of the capacitances on the output resistance diminishes as the switching frequency increases. Accordingly, the parasitic resistances will tend to dominate the output resistance at high frequency. It is reasonable to assume that the on-state resistances of the switches are much larger than the equivalent series resistances (ESRs) of the capacitors. The capacitors will be constructed from many parallel devices in order to carry large RMS currents. The ESRs can be included in the analysis, but considering ideal capacitors is sufficient and simplifies the analysis greatly.

Define a charge vector for the switches, $a_s = [a_{s1} \ a_{s2} \ a_{s3} \ a_{s4} \ a_{s5} \ a_{s6} \ a_{s7} \ a_{s8}]^T$, which describes the charge transferred through the switches in each phase. This vector has components that are functions of the capacitor charge vector components. By consistently defining the polarity of the switches in each phase, the switch charge vector for the switched capacitor divider is found to be

$$a_s = \begin{bmatrix} a_{s1-s4} & a_{s5-s8} \end{bmatrix}^T$$

$$a_{s1-s4} = [a_{c5} \ a_{c5} \ (a_{c1} - a_{c2}) \ (a_{c6} - a_{c5})]^T \quad (13)$$

$$a_{s5-s8} = [(a_{c2} - a_{c3}) \ (a_{c7} - a_{c6}) \ -a_{c7} \ -a_{c7}]^T$$

$$a_s = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & -\frac{3}{4} & -\frac{3}{4} \end{bmatrix}^T$$

The average value of the switch currents during the on-interval of the respective switches are derived from the switch charge vector. Consider that each switch is operated at a 50% duty cycle, D_i . The switch currents are equal to the charge transferred per phase interval divided by the duration of the

interval. The phase interval is the product of the duty cycle and the switching period. Explicitly,

$$I_{si} = \frac{q_{si} f_{sw}}{D_i} = 2q_{si} f_{sw} = 2a_{si} I_o \quad (14)$$

where $I_o = q_o f_{sw}$ and $a_{si} = q_{si} / q_o$. The average switch currents are

$$I_s = \left[\frac{1}{2} \ \frac{1}{2} \ \frac{1}{2} \ \frac{1}{2} \ \frac{1}{2} \ \frac{1}{2} \ -\frac{3}{2} \ -\frac{3}{2} \right]^T I_o \quad (15)$$

The RMS switch currents are calculated from the average currents. Note that the average current is carried in the duration of the switching period when each switch is turned on. At high switching frequencies, the switch currents can be approximated to square waveforms switching between zero and I_{si} . For a 50% duty cycle the switch RMS currents are

$$I_{SRMS} = (\sqrt{D_i}) |I_{si}| = \frac{|I_{si}|}{\sqrt{2}} = \sqrt{2} a_{si} I_o \quad (16)$$

This approximation is valid only for fast switching frequencies. The actual RMS currents will be larger for slow switching frequencies, since the peak currents will be larger. A typical design will select an adequately large switching frequency such that efficiency is preserved at heavy loads. Therefore the maximum switch RMS current can be calculated from the above approximations.

The fast switching limit for the output resistance is derived from the switch RMS currents. Consider the power losses due to the R_{don} of each switch. The sum of these losses is the total switching loss.

$$P_{FSL} = \sum_i (R_{don} I_{SRMS}^2) \quad (17)$$

$$= 2R_{don} \left(\sum_i (a_{si}^2) \right) I_o^2$$

The fast switching limit resistance is equal to the total loss divided by the square of the load current.

$$R_{FSL} = 2R_{don} \left(\sum_i (a_{si}^2) \right) = 3R_{don} \quad (18)$$

This is a very simple relationship and quite useful for selecting switches. Certainly switches with low on-state resistance should be selected, otherwise efficiency will be low at heavy loads.

The capacitor RMS currents at high switching frequencies can be calculated following an approach similar to the switch currents. In short, the capacitor current waveforms are square waves with peaks $\pm 2a_{ci} I_o$. Thus RMS currents are

$$I_{CRMS} = |2a_{ci} I_o| = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 1 & \frac{3}{2} \end{bmatrix}^T I_o \quad (19)$$

As mentioned earlier, the switching capacitors ($C5$, $C6$, $C7$) carry more RMS current than the stationary capacitors ($C1$, $C2$, $C3$, $C4$). This signifies that the charge is moved from the input source to the load via the switching capacitors. The stationary capacitor stack does not transfer any net charge per cycle to the load. The components with values of zero in the RMS current vector are not exactly zero in an actual switching circuit. A zero value indicates that the RMS current of a

device does not charge with the load current. The importance of the particular values in the RMS current vector is that they can be used to select suitable capacitors. The RMS current ratings of the capacitors are the critical factors in the design, more so than the capacitance values. Supporting large RMS currents requires using many devices in parallel. The RMS current vector indicates the relative number of parallel capacitors that should be used.

Charge transfer analysis is a simple methodology that completely characterizes a switch capacitor network. The formulation yields intuitive relationships, which are well suited as design equations. The following section, regarding load adaptation, exemplifies such a design process, which would be difficult with more complicated models. The models developed with charge transfer analysis are used to determine the optimal switching frequency for a particular load current. A load adaptation scheme is proposed to track the optimal efficiency as the load varies.

III. LOAD ADAPTATION

A load adaptation scheme is devised that maximizes the efficiency for all load currents. It will be shown that the optimal switching frequency is proportional to the load current. In a hardware implementation, optimality can be maintained by using a voltage control oscillator (VCO) for generating the MOSFET switching signal. A sensor measurement of the load current is scaled and applied as the input to the VCO. Hence, the switching frequency adjusts dynamically to changes in the load current. The adaptation gain, g , of the sensed current is set equal to the optimal proportionality constant, which is derived in this section. The load adaptive control action tracks the maximum efficiency point of the converter.

The efficiency of the switched capacitor divider decreases as the load current increases. The switched cap losses, which are lumped in the output resistance (R_o) derived in the previous section, increase with the square of the load current. Therefore it is favorable to increase the switching frequency at heavy loads. Conversely, there are additional losses due to charging and discharging of the input and output capacitances of the MOSFET switches. These losses dominate the efficiency at light loads and are proportional to the switching frequency. Thus it is best to decrease the switching frequency at light loads. The general trends of switched capacitor losses, P_{SC} , and MOSFET capacitor losses, P_{MC} , are shown in Fig. 5.

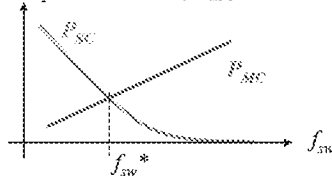


Figure 5 The switched capacitor losses, P_{SC} , decrease with frequency; whereas, the MOSFET capacitor losses, P_{MC} , increase with frequency.

The MOSFET capacitor losses are determined by the charge stored in the gate and output capacitances, Q_g and Q_{oss} . These values are typically provided in MOSFET datasheets or can be calculated from the parasitic capacitances and the

applied voltages. The losses are proportional to the switching frequency and the applied voltages. The total losses are summations of the losses from all switches. In total, there are $2N$ switches where N is the number of stationary caps in the divider stack. In this case, $N=4$, so there are 8 switches. Explicitly the MOSFET cap losses are

$$P_{MC} = 2(Q_g + Q_{oss})V_{in}f_{sw} = 2Q_M V_{in}f_{sw} \quad (20)$$

which assumes that the MOSFETs gates are driven by a (V_{in}/N) voltage. For compactness, the two charges are summed to form the quantity, $Q_M = Q_g + Q_{oss}$. Note in (20) that the MOSFET cap losses are independent of the number, N , of caps in the stack, if the gate drive voltage is equal to V_{in}/N . This is significant since the MOSFET losses would not increase if the topology was scaled to divide by $N>4$. Hence efficiency does not decrease by adding more switches.

The switched capacitor losses in the network are modeled using the output resistance and are proportional to the square of the load current.

$$P_{SC} = R_o I_o^2 \quad (21)$$

Ideally, the power transferred to the load, P , is determined by the input voltage, the load current, and the number of stationary caps in the divider stack.

$$P = \frac{V_{in}}{N} I_o \quad (22)$$

The actual power delivered to the load is P less the switched cap losses. The input power is P more the MOSFET cap losses. The efficiency, η , is calculated using (20) (21), and (22):

$$\eta = \frac{P_o}{P_{in}} = \frac{P - P_{SC}}{P + P_{MC}} \quad (23)$$

F. Optimal Policy

The efficiency is maximized by selecting a particular switching frequency. This optimal switching frequency, f_{sw}^* , varies with the load current. Clearly, the efficiency as a function of the switching frequency has extrema where its first derivative with respect to frequency is zero. The maximum occurs at the extrema where the second derivative is negative. The optimal policy is

$$f_{sw}^* = \arg \max \eta = \left\{ f_{sw} : \frac{\partial \eta}{\partial f_{sw}} = 0, \frac{\partial^2 \eta}{\partial f_{sw}^2} < 0 \right\} \quad (24)$$

Solving this optimization problem symbolically is rather involved and actually not necessary. One can argue that the efficiency is maximized when the switched cap losses are equal to the MOSFET cap losses, as illustrated in Fig. 5. Intuitively this is reasonable, because P_{SC} varies inversely with frequency and P_{MC} varies directly. In other words, minimizing the sum of the losses maximizes the efficiency. The minimum of the sum occurs when the two losses are equal. Specifically, the problem simplifies to finding the switching frequency such that

$$f_{sw}^* = \arg \max \eta = \{ f_{sw} : P_{SC}(f_{sw}) = P_{MC}(f_{sw}) \} \quad (25)$$

Equating the losses (20) and (21), the following relationship must hold

$$I_o^4 + (3R_{don} C_{eq})^2 f_{sw}^*{}^2 = (2C_{eq} Q_M V_{in})^2 f_{sw}^*{}^4 \quad (26)$$

The roots of this quartic equation are rather complicated. The positive real root is

$$f_{sw}^* = \sqrt{\frac{2C_{eq} \sqrt{9R_{don}^2 C_{eq}^2 + (81R_{don}^4 C_{eq}^2 + 16Q_M^2 V_{in}^2 I_o^4)}}{4C_{eq} Q_M V_{in}}} \quad (27)$$

This solution is not very practical for implementing an adaptive control. Instead an approximation can be made. One can either argue that $R_{don} \ll 1$ and so the resistive terms in (27) can be omitted, or that the quartic terms dominate (26) such that

$$I_o^4 \approx (2C_{eq} Q_M V_{in})^2 f_{sw}^*{}^4 \quad (28)$$

Solving this equation yields a more practical result. Though the solution is strictly suboptimal, the approximation is reasonably accurate for use in the load adaptive scheme. The suboptimal solution is

$$f_{sw}^* \approx \frac{I_o}{\sqrt{2C_{eq} Q_M V_{in}}} = g I_o \quad (29)$$

where g is the adaptation gain that maps the load current to the optimal switching frequency. The load adaptation scheme maintains this optimal relationship between the load current and the switching frequency. Hence by applying this simple load adaptive control, the efficiency is maximized.

IV. CURRENT SPLITTING TECHNIQUE

The RMS current ratings of the switching capacitors pose a significant design challenge. The capacitor currents for the proposed network, (19), have ratings up to 150% of the load current. A novel circuit technique can be introduced that reduces the RMS rating of certain capacitors. Using this technique the max RMS current can be reduced to 50% of the load current. The technique involves adding additional capacitors across several capacitors in the network in Fig. 1. The method presented in this paper discusses adding caps across the switching caps ($C5, C6, C7$), since these carry the largest RMS currents. But the technique could be equally applied to the stationary caps ($C1, C2, C3, C4$). The proposed embodiment is shown in Fig. 6.

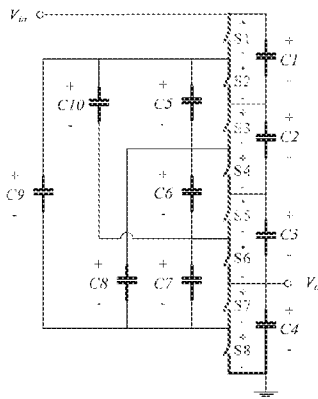


Figure 6 Proposed topology using current splitting technique to reduce the RMS current ratings of the switching capacitors. The additional capacitors, $C8$, $C9$, and $C10$, reduce the original $C6$ and $C7$ currents. The currents through the

switches are split amongst the additional capacitor branches. This potentially reduces the current through a single branch. Using the proposed configuration, the maximum RMS current of all capacitors in the network is reduced to 50% of the load current. Other variations could be used, such as including only $C8$ or $C9$, but using the topology in Fig. 6 is most effective. The charge transfer vectors and the RMS currents for using different splitting configurations are summarized in Table 1.

Table 1 Charge transfer vectors and capacitor RMS currents for different configurations of splitting capacitors.

Splitting Caps Used	a_c	I_{CRMS}/I_o
none	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 1/4 & 1/2 & 3/4 \\ 1/4 & 0 & -1/4 & 0 & 1/4 & 1/2 & 3/4 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 1/2 & 1 & 3/2 \\ 1/2 & 0 & 1/2 & 0 & 1/2 & 1 & 3/2 \end{bmatrix}^T$
$C8, C9$	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 \\ 1/4 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 \end{bmatrix}^T$
$C8, C9, C10$	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 & 0 \\ 1/4 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 & 0 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}^T$

The data in Table 1 indicate that including $C8$ and $C9$ greatly reduces the current ratings of all capacitors. Moreover it is not necessary to add $C10$, including only ($C8, C9$) yields the same currents as ($C8, C9, C10$). The current splitting technique reduces the maximum RMS current of all caps to half the load current. This facilitates an even distribution of capacitors. The basic network in Fig. 1 would have an uneven distribution, since two capacitors have a really large RMS current rating while the rest have much smaller ratings. One might suspect that adding more capacitors for this splitting technique would increase the converter cost and complicate the board layout. Actually since the current distribution is even among all capacitors, the layout can be more compact and hence possibly less complicated. If the equivalent capacitance is examined, the converter cost will not increase. Table 2 provides the equivalent capacitance for each configuration.

Table 2 Equivalent capacitance for different configurations of splitting capacitors.

Splitting Caps Used	C_{eq}
none	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + 4 C_2 C_3 C_4 C_5 + 9 C_1 C_2 C_3 C_4)}$
$C8, C9$	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + C_2 C_3 C_4 C_5 + C_1 C_2 C_3 C_4)}$
$C8, C9, C10$	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + C_2 C_3 C_4 C_5 + C_1 C_2 C_3 C_4)}$

Suppose all capacitors have equal capacitance, C (i.e. $C1 = C2 = C3 = \dots = C$). Substitute C for all C_i in Table 2. The equivalent capacitance using $C8$ and $C9$ is $3.2C$; whereas the equivalent capacitance with no splitting caps equals only C . Therefore the total capacitance in the network could be reduced 70% using splitting caps and still maintain the same equivalent capacitance as network with out the splitting caps. Thus the converter cost could actually decrease if splitting capacitors are used.

V. BUS STABILITY

Output impedance of a bus converter is important to the stability of a cascaded system, Fig. 7. The stability depends on the interaction between the output impedance of the bus

converter and the input impedance of the down stream point of load (POL) converters.

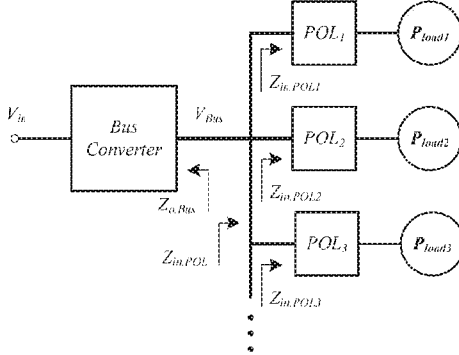


Figure 7 Cascaded system of an Intermediate Bus Architecture.

The output impedance of the bus converter and the input impedance of the POLs form voltage dividers. The transfer function from the input voltage to the bus voltage is

$$\frac{V_{bus}(s)}{V_{in}(s)} = \frac{1}{1 + \frac{Z_{o,Bus}(s)}{Z_{in,POL}(s)}} \quad (30)$$

where $Z_{in,POL}$ is the parallel combination of all POL input impedances.

$$Z_{in,POL} = Z_{in,POL1} // Z_{in,POL2} // Z_{in,POL3} // \dots \quad (31)$$

The POL converters are regulated and thus exhibit negative impedance at low frequency. This negative impedance can lead to instability at the bus. A necessary condition for stability is the output impedance magnitude of the bus converter must be less than the input impedance of all POLs.

$$|Z_{o,Bus}(j\omega)| < |Z_{in,POL}(j\omega)| \quad \forall \omega \quad (32)$$

A sufficient condition for stability is the maximum output impedance magnitude of the bus converter is less than minimum input impedance of all POLs. It can be shown that the input impedance of a POL has its minimum at low frequency. An approximation for the minimum is the DC impedance, which is determined by the power delivered to the load.

$$\min(Z_{in,POL1}) = \frac{V_{Bus}^2}{\max(P_{load1})} \quad (33)$$

Further, the effective input impedance of all parallel POLs is determined by the sum of all loads.

$$\min(Z_{in,POL}) = \frac{V_{Bus}^2}{\sum_i \max(P_{loadi})} = \frac{V_{Bus}^2}{\max(P_{Bus})} = \frac{V_{Bus}}{\max(I_{Bus})} \quad (34)$$

The output impedance of the bus converter can be approximated by the parallel combination of the output resistance, R_o , and the equivalent capacitance, C_{eq} , of the network. (11) is a valid approximation of the equivalent capacitance for impedance frequencies less than the switching frequency. At frequencies exceeding f_{sw} , the equivalent capacitance of the output impedance is determined by the stationary capacitors, $C_{eq} \rightarrow (C1//C2//C3)+C4$. The complete average model for the switched capacitor bus converter is illustrated in Fig. 8, including power losses and the approximate output impedance model.

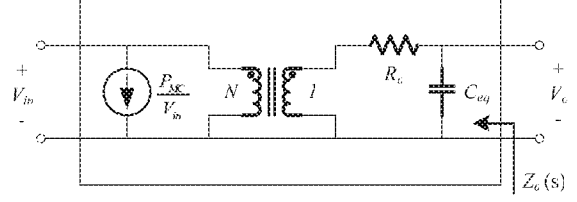


Figure 8 Complete average model for the switched capacitor bus converter.

The complete average model describes steady state and transient behavior and captures the efficiency of the converter. Notably, the current source at the primary side of the ideal transformer models the power losses in the MOSFET capacitors. The output resistance models the switched-capacitor losses. Both of these depend on the switching frequency.

The first order model for the bus converter output impedance is

$$Z_{o,Bus}(s) = \frac{R_o}{sR_oC_{eq} + 1} \quad (35)$$

This has a maximum at DC,

$$\max(Z_{o,Bus}) = R_o \quad (36)$$

Using (34) and (36), bus stability, (32), is ensured if

$$R_o < \frac{V_{Bus}}{\max(I_{Bus})} \quad (37)$$

Since R_o is limited by the $R_{ds(on)}$ of the switches, (37) is a useful design equation for selecting suitable MOSFETs.

VI. DESIGN PROCEDURE

An appropriate design approach is to specify a desired efficiency at a particular load. This procedure describes how the capacitors in the network must be sized in order to achieve a specified efficiency. Assume C8 and C9 are used.

1. Use a_C (Table 1) to determine the relative ratios of the capacitances. Define a vector of the network capacitances, $C_{Network}$. The capacitances are defined in proportion to a nominal value, C , which is the capacitance of each MLCC used.

$$C_{Network} = [\max(k a_C N_C, 1)] C \quad (38)$$

The maximum operator is used so that $C_{Network}$ does not have zero elements, since a_C may contain zero elements. As such one MLCC will be used for each capacitor with a zero element in a_C . k is a normalizing factor that scales all capacitors relative to the largest capacitance.

$$k = \frac{1}{\max(a_C)} \quad (39)$$

2. Use the I_{CRMS} vector to determine the number, N_C , of parallel MLCCs for the largest capacitor to meet the maximum current rating. Assume that each MLCC can carry 2A to 3A RMS, $I_{MLCC} = 2A-3A$.

$$N_C = \frac{\max(I_{CRMS})}{I_{MLCC}} \quad (40)$$

3. Solve for the optimal equivalent capacitance, C_{eq}^* . Assume optimal switching frequency, (30).

$$C_{eq}^* = \frac{2V_m N^2 Q_m (\eta - 1) \left(V_m^2 (\eta^3 - \eta^2 + \eta - 1) + V_{R_s}^2 (\eta + 1) - 2\eta V_m \sqrt{(\eta - 1) (\eta (V_{R_s}^2 + V_m^2) + V_{R_s}^2 - V_m^2)} \right)}{(V_m (1 - \eta) + V_{R_s})^2 (V_m (\eta - 1) + V_{R_s})^2} \quad (41)$$

where

$$V_{R_s} = I_o (3R_{ds(on)})N \quad (42)$$

4. Determine the equivalent capacitance as a function of the network capacitance, $C_{eq}(C_{Network})$. Solve for the nominal capacitance C .

$$C_{eq} = \frac{16}{5} N_c C \quad (43)$$

5. Select an appropriate MLCC by equating (41) = (43).

VII. PROTOTYPE

A 300W prototype was designed and constructed using the proposed design procedure. The design achieves 96% efficiency at full load, $I_{o,max}=25A$. The MOSFETs selected are IRLU7843, which have $R_{ds(on)}=3.3m$ and $Q_m=110nC$ for $V_{gs}=4.5V$ and $V_{ds}=12V$. Four floating ISL6605 drivers drive the gates of adjacent MOSFET pairs (e.g. S1 and S2). The voltage across each stationary capacitor provides a 12V supply voltage for the drivers. The 12V is stepped down to 4.5V using ADP3331 LDOs. The efficiency of the converter is greatly improved using this 4.5V gate drive voltage rather than using the full 12V. The gate charge is more significant than driving the gate at a higher voltage to reduce the $R_{ds(on)}$. The PWM signal is AC coupled to the floating drivers. A MM74HC4046 PLL is used as a VCO to generate the PWM signal. A 0.002 current sense resistor is in series with the output current. The sensed current is applied as the input to the VCO to implement the load adaptive scheme.

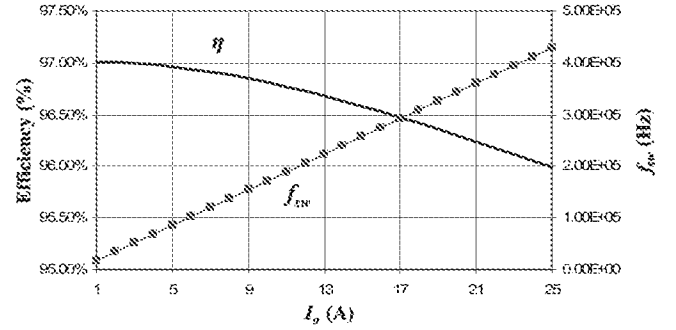
Using C8 and C9 splitting caps, the design values are compiled in Table 3. Design 1 is the exact calculation using the proposed procedure from section VI. The second is an equivalent solution that uses standard 10uF MLCC capacitors.

Table 3 Design values for a 48V to 12V 300W load adaptive switched capacitor bus converter using C8 and C9 splitting capacitors.

	Design 1	Design 2
$C_{Network} / (N_c C)$	$[1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1]^T$	$[1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1]^T$
k	4	4
I_{MLCC}	2A RMS	2A RMS
N_c	7	10
C_{eq}^*	320uF	320uF
C	14.3uF	10uF
Total MLCCs	51	72
g	17.2 kHz/A	17.2 kHz/A

Using the load adaptive scheme, the efficiency is 96% at full load and peaks to 97% at light load. The efficiency and optimal switching frequencies are plotted in Fig. 9.

Table 4 Efficiency and optimal switching frequency for a Load Adaptive Switched Capacitor Bus Converter using C8 and C9 splitting caps.



VIII. CONCLUSION

This paper presented a new approach to Intermediate Bus Architectures. A 300W 48V to 12V load adaptive switched capacitor bus converter was shown to achieve 96%-97% efficiency over the entire load range, using an efficiency-specified design procedure. A charge transfer analysis was described in detail and used to derive simple steady-state average models for output impedance, capacitor and switch RMS currents, and efficiency. A novel current splitting techniques was introduced that reduces the capacitors' RMS currents, and criterion for bus stability was examined.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	13/044,063	Examiner :	Jye-June Lee
Filed :	March 9, 2011	Conf. No. :	1081
Title :	FAULT TOLERANT POWER CONVERTER		

Mail Stop Amendment

Commissioner for Patents
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REPLY TO ACTION OF MAY 23, 2014

Please consider the following reply.

Amendment to the Abstract:

Please replace the abstract at page 66 with the following amended abstract:

ABSTRACT

A power converter provides a low-voltage output using a full-bridge fault-tolerant rectification circuit. The output circuit uses controlled switches as rectifiers. A fault detection circuit monitors circuit conditions. Upon detection of a fault, the switches are disabled decoupling the power converter from the system.

A common-source dual MOSFET device includes a plurality of elements arranged in alternating patterns on a semiconductor die. A common-source dual synchronous rectifier includes control circuitry powered from the drain to source voltage of the complementary switch.

A DC-to-DC transformer converts power from an input source to a load using a fixed voltage transformation ratio. A clamp phase may be used to reduce power losses in the converter at light loads, control the effective output resistance of the converter, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage for example during start up. ~~One or more of the transformer windings may be clamped. The converter may use the sine amplitude converter topology. The converter may use common-source dual MOSFET devices and fault detection.~~

~~The density of point of load power conversion may be increased and the associated power dissipation reduced by removing the input driver circuitry from the point of load where it is not necessary. An output circuit may be located at the point of load providing fault tolerant rectification of the AC power from the secondary winding of a power transformer which may be located nearby the output circuit. The resonant voltage and current waveforms on the primary side of the transformer are readily communicated via an AC bus between the driver circuit and the primary winding of the power transformer. The driver circuit may drive a plurality of transformer output circuit pairs. The transformer and output circuit may be combined in a single module at the point of load. Alternatively, the output circuit may be integrated into point of load circuitry such as a processor core. The transformer may be deployed near the output circuit.~~

List of claims (replacing prior versions).

1. (Currently Amended) A method of converting power via a transformer for supplying a low voltage output comprising:

- providing a secondary winding having a first terminal and a second terminal;
- providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;
- forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;
- connecting the first terminal of the secondary winding to the first junction;
- connecting the first series circuit across ~~[[the]]~~ an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;
- preventing the first switch from turning ON if the second switch fails to open; and
- preventing the second switch from turning ON if the first switch fails to open.

2. (Currently Amended) The method of claim 1 further comprising:

- sensing across each of the first and second switches, a respective voltage polarity; and
- turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

3. (Original) The method of claim 1 further comprising:

- providing a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction;
- forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
- connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

4. (Original) The method of claim 3 further comprising:
preventing the third switch from turning ON if the fourth switch fails to open;
preventing the fourth switch from turning ON if the third switch fails to open.

5. (Currently Amended) The method of claim 4 further comprising:
sensing across each of the third and fourth switches, a respective voltage polarity; and
turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

6. (Original) The method of claim 5 further comprising:
providing a first control circuit to operate the first switch;
providing power to operate the first control circuit and first switch from across the third switch;
providing a third control circuit to operate the third switch; and
providing power to operate the third control circuit and third switch from across the first switch.

7. (Withdrawn) The method of claim 6 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and

using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

8. (Withdrawn) The method of claim 7 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

9. (Withdrawn) The method of claim 8 further comprising:
providing a plurality of individual first and third MOSFET elements on the semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET; and
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

10. (Withdrawn) The method of claim 9 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

11. (Original) The method of claim 6 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

12. (Withdrawn) The method of claim 11 further comprising:
providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device;

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

13. (Withdrawn) The method of claim 12 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

14. (Withdrawn) The method of claim 13 further comprising:

providing a plurality of individual first and third MOSFET elements on the first semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET;

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;

providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;

arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;

connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second
and fourth MOSFET elements.

15. (Withdrawn) The method of claim 14 wherein the first and second common-source
dual MOSFET semiconductor devices each have exactly three terminals.

16. (Original) The method of claim 1 further comprising:
forming a second series circuit with a first capacitor connected to a second capacitor at a
second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output with each of the two end terminals
connected to a respective terminal of the output.

17. (Original) The method of claim 16 further comprising:
sensing across each of the switches, a respective voltage polarity; and
turning each of the switches ON when the sensed voltage polarity across the respective
switch comprises a first polarity and OFF when the sensed voltage polarity across the respective
switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

18. (Currently Amended) A method of providing fault tolerance in a power converter
comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge
rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating
current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power
converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the second switch connected to a second terminal of the output of the full-bridge rectifier circuit.

19. (Original) The method of claim 18 further comprising:
sensing across each of the first and second switches, a respective voltage polarity; and
turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

20. (Currently Amended) The method of claim 18 further comprising:
providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction;
forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output of the full-bridge rectifier circuit with each of the two end terminals connected to a respective terminal of the output of the full-bridge rectifier circuit and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output of the full-bridge rectifier circuit, wherein a short across one of the third or fourth switches does not short the output of the full-bridge rectifier circuit.

21. (Original) The method of claim 20 further comprising:
preventing the third switch from turning ON if the fourth switch fails to open; and

preventing the fourth switch from turning ON if the third switch fails to open.

22. (Original) The method of claim 20 further comprising:
sensing across each of the third and fourth switches, a respective voltage polarity; and
turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

23. (Original) The method of claim 20 further comprising:
providing a first control circuit to operate the first switch;
providing power to operate the first control circuit and first switch from across the third switch;
providing a third control circuit to operate the third switch; and
providing power to operate the third control circuit and third switch from across the first switch.

24. (Withdrawn) The method of claim 20 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and
using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

25. (Withdrawn) The method of claim 24 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

26. (Withdrawn) The method of claim 24 further comprising:

providing a plurality of individual first and third MOSFET elements on the semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET; and

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

27. (Withdrawn) The method of claim 24 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

28. (Original) The method of claim 20 further comprising:

providing a second control circuit to operate the second switch;

providing power to operate the second control circuit and second switch from across the fourth switch;

providing a fourth control circuit to operate the fourth switch; and

providing power to operate the fourth control circuit and fourth switch from across the second switch.

29. (Withdrawn) The method of claim 20 further comprising:

providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device; and

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

30. (Withdrawn) The method of claim 29 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

31. (Withdrawn) The method of claim 29 further comprising:

providing a plurality of individual first and third MOSFET elements on the first semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET;

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;

providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;

arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;

connecting the second MOSFET elements together to form the second MOSFET;

connecting the fourth MOSFET elements together to form the fourth MOSFET; and

using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

32. (Withdrawn) The method of claim 29 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

33. (Withdrawn) A semiconductor device comprising:

a first electronic switch having a first control terminal, a first switched terminal, and a first common terminal,

a second electronic switch having a second control terminal, a second switched terminal, and a second common terminal,

a first control circuit having an input connected to the first switched terminal and the first common terminal and an output connected to the first control terminal, the first control circuit being adapted to turn the first switch ON and OFF in response to a first voltage across the first switched terminal and the first common terminal;

a second control circuit having an input connected to the second switched terminal and the second common terminal and an output connected to the second control terminal, the second control circuit being adapted to turn the second switch ON and OFF in response to a second voltage across the second switched terminal and the second common terminal; and

a package having at least three terminals including one terminal connected to the first switched terminal, another terminal connected to the second switched terminal, and a common terminal connected to the first common terminal and the second common terminal;

wherein the first control circuit is connected to receive power from the second switched terminal; and the second control circuit is connected to receive power from the first switched terminal.

34. (Withdrawn) The apparatus of claim 33 wherein

the first and second electronic switches are formed in a common monolithic semiconductor die;

the first and second electronic switches respectively comprise a plurality of first and second switch elements;

the first and second switch elements each have a respective G terminal, S terminal, and D terminal; and

the S terminals of the first and second switch elements are connected together on the semiconductor die.

35. (Withdrawn) The apparatus of claim 33 wherein the first and second switch elements are arranged in an alternating pattern on the semiconductor die.

36. (Withdrawn) The apparatus of claim 34 wherein the first and second control circuits are also formed in the common monolithic semiconductor die.

37. (Withdrawn) The apparatus of claim 33 wherein each of the first control circuit has an offset and a gain that are configured to ensure that the first switch is turned ON when the voltage across the first switched terminal and the first common terminal is above a first predetermined threshold, and fully ON when the voltage across the first switched terminal and the first common terminal is above a second predetermined threshold.

38. (Withdrawn) A semiconductor device comprising:
a semiconductor die comprising a multiplicity of switch elements each having a respective G terminal, S terminal, and D terminal;
a first electronic switch and a second electronic switch respectively comprising a first plurality and a second plurality of the switch elements,
the switch elements of the first plurality being arranged in an alternating pattern with the switch elements of the second plurality on the semiconductor die;
the S terminals of the switch elements of the first and second pluralities being connected together on the semiconductor die;
the first electronic switch having a first control terminal connected to the G terminals of the switch elements of the first plurality and a first switched terminal connected to the D terminals of the switch elements of the first plurality;
a second electronic switch having a second control terminal connected to the G terminals of the switch elements of the second plurality and a second switched terminal connected to the D terminals of the switch elements of the second plurality;

a first control circuit having an input connected to the first switched terminal and the common terminal and an output connected to the first control terminal, the first control circuit being adapted to turn the first switch ON and OFF in response to a first voltage across the first switched terminal and the common terminal; and

a second control circuit having an input connected to the second switched terminal and the common terminal and an output connected to the second control terminal, the second control circuit being adapted to turn the second switch ON and OFF in response to a second voltage across the second switched terminal and the common terminal.

39. (Withdrawn) The apparatus of claim 38 wherein the first and second control circuits are also formed in the common monolithic semiconductor die.

40. (Withdrawn) The apparatus of claim 38 further comprising:

a package including one terminal connected to the first switched terminal, another terminal connected to the second switched terminal, and a third terminal connected to the common terminal.

41. (Withdrawn) The apparatus of claim 40 wherein the package further comprises a fourth terminal connected to supply power to the first and second control circuits.

42. (Original) The apparatus of claim 40 wherein

the first control circuit is connected to receive power from the second switched terminal; and the second control circuit is connected to receive power from the first switched terminal.

43. (Withdrawn) The apparatus of claim 42 wherein the package comprises exactly three terminals.

44. (Original) A fault-tolerant converter comprising:

a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;

a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and

a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open.

45. (Original) The apparatus of claim 44 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

46. (Original) The apparatus of claim 44, comprising a second series circuit that comprises a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals, the second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

47. (Original) The apparatus of claim 46 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, and the third control circuit and third switch are powered using a voltage across the first switch.

48. (Withdrawn) The apparatus of claim 46, comprising a common-source dual MOSFET semiconductor device having a first MOSFET and a third MOSFET that are integrated on a semiconductor die, the first MOSFET is used for the first switch, the third MOSFET is used for the third switch, and the common-source dual MOSFET semiconductor device has a source terminal connected to the source of the first MOSFET and the source of the third MOSFET.

49. (Withdrawn) The apparatus of claim 48 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, the third control circuit and the third switch are powered using a voltage across the first switch, and the first and third control circuits are integrated in the semiconductor die of the common-source dual MOSFET semiconductor device.

50. (Withdrawn) The apparatus of claim 49 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the semiconductor die, and a common source metallization interconnects the source terminals of the first and third MOSFET elements.

51. (Withdrawn) The apparatus of claim 50 wherein the common-source dual MOSFET semiconductor device has exactly three terminals that includes the source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET.

52. (Original) The apparatus of claim 47 wherein the switch control circuit comprises a second control circuit and a fourth control circuit, the second control circuit operates the second switch, the fourth control circuit operates the fourth switch, the second control circuit and the second switch are powered using a voltage across the fourth switch, and the fourth control circuit and fourth switch are powered using a voltage across the second switch.

53. (Withdrawn) The apparatus of claim 52, comprising:

a first common-source dual MOSFET semiconductor device that comprises a first MOSFET and a third MOSFET integrated on a first semiconductor die, the first common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the first and third MOSFETs, wherein the first MOSFET and the third MOSFET are used for the first and third switches; and

a second common-source dual MOSFET semiconductor device that comprises a second MOSFET and a fourth MOSFET integrated on a second semiconductor die, the second common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the second and fourth MOSFETs, wherein the second MOSFET and the fourth MOSFET are used for the second and fourth switches.

54. (Withdrawn) The apparatus of claim 53 wherein the first and third control circuits are integrated on the first semiconductor die, and the second and fourth control circuits are integrated on the second semiconductor die.

55. (Withdrawn) The apparatus of claim 54 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the first semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the first semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the first semiconductor die, a common source metallization interconnects the source terminals of the first and third MOSFET elements, the second MOSFET comprises a plurality of individual second MOSFET elements on the second semiconductor die, the individual second and fourth MOSFET

elements are arranged in an alternating pattern on the second semiconductor die, and a common source metallization interconnects the source terminals of the second and fourth MOSFET elements.

56. (Withdrawn) The apparatus of claim 55 wherein the first common-source dual MOSFET semiconductor device has exactly three terminals that includes the common source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET, and

the second common-source dual MOSFET semiconductor device has exactly three terminals that include the common source terminal, a terminal connected to the drain of the second MOSFET, and a terminal connected to the drain of the fourth MOSFET.

57. (Original) The apparatus of claim 44, comprising a second series circuit having a first capacitor connected to a second capacitor at a second junction that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter.

58. (Currently Amended) The apparatus of claim 57 wherein the switch control circuit is configured to sense across each of the first and second ~~first, second, third, and fourth~~ switches, a respective voltage polarity, and to turn each of the first and second ~~first, second, third, and fourth~~ switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

59. (Withdrawn) A multi-cell power converter comprising:
a plurality of input cells having inputs connected in parallel with an input, and connected to drive parallel respective primary windings of a transformer with power received from the input, the input cells forming a power-sharing array of cells in which if one of the input cells fails, the converter continues to process power through remaining functional input cell or cells;

a plurality of output circuits to receive power from respective secondary windings of the transformer, the output circuits having outputs that are connected in parallel to provide an output voltage to a load, the output circuits forming a power-sharing array in supplying power to the output in which if one of the output circuits fails, the converter continues to process power through remaining functional output circuit or circuits; and

a switch control circuit;

wherein each of the output circuits comprises a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch is connected to the second switch at a first junction, the first series circuit has two end terminals, the first junction is connected to a first terminal of the respective secondary winding, the first series circuit is connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, and a short across one of the first or second switches does not short the output; and

wherein the switch control circuit is configured to prevent the first switch in each output circuit from turning ON if the second switch in the output circuit fails to open, and to prevent the second switch in the output circuit from turning ON if the first switch in the output circuit fails to open.

60. (Withdrawn) The apparatus of claim 59 wherein the switch control circuit is configured to sense across each of the first and second switches in each of the output circuits, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

61. (Withdrawn) A method of converting power received from an input source at an input voltage via a transformer for delivery via an output at an output voltage to a load, where the load may vary over a normal operating range, comprising:

using an input circuit having a plurality of primary switches to drive a primary winding of the transformer with power received from the input source;

using output circuitry having a plurality of secondary switches for rectifying power received from a secondary winding of the transformer to conduct current from the secondary winding to the output;

operating the primary switches in a series of converter operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration during which (i) one or more of the primary switches are ON, (ii) a primary winding of the transformer is driven with power received from the input, (iii) a magnetizing current flows through the primary winding and increases in magnitude, and (iv) power is transferred between the input and the output via the transformer; and

(b) a clamp phase during which a winding of the transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

62. (Withdrawn) The method of claim 61 further comprising:

forming a resonant circuit including the transformer and having a characteristic resonant frequency and period; wherein a resonant current flows in the primary winding during the power transfer phases; and

turning one or more primary switches OFF essentially at times when the resonant current returns to zero.

63. (Withdrawn) The method of claim 61 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

64. (Withdrawn) The method of claim 61 further comprising using an inductance to limit the rate of change of current in the transformer; and wherein

the primary and secondary switches are operated in the series of converter operating cycles and each converter operating cycle further comprises:

a freewheel phase during which a circuit including a winding of the transformer is clamped, the freewheel phase being characterized by (i) essentially zero voltage across the clamped circuit and (ii) an average value of current flowing in the clamped circuit, the average value of current having a polarity, wherein the average is taken over the duration of the initial clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer interval immediately preceding the freewheel phase, (iii) a current flowing in the inductance declines in magnitude to essentially zero current, and (iv) the secondary switches conduct current between the secondary winding and the output.

65. (Withdrawn) The method of claim 64 wherein each clamp phase occurs after a respective freewheel phase; and the freewheel phase ends, and the secondary switches are turned OFF, essentially when the current between the secondary winding and the output returns to zero.

66. (Withdrawn) The method of claim 65 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

67. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to control an effective output resistance of the power conversion.

68. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to regulate conversion factor defined as the ratio of input voltage to output voltage.

69. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to reduce power losses at light loads.

70. (Withdrawn) The method of claim 61, 62 or 65 further comprising:
varying the duration of the clamp phase to control the rate of change of the output voltage.

71. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to regulate the output voltage.

72. (Withdrawn) A method of converting power received from an input source at an input voltage via transformer for supplying a load at a DC output voltage, comprising:
providing a driver circuit having an input for receiving power from the source and a driver output,
providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC output voltage to the load,
providing a power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit;
forming a resonant circuit including a winding of the transformer, the resonant circuit being characterized by a characteristic resonant period;
providing a power bus between the driver circuit and the power transformer;
operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and

deploying the driver circuit as a self contained unit at a first location separated from the output circuit.

73. (Withdrawn) The method of claim 72, comprising deploying the output circuit as a self-contained unit that is separate from the self-contained unit of the driver circuit.

74. (Withdrawn) The method of claim 72 wherein providing an output circuit comprises providing an output circuit that comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

75. (Withdrawn) The method of claim 74 wherein providing an output circuit comprises providing an output circuit that comprises the first series circuit and a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

76. (Withdrawn) The method of claim 74, comprising deploying the first and second switches of the output circuit on a die having the load, wherein the driver circuit is located outside of the die.

77. (Withdrawn) The method of claim 76 wherein the load comprises a processor core.

78. (Withdrawn) The method of claim 76, comprising providing control circuitry to operate the first and second switches of the output circuit, and deploying the control circuitry on the die having the load.

79. (Withdrawn) The method of claim 72 wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and

the method comprises turning one or more of the primary switches OFF essentially at times when the resonant current returns to zero.

80. (Withdrawn) The method of claim 79 wherein a magnetizing current flows through the primary winding during the power transfer phases, and each converter operating cycle comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF, and

the method comprises using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

81. (Withdrawn) An apparatus comprising:

a first package comprising a driver circuit having an input for receiving power from an input source and a driver output;

an output circuit having an AC input and an output for supplying a DC output voltage to a load, the output circuit being disposed outside of the first package;

a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit;

a resonant circuit formed in part by one of the windings of the transformer, the resonant circuit having a characteristic resonant period; and

a power bus for electrical connection to the driver circuit and the power transformer;

wherein the driver circuit is configured to drive the transformer in a series of converter operating cycles, each converter operating cycle comprising two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the transformer.

82. (Withdrawn) The apparatus of claim 81 wherein the output circuit and the power transformer are provided in a second package that is separate from the first package.

83. (Withdrawn) The apparatus of claim 81 wherein the output circuit comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

84. (Withdrawn) The apparatus of claim 83 wherein the output circuit comprises a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

85. (Withdrawn) The apparatus of claim 83 wherein the first and second switches of the output circuit are provided on a die having the load.

86. (Withdrawn) The apparatus of claim 85 wherein the load comprises a processor core.

87. (Withdrawn) The apparatus of claim 85, comprising control circuitry to operate the first and second switches of the output circuit, wherein the control circuitry is provided on the die having the load.

88. (Withdrawn) The apparatus of claim 81, wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and wherein the apparatus comprises a control circuit to operate the primary switches, and the control circuit is configured to turn OFF one or more of the primary switches essentially at times when the resonant current returns to zero.

89. (Previously Presented) A method of delivering power to an output, comprising:

- providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;
- providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;
- forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;
- forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
- connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;
- connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;
- connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output;
- converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

90. (Previously Presented) The method of claim 89 wherein the unipolar output is approximately 1 Volt.

First Named Inventor : Patrizio Vinciarelli
Serial No. : 13/044,063
Filed : March 9, 2011
Page : 27 of 33

Attorney's Docket No.: 00614-0166001

Amendments to the Drawings:

The attached replacement sheets of drawings include changes to FIGS. 1, 2A, 2B, 5, 6, and 9 and replaces the original sheets including FIGS. 1, 2A, 2B, 5, 6, and 9.

In Figures 1, 2A, 2B, and 9, the legend "Prior Art" has been added as suggested by the Examiner.

In Figures 1, 5, and 6, descriptive text labels have been added as suggested by the Examiner.

Attachments following last page of this Amendment:

Replacement Sheet (5 pages)

REMARKS

The comments of the Applicant below are each preceded by related comments of the Examiner (in small, bold type).

Drawings

**Figures 1, 2A, 2B and 9 should be designated by a legend such as -
-Prior Art -- because only that which is old is illustrated.**

**...
The drawings are objected to because of the following
informalities. Regarding Fig. 1, 5, and 6, the unlabeled rectangular box(es)
shown in the drawings should be provided with descriptive text labels.**

Figures 1, 2A, 2B, 5, 6, and 9 have been amended.

Specification

**The abstract of the disclosure is objected to due to its undue
length. Applicant is reminded of the proper language and format for an
abstract of the disclosure**

The Abstract has been amended.

Claim Objections

**Claims 1, 2 and 5 are objected to because of the following
informalities:
 regarding claim 1, in line 9, "the output" should read as "an
output";
 regarding claim 2, in line 5, "the second polarity is opposite the ...
" should read as "the second polarity is opposite of the ... ";
 regarding claim 5, in line 5, "the second polarity is opposite the ...
" should read as "the second polarity is opposite of the ... ";**

Claims 1, 2, and 5 have been amended.

Claim Rejections – 35 USC § 112

**Claims 18, 20-23, 28, and 58 are rejected under 35 U.S.C. 112(b)
or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for
failing to particularly point out and distinctly claim the subject matter
which the inventor or a joint inventor, or for pre-AIA the applicant
regards as the invention.**

Claims 18, 20, and 58 have been amended.

Claim Rejections – 35 USC §103

Claims 1, 3, 4, 16, 44, 46, and 57 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen et al. (US Patent 7,768,807 B2, hereafter referred to as "Chen") in view of ISL621 0 (Datasheet ISL621 0, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008).

Claim 1

The Examiner acknowledges that Chen does not disclose preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open. The Applicant notes this feature is also not disclosed or suggested by ISL6210.

ISL6210 describes an adaptive shoot-through circuit that monitors the LGATE voltage, such that when LGATE has fallen below 1V, UGATE is allowed to turn ON (page 6). The LGATE voltage is the voltage of the lower gate drive output, which is connected to the gate of the low-side power N-Channel MOSFET (page 5). However, monitoring the LGATE voltage alone does not provide information on whether the MOSFET failed. For example, if the MOSFET malfunctions, the MOSFET may be shorted even when the LGATE voltage has fallen below 1V, and upon detecting that the LGATE voltage has fallen below 1V, the driver in ISL6210 will allow UGATE to turn on. The adaptive shoot-through circuit of ISL6210 prevents the lower and upper MOSFETs from conducting simultaneously if both lower and upper MOSFETs are functioning properly. The adaptive shoot-through circuit of ISL6210 does not prevent the lower MOSFET from turning ON if the upper MOSFET fails to open (e.g., due to failure of the device), and does not prevent the upper MOSFET from turning ON if the lower MOSFET fails to open (e.g., due to failure of the device). Therefore, ISL6210 does not disclose or suggest "preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open," as recited in claim 1.

Because neither Chen nor ISL6210 discloses the feature of claim 1 recited above, claim 1 would not have been made been obvious by Chen and ISL6210. Claim 1 is patentable over Chen and ISL6201 for at least the above reasons.

Claim 44

The Examiner acknowledges that Chen does not disclose a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open. The Applicant notes this feature is also not disclosed or suggested by ISL6210.

ISL6210 describes an adaptive shoot-through circuit that monitors the LGATE voltage, such that when LGATE has fallen below 1V, UGATE is allowed to turn ON (page 6). The LGATE voltage is the voltage of the lower gate drive output, which is connected to the gate of the low-side power N-Channel MOSFET (page 5). However, monitoring the LGATE voltage alone does not provide information on whether the low-side MOSFET failed. For example, if the MOSFET malfunctions, the MOSFET may be shorted even when the LGATE voltage has fallen below 1V; upon detecting that the LGATE voltage has fallen below 1V, the driver in ISL6210 would allow UGATE to turn on. The adaptive shoot-through circuit of ISL6210 prevents the lower and upper MOSFETs from conducting simultaneously if both lower and upper MOSFETs are functioning properly. The adaptive shoot-through circuit of ISL6210 is not configured to prevent the lower MOSFET from turning ON if the upper MOSFET fails to open (e.g., due to failure of the device), and is not configured to prevent the upper MOSFET from turning ON if the lower MOSFET fails to open (e.g., due to failure of the device). Therefore, ISL6210 does not disclose or suggest “a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open,” as recited in claim 44.

Because neither Chen nor ISL6210 discloses the feature of claim 44 recited above, claim 44 would not have been made been obvious by Chen and ISL6210. Claim 44 is patentable over Chen and ISL6201 for at least the above reasons.

Claims 18, and 20 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009).

Claim 18

The Examiner acknowledges that Chen does not disclose monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that that predetermined fault is present.

AN2738 discloses monitoring the supply voltage VCC using an under-voltage lockout circuitry that turns off the IC outputs when the supply voltage goes below a threshold and turns on the device when the supply voltage goes above the threshold (page 7). A person of ordinary skill in the art would have no reason to modify Chen's switching power conversion circuit to use the under-voltage lockout circuitry of AN2738. In Chen, the switches are not powered by a supply voltage, so there is no need to detect whether a supply voltage is below a threshold and then turn off the switches when the supply voltage is below the threshold.

Furthermore, in AN2738, L6390 is a half-bridge gate driver. The conditions for affecting a half-bridge gate driver are quite different from the conditions affecting the full-bridge gate driver described in Chen. It is unclear how the under-voltage lockout circuitry of the half-bridge gate driver of AN2738 can be used in Chen's full bridge switching power conversion circuit. It is also unclear from AN2738 what circuit conditions in Chen's full bridge switching power conversion circuit should be monitored to determine a predetermined fault, such that the switches should be disabled upon determining that the predetermined fault is present.

Because a person of ordinary skill in the art would not have modified Chen full bridge switching power conversion circuit based on the teachings of AN2738, claim 18 would not have been made obvious by the combination of Chen and AN2738.

Claim 18 is patentable over Chen and AN2738 for at least the above reasons.

Claims 89 and 90 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen.

Claim 89

Chen does not disclose and would not have made obvious “each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output,” as recited in claim 89.

In claim 89, each of the first and second switches is arranged in the OFF state, wherein a short across one of the first or second switches does not short the output. This condition is not disclosed in Chen. Chen discloses operation conditions in which the switches are all functioning properly, such that when a switch is arranged in the OFF state, there would not be a short across the switch. Chen does not describe a situation (e.g., where a switch fails) such that when the switch is arranged in the OFF state, there is still a short across the switch.

Similarly, Chen does not disclose or suggest “each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output,” as recited in claim 89.

Because Chen does not disclose or suggest the above features of claim 89, claim 89 is patentable over Chen.

Allowable Subject Matter

Claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The Applicant acknowledges that claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58 are allowable, but notes that the other claims are also allowable for the reasons stated above.

All of the dependent claims are patentable for at least the reasons for which the claims on which they depend are patentable.

First Named Inventor : Patrizio Vinciarelli
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Any circumstance in which the Applicant has addressed certain comments of the Examiner does not mean that the Applicant concedes other comments of the Examiner. Any circumstance in which the Applicant has made arguments for the patentability of some claims does not mean that there are not other good reasons for patentability of those claims and other claims. Any circumstance in which the Applicant has amended a claim does not mean that the Applicant concedes any of the Examiner's positions with respect to that claim or other claims.

The fees for excess claims and extension of time are being paid with this reply on the Electronic Filing System. Apply those fees and any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: October 23, 2014_____

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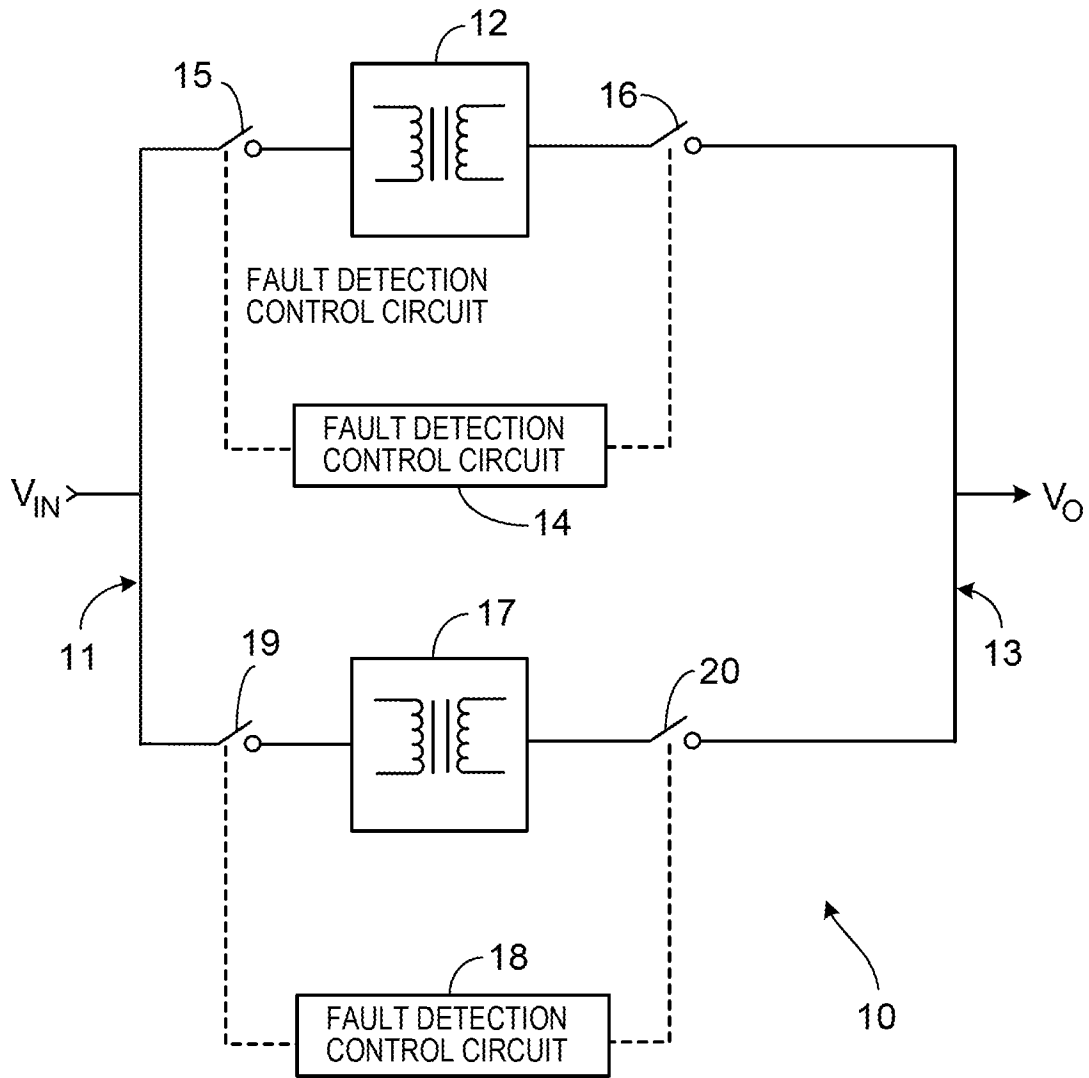
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ABSTRACT

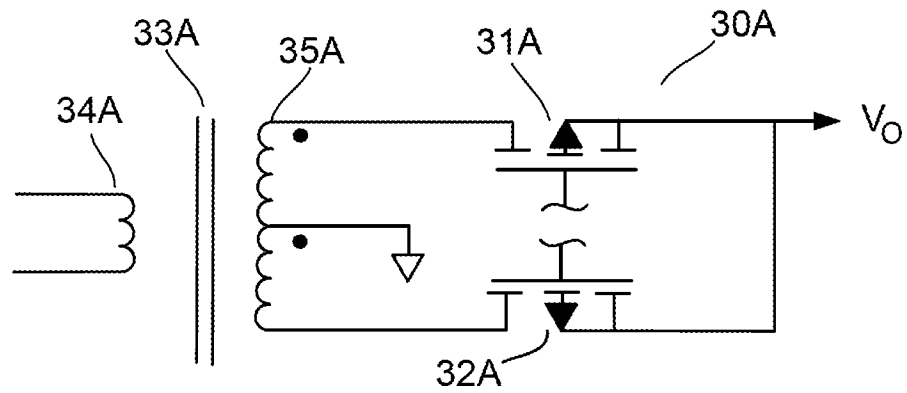
A power converter provides a low-voltage output using a full-bridge fault-tolerant rectification circuit. The output circuit uses controlled switches as rectifiers. A fault detection circuit monitors circuit conditions. Upon detection of a fault, the switches are disabled decoupling the power converter from the system.

A common-source dual MOSFET device includes a plurality of elements arranged in alternating patterns on a semiconductor die. A common-source dual synchronous rectifier includes control circuitry powered from the drain to source voltage of the complementary switch.

A DC-to-DC transformer converts power from an input source to a load using a fixed voltage transformation ratio. A clamp phase may be used to reduce power losses in the converter at light loads, control the effective output resistance of the converter, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage for example during start up.

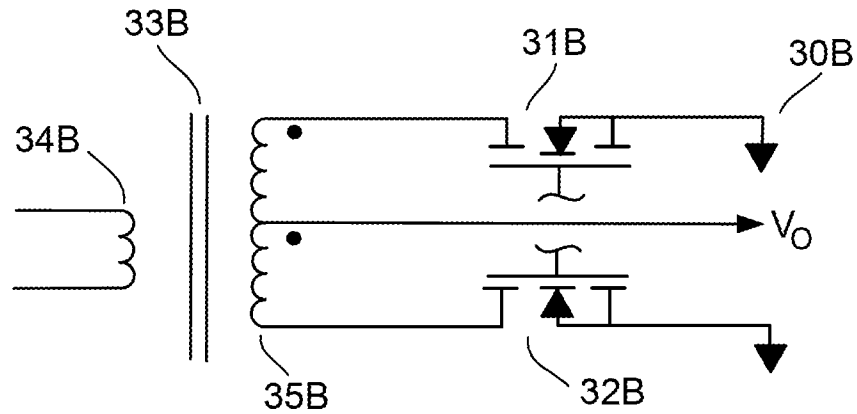


(PRIOR ART)
FIG. 1



(PRIOR ART)

FIG. 2A



(PRIOR ART)

FIG. 2B

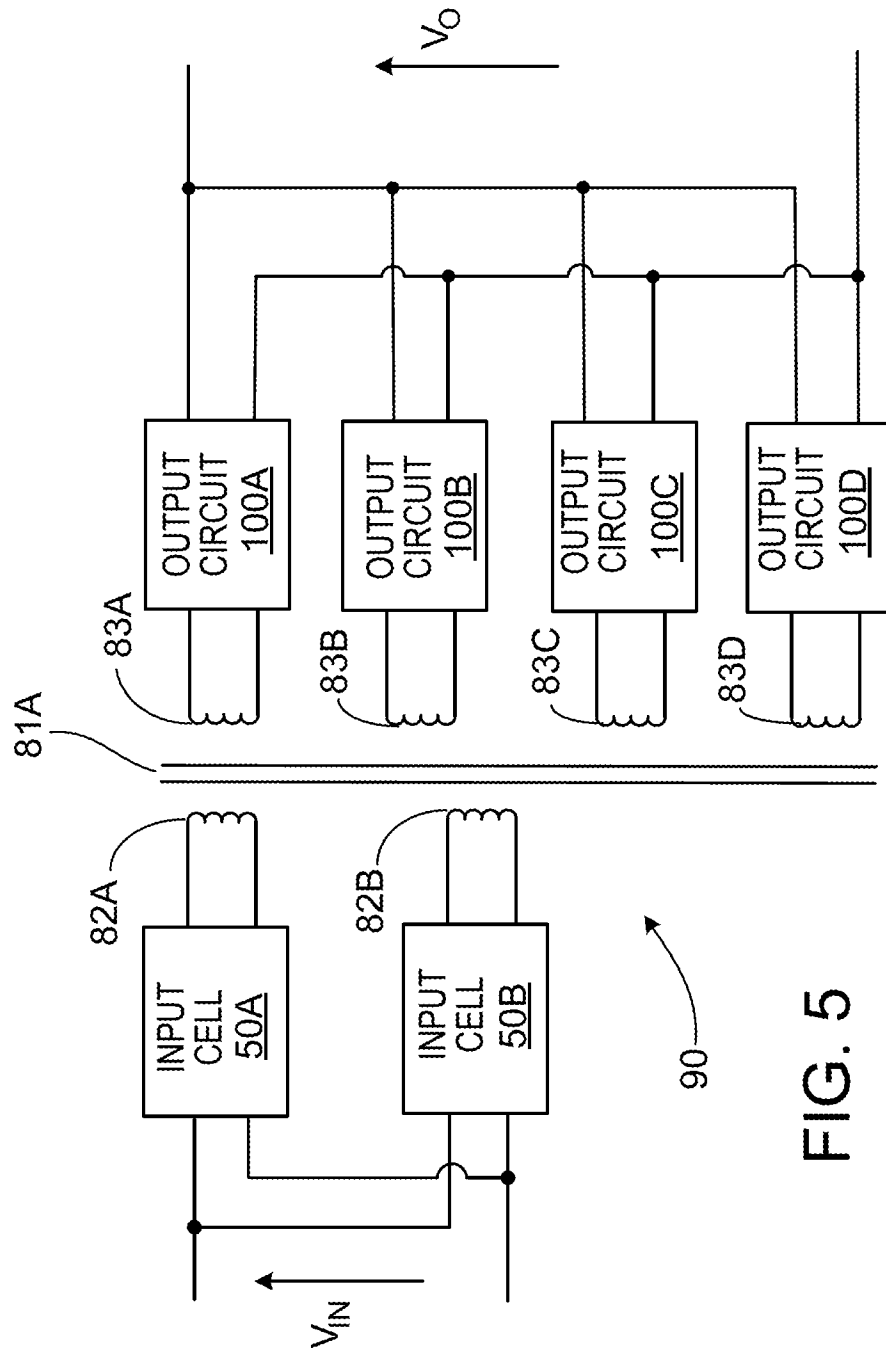


FIG. 5

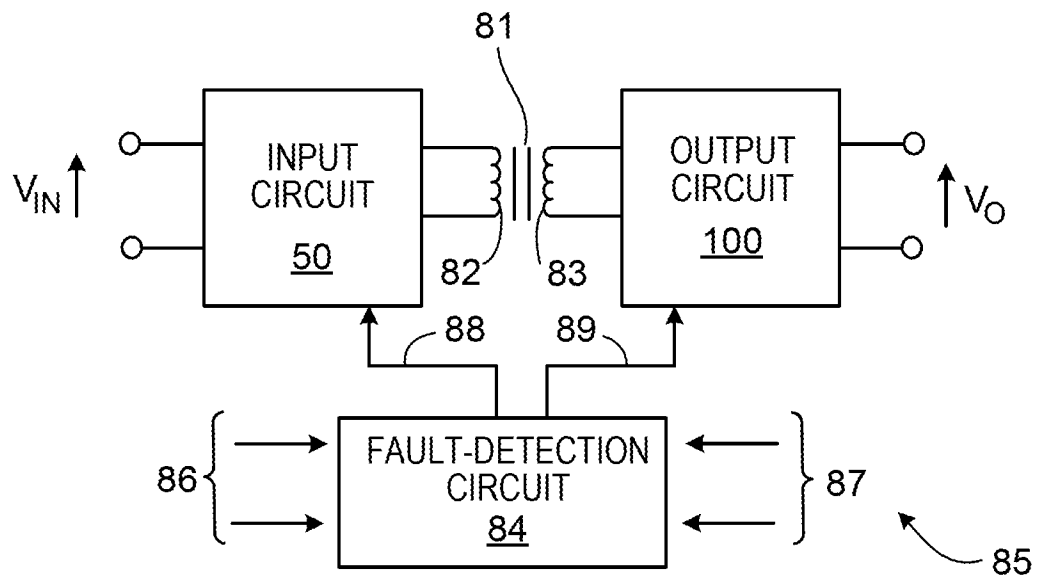
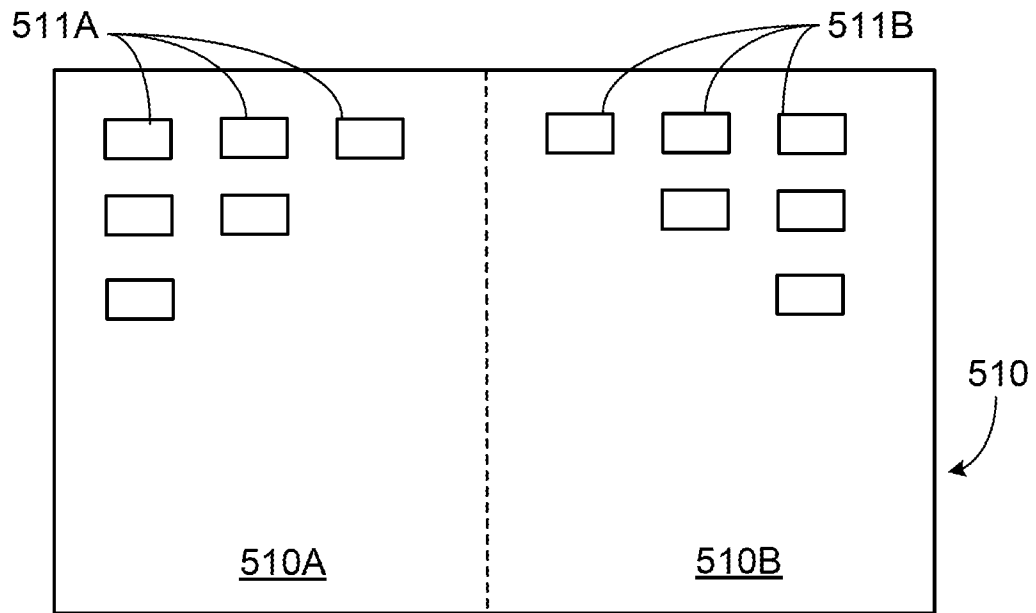


FIG. 6



(PRIOR ART)
FIG. 9

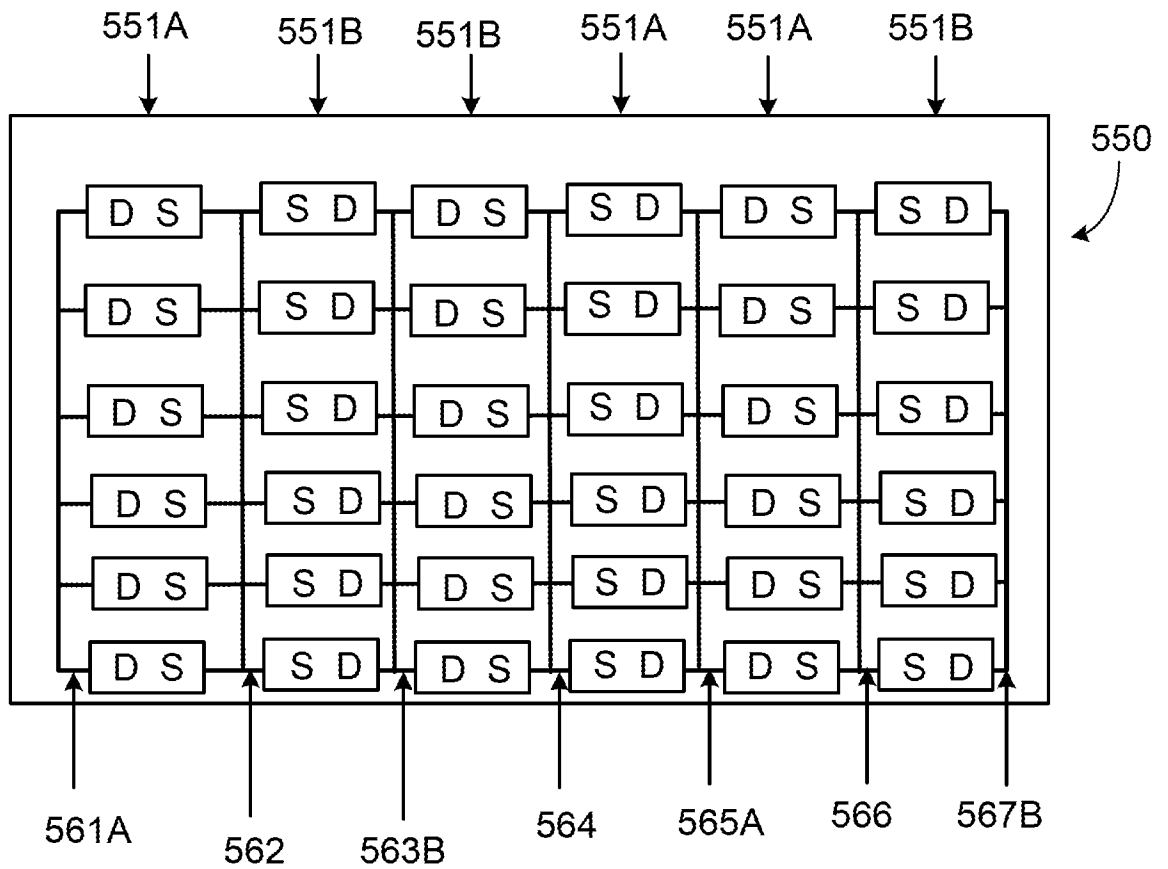


FIG. 10

Analysis and Optimization of Switched-Capacitor DC–DC Converters

Michael D. Seeman, *Student Member, IEEE*, and Seth R. Sanders, *Member, IEEE*

Abstract—Analysis methods are developed that fully determine a switched-capacitor (SC) dc–dc converter's steady-state performance through evaluation of its output impedance. This analysis method has been verified through simulation and experimentation. The simple formulation developed permits optimization of the capacitor sizes to meet a constraint such as a total capacitance or total energy storage limit, and also permits optimization of the switch sizes subject to constraints on total switch conductances or total switch volt-ampere (V-A) products. These optimizations then permit comparison among several switched-capacitor topologies, and comparisons of SC converters with conventional magnetic-based dc–dc converter circuits, in the context of various application settings. Significantly, the performance (based on conduction loss) of a ladder-type converter is found to be superior to that of a conventional magnetic-based converter for medium to high conversion ratios.

Index Terms—Analysis, dc–dc converter, output impedance, switched-capacitor.

I. INTRODUCTION

THIS paper develops analysis methods that fully determine a switched-capacitor (SC) dc–dc converter's steady-state performance through evaluation of its output impedance. This resistive impedance is a function of frequency and has two asymptotic limits: one where resistive paths dominate the impedance, and another where charge transfers among idealized capacitors dominate the impedance. This work develops a network theoretic analysis of these two asymptotic limits, which can be used to evaluate both the converter efficiency and output regulation as a function of load for a broad class of SC converters. Simulations and experiments have been performed to verify the analysis methods.

The comprehensive analysis and design calculations given here are new, but connect with the analysis framework developed in the pioneering work of [1]. The work in [1], [2] offered a network theoretic formulation for computation of open-circuit dc–dc conversion ratios, and a rather involved method for computation of output impedance. Reference [1] and other previous analysis work [3]–[7] mainly focused on the performance analysis (i.e., output impedance computation) for a single converter topology.

The simple formulation developed permits optimization of the capacitor sizes to meet a constraint such as a total capacitance or total energy storage limit, and also permits optimization

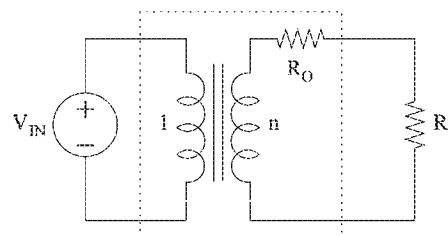


Fig. 1. Model of an idealized switched-capacitor converter.

of the switch sizes subject to constraints on total switch conductances or total switch volt-ampere (V-A) products. These optimizations are carried out for a set of representative switched-capacitor topologies. These optimizations then permit comparison among several switched-capacitor topologies, and comparisons of SC converters with conventional magnetic-based dc–dc converter circuits. The performance (based on conduction loss) of a ladder-type converter is found to be superior to that of a conventional boost converter for medium to high conversion ratios.

II. SWITCHED-CAPACITOR CONVERTER IMPEDANCE ANALYSIS

With the model in Fig. 1, the converter provides an ideal dc voltage conversion ratio under no load conditions, and all conversion losses are manifested by voltage drop associated with non-zero load current through the output impedance [1], [5]. The resistive output impedance accounts for capacitor charging and discharging losses and resistive conduction losses. Additional losses due to short-circuit current and parasitic capacitances, in addition to gate-drive losses, can be incorporated into the model. However, they will not be considered here since these effects are generally application and implementation dependent. Insight gained can be used to model effects of parasitic capacitances [8]. For the present, our aim is to provide a general analysis and design framework.

The low-frequency output impedance R_O in Fig. 1 sets the maximum converter power, constrained by a minimal efficiency objective, and also determines the open-loop load regulation properties. There are two asymptotic limits to output impedance, the slow and fast switching limits, as related to switching frequency. The slow switching limit (SSL) impedance is calculated assuming that the switches and all other conductive interconnects are ideal, and that the currents flowing between input and output sources and capacitors are impulsive, modeled as charge transfers. The SSL impedance is inversely proportional to switching frequency. The fast switching limit (FSL) occurs when the resistances associated with switches, capacitors and interconnect dominate, and the capacitors act

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effectively as fixed voltage sources. In the FSL, current flow occurs in a frequency-independent piecewise constant pattern.

The set of converters considered in this paper is limited to two-phase converters made solely of ideal capacitors, resistive switches, and input and output voltage sources. Two-phase converters switch alternately between two configurations. Multi-phase converters [4] are outside the scope of this paper, but can be considered using similar methods. This paper does not address the more fundamental topological conditions needed to determine whether or not a specific circuit constitutes a well-formulated two-phase converter. Rather, the paper assumes that the circuits under consideration all have well-defined two-phase operation. References [1], [2] begin to address the topological question of what constitutes a well-formulated two-phase SC dc-dc circuit, though the characterization given is not complete.

A. Slow-Switching Limit Impedance

For the slow-switching limit (SSL) impedance analysis, the finite resistances of the switches, capacitors, and interconnect are neglected. A pair of charge multiplier vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ can be derived for any standard non-degenerate two-phase SC converter. The charge multiplier vectors correspond to charge flows that occur immediately after the switches are closed to initiate each respective phase of the SC circuit. Each element of a charge multiplier vector corresponds to a specific capacitor or independent voltage source, and represents the charge flow into that component, normalized with respect to the output charge flow. As outlined in [1], the charge multiplier vectors can be uniquely computed using the KCL constraints in each topological phase and the constraint that the two charge multiplier quantities on each capacitor are equal and opposite.

The charge multiplier vector $\mathbf{a}^1$ is defined as

$$\mathbf{a}^1 = [q_{out}^1 \ q_1^1 \ \cdots \ q_n^1 \ q_{in}^1]^T / q_{out} \quad (1)$$

where each component is the ratio of charge transfer in each element during phase 1 of the switching period to the charge delivered to the output during a full period. If charge flows into the element during phase 1, the corresponding entry in the $\mathbf{a}^1$ vector is positive. Vector $\mathbf{a}^2$ is defined analogously, for phase 2. The charge multiplier vector can be partitioned into output, capacitor and input components, respectively

$$\mathbf{a}^1 = [a_{out}^1 \ \mathbf{a}_c^1 \ a_{in}^1]^T. \quad (2)$$

For the ladder network example of Fig. 2, the charge multiplier vectors can be obtained through network analysis using Kirchhoff's Current Law (KCL) [1]. In this example, and in all other examples encountered by the authors, the charge multiplier vectors can be obtained by inspection (in Fig. 3). The charge from the input source flows into C4 during phase 2. In phase 1, that charge is transferred into C3. By considering alternating phases, the charge flow in each component can be found

$$\mathbf{a}^1 = [2/3 \ -2/3 \ 1/3 \ -1/3 \ 0]^T \quad (3)$$

$$\mathbf{a}^2 = [1/3 \ 2/3 \ -1/3 \ 1/3 \ -1/3]^T. \quad (4)$$

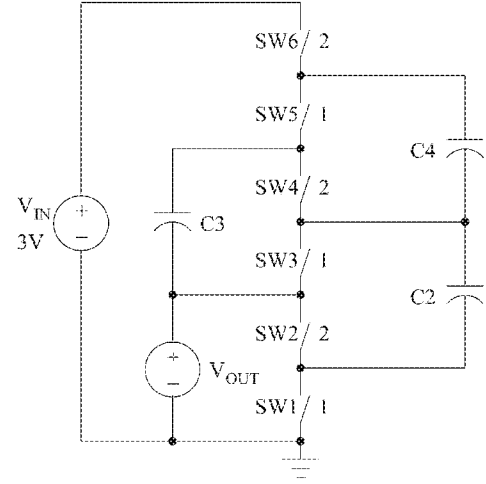


Fig. 2. 3 V to 1 V ladder circuit.

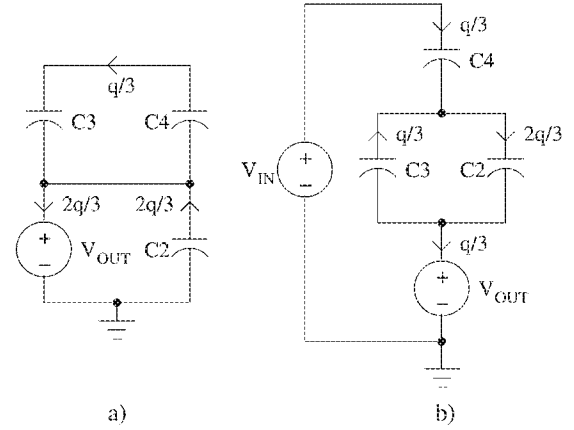


Fig. 3. Charge flow in ladder converter. (a) phase 1 (b) phase 2.

In each of these charge multiplier vectors, the first component corresponds to the output charge flow, thus these two components must sum to one. The last component of each charge multiplier vector corresponds to the charge flow into the input source, and is non-zero during only phase 2 in this example.

The charge multiplier vectors, the capacitor characteristics, and the switching frequency are the only data needed to determine the output impedance under the asymptotic SSL condition. The calculation, developed here, is based on Tellegen's Theorem [9] which states that for any network, any vector of branch voltages that satisfies KVL is orthogonal to any vector of branch currents (or equivalently charge flows) that satisfies KCL. This theorem is applied in each of the two configurations for a two-phase switched capacitor converter operating in periodic steady state, where the input is short-circuited and the output is connected to an independent dc voltage source. The charge flow per period (or average current flow) into the single independent source then defines the output impedance.

Application of Tellegen's theorem to the switched capacitor converter, in each of its two configurations, yields $\mathbf{a}^1 \cdot \mathbf{v}^1 = 0$ and $\mathbf{a}^2 \cdot \mathbf{v}^2 = 0$, where $\mathbf{v}^1$ and $\mathbf{v}^2$ are the respective steady

state network voltage vectors in phases 1 and 2. Additively combining these two applications of Tellegen's theorem, and noting that the input voltage source has value zero, yields

$$v_{\text{out}} (a_{\text{out}}^1 + a_{\text{out}}^2) + \sum_{\text{capacitors}} (a_{c,i}^1 v_{c,i}^1 + a_{c,i}^2 v_{c,i}^2) = 0 \quad (5)$$

where the first term corresponds to the constant output voltage source and the terms under the summation correspond to the capacitor branches. Recall that $a_{\text{out}}^1 + a_{\text{out}}^2 = 1$ and that $a_{c,i}^1 = -a_{c,i}^2$ for each capacitor branch (due to charge conservation in periodic steady-state). By defining $a_{c,i} = a_{c,i}^1 = -a_{c,i}^2$ and $q_i = a_{c,i} q_{\text{out}}$ and multiplying (5) by q_{out} , the net charge delivered to the output in a period, we obtain

$$q_{\text{out}} v_{\text{out}} + \sum_{\text{capacitors}} q_i \Delta v_i = 0 \quad (6)$$

where $\Delta v_i = v_{c,i}^1 - v_{c,i}^2$. In (6), the first term corresponds to the product of the constant output voltage and the total charge flow into this independent voltage source, and each term in the summation corresponds to energy loss associated with a specific capacitor. It is of direct interest here that none of the capacitor voltages need to be explicitly calculated for this analysis. Rather, Δv_i can be computed from

$$\Delta v_i = q_i / C_i \quad (7)$$

where C_i is the capacitance value of the i th capacitor, assuming linear capacitors. Introducing (7) into (6), and then dividing the result by q_{out}^2 yields

$$\frac{v_{\text{out}}}{q_{\text{out}}} + \sum_{\text{capacitors}} \left(\frac{q_i}{q_{\text{out}}} \right)^2 \frac{1}{C_i} = 0. \quad (8)$$

We note that q_i/q_{out} corresponds to the i th entry of the charge multiplier vector $\mathbf{a}_c$, since these entries are for the capacitors. Dividing (8) by the switching frequency then directly yields the average output impedance for the slow-switching asymptotic limit

$$R_{\text{SSL}} = -\frac{v_{\text{out}}}{i_{\text{out}}} = \sum_i \frac{(a_{c,i})^2}{C_i f_{\text{sw}}}. \quad (9)$$

The converter's loss in terms of the series output impedance R_{SSL} can be expressed in terms of capacitor loss. The product $q_i \Delta v_i$ in (6) represents the energy loss by charging and discharging capacitor i in each cycle, and could be used to calculate the converter's loss even with a nonlinear capacitor. In the following discussion, attention is restricted to the case of linear capacitors. The sum of the energy lost through the capacitors is equal to the calculated loss associated with the output impedance for a given load.

This powerful result yields a simple calculation of this asymptotic output impedance and some intuition into the operation of SC converters. The output impedance directly models the losses in the circuit due to capacitor charging and discharging. This

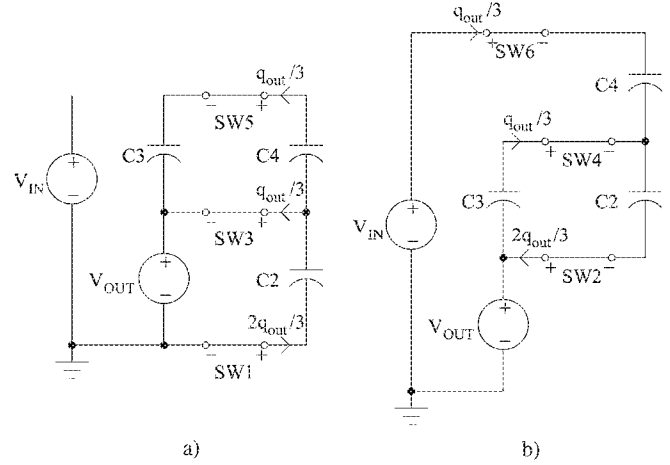


Fig. 4. Switch charge flow in ladder converter. (a) Phase 1, (b) Phase 2.

impedance can be determined by simply examining the charge flow in the converter without simulation or complicated network analysis.

B. Fast Switching Limit Impedance

The other asymptotic limit, the fast switching limit (FSL), is characterized by constant current flows between capacitors. The switch on-state impedances and other resistances are sufficiently large such that during each phase, the capacitors do not approach equilibrium. In the asymptotic limit, the capacitor voltages are modeled as constant. The circuit loss is related only to conduction loss in resistive elements. The concept of the FSL impedance is introduced informally in [5].

The duty cycle of the converter is important when considering the FSL impedance since currents flow during the entirety of each phase. For this analysis, a duty cycle of 50% is assumed for simplicity. Duty cycle differing from 50% can be included in the following analysis without much difficulty if another duty cycle is used. Additionally, only the on-state switch resistance is considered; other parasitic resistance [e.g., capacitor equivalent series resistance (ESR)] can be similarly incorporated into the model if desired.

The $a_{r,i}$ values are defined as the charge flow through each switch during the phase in which the switch is on. Even in the FSL, the charge flows must follow the same pattern as in the SSL, constrained by $\mathbf{a}^1$ and $\mathbf{a}^2$. For the switches that are on during phase 1, the corresponding $a_{r,i}$ values can be determined from the $\mathbf{a}^1$ components. Analogously, corresponding $a_{r,i}$ values for switches that conduct during phase 2 can be determined from the $\mathbf{a}^2$ components. The values of $a_{r,i}$ are independent of duty cycle as they simply represent the charge flow through the switches that ensure charge conservation on the circuit's capacitors. The $a_{r,i}$ values for the switches in the ladder converter in Fig. 2 can be determined directly. The charge flows in the switches during both phases are shown in Fig. 4, resulting in an $\mathbf{a}_r$ vector of

$$\mathbf{a}_r = [-2/3 \quad -2/3 \quad 1/3 \quad 1/3 \quad 1/3 \quad 1/3]. \quad (10)$$

For positive power flow (i.e., from the input to output source), the sign of each component of the $\mathbf{a}_r$ vector indicates the direction of current flow with respect to the blocking voltage of a switch. A positive quantity indicates the switch conducts positive current while on and blocks positive voltage while off. This switch must be implemented using an active transistor. A negative quantity indicates the switch conducts negative current and blocks positive voltage and is suitable for diode implementation (if the forward voltage drop is tolerable). For power flow in the opposite direction, the switch types reverse.

In the FSL, the current through the on-state switches is assumed to be constant. Given the charge flow vector, the current in each switch is easily determined

$$i_{r,i} = 2q_{r,i}f_{sw} \quad (11)$$

where $q_{r,i}$ is the charge flow through switch i during a single period. The factor of two appears because of the 50% duty cycle. Substituting $q_{r,i} = a_{r,i}q_{out}$ and $q_{out} = i_{out}/f_{sw}$ into (11) yields

$$i_{r,i} = 2a_{r,i}i_{out}. \quad (12)$$

The current through the switches is only dependent on the vector $\mathbf{a}_r$, which is obtainable by inspection. The network voltages never need to be found in this analysis, simplifying computation significantly.

The average power loss due to each individual switch is equal to the instantaneous on-state power loss multiplied by its duty cycle. Since the total loss of the SC converter in the FSL is just the sum of the switch losses, the total circuit loss is given by

$$P_{FSL} = \sum_{\text{switches}} \frac{1}{2} R_i (2a_{r,i}i_{out})^2 \quad (13)$$

where R_i is the on-state resistance of switch i .

Since the input and output charge flow in the SC converter is constrained by the conversion ratio n , all the power loss in an ideal SC converter (as analyzed here) is modeled by the output voltage drop. Thus the output impedance can be determined by equating the actual power loss of the circuit with the apparent power loss due to the output impedance. Since this power loss is proportional to the square of the output current, the FSL output impedance can be obtained by inspection

$$R_{FSL} = 2 \sum_i R_i (a_{r,i})^2. \quad (14)$$

Similar to the SSL output impedance in (9), the FSL output impedance is given simply in terms of component parameters and the switch charge multiplier coefficients of each switch. The power loss due to these conduction losses is equal to the equivalent power loss through the output impedance. These two simple forms of the output impedance (given in (9) for the SSL and (14) for the FSL) can be used to provide strong guidance for the design of switched-capacitor power converters.

III. COMPONENT OPTIMIZATION

Given that all converter losses attributed to the capacitors and resistive switches can be reflected in the computation of a

single real output resistance, it is now possible to optimize the components in order to minimize that output impedance. Minimal output impedance corresponds to maximum efficiency for a given power delivered, and dually, corresponds to maximum power delivery for a given loss. This section develops optimality computations for the slow switching limit (SSL) and the fast switching limit (FSL) impedance. When optimizing over capacitances, one should minimize the output impedance that is associated only with the capacitances, namely the SSL impedance. Analogously, when optimizing over switch sizes, one should minimize the FSL output impedance. The final design step is to choose a maximum operating frequency for which the parasitic losses are acceptable. The total capacitance and switch conductance should be adjusted such that the total impedance meets the design goal and the SSL and FSL impedances are balanced. The last step ensures that the total switch and capacitor sizes (and costs) are minimized for the intended power level.

The optimization procedure requires knowledge of the component working voltages, unlike the output impedance analysis. The working voltage for a capacitor is the maximum voltage on the capacitor during steady-state converter operation. For a transistor (switch), the working voltage is the voltage it blocks during steady-state converter operation. For open-circuit operation, these working voltages can be found by inspection in most examples, or by the process outlined in [1]. This analysis is based on combining KVL constraints for the two phase topologies, in combination with a known source voltage. The result is the computation of vectors denoted $\mathbf{v}_c$ and $\mathbf{v}_r$ for the working voltages of the capacitors and switches, respectively, ratioed to the converter output voltage.

The optimization is based on a physical size (or cost) constraint for the devices. When capacitors are optimized, their total energy storage capability is held constant. Or, in the case when all capacitors must be rated for the same voltage, the total capacitance is held constant. Likewise, when the switch sizes are optimized, the total V-A capacity product is held constant. This V-A metric translates to a constraint on the G-V² products summed over the switches (G refers to switch conductance). If all the switches are rated for the same voltage, the constraint reduces to holding the sum of the switch conductances constant.

A. SSL Capacitor Optimization

The capacitor optimization uses a constraint that holds the total energy storage capability, summed over all capacitors, fixed to a constant E_{tot} . This constraint can be mathematically expressed as

$$\sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i = E_{tot} \quad (15)$$

where C_i represents the value of capacitor i and $v_{c,i(\text{rated})}$ represents the rated voltage of capacitor i . The energy storage capability of a capacitor is related to its rated voltage, as that dictates its size and cost, not the maximum voltage it sees during operation. However, the capacitor's working voltage must be less than the rated voltage to avoid damaging the component, and should be close to the rated voltage to achieve good utilization of the device.

A function $\mathcal{L}$ is defined to perform the constrained optimization

$$\mathcal{L} = \sum_i \frac{(a_{c,i})^2}{C_i} + \lambda \left(\sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i - E_{\text{tot}} \right) \quad (16)$$

where the first term represents the SSL output impedance (scaled by switching frequency as it does not effect the minimization) and the second term incorporates the constraint in (15). The impedance is minimized by equating the partial derivatives of $\mathcal{L}$ with respect to C_i and λ with zero

$$\frac{\partial \mathcal{L}}{\partial C_i} = -\frac{(a_{c,i})^2}{C_i^2} + \lambda \frac{1}{2} (v_{c,i(\text{rated})})^2 = 0 \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i - E_{\text{tot}} = 0. \quad (18)$$

Equation (18) simply repeats the constraint in (15).

The relationship in (17) sets up a proportionality between C_i , $a_{c,i}$, and $v_{c,i(\text{rated})}$. The energy constraint can be used to find an expression for the value of each capacitor

$$C_i = \left| \frac{a_{c,i}}{v_{c,i(\text{rated})}} \right| \frac{2E_{\text{tot}}}{\sum_k |a_{c,k} v_{c,k(\text{rated})}|}. \quad (19)$$

The optimal energy storage of each capacitor is proportional to the V - Q product of each capacitor

$$E_i = \frac{|a_{c,i} v_{c,i(\text{rated})}|}{\sum_k |a_{c,k} v_{c,k(\text{rated})}|} E_{\text{tot}}. \quad (20)$$

When the total energy is constrained, the optimal capacitor energies are proportional to the product of their rated voltage and their charge multiplier coefficients. In addition, the ripple voltage on each capacitor is directly proportional to that capacitor's rated voltage.

The optimized output impedance can be calculated by combining (9) and (19)

$$R_{\text{SSL}}^* = \frac{1}{2E_{\text{tot}} f_{\text{sw}}} \left(\sum_i |a_{c,i} v_{c,i(\text{rated})}| \right)^2. \quad (21)$$

By optimizing the capacitors, the output impedance becomes proportional to the square of the sum of the products of voltages and charge flows (V - A product) of each capacitor. The optimization can improve the performance of an SC converter designed in an ad-hoc manner significantly, especially one with a large conversion ratio.

If all capacitors in a SC converter are rated for the same voltage, as in the ladder topology or in applications with integrated capacitors, the optimization results can be simplified. In this case, we constrain total capacitance to a value of C_{tot} , and the value of each individual capacitor is given by

$$C_i^* = \frac{|a_{c,i}|}{\sum_k |a_{c,k}|} C_{\text{tot}}. \quad (22)$$

Each capacitor is sized proportionally to its charge multiplier coefficient. With optimized capacitors, the voltage ripple on each capacitor is set equal in magnitude.

The optimized SSL output impedance (from (21)) thus simplifies to

$$R_{\text{SSL}}^* = \frac{1}{C_{\text{tot}} f_{\text{sw}}} \left(\sum_i |a_{c,i}| \right)^2. \quad (23)$$

These optimization results for the single-voltage technology are very simple to utilize in switched-capacitor converter design.

B. FSL Switch Optimization and Sizing

Like capacitors, the switches in a SC converter can be optimized, yielding dramatic performance increases. This optimization is carried out in the asymptotic fast switching limit where output impedance is directly related to switch conductance. This optimization assumes a duty cycle of 50%.

The switch VA product, summed over all switches, is used as the cost-based metric in this optimization. This V - A metric corresponds to a constraint on the G - V^2 product summed over the switches, for both discrete and integrated switches. Parallel discrete switches increases total conductance, whereas placing switches in series increases voltage blocking while decreasing conductance. To increase voltage blocking without reducing conductance, the number of devices used scales quadratically, motivating the G - V^2 metric.

In an integrated application, the same total G - V^2 constraint applies. The transistor length and nominal voltage scale linearly with process feature size. In addition, switch conductance scales proportionally with transistor width and inversely with transistor length. A cost metric A_{sw} , related to the area (or width multiplied by length) of a specific transistor, can be written as $A_{\text{sw}} = GV^2$ (in units of GV^2 , i.e., S- V^2).

This constraint, applicable to both discrete and integrated transistors, can be expressed as

$$A_{\text{tot}} = \sum_{\text{switches}} G_i (v_{r,i(\text{rated})})^2 \quad (24)$$

where G_i is the conductance of switch i and $v_{r,i(\text{rated})}$ is the rated voltage of switch i . As in the capacitor optimization, $v_{r,i(\text{rated})}$ is the voltage the device can support, not necessarily the voltage it blocks in normal operation. Naturally, the rated voltage must be larger than the nominal blocking voltage.

A Lagrange optimization function $\mathcal{L}$ is formed to minimize the FSL output impedance while satisfying the constraint in (24)

$$\mathcal{L} = \sum_i \frac{(a_{r,i})^2}{G_i} + \lambda \left(\sum_{\text{switches}} G_i (v_{r,i(\text{rated})})^2 - A_{\text{tot}} \right). \quad (25)$$

The first term corresponds to the FSL output impedance (the factor of two in (14) does not affect the optimization) and the second term corresponds to the constraint in (24). The minimization is performed by taking the partial derivative of (25) with respect to G_i and setting it to zero

$$\frac{\partial \mathcal{L}}{\partial G_i} = -\frac{(a_{r,i})^2}{G_i^2} + \lambda (v_{r,i(\text{rated})})^2 = 0. \quad (26)$$

Again, differentiating with respect to λ yields the constraint in (24).

Equation (26) yields a proportionality between G_i and the ratio between the switch's charge multiplier coefficient and its voltage rating. This proportionality, when combined with the G - V^2 constraint in (24), yields an expression for the optimal conductance of each switch

$$G_i^* = \frac{1}{R_i^*} = \left| \frac{a_{r,i}}{v_{r,i}(\text{rated})} \right| \frac{A_{\text{tot}}}{\sum_k |a_{r,k} v_{r,k}(\text{rated})|}. \quad (27)$$

Comparing the optimal conductance G_i to the optimal capacitance in (19) makes it evident that the two optimizations are analogous.

The optimal FSL output impedance is obtained by substituting (27) into (14)

$$R_{\text{FSL}}^* = \frac{2}{A_{\text{tot}}} \left(\sum_i |a_{r,i} v_{r,i}(\text{rated})| \right)^2. \quad (28)$$

Similar to the optimal SSL impedance, the optimal FSL output impedance is related to the square of the sum of the V-A products. This simple form of the optimal output impedance allows the comparison of various SC converter topologies. Several SC converter topologies are compared in Section IV.

Many SC converters use switches with a single voltage rating. For instance, many IC-based converters only use the native NMOS transistors of the process since these transistors perform the best. In addition, topologies such as the ladder converter utilize switches that must all block the same voltage. The switch-cost constraint discussed in the previous section simplifies into a constraint on total switch conductance G_{tot} . The optimal conductance of each switch simplifies to

$$G_i^* = \frac{|a_{r,i}|}{\sum_k |a_{r,k}|} G_{\text{tot}} \quad (29)$$

likewise, when all switches are rated for an identical voltage, the optimal FSL output impedance simplifies to

$$R_{\text{FSL}}^* = \frac{2}{G_{\text{tot}}} \left(\sum_i |a_{r,i}| \right)^2. \quad (30)$$

The performance of a converter is related to the square of the sum of the charge multiplier coefficients. Topologies with a small sum of these coefficients perform better for a given switch conductance than a topology with a large sum of coefficients. In integrated applications or other applications where single-voltage switches must be used, this optimization can be used. A comparison of SC converters based on single-voltage devices is performed in Section IV.

IV. COMPARISON OF SC CONVERTER TOPOLOGIES

A number of SC converter topologies exist in the literature [1], [3]–[5], [10], [11] but the merits of each have never been compared in a methodical way. The optimizations in Sections III-A and III-B can be used to provide a performance comparison among different common SC converter topologies. Fig. 5 shows five converter topologies discussed in the literature.

The first comparison uses the cost metrics in Section III and assumes that devices of every voltage rating are available.

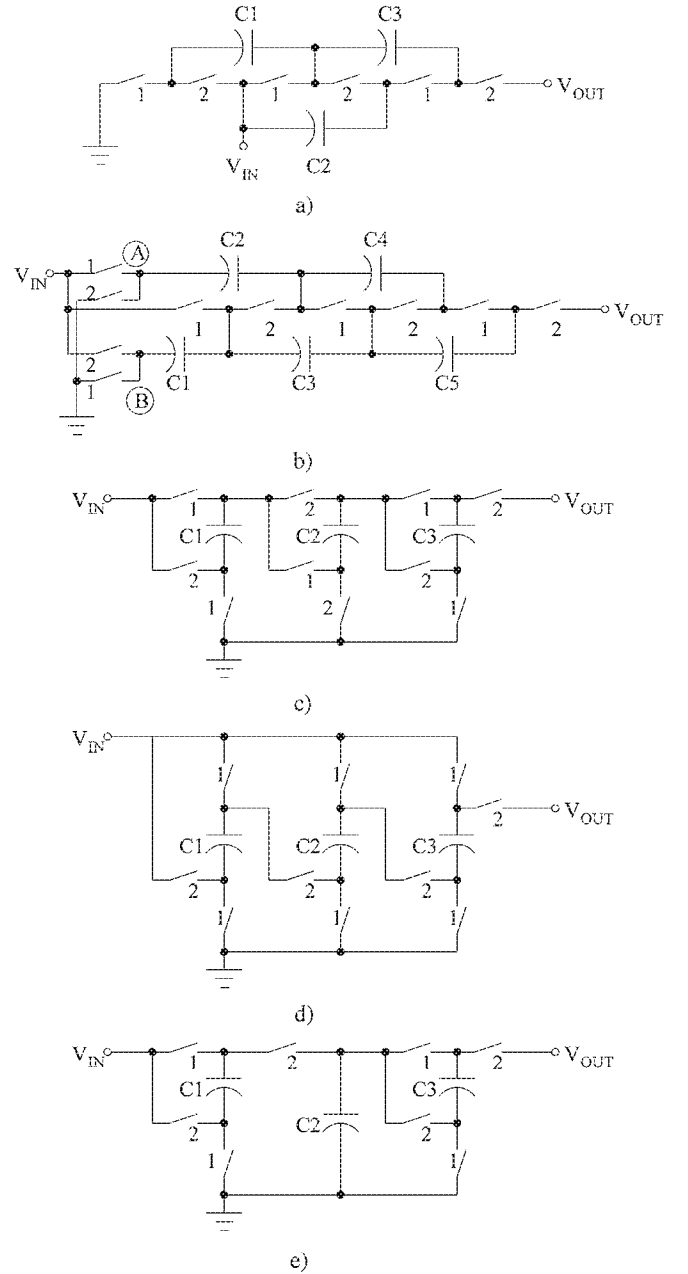


Fig. 5. Five step-up SC converter topologies. (a) Ladder. (b) Cockcroft–Walton Multiplier. (c) Fibonacci. (d) Series-Parallel. (e) Doubler.

Step-up versions of the topologies are considered, as shown in Fig. 5, although step-down versions would yield identical results. The commonly-used Dickson Charge Pump is a simple transformation of the Cockcroft–Walton multiplier, constructed by connecting the negative plate of each capacitor to either node A or B. The capacitors in the Dickson charge pump form two star networks while the capacitors in the Cockcroft–Walton multiplier form two linear strings. The optimal output impedance for all topologies in both asymptotic limits is evaluated for a range of conversion ratios (represented by n).

When evaluating the FSL output impedance, the converters are evaluated on the ratio $(V_{\text{OUT}}^2 / R_{\text{FSL}} A_{\text{tot}})$ (the ratio between the G - V^2 product of the converter and the switch G - V^2 product

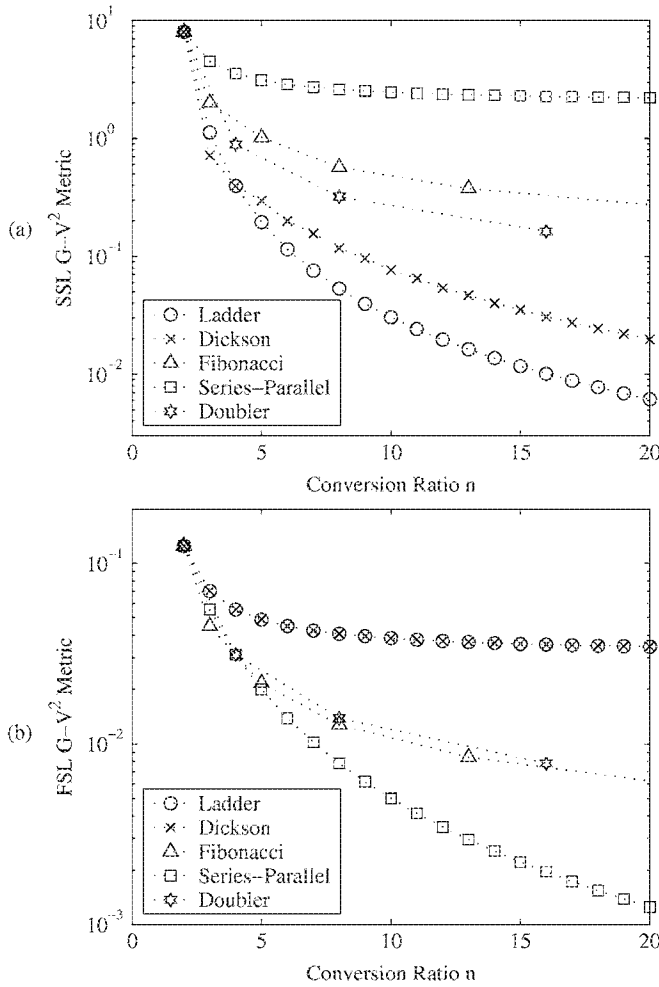


Fig. 6. (a) SSL and (b) FSL performance metrics with optimal-voltage devices.

summed over all switches). For a given cost constraint and conversion ratio, the converter with the highest metric is the one with the lowest FSL output impedance. Likewise, when the SSL output impedance is considered, the converters are evaluated on the ratio $V_{OUT}^2 / (R_{FSL} / E_{tot} f_{sw})$, where a larger metric corresponds to a smaller SSL impedance.

After performing the optimization and comparison, the five topologies are compared in Fig. 6. At a conversion ratio of two, all topologies perform identically. Upon further inspection, for $n = 2$ only, these five topologies are actually identical. Converters that do well in the SSL comparison, such as the series-parallel topology, do poorly in the FSL comparison. Conversely, topologies such as the Cockcroft-Walton multiplier and the Ladder topology that perform well in the FSL comparison typically perform poorly in the SSL comparison. Some converters use capacitors efficiently and others use switches efficiently, but none of these converters are superior in both asymptotes. For converters designed using a capacitor-limited process, a series-parallel topology would work best, while switch-limited designs should use a topology such as the Cockcroft-Walton multiplier or ladder topology.

The exponential converters (where the conversion ratio is exponentially related to the number of capacitors), such as

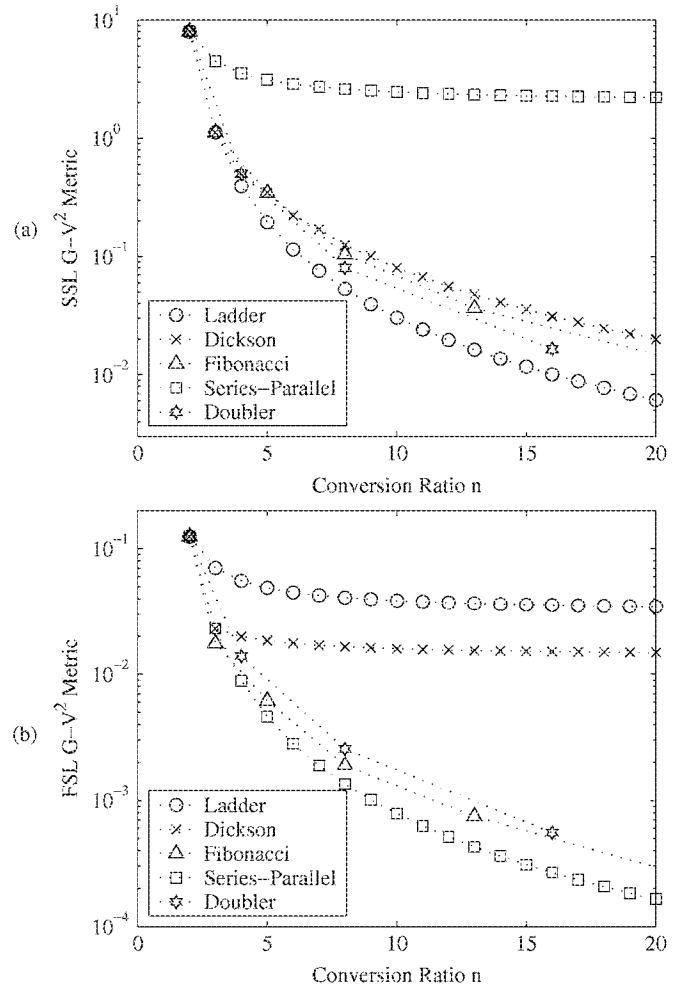


Fig. 7. (a) SSL and (b) FSL performance metrics with single-voltage devices.

the Fibonacci and Doubler topologies, exhibit mediocre performance in both the SSL and FSL comparisons. However, since the switches and capacitors used in their implementations support a range of different voltages and most of the switches are not ground-referenced, practical implementation would be difficult if not impossible.

The second comparison performed assumes that all devices must be of the same voltage rating. In integrated applications using standard CMOS processes, the switches and capacitors are usually all rated for the same voltage. The process is chosen such that this voltage rating corresponds to the maximum voltage seen on any device. However, the switches and capacitors can be rated differently from each other, i.e. if the highest-voltage switch is rated for 1 V, a 1 V process would be used, even if some capacitors support a higher voltage.

The comparative results using identically-rated switches and transistors are shown in Fig. 7. The series-parallel topology is still optimal in the SSL comparison, as all capacitors in that topology also support the same voltage. Likewise, the ladder topology is optimal in the FSL comparison, as all switches in that topology support the same voltage. However, the exponential converters are now relatively poor in both comparisons because they involve a wide range of device stresses, which is impractical in implementation. These comparisons can be used to

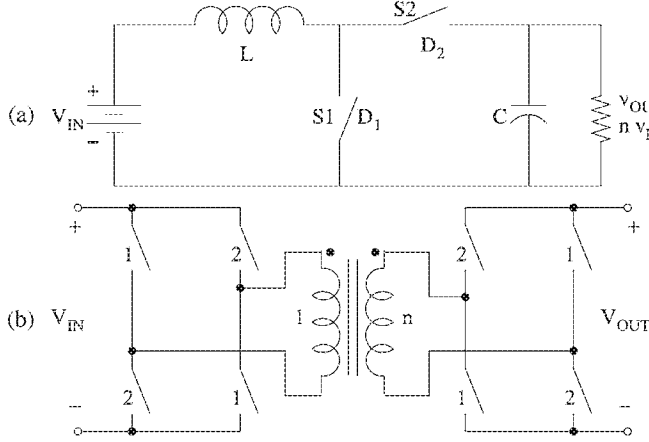


Fig. 8. (a) Standard boost converter. (b) Transformer-bridge converter.

select the best topology for any given application. Reference [8] includes more computational details for these converters.

V. COMPARISON WITH CONVENTIONAL DESIGNS

Switched-capacitor converters have several advantages over conventional inductor-based dc-dc converters. With a switched-capacitor converter, conduction and switching losses are not additional losses, but are already incorporated in the output impedance based losses calculated in Sections II-A and Sections II-B. The only losses that are not included in the output impedance are the gate drive losses, losses associated with parasitic capacitances and control power. Short-circuit (shoot-through) power can be eliminated by the use of sufficiently non-overlapping clocks. Stray capacitances from dynamic nodes must be minimized and their losses incorporated into the efficiency of the converter if the strays are not eliminated. These losses are further considered in [8].

A SC converter and a conventional dc-dc converter can be compared directly when conduction loss is considered. The silicon area (for the switches and control functions) is the dominant cost in many dc-dc converters. A converter with a significantly-lower switch conductance loss may have a cost advantage over a converter with a higher switch loss. For the SC converter, the conduction loss is equal to the loss corresponding to the FSL output impedance. The switch-related loss of an inductor-based converter is made up of conduction losses, due to switch on-state resistance, and switching losses during the switch state transitions. Only the conduction loss will be considered here, using the FSL performance metric $V_{OUT}^2 / (R_{FSL} / A_{tot})$ developed in Section IV.

A ladder-type step-up converter [such as the one in Fig. 5(a)] is considered, as it uses switches most efficiently in the FSL. Two magnetic-based converters are compared, the boost converter and transformer-bridge converter, both shown in Fig. 8. Total switch $G \cdot V^2$ product is held constant for all converters, and the SC converter is assumed to operate in the FSL. Finally, all switches are sized optimally based on the optimization methods presented in this paper. All converters are designed and optimized for a given conversion ratio n , and without loss of generality, an input voltage of 1 V is assumed.

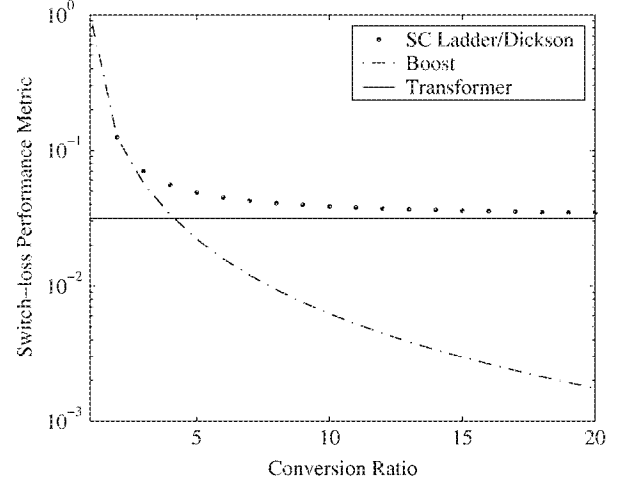


Fig. 9. Performance metric comparison.

The step-up ladder-type SC converter is considered first. All switches in the ladder topology must be rated for 1 V. The lowest two switches in the ladder structure have an a_r component of $(n - 1)$ while the other $2(n - 1)$ switches simply have an a_r component of 1. Thus, the sum of the a_r components is

$$\sum_i |a_{r,i}| = 2(n - 1) + 2(n - 1) = 4(n - 1). \quad (31)$$

The optimal FSL output impedance of this converter (constrained such that $\sum_{\text{switches}} G_i V_i^2 = A_{tot} = 1$) is thus

$$R_{out} = 2 \left(\sum_i |a_{r,i} v_{r,i(\text{rated})}| \right)^2 = 32(n - 1)^2. \quad (32)$$

Computing the ratio of this output resistance to the square of the output voltage yields the performance metric of the ladder circuit, $(n^2) / (32(n - 1)^2)$. This metric is also plotted in Fig. 9.

The boost converter in Fig. 8(a) is operated at duty cycle $D = 1/n$ to achieve a step-up ratio of n . The duty cycle of switch S1 is $D_1 = D = 1/n$ and the duty cycle of switch S2 is $D_2 = 1 - D = (n - 1)/n$. The conduction loss in this circuit is directly computed as

$$P_{cond} = \left(\frac{D}{G_1} + \frac{1 - D}{G_2} \right) I_{in}^2 = R_{out} I_{out}^2. \quad (33)$$

The equivalent loss impedance R_{out} can be directly compared to the output impedance of the SC converter. Optimizing the ratio of the two switch conductances for a given duty cycle, the following constraint can be derived

$$\left(\frac{G_1}{G_2} \right)^2 = \frac{D}{1 - D}. \quad (34)$$

Since the total $G \cdot V^2$ product of the switches is again constrained at one and each switch in the boost converter must be rated for the output voltage of n , the total conductance is restricted to $G_{tot} = 1/n^2$. From this constraint, the equivalent

loss impedance can be determined (note that $D = 1/n$ to achieve the correct conversion ratio):

$$R_{\text{out}} = n^4 \left(\frac{D}{G_1} + \frac{1-D}{G_2} \right) = n^4 \left(1 + 2\sqrt{\frac{n-1}{n^2}} \right). \quad (35)$$

Computing the ratio of the square of the output voltage to this optimal output resistance yields the performance metric of the boost circuit, $1/(n^2(1 + 2\sqrt{(n-1)/(n^2)}))$. This metric is plotted in Fig. 9.

Finally, the transformer-based direct converter in Fig. 8(b) is considered. The transformer is assumed to be ideal and to have an up-conversion ratio of n . The output switches are all identical and must be rated for the output voltage of n V. The on-current of these switches is equal to the output current I_{out} . Likewise, the input switches must be rated for 1 V and conduct a current of nI_{out} . To constrain the total $G \cdot V^2$ product equal to one, the output switches must have conductances of $1/8n^2$ and the input switches must have conductances of $1/8$. The conduction loss can then be calculated as

$$P_{\text{cond}} = 2(8n^2)I_{\text{out}}^2 + 2(8)(nI_{\text{out}})^2 = 32n^2I_{\text{out}}^2. \quad (36)$$

The resulting performance metric for this converter is then constant at $1/32$ for any conversion ratio. This makes intuitive sense as only the transformer turns ratio is changed to achieve different conversion ratios.

The conduction losses of the three converters, represented by the performance metric $V_{\text{OUT}}^2/(R_{\text{FSL}}/A_{\text{tot}})$ developed in Section IV, are compared in Fig. 9. The SC and boost converters' performance metrics decrease as the conversion ratio increases, but the SC converter approaches an asymptotic limit at $1/32$ (the same as the transformer-based converter), but the boost converter's performance continues to decrease.

At large conversion ratios, the step-up ladder-type SC converter is significantly superior to the boost converter as the switches in the ladder topology block only the input voltage and most switches carry less than the input current. However, the boost converter's switches carry the full input current and block the full output voltage. Even though the SC converter has many more switches, the low working V-A product of these switches yields a lower conduction loss than that of the boost converter, with its much higher working V-A product switches.

In an application where switches are the limiting factor in performance or cost, switched-capacitor converters are evidently advantageous over conventional magnetics-based dc-dc converters at high or moderate conversion ratios.

VI. VERIFICATION BY SIMULATION

As this analysis method is based on idealized devices, circuit-level simulation, through SPICE or spectre is appropriate for verification of this analysis. Ideal capacitor and voltage-controlled resistances are used in the simulation. Parasitics, while an important consideration in real-world implementations, are not considered in this paper, and are not considered in the verification simulation.

Five step-up switched-capacitor converters have been simulated over a range of switching frequencies. By varying the

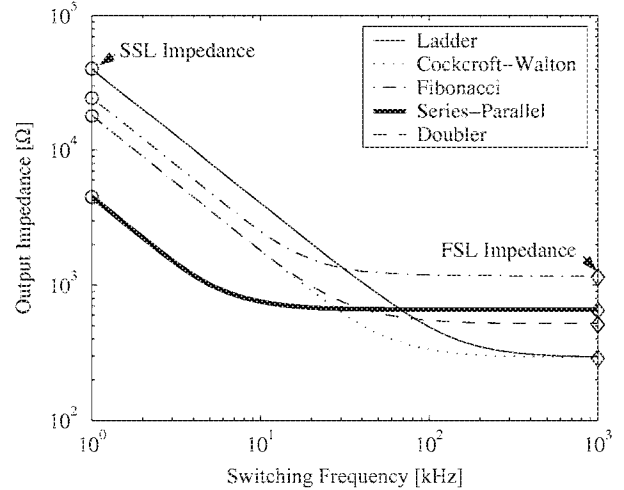


Fig. 10. Simulated output impedance versus switching frequency.

TABLE I
SSL AND FSL VA-PRODUCTS AND IMPEDANCES
FOR FIVE CONVERTERS (R_{SSL} AT 1 KHz)

	$\sum a_c v_c $	$\sum a_r v_r $	R_{SSL}	R_{FSL}
1:4 Ladder	9	12	40.5 kΩ	288 Ω
1:4 Cockcroft-Walton	6	12	18 kΩ	288 Ω
1:5 Fibonacci	7	24	24.5 kΩ	1152 Ω
1:4 Series-Parallel	3	18	4.5 kΩ	648 Ω
1:4 Doubler	6	16	18 kΩ	512 Ω

switching frequency, both the SSL and FSL output impedances can be determined and compared with those from the mathematical analysis. The converters simulated are the 1:4 (i.e., 1 V input to 4 V output) ladder converter, 1:4 Cockcroft-Walton multiplier, 1:4 doubler, 1:5 Fibonacci converter, and a 1:4 series-parallel converter, all shown in Fig. 5. The capacitors and switches in all converters are optimized using the methods in this paper using a switch GV^2 product of 1 VA and a total capacitor energy of 1 μJ . The output impedance was determined by measuring the current transfer between the input and output voltage sources in a transient simulation.

The charge-multiplier-voltage products and output impedances of the five converters found via the methods in Section II are shown in Table I. These calculations and detailed converter analysis are given in [8]. Fig. 10 shows the simulated impedance of the converters between 100 Hz and 1 MHz. The symbols in the plots indicate the calculated FSL and SSL impedances, showing a match between the theoretical and simulated values.

Five very different switched-capacitor converters were simulated for a range of frequencies encompassing both the SSL asymptote and the FSL asymptote. The simulation results verify that the analysis methods developed in Section II determine the correct output impedance in both asymptotes for all five converters. This simulation verifies that the methods developed in this paper accurately predict the performance of any two-phase SC converter.

VII. EXPERIMENTAL VERIFICATION

An ultra-low-power switched-capacitor power conversion integrated circuit (IC) has been fabricated and tested with design

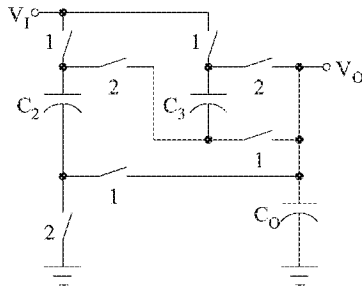


Fig. 11. 3:2 Series-parallel converter topology.

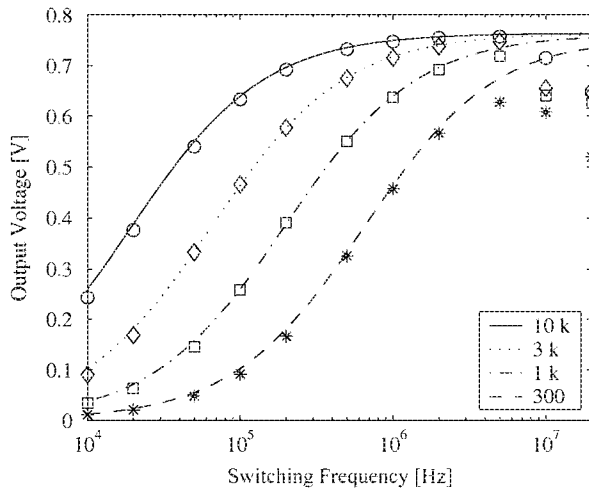


Fig. 12. Measured and calculated output voltage versus load and switching frequency.

guidance from the analysis method developed here. The design and experimental results of this IC are presented in [12]. The results from the 3:2 converter (1.2 V to 0.8 V) is presented here as an example circuit further verifying the calculations for a single converter topology.

A schematic of this converter is shown in Fig. 11. It is similar to the series-parallel circuit in that the capacitors are placed in series in one phase and parallel in the other phase. Capacitors C_1 and C_2 both have a charge multiplier of $1/3$ in this circuit and capacitor C_0 is ignored as it is in parallel with the output. Each of the seven switches also has a charge multiplier of $1/3$. In the IC, each capacitor has a value of 1.15 nF and each switch has a resistance of 4.75Ω . Thus, the predicted output impedances in both limits, calculated using (9) and (14), are

$$R_{SSL} = \frac{2(1/3)^2/1.15\text{nF}}{f_{sw}} = \frac{193 \text{ M}\Omega}{f_{sw}} \quad (37)$$

$$R_{FSL} = 2 \cdot 7(4.75(1/3)^2) = 7.39 \Omega. \quad (38)$$

Fig. 12 shows the output of the 3-2 converter for various switching frequencies and resistive loads. The plotted curves indicate the ideal output voltage calculated by considering a resistor divider between the load resistance and converter output impedance. The data indicate that the model accurately predicts the converter performance for low frequencies. At high

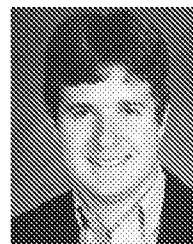
switching frequencies, parasitic losses hurt the converter's performance. The most influential parasitics, in this case the capacitor bottom-plate capacitance and drain-source capacitance, need to be carefully considered in any realization of switched-capacitor converters.

VIII. CONCLUSION

An analysis method has been presented to determine the performance of any switched-capacitor power converter using easily-determined charge multiplier vectors. The capacitors and semiconductor switches of the converter were optimized to minimize output impedance for several conditions and constraints. Five separate converter topologies were considered for their effectiveness in utilizing capacitors and switches. This comparison allows the use of an optimal topology suited to its application and implementation technology. Significantly, the performance (based on conduction loss) of a ladder-type converter was found to be superior to that of a conventional boost converter for medium to high conversion ratios.

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EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

NOTIFICATION DATE	DELIVERY MODE
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04/20/2015

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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**Supplemental
Notice of Allowability**

Application No.
13/044,063

Examiner
JYE-JUNE LEE

Applicant(s)
VINCIARELLI, PATRIZIO

Art Unit
2838

**AIA (First Inventor to
File) Status**
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 03/03/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-6,11,16-23,28,44-47,52,57,58 and 89-114. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____
3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material
4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____
5. ☐ Examiner's Amendment/Comment
6. ☒ Examiner's Statement of Reasons for Allowance
7. ☒ Other PTO-271.

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the amendment filed on 03/03/2015.

Reasons For Allowance

3. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, 90-114** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first

Art Unit: 2838

switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...”.

The aforementioned limitations in combination with all remaining limitations of the respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

4. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-

Art Unit: 2838

7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838



ON Semiconductor®



High-Efficiency 255 W ATX Power Supply Reference Design Documentation Package

Disclaimer: ON Semiconductor is providing this reference design documentation package “AS IS” and the recipient assumes all risk associated with the use and/or commercialization of this design package. No licenses to ON Semiconductor’s or any third party’s Intellectual Property is conveyed by the transfer of this documentation. This reference design documentation package is provided only to assist the customers in evaluation and feasibility assessment of the reference design. The design intent is to demonstrate that efficiencies beyond 85% are achievable cost effectively utilizing ON Semiconductor provided ICs and discrete components in conjunction with other inexpensive components. It is expected that users may make further refinements to meet specific performance goals.

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1. Overview

ON Semiconductor was the first semiconductor company to provide an 80 PLUS-certified open reference design for an ATX power supply in 2005. A second generation 80 PLUS-certified open reference design with improved efficiency was then introduced in 2007. ON Semiconductor is now introducing its third generation 80 PLUS-certified open reference design with a drastic efficiency improvement. This is a 255 W multi-output power supply for the ATX form factor. Achieving a maximum efficiency of 90% at 50% load, and at 230 and 240 Vac, this third generation reference design achieves >88% efficiency at 50% load, and 100 and 115 Vac. All efficiency measurements were obtained at the end of a 41 cm (16 in.) cable, ensuring the design can be used 'as is' in all standard desktop PC configurations.

This reference document provides the details behind this third generation design. The design manual provides a detailed view of the performance achieved with this design in terms of efficiency, performance, thermals and other key parameters. In addition, a detailed list of the bill-of-materials (BOM) is also provided. ON Semiconductor will also be able to provide technical support to help our customers design and manufacture a similar ATX power supply customized to meet their specific requirements.

The results achieved in this third generation design were possible due to the use of advanced new components from ON Semiconductor. These new ICs not only speeded up the overall development cycle for this new design, but also helped achieve the high efficiencies while at the same time keeping a check on the overall cost.

This third generation design consists of a single PCB designed to fit into the standard ATX enclosure along with a fan. Figure 1 below presents the overall architecture employed in this design. Detailed schematics are included later in this design manual.

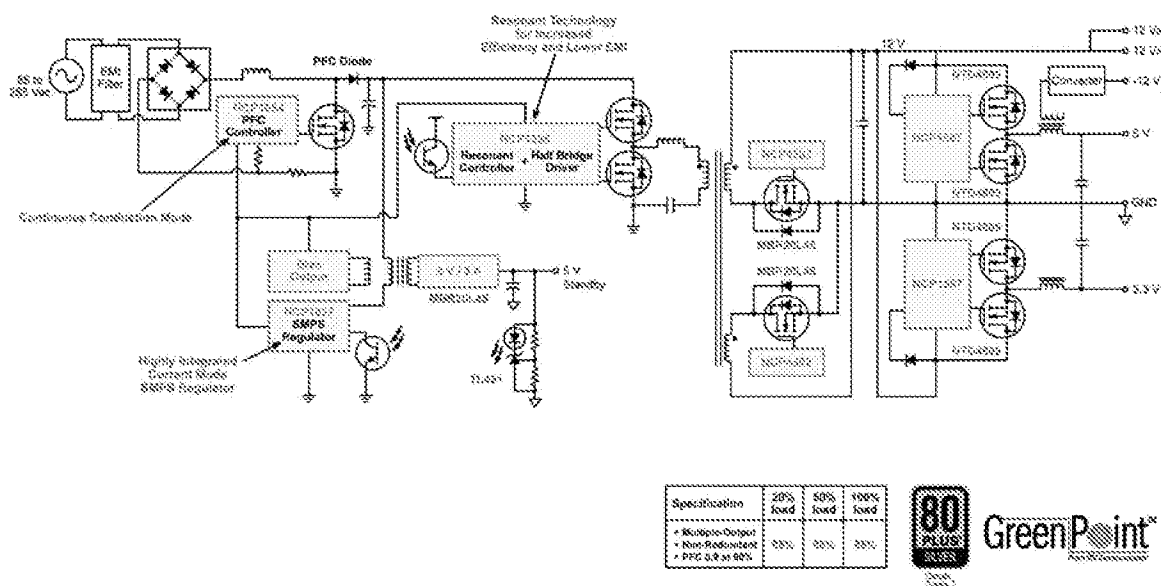


Figure 1: Reference Design Architecture Simplified Block Diagram

As seen in figure 1, the first stage, active Power Factor Correction (PFC) stage, is built around ON Semiconductor's Continuous Conduction Mode (CCM) PFC controller, NCP1654. The NCP1654 provides an integrated, robust and cost-effective PFC solution. The second stage features a resonant half-bridge LLC topology using ON Semiconductor's controller, NCP1396. This topology ensures maximum efficiency and minimizes EMI. The NCP1027, standby controller, is used to generate the 5 V standby output. The NCP1027 is an optimized integrated circuit for the ATX power supply and incorporates a high-voltage MOSFET.

On the secondary side, this architecture uses a synchronous rectification scheme built around ON Semiconductor's NCP4302 controller in order to generate a 12 V output. Finally, two identical DC-DC controllers are used to down-convert the 12 V into +5 V, +3.3 V and -12 V. The DC-DC controller is the NCP1587, a low voltage synchronous buck controller in a very small surface mount 8-pin package. Each DC-DC controller drives two NTD4809 (30 V, 58 A, single N-channel power MOSFET) in a synchronous rectification scheme.

With the introduction of this third generation, high-efficiency ATX Power Supply, ON Semiconductor has shown that with judicious choice of design tradeoffs, optimum performance is achieved at minimum cost.

2. Specifications

The design closely follows the ATX12V version 2.2 power supply guidelines and specifications available from www.formfactors.org, unless otherwise noted. This 255 W reference design exceeds the 80 PLUS Silver (www.80plus.org), ENERGY STAR® 5.0 (www.energystar.gov), and Climate Savers Computing Initiative (CSCI) Step 3 (www.climatesaverscomputing.org) efficiency targets for desktop PC multi-output power supplies. Table 1 hereafter shows a summary of the efficiency targets from these different organizations.

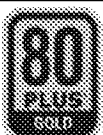
Multi-Output ATX Power Supplies	Levels	Specification	20% of rated output	50% of rated output	100% of rated output	Compliance
		• Multiple-Output • Non-Redundant • PFC 0.9 at 100% of rated output	80% efficiency	80% efficiency	80% efficiency	ENERGY STAR rev. 4.0 & CSCI step #1 Effective date: July 2007
		• Multiple-Output • Non-Redundant • PFC 0.9 at 50% of rated output	82% efficiency	85% efficiency	82% efficiency	ENERGY STAR rev. 5.0 (Effective date: July 2009) & CSCI step #2 (June 2008 thru July 2009)
		• Multiple-Output • Non-Redundant • PFC 0.9 at 50% of rated output	85% efficiency	88% efficiency	85% efficiency	CSCI step #3 (July 2009 thru June 2010)
		• Multiple-Output • Non-Redundant • PFC 0.9 at 50% of rated output	87% efficiency	90% efficiency	87% efficiency	CSCI step #4 (July 2010 thru June 2011)

Table 1: Summary of Efficiency Targets

Key specifications for this reference design are included in Table below.

Input			Output Voltage (V _{dc})	DC Output Current		
Voltage (V _{ac})	Frequency (Hz)			Min DC Current (A)	Full Load DC Current (A)	Max DC Current (A)
Min.	90	47	+12 V _A	0	9.50	13.0
Typ.	100	50	+12 V _B	0	5.12	7.0
	115	60	+5 V	0.3	9.44	15.0
	230	50	+3.3 V	0.3	5.03	8.0
	240	50	-12 V	0	0.32	0.4
Max.	264	63	+5 V _{SB}	0	2.39	3.0

Output Voltage (V _{dc})	Full Load DC Current (A)	Output Power	
		Full Load Output Power (W)	Notes
+12 V _A	9.50	114.0	Peak +12 V _{Adc} output current up to 14 A. Peak +12 V _{Bdc} output current to be 8 A. The maximum combined load on the +12 V _{Adc} and +12 V _{Bdc} outputs shall not exceed 220 W.
+12 V _B	5.12	61.4	
+5 V	9.44	47.2	The maximum continuous combined load on the +5Vdc and +3.3 Vdc outputs shall not exceed 80 W.
+3.3 V	5.03	16.6	
-12 V	0.32	3.8	
+5 V _{SB}	2.39	11.95	
Full Load Total Output Power = 255 W			

Output Voltage (V _{dc})	Output Voltage Regulation (V)			Tolerance (%)	Output Ripple / Noise (mVpp)
	Min.	Typ.	Max.		
+12 V _A	+11.4	+12	+12.6	±5	120
+12 V _B	+11.4	+12	+12.6	±5	120
+5 V	+4.75	+5	+5.25	±5	70
+3.3 V	+3.14	+3.3	+3.47	±5	50
-12 V	-10.8	-12	-13.2	±10	250
+5 V _{SB}	+4.75	+5	+5.25	±5	100

Table 2: Target Specifications

Target specifications for other key parameters of the reference design include:

- Efficiency: Minimum efficiency of 85% at 20% and at 100% of rated output power, and 88% at 50% of rated output power as defined by the 80 PLUS requirements.
- Power Factor: Power factor of 0.9 or greater at 100 % load.
- Input Voltage: Universal Mains: 90 Vac to 265 Vac, frequency: 47 to 63 Hz.
- Safety Features: As per the ATX12V version 2.2 power supply guidelines, this design includes safety features such as OVP, UVP, and OCP.

3. Architecture Overview

The architecture selected is designed around a succession of conversion stages as illustrated in Figure 1. The first stage is a universal input, active power factor boost stage delivering a constant output voltage of 385 V to the second stage, the half bridge resonant LLC converter. On the secondary side, this architecture uses a synchronous rectification scheme built around ON Semiconductor's NCP4302 controller in order to generate a +12 V output. Finally, the +12 V is down-converted +5 V, +3.3 V and -12 V by a DC-DC conversion stage, built around two identical DC-DC controllers. In addition, a small integrated flyback converter delivers 12 W of standby power to another isolated 5 V rail. All the different voltage rails are monitored by a dedicated supervisory controller.

ON Semiconductor has developed multiple power management controllers and MOSFET devices in support of the ATX program. Web-downloadable data sheets, design tools and technical resources are available to assist design optimization. The semiconductor components, supporting the ATX Generation 3 platform, are the NCP1654 PFC controller, the NCP1396 half-bridge resonant LLC controller, the NCP4302 synchronous rectification, the NCP1027 standby controller, the NCP1587 DC-DC controller with synchronous rectification, and the NTD4809 single N-channel power MOSFET driven by the NCP1587, in synchronous rectification.

3.1 Primary Side: PFC Stage

There are a variety of PFC topologies available. These include discontinuous conduction mode (DCM), critical conduction mode (CRM) and continuous conduction mode (CCM). At this power level, CCM is the preferred choice and the NCP1654 will implement an IEC61000-3-2 compliant, fixed frequency, peak current mode or average current mode PFC boost converter with very few external components.

3.2 Primary Side: Half bridge resonant LLC Converter

The heart of the half-bridge resonant LLC converter stage is the NCP1396 resonant mode controller. Thanks to its proprietary high-voltage technology, this controller includes a bootstrapped MOSFET driver for half-bridge applications accepting bulk voltages up to 600 V. Protections featuring various reaction times, e.g. immediate shutdown or timer-based event, brown-out, broken optocoupler detection etc., contribute to a safer converter design, without engendering additional circuitry complexity. An adjustable dead-time also helps lower the shoot-through current contribution as the switching frequency increases. More information about LLC structure can be found in the ON Semiconductor application note [AND8311/D](#) (Understanding the LLC Structure in Resonant Applications).

3.3 Secondary Side: Synchronous Rectification

The 12 V output generated by the half-bridge resonant LLC converter is rectified using a proprietary synchronous rectification scheme built around two NCP4302 and two external single N-channel MOSFETs.

3.4 Secondary Side: DC-DC Conversion Stage

Two identical DC-DC controllers are used to down-convert the 12 V into +5 V and +3.3 V. The DC-DC controller is the NCP1587, a low voltage synchronous buck controller in a very small surface mount 8-pin package. The NCP1587 is a low cost PWM controllers designed to operate from a 5 V or 12 V supply. This device is capable of producing an output voltage as low as 0.8 V. The NCP1587 provides a 1 A gate driver design and an internally set 275 kHz (NCP1587) oscillator. Other efficiency enhancing features of the gate driver include adaptive non overlap circuitry. The NCP1587 also incorporates an externally compensated error amplifier and a capacitor programmable soft start function. Protection features include programmable short circuit protection and under voltage lockout. Each DC-DC controller drives two NTD4809 (30 V, 58 A, single N-channel power MOSFET) in a synchronous rectification scheme.

The -12 V is generated from the +5 V using a small discrete-based converter.

3.5 Secondary Side: Monitoring and Supervisory Stage

The four dc outputs +5 V, +3.3 V, +12 V_A and +12 V_B are monitored by a dedicated monitoring controller which also provides over-current protection, over-voltage protection, under-voltage protection and generates the Power Good logic signal.

3.6 Standby Power

The NCP1027 integrates a fixed frequency current mode controller and a 700 volt MOSFET. The NCP1027 is an ideal part to implement a flyback topology delivering 15 W to an isolated 5 V output. At light loads the IC will operate in skip cycle mode, thereby reducing its switching losses and delivering high efficiency throughout the load range.

4. Performance Results

Measurements are done at three loading conditions, the load being expressed as a % of the rated output power, i.e. at 20%, 50% and at 100% of rated output power.

Measurements are also done at four AC line voltages, at 100 Vac, 115 Vac, 230 Vac and 240 Vac, at 50 Hz, 60 Hz and 63 Hz.

All measurements are taken at the end of the 41 cm-long (16 inches) cable.

The converter efficiency is measured according to the loading conditions detailed in Table 3:

Load as % of rated output power	Output Current (A)					
	+12V _A	+12V _B	+5V	+3.3V	-12V	+5V _{SB}
20%	1.90	1.02	1.89	1.01	0.06	0.48
50%	4.75	2.56	4.72	2.52	0.16	1.20
100%	9.50	5.12	9.44	5.03	0.32	2.39
Max.	13.0	7.0	15.0	8.0	0.4	3.0

Table 3: Load Matrix for Efficiency Measurements

4.1 Total Efficiency

AC Input	Total Efficiency (%)		
	20% Load	50% Load	100% Load
100 V _{AC} / 50 H _Z	85.9%	88.3%	85.9%
115 V _{AC} / 60 H _Z	86.3%	88.8%	86.8%
230 V _{AC} / 50 H _Z	86.5%	90.1%	88.7%
240 V _{AC} / 63 H _Z	86.5%	90.1%	88.8%

Table 4: Efficiency Results

The converter achieves over 85% efficiency with room to spare over all load conditions. All measurements are done at the end of the power cable.

4.2 Power Factor

The power factor exceeds 0.9 over all operating conditions as shown in Table .

AC Input	Power Factor			Specification
	20% Load	50% Load	100% Load	
100 V _{AC} / 50 H _Z	0.96	0.983	0.987	PF > 0.90 @100% & 50% of rated output power
115 V _{AC} / 60 H _Z	0.945	0.978	0.986	
230 V _{AC} / 50 H _Z	0.714	0.912	0.966	
240 V _{AC} / 63 H _Z	0.689	0.902	0.962	

Table 5: Power Factor vs Load as % of Rated Output Power

4.3 Standby Power

AC Input	Output Current on +5V _{SB} (A)	Input Power (W)	Specification
100 V _{AC} / 50 Hz	0.147	0.36	$< 1\text{W}$ • U.S. Department of Energy, FEMP (Federal Energy Management Program): http://www1.eere.energy.gov/femp/procurement/index.html • Executive Order 13221 of July 31, 2001: http://www1.eere.energy.gov/femp/pdfs/EO13221.pdf
115 V _{AC} / 60 Hz	0.171	0.31	
230 V _{AC} / 50 Hz	0.244	0.55	
240 V _{AC} / 63 Hz	0.312	0.57	

Table 6: Standby Power vs AC Line Voltage

4.4 Input Current

AC Input	Measurement (A)			Specification
	20% Load	50% Load	100% Load	
90 V _{AC} / 47 Hz	0.673	1.609	3.245	90V (max 3.6A) 180V (max.1.8A) @ 100% load
100 V _{AC} / 50 Hz	0.614	1.458	2.952	
115 V _{AC} / 63 Hz	0.544	1.271	2.549	
180 V _{AC} / 50 Hz	0.363	0.799	1.573	
230 V _{AC} / 53 Hz	0.355	0.667	1.266	
240 V _{AC} / 60 Hz	0.353	0.646	1.217	
264 V _{AC} / 63 Hz	0.339	0.597	1.092	

Table 7: Input Current vs Load and AC Line Voltage

4.5 Inrush Current

Parameter Description	Min.	Typ.	Max.	Units
Initial In-rush Current			55	A (Peak)
Secondary In-rush Current			35	A (Peak)

Table 8: Inrush Current Specification

AC Input	Output Load	Measurement (A)		Specification
90 V _{AC} /47 Hz	100% Load	Initial In-rush Current	9.9	Initial In-rush Current $< 55\text{ A}$ Secondary In-rush Current $< 35\text{ A}$
		Secondary In-rush Current	8.0	
120 V _{AC} /63 Hz	100% Load	Initial In-rush Current	17.9	
		Secondary In-rush Current	16.2	
220 V _{AC} /50 Hz	100% Load	Initial In-rush Current	25.4	
		Secondary In-rush Current	24.5	
264 V _{AC} /63 Hz	100% Load	Initial In-rush Current	34.1	
		Secondary In-rush Current	33.3	

Table 9: Inrush Current vs AC Line Voltage

4.6 Output Transient Response (Dynamic Loading)

Output Transient starting Load T1 / T2 (0.1 A/ μ sec), T1 / T2 (1 ms)

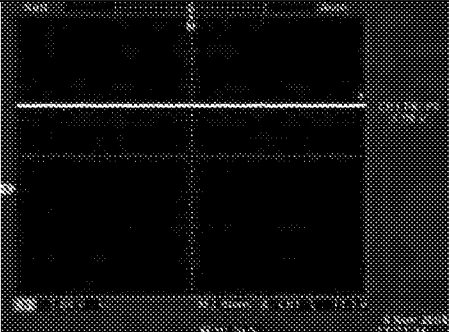
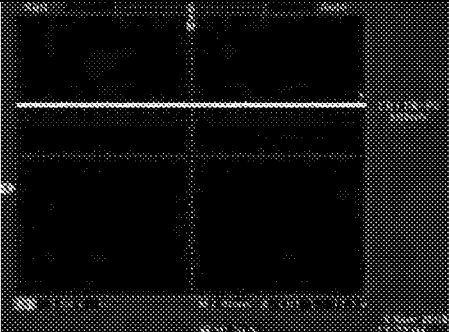
DC Output	Load (A)			Voltage Max. (V)	
	Min.	Load Step	Max.	Overshoot	Undershoot
+12 V _A	0.5	6.5	6.5	0.6	0.6
+12 V _B	0.5	3.5	3.5	0.6	0.6
+5 V	0.3	5	10	0.25	0.25
+3.3 V	0.3	2.66	5.34	0.17	0.17
-12 V	0	0.17	0.33	0.6	0.6
+5 V _{SB}	0	1.33	2.67	0.25	0.25

Table 10: Output Transient Response (Dynamic Loading)

4.7 Overshoot at Turn-On/Turn-Off

DC Output	Min. – Load Step	Load Step – Max.	Specification	Unit
+12V _A	0.34	N/A	1.2	V _{P-P}
+12V _B	0.38	N/A	1.2	
+5V	0.21	0.23	0.5	
+3.3V	0.15	0.17	0.33	
-12V	0.18	0.20	-1.2	
+5V _{SB}	0.21	0.20	0.5	

Table 11: Overshoot at Turn-On/Turn-Off

Measured and Calculated Data at 115V / 60Hz		
DC Output	Min. – Load Step	Load Step – Max.
+12 V _A		N/A
+12 V _B		N/A

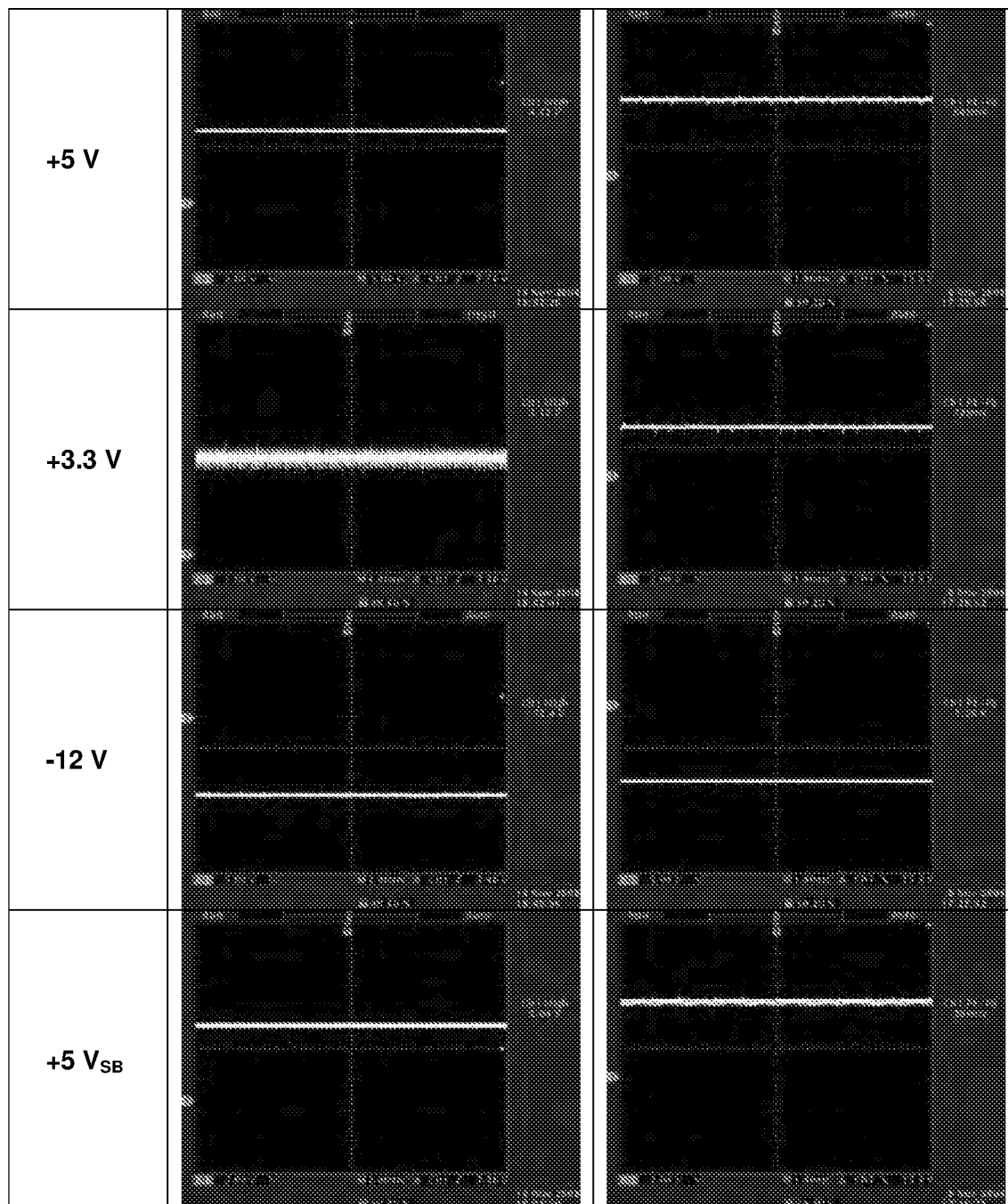


Figure 2: Dynamic Load Test Waveforms

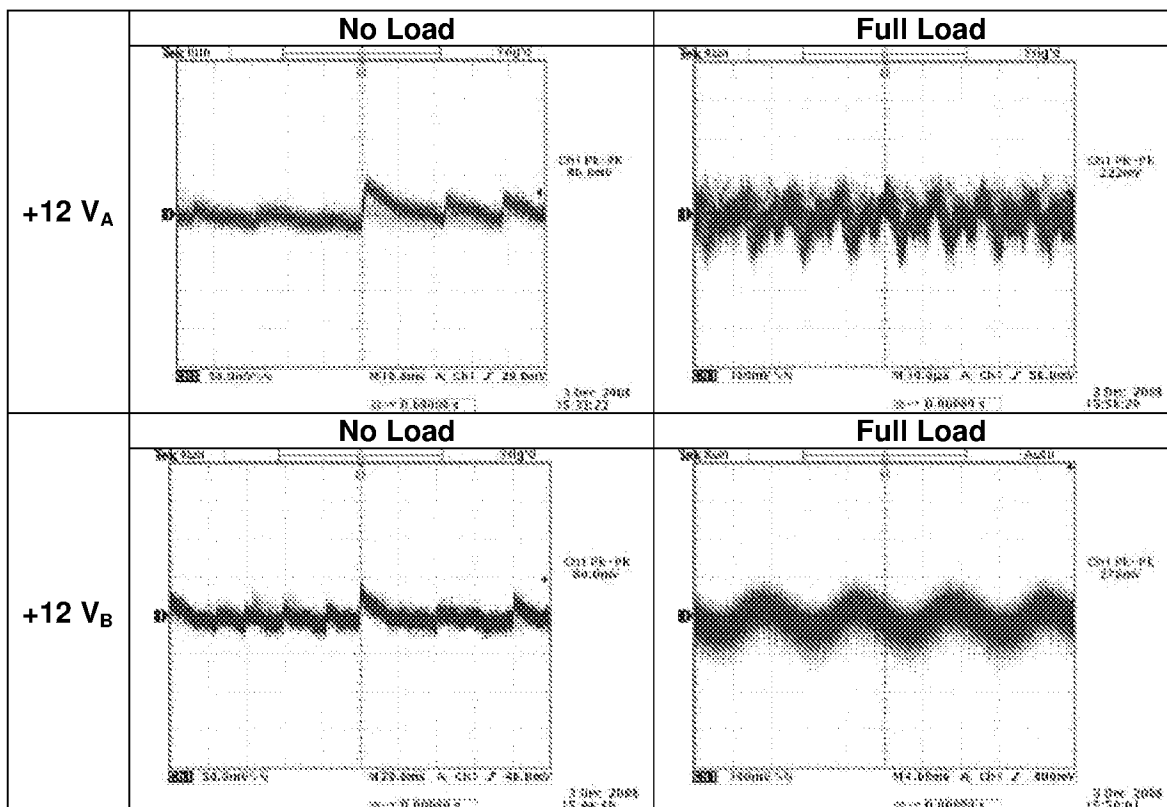
4.8 Output Ripple / Noise

The ripple voltage of each output is measured at no load and at the maximum load for each output., and at the four different line voltages. The output ripple is measured across 10 μ F/MLC parallel 1000 μ F low ESR/ESL termination capacitors. Figure 3 through 6 show the output voltage ripple measurements. All outputs meet the voltage ripple requirements.

DC Output	No Load (A)	Full Load (A)	Output Ripple / Noise Max (mVp-p)
+12 V _A	0	9.50	120
+12 V _B	0	5.12	120
+5 V	0	9.44	70
+3.3 V	0	5.03	50
-12 V	0	0.32	250
+5 V _{SB}	0	2.39	100

Table 12: Output Ripple / Noise Specification

4.8.1 100 V_{AC} / 50 Hz - Ripple / Noise Test Waveform



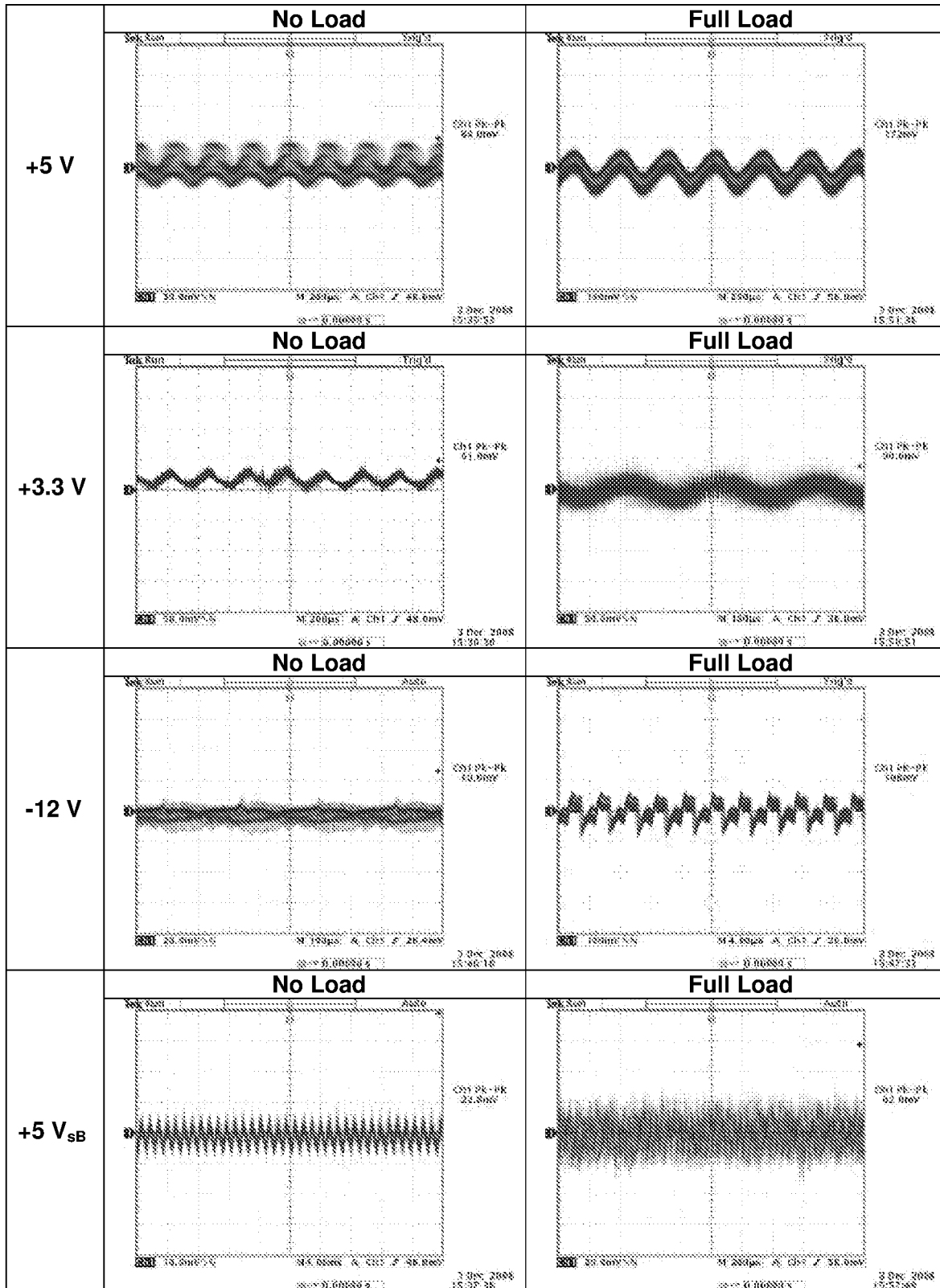
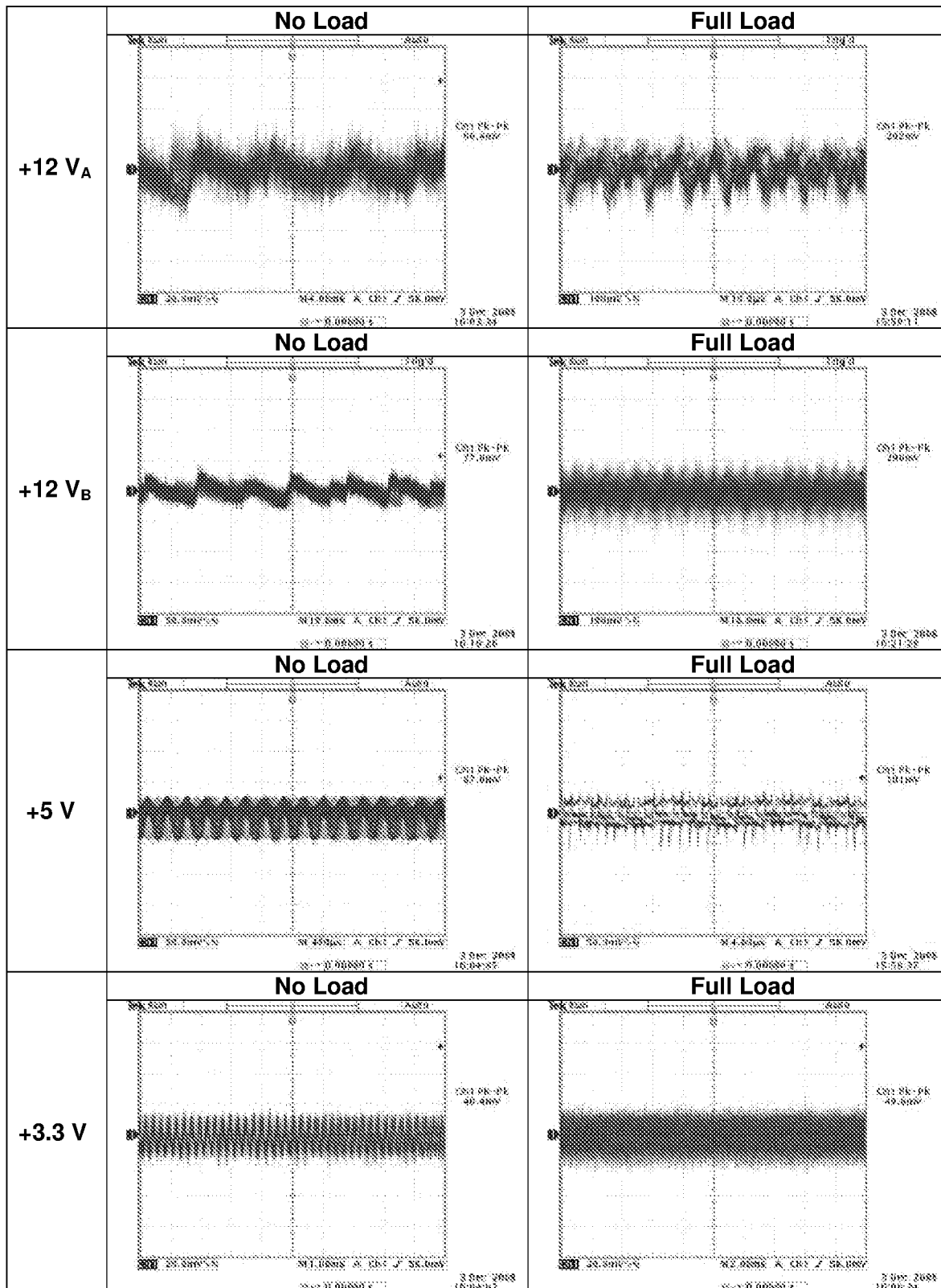


Figure 3: 100 V_{AC} / 50 Hz - Ripple / Noise Test Waveform

4.8.2 115 V_{AC} / 60 Hz - Ripple / Noise Test Waveform



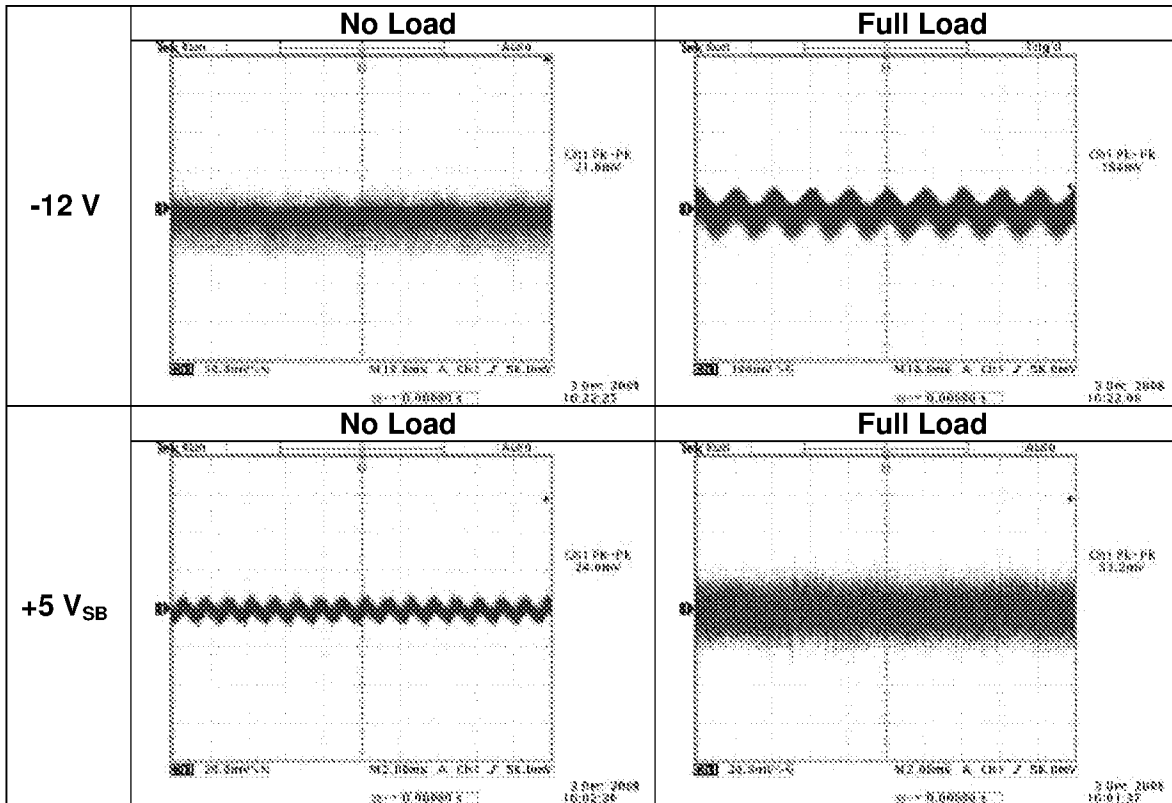
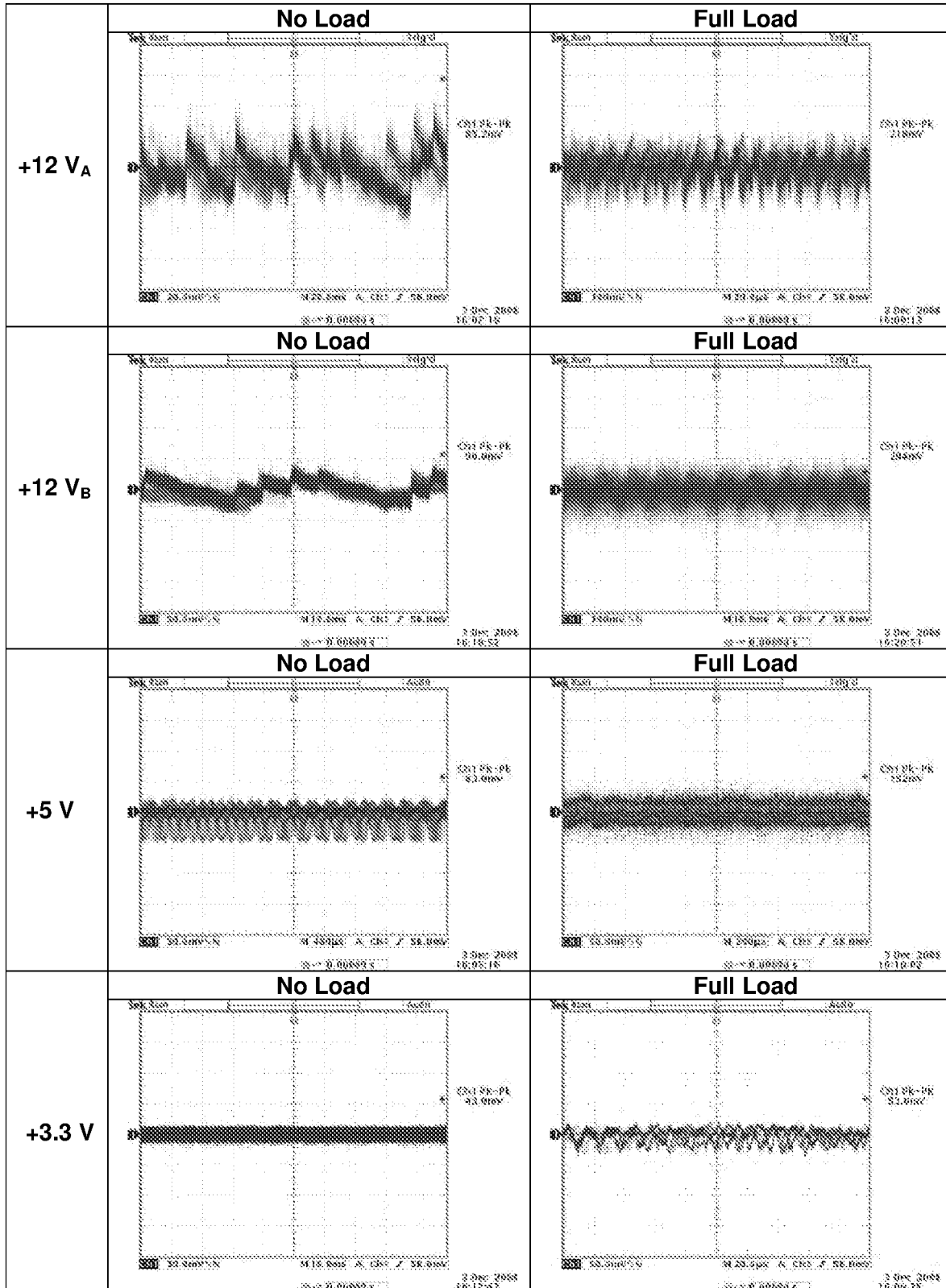


Figure 4: 115 V_{AC} / 60 Hz - Ripple / Noise Test Waveform

4.8.3 230 V_{AC} / 50 Hz - Ripple / Noise Test Waveform



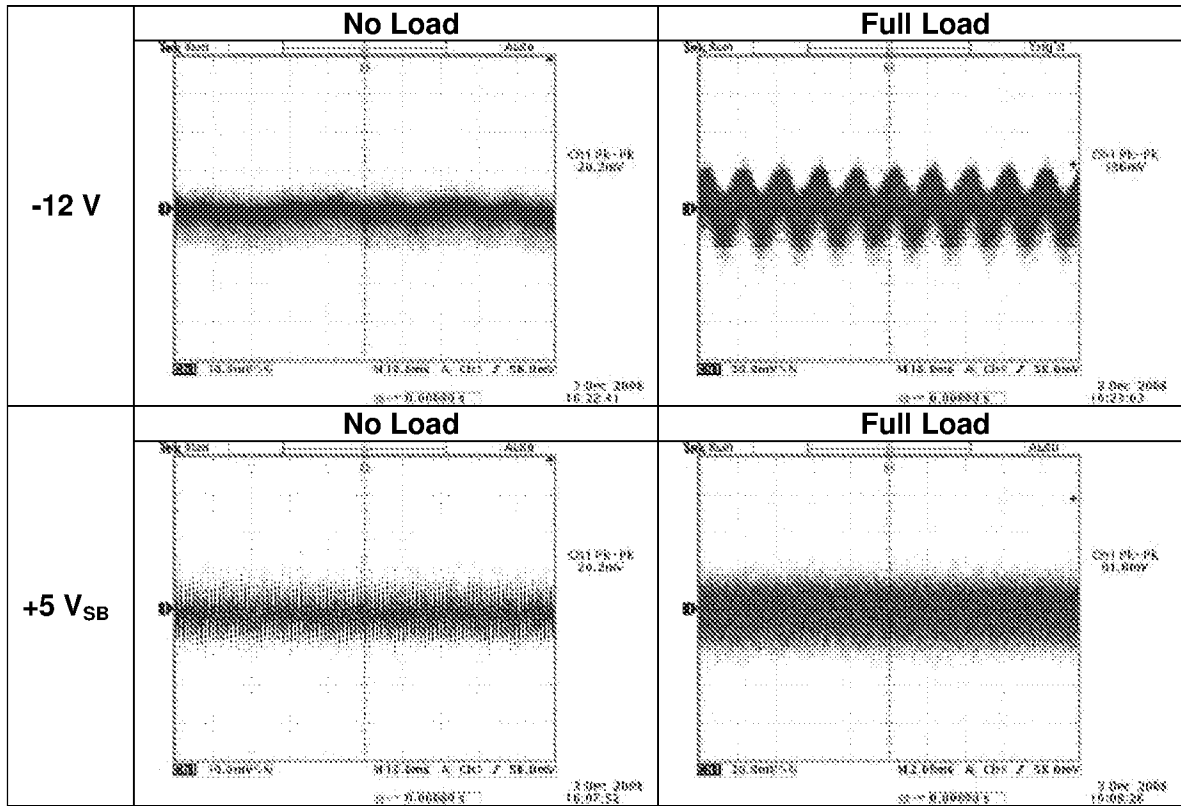
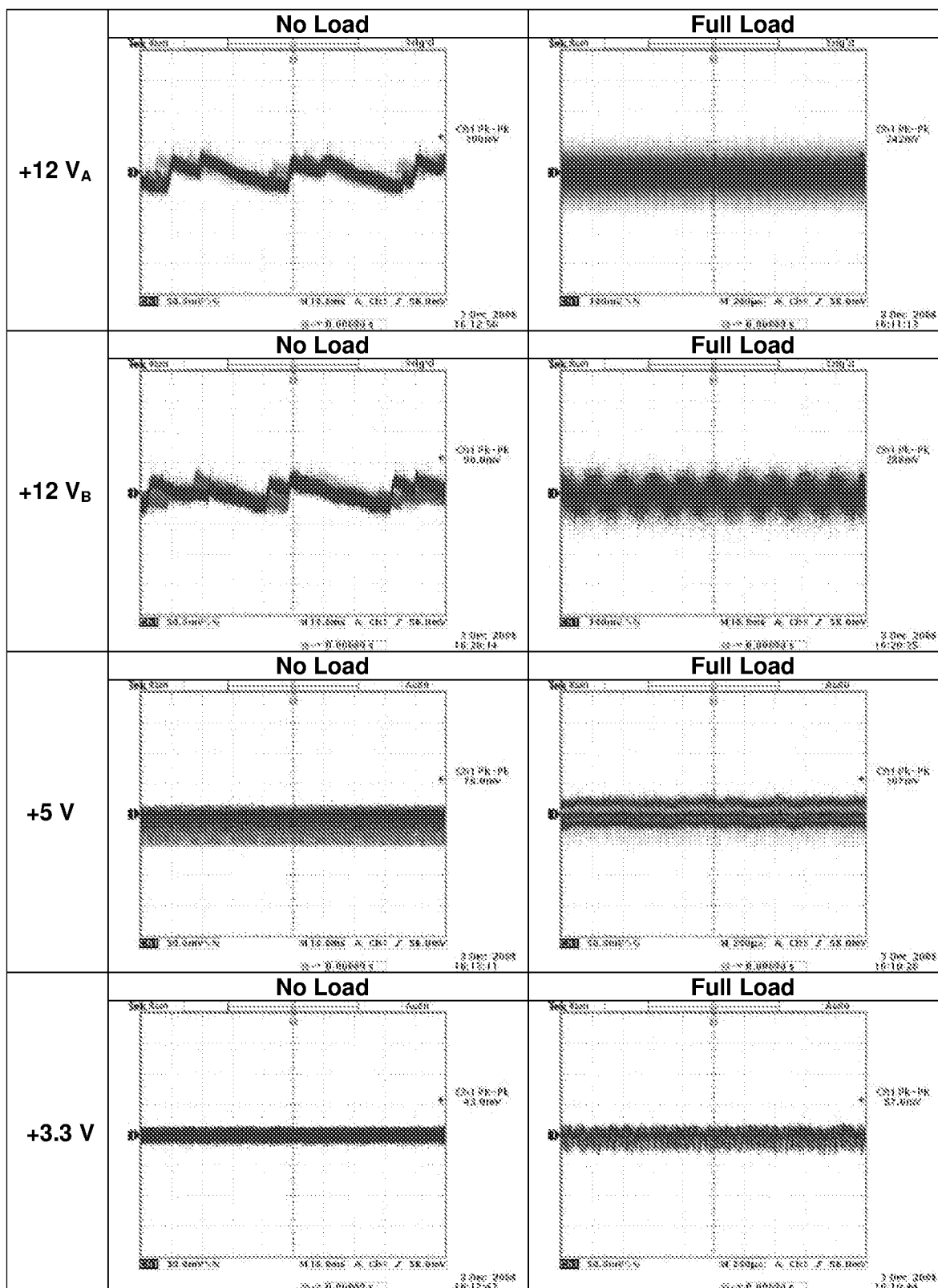


Figure 5: 230 V_{AC} / 50 Hz - Ripple / Noise Test Waveform

4.8.3 240 V_{AC} / 63 H_Z - Ripple / Noise Test Waveform



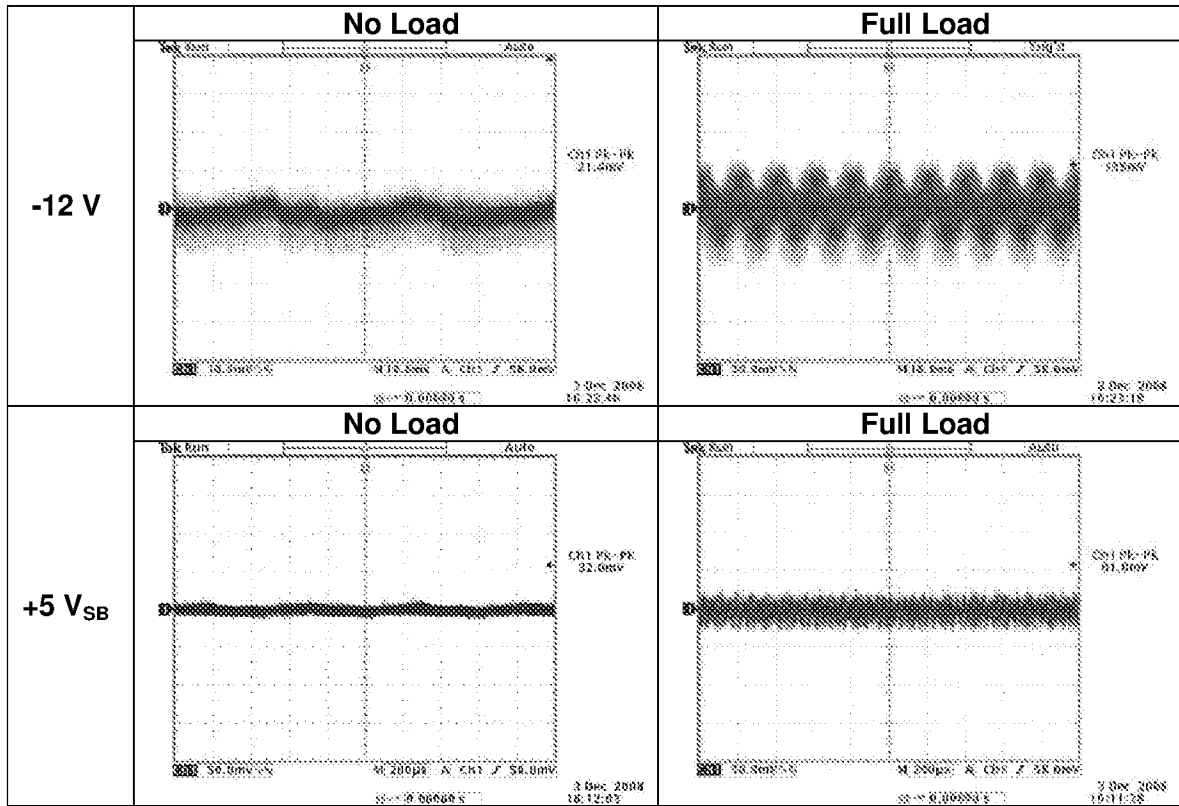


Figure 6: 240 V_{AC} / 63 Hz - Ripple / Noise Test Waveform

4.9 Hold-up Time

The required holdup time at 50% load is 16 ms. Holdup time is measured from the moment the AC power is removed to when the PWR_OK signal goes low. Figure 7 shows the holdup time at 50% load, and at 115 Vac and 230 Vac. Channel 4 is the AC power and Channel 1 is the PWR_OK signal.

AC Input	Dropout Loading Condition	Measurement (msec)	Specification
115 V_{AC} / 60 Hz	50% Load	22	> 16 ms
230 V_{AC} / 50 Hz	50% Load	23	

Table 13: Hold-up Time Specification

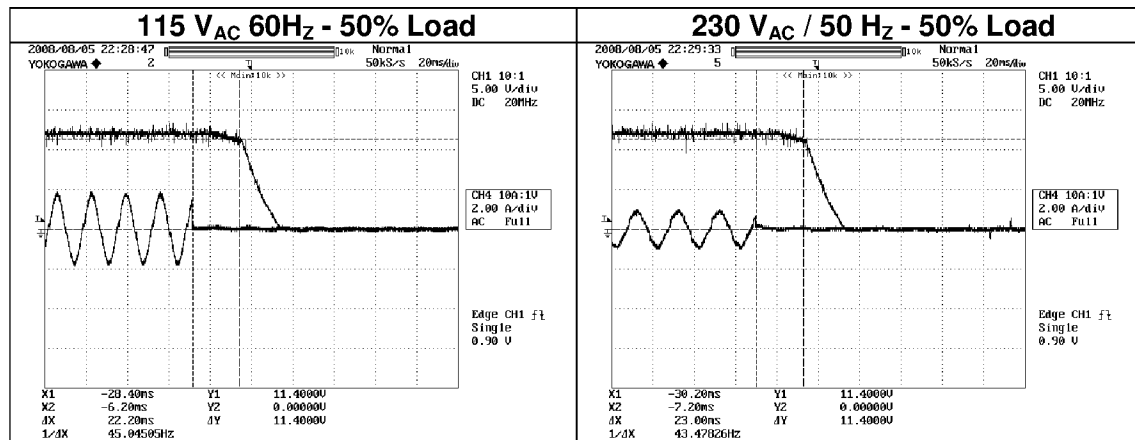


Figure 7: Hold-up Time

4.10 Timing / Housekeeping / Control

4.10.1 AC On / Off Control

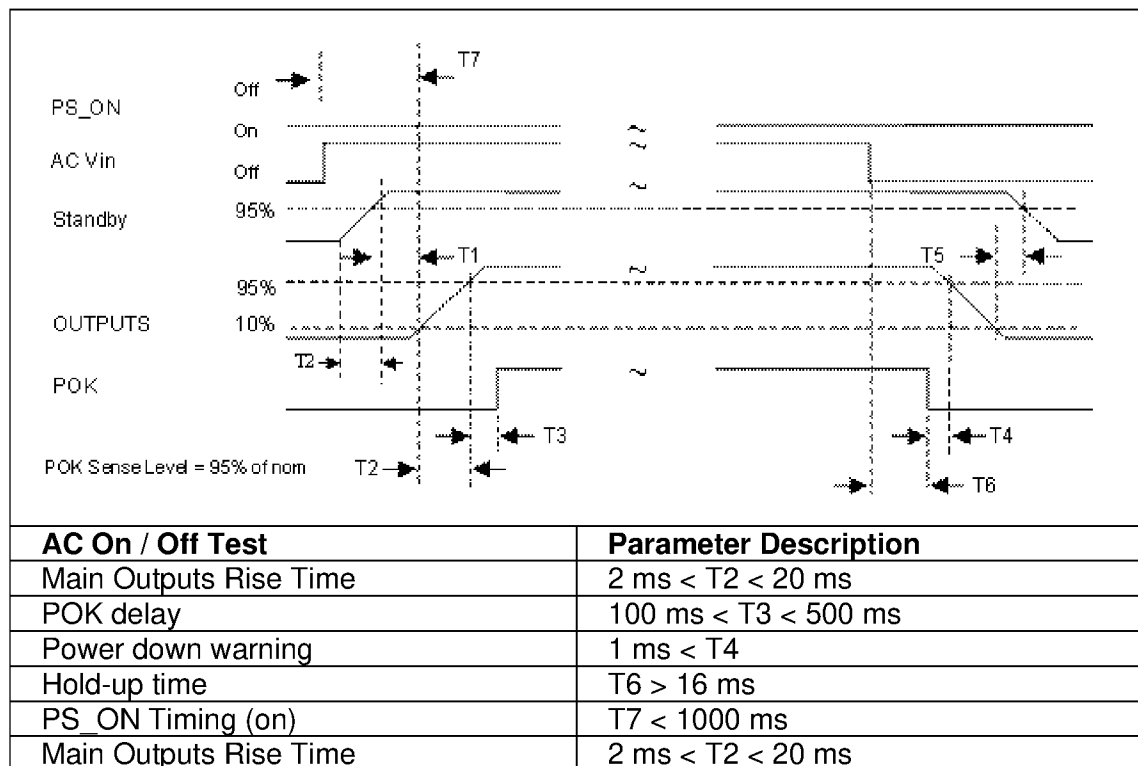


Figure 8: AC On / Off Control

4.10.2 PS_ON On / Off Control

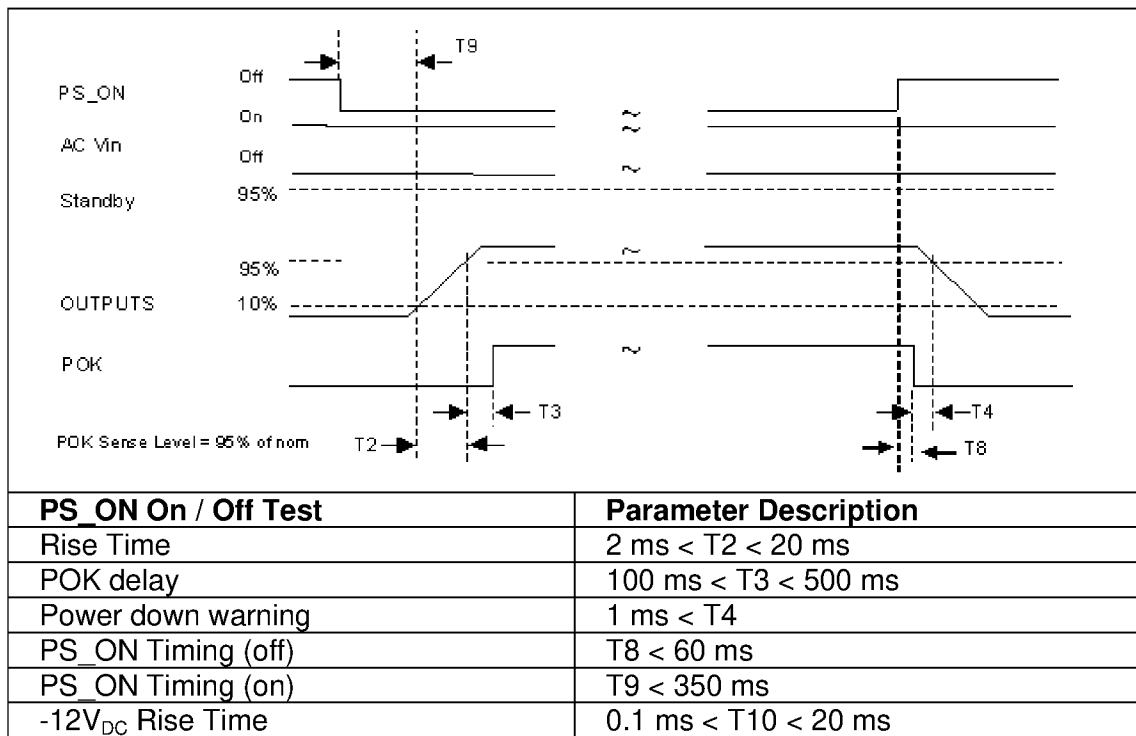


Figure 9: PS_ON On / Off Control

4.10.3 Logic Timings

Parameter Description	Description	Min.	Max.	DC Output	Measurements		Units
					AC IN	PS_ON	
T1	Delay from Standby within regulation to DC outputs turn on	5	300	N/A	148	N/A	ms
T2	Standby, +3.3V _{DC} , +5V _{DC} , and 12V _{DC} output rise time	2	20	+12V	2.8	4.2	
				+5V	2.6	2.4	
				+3.3V	2.8	2.2	
				+5Vsb	14	N/A	
T3	Delay from output voltages within regulation limits to POK asserted at turn on	100	500	+12V	250	256	
				+5V	244	250	
				+3.3V	244	252	
				-12V	6	12	
T4	Delay from POK deasserted to output voltages (3.3V, 5V, 12V, -12V) dropping out of regulation limits	1		+12V	1.6	28.4	
				+5V	4.8	30	
				+3.3V	6.8	32.4	
				-12V	6	12	
T5	Delay from DC output deasserted to Standby out of regulation at turn off.	5		+12V	608		
				+5V	600		
				+3.3V	600		
T6	Delay from loss of AC to desertions of PWOK.	16		N/A	604		
T7	PS_ON Timing (on)		1000	+12V	143		
				+5V	178		
				+3.3V	179		
T8	PS_ON Timing (off)		60	+12V		65.6	
				+5V		67.2	
				+3.3V		69.2	
T9	Delay from PS-ON reasserted to output turn on		350	+12V		80.8	
				+5V		86.8	
				+3.3V		86.8	
T10	-12V _{DC} output rise time	0.1	20	-12V	1.822	1.239	

Table 14: Logic Timings

4.10.4 PWR_OK

PWR_OK Full Load	CONTROL AND LOGIC SIGNALS RIPPLE/NOISE		
	Measurement	Max.	Unit
	35	400	mV _{P-P}

Table 15: PWR_OK Timings

4.10.5 PS_ON

PS_ON Full Load	CONTROL AND LOGIC SIGNALS RIPPLE/NOISE		
	Measurement	Max.	Unit
	30	400	mV _{P-P}

Table 16: PS_ON Timings

4.11 Output Protection

4.11.1 Output Over-Voltage Protection

DC Output	Specification		Measurements (V)
	Min. (V)	Max. (V)	
+12 V _{A/B}	13.5	15	
+5 V	5.6	7	6.36
+3.3 V	3.76	4.3	4.2
-12 V	-13.5	-15	
+5 V _{SB}	5.6	7	

Table 17: Output Over-Voltage Protection

4.11.2 Output Under-Voltage Protection

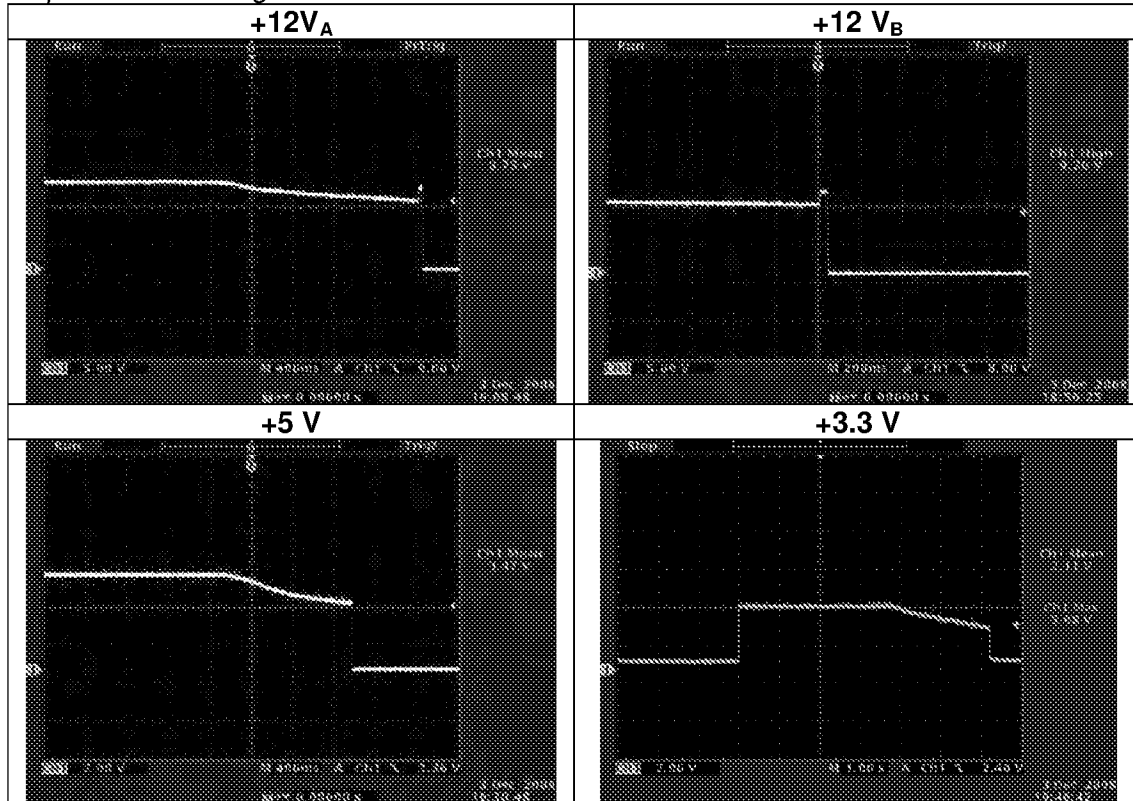


Figure 10: 115 V_{AC} / 60 Hz DC Output Under-Voltage Protection @ Full Load

4.11.3 Short Circuit Protection

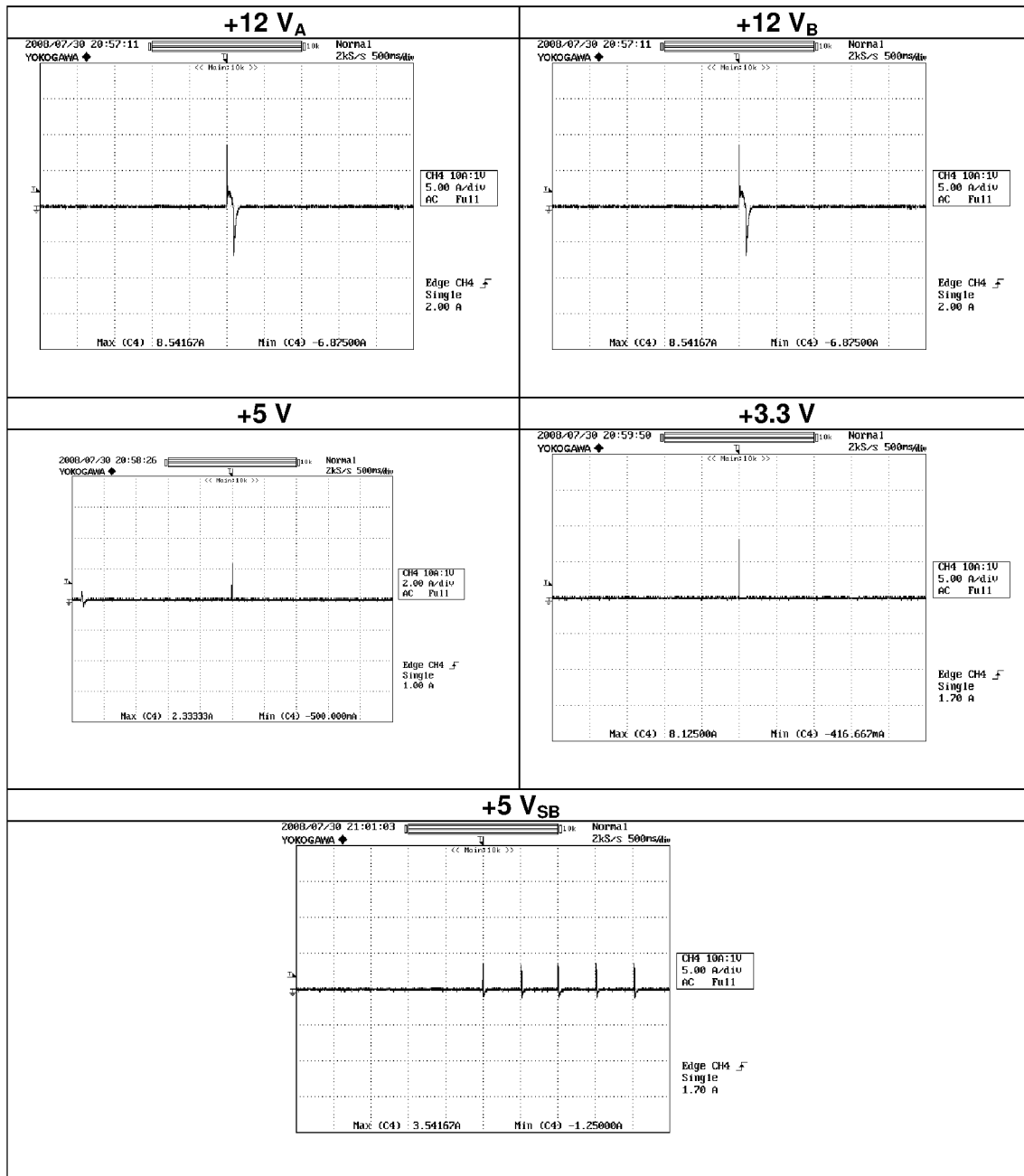


Figure 11: 115 V_{AC} / 60 Hz DC Output Short circuit protection @ Full Load

4.11.3 Over-Current Protection

DC Output	Over Current Protection		Measurements (A)
	Min. (A)	Max. (A)	
+12 V _A	15	21 (< 240 V _{AC})	19.6
+12 V _B	8.5	11.5 (< 240 V _{AC})	10.0
+5 V	18	24	20.5
+3.3 V	10	13	10.9
-12 V	N/A	N/A	N/A
+5 V _{SB}	3	6	6.0

Table 18: Over-Current Protection

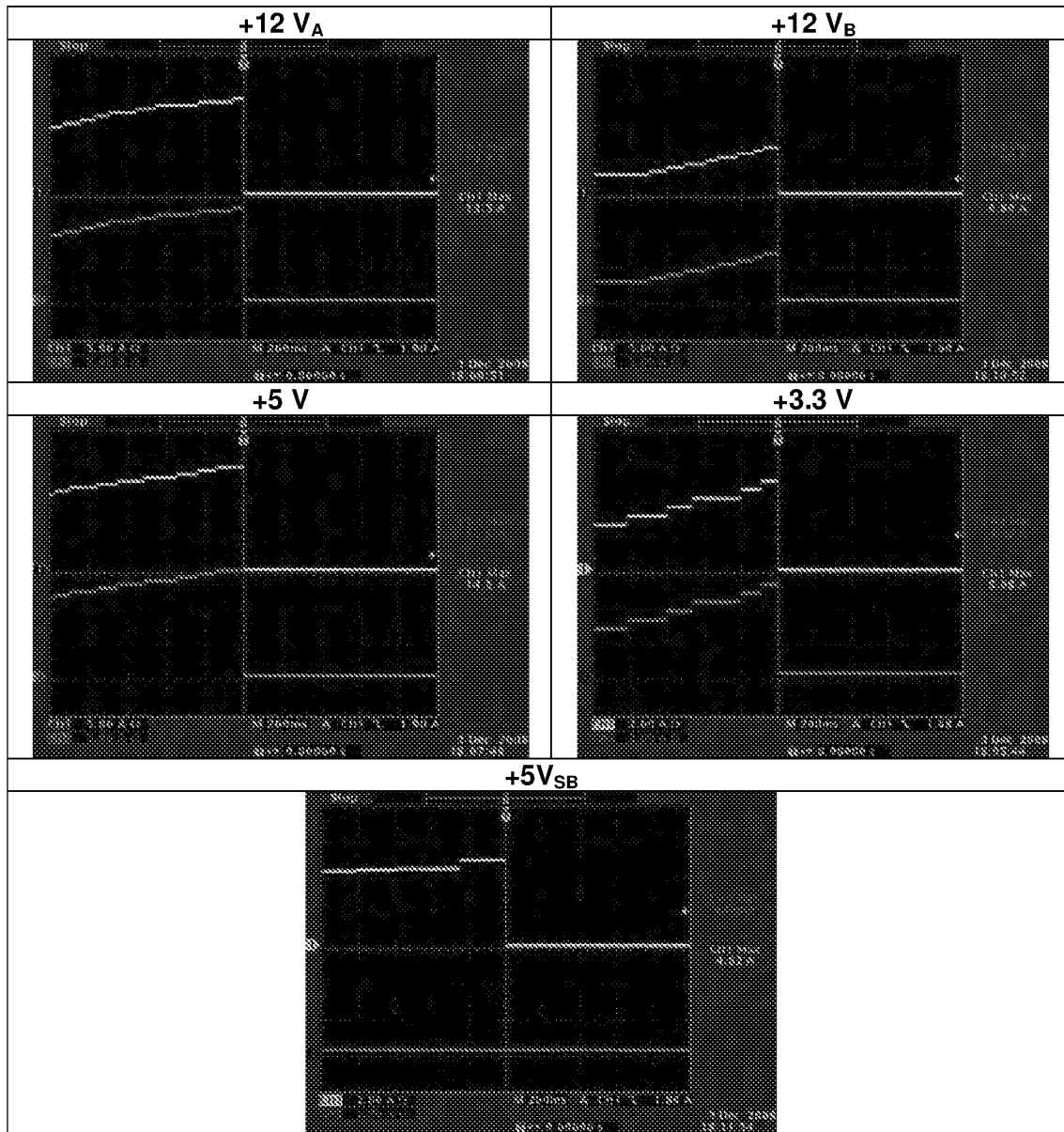


Figure 12: DC Output Short circuit protection @ Full Load

5. Evaluation Guidelines

Evaluation of the reference design should be attempted only by persons who are intimately familiar with power conversion circuitry. Lethal mains referenced voltages and high dc voltages are present within the primary section of the ATX circuitry. All testing should be done using a mains high-isolation transformer to power the demonstration unit so that appropriate test equipment probing will not affect or potentially damage the test equipment or the ATX circuitry. The evaluation engineer should also avoid connecting the ground terminal of oscilloscope probes or other test probes to floating or switching nodes (e.g. the source of the active clamp MOSFET). It is not recommended to touch heat sinks, on which primary active components are mounted, to avoid the possibility of receiving RF burns or shocks. High impedance, low capacitance test probes should be used where appropriate for minimal interaction with the circuits under investigation.

As with all sensitive switchmode circuitry, the power supply under test should be switched off from the ac mains whenever the test probes are connected and/or disconnected.

The evaluating engineer should also be aware of the idiosyncrasies of constant current type electronic loads when powering up the ATX demonstration unit. If the loads are adjusted to be close to the ATX's maximum rated output power, the unit could shut down at turn on due to the instantaneous overloading effect of the constant current loads. As a consequence, electronic loads should be set to constant resistance mode or rheostats should be used for loads. The other alternative is to start the supply at light to medium load and then increase the constant current electronic loads to the desired level.

The board is designed to fit in a traditional ATX enclosure as shown in Figure .

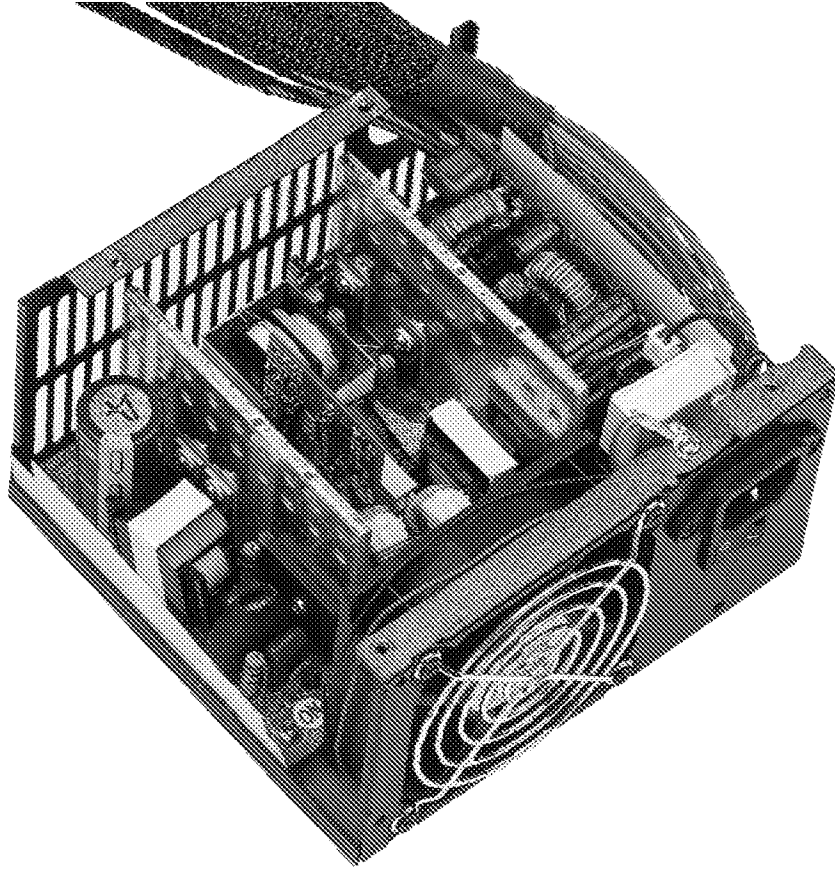


Figure 13: ON Semiconductor's 255 W Reference Design for ATX Power Supplies

6. Schematics

The power supply is implemented using a single sided PCB board. Added flexibility is provided by using daughter cards for the PFC (NCP1654) circuit and the NCP1396 resonant mode controller. The individual PCB board schematics are shown in figure 14.

7. Parts List

The bill of materials (BOM) for the design is provided in this section. To reflect the schematics shown in the previous section, the BOM have also been broken into different sections and provided in separate tables – Table 19 through 22.

It should be noted that a number of components used during the development cycle were based on availability. As a result, further cost reductions and better inventory management can be achieved by component standardization, i.e. the unique part numbers can be SIGNIFICANTLY reduced by standardization and re-use of component values and case sizes. This will result in a lower cost BOM and better inventory management.

QTY	SYMBOL	DESCRIPTION	VENDOR	VENDOR P/N
MAIN BOARD (PFC Stage, Synchronous Rectification Output Stage)				
1	HS1	HEAT SINK 93*60*4 AL4.0t	Taizhi	
1	BD1	DIO.BRI 10A 600V / TS10P05G	PANJIT	
1	D2	DIO.NR LQA08TC600 8A 600V TO-220AC	QSPEED	
1	Q1	FET.NCH SPP15N60C3 TO220	INFINEON	
2	Q3, Q4	FET.NCH STP12NM50 TO-220	ST	
1	SCREW(BD1)	SCREW PAN-HEAD M3*10	LONGFEI	
4	SCREW(D2, Q1, Q3, Q4)	SCREW PAN-HEAD M3*6	LONGFEI	
4	SL(D2, Q1, Q3, Q4)	SLTO-220 13*19*0.3mm	JUNHO	
4	B(D2, Q1, Q3, Q4)	BUSHING TO-220	JUNHO	
1	HS2	HEAT SINK 93*60*4 AL4.0t	Taizhi	
2	Q30, Q31	STP80NF55-06 80A55V TO220	ST	
1	D52,	MBR20L45CT 20A 45V TO-220	ON	
2	D70, D71	MBR60L45CTG 60A 45V TO-220	ON	
5	SCREW(D52, Q32, Q31, Q70, Q71)	SCREW PAN-HEAD M3*6	LONGFEI	
5	SL(Q30, Q31, D70, D71, D52)	SLTO-220 13*19*0.3mm	JUNHO	
5	B(Q30, Q31, D70, D71, D52)	BUSHING TO-220	JUNHO	
1	R3	RES.WW. 0.1R 3WS NKNP	Synton-Tech	
1	R13	RES.MO. 10- 2W	Synton-Tech	
3	R130, R131, R111	RES.CR. 10- 2WS	Synton-Tech	
1	D48	P6KE200A DO-15	PANJIT	
1	D50	UF4007 DO-41	PANJIT	
1	D1	STTH310 3A 1000V / UF5407G	DII	
1	C109	CAP.ELE 1000uF 10V 10*16KY	SU'SCON	
1	R1	RES.CR. 1M 1/4W		
3	RS12A, RS12A-1, RS12B	COPPER 0.002-		
1	F1	MST 5A/250V	CONQUER	
2	LF1.LF3	0.8*25t L=3mH+30% L=2.5mm	MEIHUA	(ASC-2203V-GH)25T
1	L2	POT3319 0.1Φ*50t*61t L=0.68uH	MEIHUA	
1	L3	T80-26+UL L=120uH±20%	MEIHUA	
1	L5	DR6*8 L=3.6uH	MEIHUA	
2	L6, L9	R8*20+UL L=0.8uH 2.4Φ*5.5t	J.X.E.	J-YH-R8*20-789

1	T1	YC3501 L=0.63mH Ls=80uH	MEIHUA	
1	T2	EE25	MEIHUA	
1	MOV	TVR10471KSY	TKS	
1	IC10	NCP1027P065G DIP-8	ON	
4	IC23, IC12, IC20, IC21	PHOTO PC817B DIP-4P	SHARP	
1	D54	1N4734A 1W 5.32-5.88V		
2	D122, D123	1N4006		
1	C4-1	CAP.PEI 0.022uF 630V		
1	C4	CAP.PEI 0.033uF 630V	PAC	R75PI2330JEMEJ
2	C1, C6	CAP.MEF 0.47uF 400V P=10	UTX	
1	CX3	CAP.MPP 0.33uF 275vac p=15	UTX	
1	CX2	CAP.MPP 0.47uf 275V HQX P=15	UTX	
1	C100	CAP.ELE 22uF 450V 8*11	SU'SCON	
1	C55	CAP.ELE 2.2uF 50V 5*11		
1	C2	CAP.ELE 220uf 450V	NDB	
1	C56	CAP.ELE 47uF 25V 5*11	NDB	
1	C105	CAP.ELE 220uF 25V 6.3*11	NDB	
1	C103	CAP.ELE 10uF 50V 5*11	NDB	
1	C104	CAP.ELE 100uF 35V 6*11	NDB	
5	C136, C137, C138, C139, C140	CAP.ELE 2200uF 16V 10*25 KZE	NDB	
1	C108	CAP.ELE 2200uF 10V 10*16	NDB	
1	C110	CAP.CER 1000PF 1KV Y5P P=5	SEC	2Y5P102K102C56E
2	C130, C131	CAP.CER 560PF 1KV Y5P	SEC	
1	C146	CAP.ELE 220uF 16V 6.3*11		
2	CY3, CY4	CAP.CER 4700PF Y2	SEC	
1	CY6	CAP.CER 2200PF Y1	SEC	
3	J4, J5, J6	JUMP 0.8Φ P=12.5mm		
3	J7, J19, J22	JUMP 0.8Φ P=20mm		
8	J1, J2, J3, J8, J11, J15, J18, J24	JUMP 0.8Φ P=10mm		
1	J23	JUMP 0.8Φ P=5mm		
1	MB CONNECT	MB CONNECT 24-PORT	EVERBIZ	
1	CPU CONNECT	CPU CONNECT 4-PORT(2*2)	EVERBIZ	
3	P3, P4, P5	SATA+SATA CONNECT	EVERBIZ	
2	P6, P7	SATA CONNECT	EVERBIZ	
2	P8, P9	FDD CONNECT 4-PORT	EVERBIZ	
2	HS3, HS4	HEAT SINK 28*38*5 CU1.2t	Taizhi	
1	CON1	WAFER 3.96(3-1)PIN 180°	SUNDA	
8	J10, J12, J13, J14, J16, R9, J20, R105	RES.SMD 0- 5% 0805		
1	J9	RES.SMD 33- 5% 0805		
1	J21	RES.SMD 0- 5% 1206		
1	R3-1	RES.SMD 0.5- 1% 2512		
5	R5, R10, R11, R133, R140	RES.SMD 10K 5% 0805		
2	R7, R8	RES.SMD 1.8M 5% 0805		
2	R12, R12A	RES.SMD 24K 5% 0805		

1	R54	RES.SMD 47K 1% 0805		
1	R55	RES.SMD 82.5K 1% 0805		
1	R56	RES.SMD 12K 5% 0805		
1	R57	RES.SMD 23.2K 5% 0805		
2	R51, R6	RES.SMD 3.3M 5% 0805		
1	R52	RES.SMD 3.6K 1% 0805		
2	R81, R82	RES.SMD 499K 1% 0805		
1	R83	RES.SMD 6.8K 1% 0805		
2	R99, R100	RES.SMD 2.2M 5% 0805		
1	R101	RES.SMD 27K 5% 0805		
1	R102	RES.SMD 78.7K 1% 0805		
1	R103	RES.SMD 3.3K 5% 1206		
6	R142, R147, R148, R250, R253, R150	RES.SMD 1K 5% 0805		
2	R108, R109	RES.SMD 10K 1% 0805		
1	R110	RES.SMD 100- 5% 0805		
2	R132, R141	RES.SMD 2.2- 5% 0805		
1	R134	RES.SMD 6.2K 5% 0805		
1	R135	RES.SMD 3.3K 5% 0805		
2	R137, R138	RES.SMD 10- 5% 0805		
1	R53	RES.SMD 10- 5% 1206		
1	R143	RES.SMD 68K 5% 0805		
1	R144	RES.SMD 18K 1% 0805		
1	R145	RES.SMD 4.7K 1% 0805		
2	R146, R149	RES.SMD 22- 5% 0805		
1	R251	RES.SMD 4.7K 5% 0805		
1	R252	RES.SMD 470- 5% 0805		
1	R269	RES.SMD 330- 5% 0805		
1	C5	CAP.MON 220P 50V 0805 X7R		
1	C50	CAP.MON 0.47uF 50V X7R 0805		
2	C101, C102	CAP.MON 0.01uF 50V X7R 0805		
2	C107, C54	CAP.MON 0.22uF 50V X7R 0805		
3	C53, C144, C145	CAP.MON 1000PF 50V X7R 0805		
3	C134, C132, C52	CAP.MON 0.1uF 50V X7R 0805		
1	C141	CAP.MON 0.022uF 50V X7R 0805		
1	D51	ES1D 1A 200V SMA	TSC	
4	D5, D72, D73, D49	DIO.NR LL4148	TSC	
2	D77, D78	DIO.ZEN RLZ7.5B	ROHM	
2	Q16, Q17	MMBT2222 SOT23		
1	IC1	NCP1654A 65KHz SO-8	ON	
2	IC11, IC16	TL431	ON	
2	IC14, IC15	NCP4302 SO-8	ON	
1	PCB	CEM-1 1oz 1.6t 145*108.5 NO.SPB011V5		

Table 19: Main Board (PFC Stage, Synchronous Rectification Output Stage)

QTY	SYMBOL	DESCRIPTION	VENDOR	VENDOR P/N
DC-DC Converter Stage, Supervisory Stage (referred to as CTL2 in schematics of figure 14)				
1	C220	CAP.ELE 2200uF 16V 10*25	NDB	
1	L11	BEAD RH035100ST-A8		
1	RS33	COPPER 0.002-	Tzai Yuan	
1	C208	CAP.ELE PSC 680uF 10V 10*11.5	NDB	
5	C206, C207, C215, C218, C219	CAP.ELE 2200uF 16V 10*25	NDB	
1	C260	CAP.ELE 10uF 50V 5*11 KY	NDB	
1	C142	CAP.ELE 330uF 25V 8*15 KY	NDB	
1	C143, C200, C200A	CAP.ELE 220uF 25V KY 6.3*11	NDB	
1	C209	CAP.PEI 0.47uF 100V P=10	欣統	
1	C262	CAP.ELE 100uF 25V 6.3*11	NDB	
1	D110	1N4744		
1	Q21	S8050L-C EBC TO92	UTC	
1	L7	HKH080 L=6.8uH		
1	L8	HKH080 L=6.8uH	MEIHUA	HKH-080CE/034
1	L9	DR6*8 L=3.6uH	MEIHUA	
2	L10, L13	R6*20+UL L=3.0uH±20% 1.2Φ*11.5t	MEIHUA	R6*20-0003
1	RS5	R8*20+UL L=1uH±20%		
1	TM1	T10K	TKS	TTC05103KSY
1	TM2	T204K	TKS	TTC05204KSY
1	HS7	HEAT SINK 25.5*12.5 Cu1.2t	Taizhi	
2	HS8, HS9	HEAT SINK 40.5*12.5 Cu1.2t	Taizhi	
1	CON3	WAFER P=2.5*2 90°	SUNDA	
1	P4(CTL3-PCB)	Pin Header 3pin 90° P=2.54mm	SUNDA	
1	P5(CTL3-PCB)	Pin Header 6pin 90° P=2.54mm	SUNDA	
1	PPTC	Fuse 1A 1206	CONQUER	nSMD100
3	R266, R300, R302	RES.SMD 10K 5% 0805		
1	R203	RES.SMD 27K 1% 0805		
2	R204, R210	RES.SMD 5.11K 1% 0805		
4	R212, R214, R213, R215	RES.SMD 10- 5% 0805		
1	R206	RES.SMD 75K 5% 1206		
1	R216	RES.SMD 18K 5% 1206		
1	R209	RES.SMD 16K 1% 0805		
5	R254, R259, R260, R261, R262	RES.SMD 47- 5% 0805		
1	R255	RES.SMD 470- 1% 0805		
1	R256	RES.SMD 240- 1% 0805		
1	R257	RES.SMD 100- 1% 0805		
1	R258	RES.SMD 200- 1% 0805		
1	R263	RES.SMD 2K 5% 0805		
1	R264	RES.SMD 2.2K 5% 0805		
5	R265, R267, R301, R202, R208	RES.SMD 1K 5% 0805		
1	R268	RES.SMD 62K 1% 0805		
1	R303	RES.SMD 15- 5% 0805		
2	C209, C209-1	CAP.MON 0.22uF 50V X7R 1812		
14	C204, C213, C216, C217, C250, C252, C253, C254, C255, C256, C257, C258, C259, C261	CAP.MON 0.1uF 50V X7R 0805		
2	C202, C211	CAP.MON 100P 50V X7R 0805		
2	C203, C212	CAP.MON 0.056uF 50V X7R 0805		

1	C251	CAP.MON 0.47uF 16V X7R 0805		
2	D100, D103	ES1D 1A 200V SMA	TSC	
2	D101, D102	DIO.SB SR24 2A 40V SMA	PANJIT	
6	D104, D105, D112, D113, D106, D107	DIO.ZEN LL4148	TSC	
2	D120, D121	DIO.ZEN RLZ6.2B	ROHM	
1	Q20	MMBT2222 SOT23		
4	Q32, Q33, Q34, Q35	NTD4809N-D 58A 30V DPAK	ON	
1	IC17	PS223 SOP-16	SITI	
2	IC18, IC19	NCP1587 SO-8	ON	
1	IC20	TL431	ON	
1	PCB	FR4 1oz 1.6t 62*83.5 NO.PB011V5-CTL3		

Table 20: DC-DC Converter Stage, Supervisory Stage (referred to as CTL2 in schematics of figure 14)

P/N	QTY	SYMBOL	DESCRIPTION	VENDOR	VENDOR P/N
HB Resonant LLC-Stage (referred to as CTL1 in schematics of figure 14)					
	1	C58	CAP.ELE 4.7uF 50V KMG 5*11	NDB	
	1	C60	CAP.ELE 47/25V KMG 5*11	NDB	
	1	C64	CAP.ELE. 10uF 25V	NDB	
	1	P2 (CTL2-PCB)	Pin Header 3pin 90° P=2.54mm	SUNDA	
	1	P3 (CTL2-PCB)	Pin Header 13pin 90° P=2.54mm	SUNDA	
	1	HS	25*16*8		
	1	PCB HOLD	RCC-5	KANGYANG	
	4	R55, R69, R70, R84	RES.SMD 1K 5% 0805		
	2	R56, R57	RES.SMD 30K 5% 0805		
	1	R58	RES.SMD 15K 5% 0805		
	1	R59	RES.SMD 6.8K 5% 0805		
	1	R60	RES.SMD 8.2K 1% 0805		
	1	R61	RES.SMD 150K 5% 0805		
	1	R62	RES.SMD 2K 5% 0805		
	2	R64, R73	RES.SMD 1.1K 1% 0805		
	6	R63, R67, R72, R79, R80, R86	RES.SMD 10K 5% 0805		
	1	R65	RES.SMD 18- 5% 1206		
	1	R66	RES.SMD 5.1K 5% 0805		
	2	R68, R71	RES.SMD 10- 5% 0805		
	1	R74	RES.SMD 3.3K 5% 0805		
	1	R76	RES.SMD 1.8K 5% 0805		
	1	R77	RES.SMD 33- 5% 0805		
	1	R88	RES.SMD 100- 5% 0805		
	1	R85	RES.SMD 470K 5% 0805		
	1	R87	RES.SMD 22K 5% 0805		
	1	R89	RES.SMD 4.7K 5% 0805		
	1	C59	CAP.MON 2200PF 50V X7R 0805		
	2	C61, C65	CAP.MON 0.1uF 50V X7R 0805		
	1	C62	CAP.MON 0.068uF 50V X7R 0805		
	1	C63	CAP.MON 0.01uF 50V X7R 0805		
	3	C66, C67, C57	CAP.MON 1uF 25V X7R 0805		
	1	D20	UF 1A600V / US1J	PANJIT	
	4	D21, D24, D25, D29	DIO.ZEN. LL4148	TSC	
	1	D22	DIO.ZEN. RLZ7.5B	ROHM	
	1	D23	DIO.ZEN. RLZ3.6B	ROHM	
	2	Q5, Q12	MMBT2907	PANJIT	
	1	Q11	2SC4672	ROHM	
	1	Q14	2SA1797	ROHM	
	1	IC2	NCP1396A SO-16	ON	
	2	IC3, IC4	TL431	ON	
	1	IC5	LM393 SO-8	ON	LM393D
	1	PCB	FR4 1oz 1.6t 44*56 NO.SPB011V4-CTL2		

Table 21: HB Resonant LLC-Stage (referred to as CTL1 in schematics of figure 14)

QTY	SYMBOL	DESCRIPTION	VENDOR	VENDOR P/N
Mechanical and Miscellaneous Items				
1	SW1	0-1 4P 10A Look SW	SWEETA	SS21-BBIWG-R
1	SW-INLET	UL1015#18 L=115mm	CHARNG MIN	
1	SW-INLET	UL1015#16 L=60mm White	CHARNG MIN	
1	SW-INLET	UL1015#16 L=60mm Black	CHARNG MIN	
1	INLET(G)-GND	UL1015#18 L=125mm	CHARNG MIN	
1	CORE(INLET-GND)	RH16*9*17		
1	DTOD-BOARD			
1	FAN	80*80*25mm 12V	SUNON	
2	CY1, CY2	CAP.CER 4700PF Y2	SEC	
1	CX1	CAP.MPP 1uF 275VAC HQX P=22.5	UTX	
1	AC INLET	10A/15A 250V	SWEETA	SC-9-1
1	CASE	CASE 150.2*140*84mm		
1	CASE	CASE 140*148.2*85.5mm		
1	MYLAR	MYLAR FILM 165*110*0.35mm	JUNHO	
2	SCREW(AC INLET)	SCREW F3*10 ISO(BLACK)	LONGFEI	
4	SCREW(CASE)	SCREW F3*6 ISO(BLACK)	LONGFEI	
1	SCREW(INLET-GND))	SCREW F3*5 ISO(BLACK)	LONGFEI	
4	SCREW(S1~S4)	SCREW MAIN BOARD M3*4 ISO	LONGFEI	
4	SCREW(FAN)	SCREW I5*10 TAP (BLACK)	LONGFEI	
1	SCREW(GND)	SCREW K/NUT 8#32T	LONGFEI	
1		FAN GUARD 80*80mm COLOR-GOLD	PRO-CROWN	
1		SANP BUSHING NB-27A	KANGYANG	

Table 22: Mechanical and Miscellaneous Items

8. Resources/Contact Information

Data sheets, applications information and samples for the ON Semiconductor components are available at www.onsemi.com. Links to the datasheets of the main components used in this design are included in the Appendix.

Authors of this document are: Edward Weng, Patrick Wang, Roman Stuler, and Laurent Jenck.

9. Appendix

Link to ON Semiconductor's web site:

- * [ON Semiconductor Home Page](http://www.onsemi.com)

Industry information links:

- * [ENERGY STAR](http://www.energystar.gov)
- [80 PLUS Efficiency Requirements](http://www.energystar.gov)
- [Climate Savers Computing Initiative](http://www.energystar.gov)
- [IEC61000-3-2 Requirements](http://www.energystar.gov)
- [ATX 12 V Form Factor](http://www.energystar.gov)
- [European Union \(EU\) Energy Star Page](http://www.energystar.gov)

Additional collateral from ON Semiconductor:

- [NCP1654](#) Continuous conduction mode PFC controller
- [NCP1396](#) Resonant mode controller with high voltage drivers
- [NCP4302](#) Synchronous rectification controller
- [NCP1587](#) Low voltage synchronous buck controller
- [NCP1027](#) High voltage integrated switcher
- [NTD4809](#) Single N-Channel MOSFET 30 V, 58 A
- [MBR20L45](#) 20 A, 45 V dual schottky rectifier

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Jason Lai	Committee Member												

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Abstract

Power conversion system design issues are expanding their role in technology equipment design philosophies. These issues include not only power conversion efficiency, but also increased concerns regarding the complexity of the power conversion design techniques utilized to satisfy a system's total performance requirements. In particular, in computer (desktop computers, workstations, and servers) designs, the power "supplies" are becoming a limiting factor in meeting overall design objectives.

This dissertation addresses the issue of simplifying the architecture of power systems incorporated into computing equipment. In the first half, the subject of the design of the distributed power system's front end is investigated from the perspective of simplifying the conversion process while simultaneously improving efficiency. This is initially accomplished by replacing the second-stage DC/DC converter in the standard two-stage front-end (consisting of a first-stage AC/DC converter followed by DC/DC conversion) through the incorporation of second-order active power factor control. Unique modifications are then made to two basic topologies (buck and boost converter topologies) that enable the two-stage front-end to be reduced to an isolated PFC conversion stage, resulting in a front-end architecture with features of reduced complexity and higher efficiency.

In the dissertation's second half, the overall DC distributed power system concept is simplified through the elimination of power processing components, the result being the creation of a high-frequency (HF) AC distributed power system. Design techniques for generating, distributing, and processing HF AC power are developed and experimentally verified. Also, an experimental comparison between both DC and AC distributed power systems is

illustrating in a succinct fashion the merits and limitations of both a

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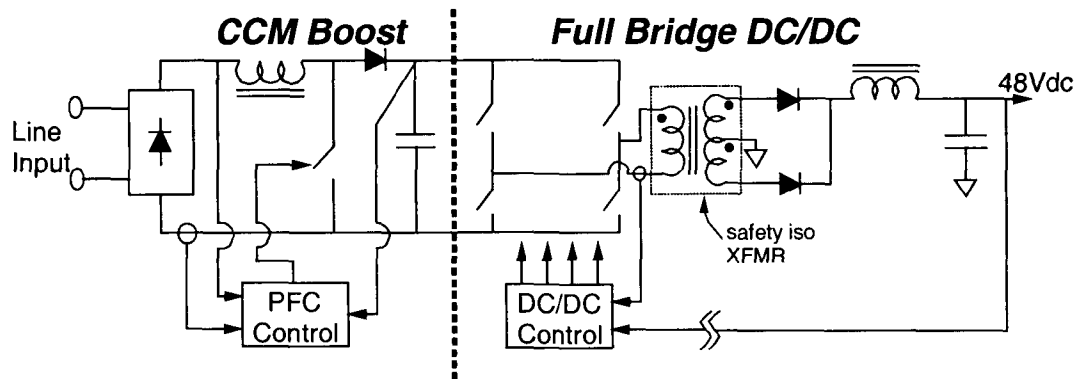
4. ACTIVE-CLAMP BOOST AS AN ISOLATED PFC FRONT-END CONVERTER

4.1 Introduction

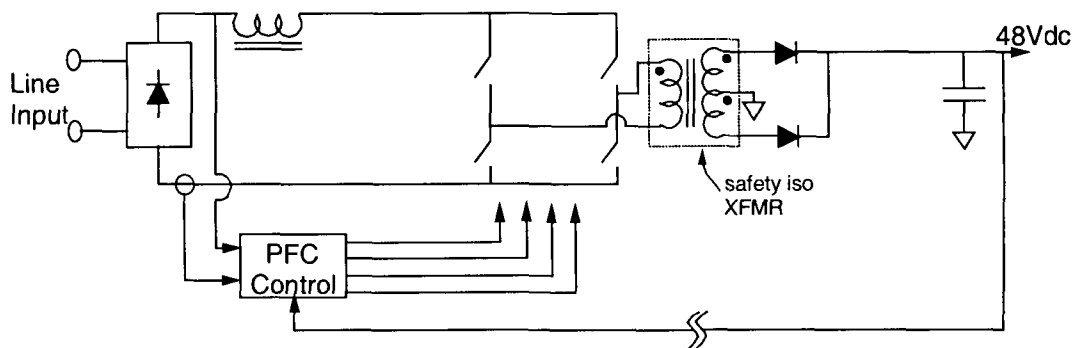
This chapter continues the theme set by Chapter 3 - simplifying the standard two-stage front-end implementation to one that consists of just an isolated PFC converter. However, this chapter addresses this issue for higher power level system requirements.

As a starting point, shown in Fig. 4.1(a) is a typical, higher power output, two-stage front-end design topology. In Chapter 2, this design was simplified by incorporating secondary-side switching in the DC/DC converter. However, just as in Chapter 3, the "bulk" capacitor can be moved to the secondary, as shown in Fig. 4.1(b). Because kilowatt applications are being targeted, the full-bridge structure is maintained, and essentially replaces the PFC boost converter switch. Therefore, whatever the preferred PFC control method that is used in the two-stage approach is preserved in the implementation of Fig. 4.1(b). As with the front-end PFC flyback converter described in Chapter 3, the input line current's second harmonic is processed by the isolation transformer in the full-bridge boost converter shown in Fig. 4.1(b) and appears as voltage ripple on the output DC bus. However, the overall power stage design is much simpler, and, as will be experimentally demonstrated later in this chapter, this PFC implementation is more efficient than the two-stage circuit shown in Fig. 4.1(a).

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(a) Topological concept of a traditional two-stage front-end design.



(b) Front-end consisting of an isolated boost PFC converter only.

Fig. 4.1 Two-stage and isolated boost PFC front-end conceptual designs.

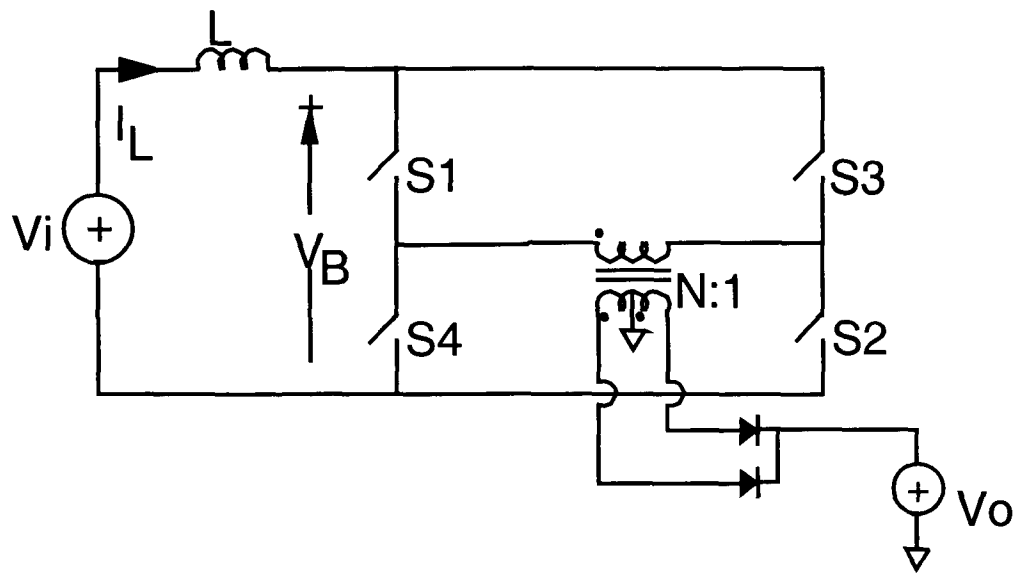
The full-bridge structure of the PFC boost converter's "switch" introduces the possibility of phase-shift control, in an analogous fashion to the phase-shift control used in voltage fed full-bridge converters. As will be discussed in the next section, phase-shifted control for the design shown in Fig. 4.1(b) can realize zero-current-switching (ZCS) for two of the four bridge switches, and a solution to the "voltage-spike" problem - so apparent in transformer isolated current fed designs, leads to ZVS for all four of the bridge switches [51].

4.2 Theory of Operation

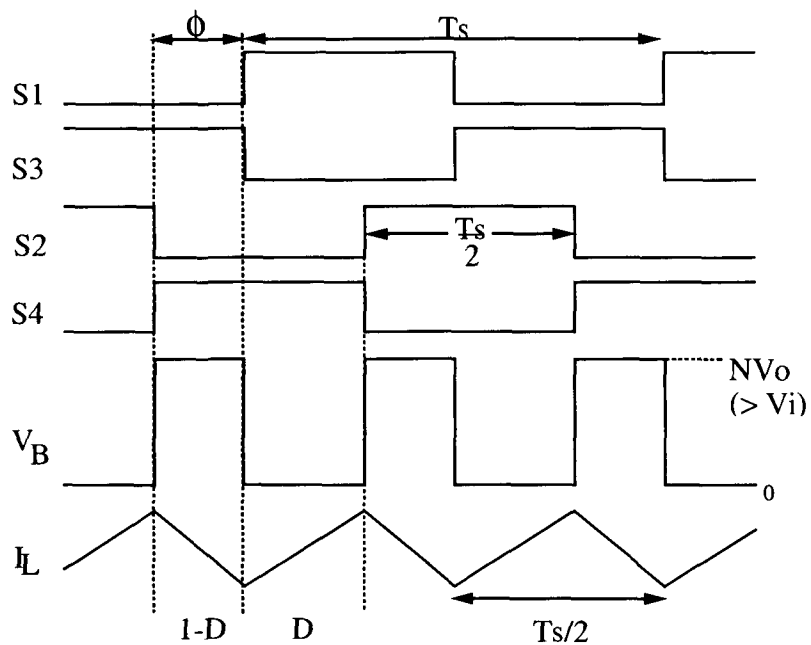
Figure 4.2 shows a simplified schematic of the active-clamp full-bridge boost topology. CCM operation for the boost inductor is assumed. Phase-shifting of the bridge switching sequence is utilized to control the boost function. The full-bridge topology, coupled with phase-shift control, offers a couple of advantages: 1) the boost inductor ripple frequency is twice the switching frequency, reducing its size, and, 2) with a slight modification to the phase-shift control ZCS becomes possible for the two of the four bridge switches [52]. Figures 4.2 and 4.3 illustrate these two concepts. For ZCS it is necessary to utilize the leakage inductance in the transformer (or possibly an external resonant inductor). Referring to Fig. 4.3, turning on S4 before S2 is turned off forces the current in L_{lk} to decrease linearly at a rate determined by:

$$\frac{di_{L_{LK}}}{dt} = \frac{NV_O}{L_{LK}}. \quad (4.1)$$

Current is increasing in S4 at the same rate (it's assumed the boost inductor can be

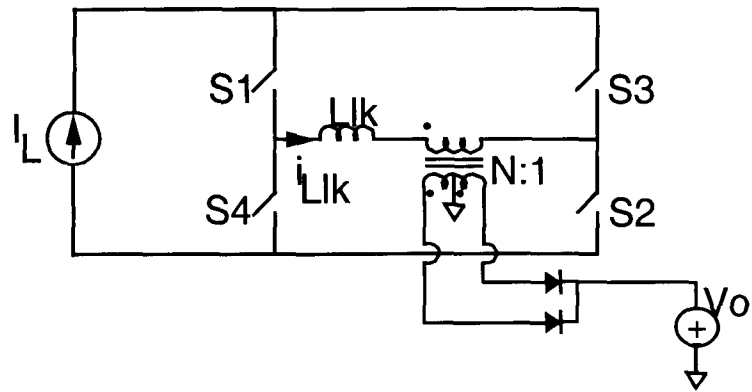


(a) Simplified circuit.

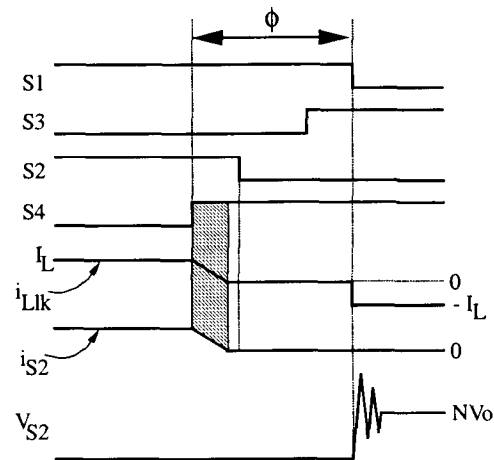


(b) Idealized waveforms.

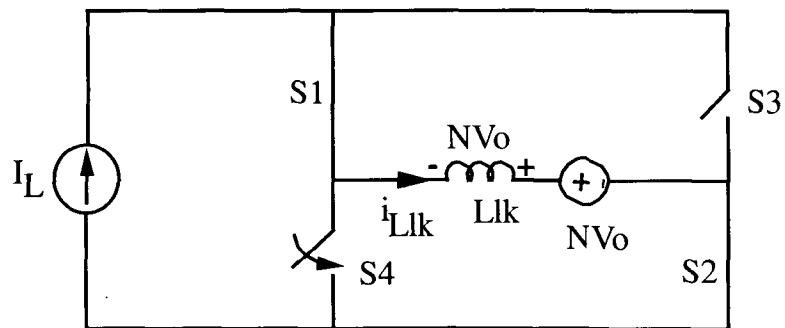
Fig. 4.2 Simplified ϕ -shifted active-clamp full-bridge boost converter operation.



(a) Simplified circuit with leakage inductance.



(b) Circuit waveforms.



(c) Converter topology for ZCS.

Fig. 4.3 Basic concept of boost full-bridge phase-shift operation [52].

idealized as a current source). When i_{s2} reaches zero the switch can be turned off with zero current.

As a practical matter, the isolated boost converter suffers from a problem at switch turn-off due to the dissimilarity of the values of the current source feeding the bridge (assuming the boost inductor can be idealized as a current source) and the current in the isolation transformer's leakage inductance. Unless an alternative path for the boost inductor current is temporarily provided, a large voltage spike, capable of destroying devices, will appear across the opened bridge switches. Some form of snubbing or clamping is necessary. For high-power applications RC snubbers are not very efficient. Therefore, an active-clamp circuit [51] (as shown in Fig. 4.4) is proposed for this application. In addition to the clamping function, the active-clamp circuit enables the isolation transformer's leakage energy to be used for zero-voltage switching (ZVS) of the bridge switches. Zero-voltage-switching is particularly important for S1 and S3, since snubber capacitors are used to reduce their turn-off losses. The ZVS mechanism recovers this turn-off energy before it is dissipated by switch turn-on. However, it should be pointed out that the circuit proposed here differs from that discussed in [51] in three fundamental ways: 1) ϕ -shift operation of the bridge switches enables zero-current switching (ZCS), 2) individual switches in the bridge operate with a fixed 50 % duty cycle, and 3) the operation of the boost inductor is in continuous conduction mode - necessary for single-phase high power operation.

Figure 4.4 shows the basic waveforms of operation with the active-clamp circuit. These waveforms are for a switching cycle and assume sufficient energy is stored in the leakage inductance to discharge C_{snub} completely. Also, it's assumed the clamp capacitor is infinite in value and the boost inductor can be replaced by a current source. The sequence of topological states is shown in Fig. 4.5, and is described below:

T_0-T_1 : At T_0 , S3 is turned off and the boost inductor current charges the switch capacitance. In order to control the switch turn-off losses, an external capacitor may be

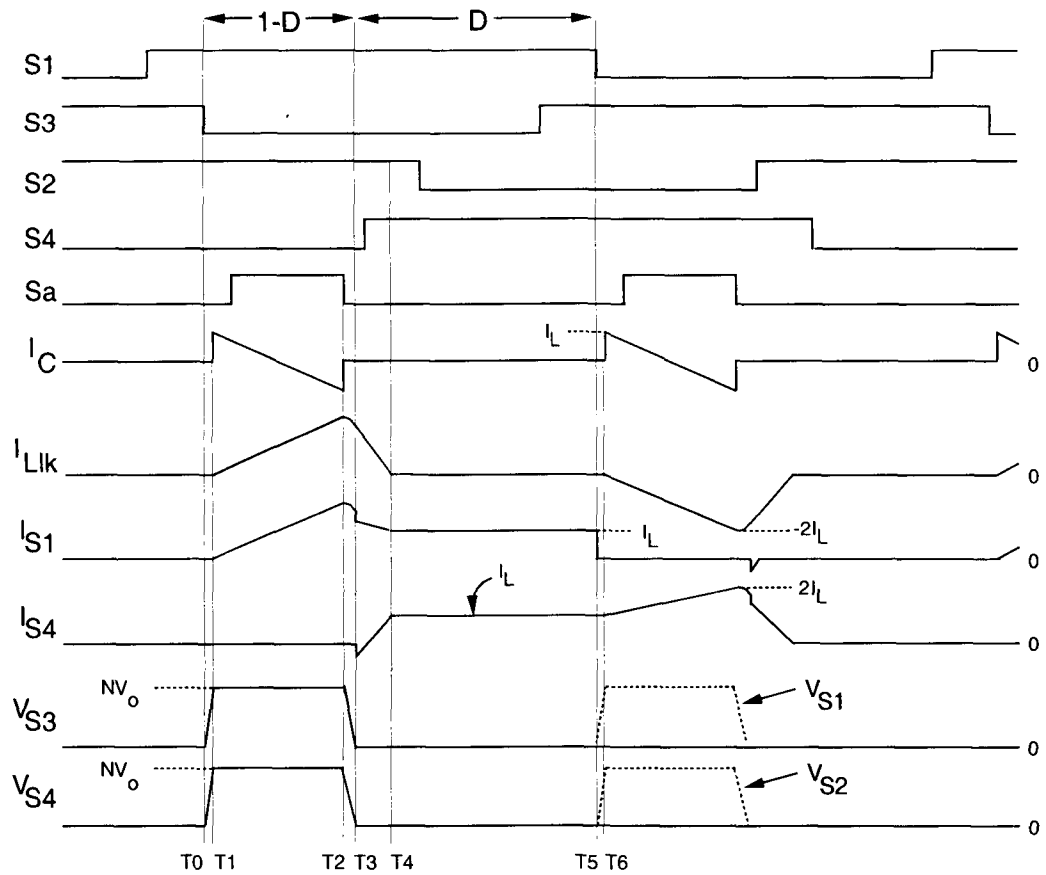
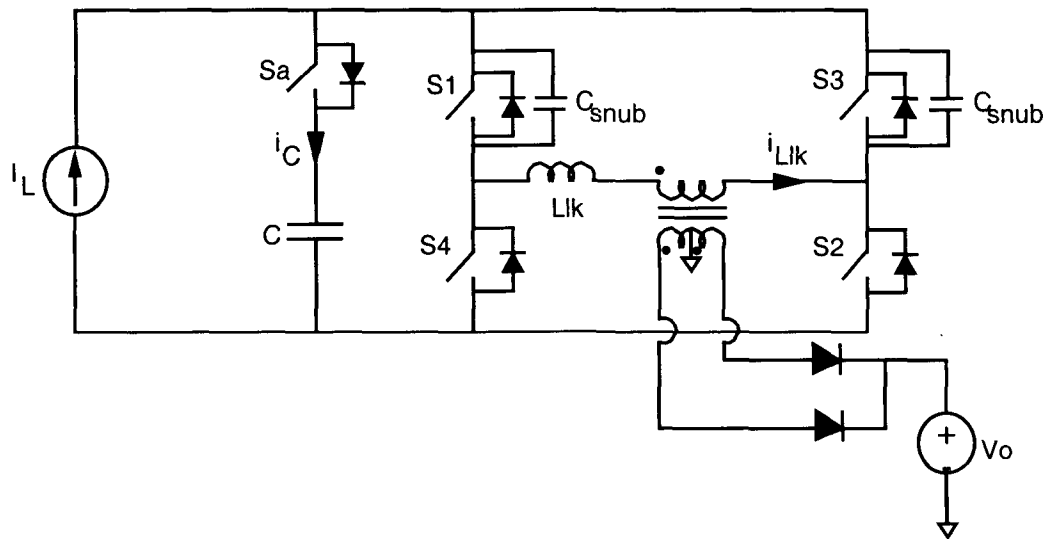


Fig. 4.4 Active-clamp FB boost converter ideal waveforms.

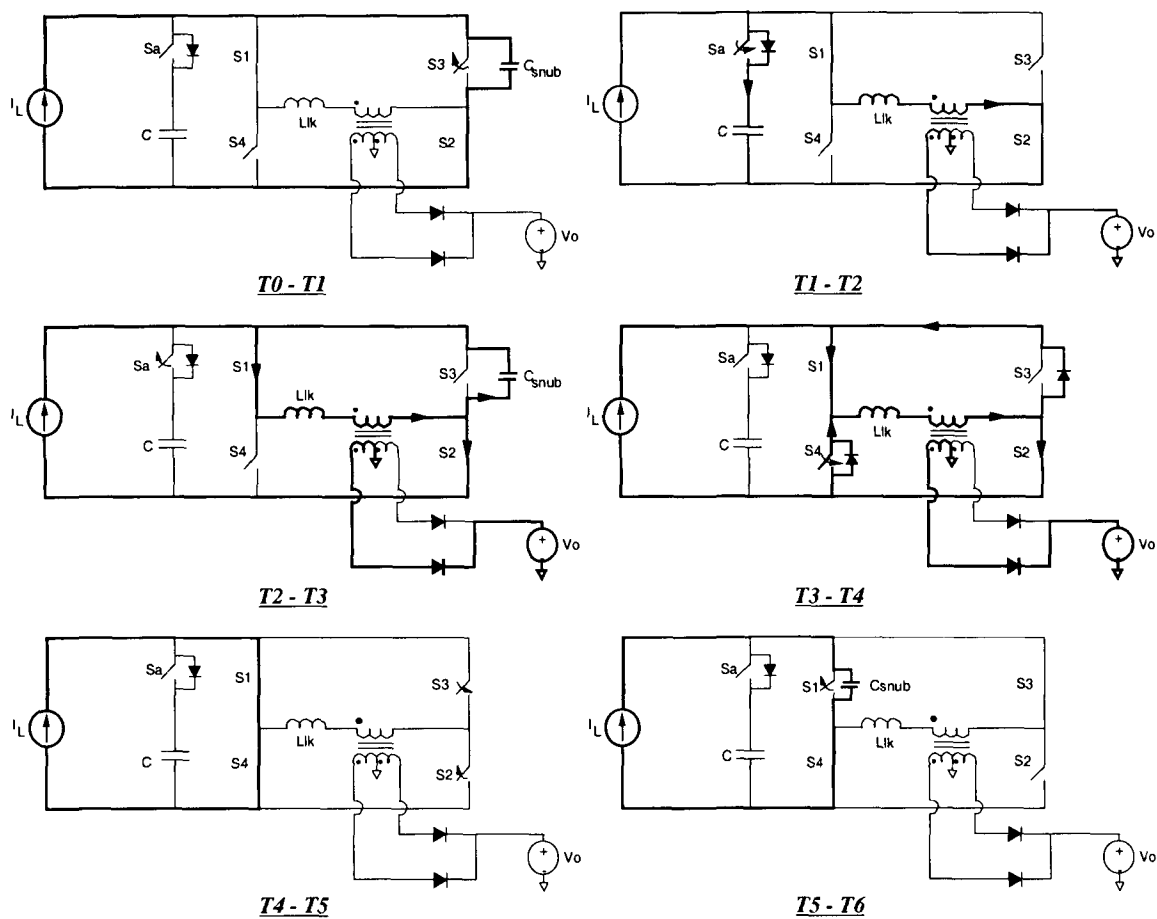


Fig. 4.5 Active-clamp full-bridge boost converter topological states.

added across the switch to form a turn-off snubber. This would be done, however, at the expense of decreasing the ZVS range.

T_1 - T_2 : At T_1 , V_{S3} reaches the clamp capacitor voltage ($\approx NV_o$), and the clamp diode conducts the boost inductor current. The current through the leakage inductance increases as L_{LK} and C start to resonate. Current starts flowing to the output. To facilitate ZVS for S_a , the switch is turned on before the clamp current reverses.

T_2 - T_3 : At T_2 the active-clamp switch is turned off. Under the assumption of an infinite value of clamp capacitance, the turn-off current is equal to the boost inductor current (which implies $i_{Llk} = 2I_L$ at T_2). The difference between i_{Llk} and I_L discharges C_{snub} (C_{snub} and L_{LK} resonate). The voltage across S_3 and S_4 starts decreasing.

T_3 - T_4 : Assuming sufficient energy has been stored in L_{LK} to discharge C_{snub} , at T_3 the body diodes of S_3 and S_4 begin to conduct. S_4 can then be turned on with zero voltage. I_{Llk} (and the current through S_2) now decrease at a linear rate determined by the reflected output voltage and the value of L_{LK} . When i_{Llk} decreases to the value of the boost inductor current, S_4 begins conducting the difference between the two currents.

T_4 - T_5 : When i_{Llk} decreases to zero, S_2 can be turned off with zero current. Sometime during this interval S_3 can be turned on with zero voltage, enabling C_{snub} to perform a lossless snubber action for turn-off of S_3 .

T_5 - T_6 : At T_5 , S_1 is turned off, ending the inductor charging interval. The sequence of topologies is then repeated for the other half of the bridge.

4.3 Design Considerations and Calculations

4.3.1 Switch timing

Other than the timing required to modulate the duty cycle through phase-shift control, it is necessary to derive the timing constraints to realize ZVS and ZCS. For ZVS, the timing between the turn off of S_a and the turn-on of S_2 or S_4 must allow the snubber capacitor voltage to reach its minimum. This is simply given by one-quarter of the L_{LK} - C_{snub} resonant period (during the T_2 - T_3 interval):

$$t_{Sa-S2} = t_{Sa-S4} = \frac{\pi}{2} \sqrt{C_{snub} L_{Lk}} . \quad (4.2)$$

For ZCS, sufficient time must be given between the turn-on of S_2 or S_4 and the turn-off of the switch on the opposite side of bridge (S_4 or S_2 , respectively). The required time is determined by the value of transformer leakage inductance (referred to either the primary or the secondary), the output voltage (or reflected output voltage if leakage is considered from the primary), and the initial leakage inductance current. The worst case maximum time occurs with maximum leakage current (during the T_3 - T_4 interval):

$$t_{S2-S4} = t_{S4-S2} = \frac{2I_L^{peak} L_{Lk}}{\frac{N_p}{N_s} V_o} . \quad (4.3)$$

I_L^{peak} is the peak boost inductor current.

4.3.2 Circulating energy

Due to the action of the active-clamp circuit, circulating energy is increased in the power stage components. To some degree this is beneficial, since it is desirable to operate under zero-voltage switching conditions. However, for ZVS it is not necessary for the clamp switch to conduct for almost the entire $(1-D)T_s$ time period. As long as the leakage inductance current is larger than I_L at the instant the clamp switch is turned off, ZVS may be possible. Therefore, the timing for S_a can be modified, as shown in Fig. 4.6, to reduce conduction losses. The drawback to this modified timing scheme is that ZVS is lost for S_a and some ringing will be introduced across the bridge when neither S_a nor its diode is conducting.

4.3.3 Design calculations

The design calculations developed in this section primarily revolve around determining various component losses. It is assumed insulated gate bipolar transistors (IGBTs) are used for the bridge switches. In fact, the experimental results reported in Section 4.4 are based on the use of IGBT bridge switches.

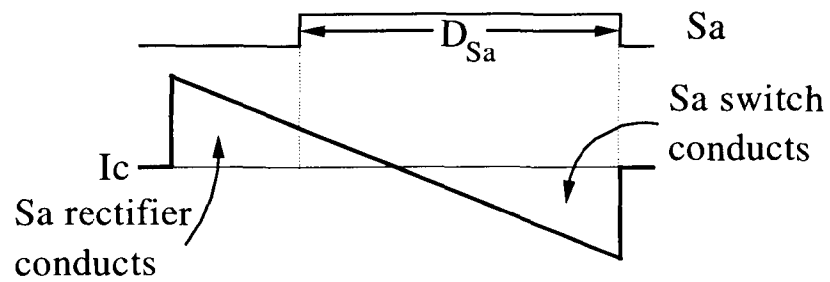
4.3.3.1 Full-bridge switch calculations

4.3.3.1.1 Conduction loss

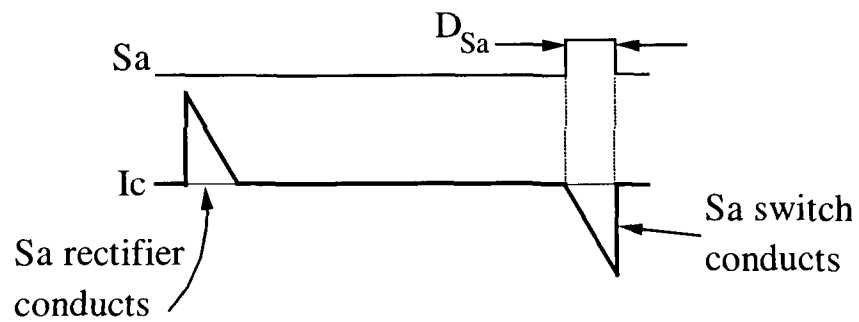
As a function of the 120 Hz line cycle, the average (switching cycle) current seen by individual switches in the bridge is given by:

$$\langle I_S \rangle(\phi) = \frac{\sqrt{2}P_o}{\eta 2V_{RMS}} \sin \phi, \quad (4.4)$$

where ϕ is the instantaneous operating point on the input line (normalized over the interval $[0, \pi]$) and η is the conversion efficiency. Due to the phase-shifted control of the



(a) Original timing.



b) Modified timing.

Fig. 4.6 Active-clamp switch timing to reduce conduction loss

full bridge, the duty cycle of each switch is approximately 50 % regardless of the operating point. Assuming $V_{CE(sat)}$ for IGBTs is constant as a function of collector current, the conduction loss averaged over a line cycle is:

$$P_{cond,bridge} = \frac{4}{\pi} \int_0^{\pi} \langle I_S \rangle(\phi) V_{CE(sat)} d\phi \quad (4.5a)$$

$$= \frac{4\sqrt{2}P_o}{\eta\pi V_{RMS}} V_{CE(sat)}. \quad (4.5b)$$

4.3.3.1.2 Switching loss

Calculating the switching loss is more involved than calculating the conduction loss. Assuming unity power factor, ZVS is a function of the input voltage as it changes over the line cycle. This dependence greatly complicates matters, so in order to simplify the arithmetic, it will be assumed that C_{snub} is so small as to have no effect on reducing the turn-off loss for S1 and S3. As a consequence, however, ZVS will be available over the entire line cycle. The result of these assumptions is probably an overestimation of the S1, S3 turn-off loss and an underestimation of the S2, S4 turn-on loss. For ZVS operation, all four switches are turned on losslessly, and the bottom two switches are turned off with zero current. An important aspect for the zero-current turn-off of S2 and S4 is that their collector voltages remain at approximately $V_{CE(sat)}$ for the duration of the boost inductor charge interval (see Fig. 4.10(a)). The top two switches suffer from hard turn-off, which is the reason for installing the snubber capacitors. The use of empirical methods is probably best to determine an optimum value for these capacitors.

For S1, S3 turn-off:

$$P_{off}(\phi)|_{S1 \text{ or } S3} = F_S E_{off}(\phi) \frac{V_{CE,off}}{V_{CE,baseline}}, \quad (4.6)$$

where $V_{CE,baseline}$ is the test voltage used for device turn-off measurements and

$$E_{off}(\phi) = f(I_{CE,off}(\phi)). \quad (4.7)$$

$I_{CE,off}(\phi)$ is the boost inductor current at device turn-off. Its value is twice that given by Eq. (4.4). The turn-off voltage is constant and equal to the reflected output voltage. A curve fit based on device turn-off loss data can then be used to develop a relationship between turn-off energy and collector current.

4.3.3.2 Output rectifier calculations

For the output rectifiers, assume that individual rectifier forward drops (V_{fw}) are independent of forward current, and that the forward drops in each rectifier are equal. In an average sense, each rectifier sees the one-half of the output current. Therefore:

$$P_{output \ rectifier} = \frac{P_o}{2V_o} V_{fw}. \quad (4.8)$$

For rectifier switching losses, the stored charge can be used to determine the reverse recovery energy:

$$E_{sw}|_{output \ rectifier} = V_R Q_{rr}. \quad (4.9)$$

Q_{rr} is a function of the peak rectifier forward current (just before turn-off) and the turn-off di_F/dt . The rate of turn-off current is fixed by the reflected value of the leakage (or resonant) inductance and the output voltage:

$$\frac{di_F}{dt} = \frac{V_O}{L_{LK} / \left(\frac{N_P}{N_S} \right)^2}. \quad (4.10)$$

The peak rectifier forward current is given by:

$$I_F^{peak}(\phi) = 2 \frac{N_P}{N_S} \frac{\sqrt{2} P_O}{\eta V_{RMS}} \frac{N_P}{N_S} \sin(\phi). \quad (4.11)$$

The factor of 2 appearing in Eq. (4.11) is due to the action of active-clamp and is worst-case conservative. The reverse voltage (V_R) is equal to the output voltage. Assuming all the reverse recovery energy is dissipated, the switching losses are given by:

$$P_{SW}|_{output\ rectifier} = \frac{1}{\pi} \int_0^\pi F_s \frac{V_o}{2} Q_{rr} \left(I_F^{peak}(\phi), V_o \right) d\phi. \quad (4.12)$$

Because of the nonlinear dependence of Q_{rr} on I_F and V_R , the integration called for in Eq. (4.12) is a math phobic's worst nightmare. Things can be simplified by using the average value (over a line cycle) of the peak rectifier current (Eq. (4.11)) as a "steady-

state" operating point. This (along with V_o) can be used to select a value for Q_{rr} . Therefore:

$$P_{SW}|_{output\ rectifier} \approx \frac{1}{2} F_S V_o Q_{rr} \langle I_F^{peak}(\phi) \rangle V_o. \quad (4.13)$$

4.3.3.3 Active-clamp circuit calculations

Active-clamp circuit losses consist of conduction losses for both the auxiliary diode and MOSFET and switching (turn-off) losses of the MOSFET. For the rectifier, or MOSFET body diode, conduction losses are given by:

$$P_{Sa, Rect} = \langle I_{Sa} \rangle_{line\ cycle} V_F = \frac{V_F}{\pi} \int_0^\pi \langle I_{Sa} \rangle_{switching\ cycle} d\phi, \quad (4.14)$$

where:

$$\langle I_{Sa} \rangle_{switching\ cycle} \cong I_L(\phi) D_{Sa}, \quad (4.15)$$

and D_{sa} is the switch's duty cycle. $I_L(\phi)$ is the boost inductor (and line) current. Carrying out the integration called for in Eq. (4.14) yields:

$$P_{Sa, Rect} = \frac{D_{Sa} \sqrt{2} P_o}{\eta \pi V_{RMS}} V_F. \quad (4.16)$$

Calculating conduction losses for the MOSFETs requires knowledge of the RMS current through the devices:

$$P_{Sa, MOSFET}(\phi) = (I_{Sa, RMS}(\phi))^2 R_{DS, on}. \quad (4.17)$$

Therefore:

$$\langle P_{Sa, MOSFET} \rangle_{cycle}^{line} = \frac{1}{\pi} \int_0^{\pi} P_{Sa, MOSFET}(\phi) d\phi. \quad (4.18)$$

Turn-off losses are given by:

$$P_{off, MOSFET} = \frac{1}{\pi} \int_0^{\pi} P_{off}(\phi) d\phi \quad (4.19a)$$

$$= \frac{1}{\pi} \int_0^{\pi} F_S \frac{N_P}{N_S} \frac{V_o}{3} \frac{t_f}{2} I_L(\phi) d\phi. \quad (4.19b)$$

The turn-off fall time, t_f , is assumed to be 75 ns. Turn-on losses are calculated in a similar fashion ($t_r = 50$ ns):

$$P_{on} = \frac{1}{\pi} \int_0^{\pi} P_{on}(\phi) d\phi = \frac{1}{\pi} \int_0^{\pi} F_S \frac{N_P}{N_S} \frac{V_o}{3} \frac{t_r}{2} I_L(\phi) d\phi. \quad (4.20)$$

4.4 Experimental Results

To provide experimental verification of the operating concepts incorporated into the phase-shift controlled, active-clamp FB boost converter, a breadboard was constructed to provide a 48 Vdc, 1 kW output DC bus voltage from a universal AC line input. The efficiency of this design is then compared to the efficiency obtained from two-stage front-end design consisting of a zero-voltage transition (ZVT) PFC boost converter cascaded with a ZVS FB second stage DC/DC converter.

4.4.1 Power stage design

The power stage design for the active-clamp boost converter is shown in Fig. 4.7. The bridge switches operate at 75 kHz, and are each 600 V ultra-fast IGBTs. The turn-off snubber capacitors are each 1500 pF, a value that was arrived at empirically. The clamp network consists of 600 V, 2 μ F, polypropylene capacitor. The clamp capacitor should be of high quality in order to process the HF ripple currents introduced by the action of the active clamp network with as little loss as possible. The output diodes are 100 V Schottky rectifiers. Here the single-stage boost converter has a distinct advantage over the second stage DC/DC converter in the two-stage approach, where the use of Schottky rectifiers would be prohibited due to too high of a reverse voltage. Three 4700 μ F Aluminum electrolytic capacitors form the output bulk capacitor. Peak 120 Hz voltage ripple was measured at about 2.5 V peak (± 5.2 %). In practice, the amount of output capacitance is determined by ripple current capability.

Also shown in Fig. 4.7 is the design of the boost inductor and the isolation transformer. Because of the high output current required and the additional circulating current introduced by the active clamp circuit, the design of T1's secondary required the use of Cu foil, as shown.

A block diagram of the average current mode control scheme is shown in Fig. 4.8. In order to obtain good power factor at the input, the full-bridge phase-shift is modulated.

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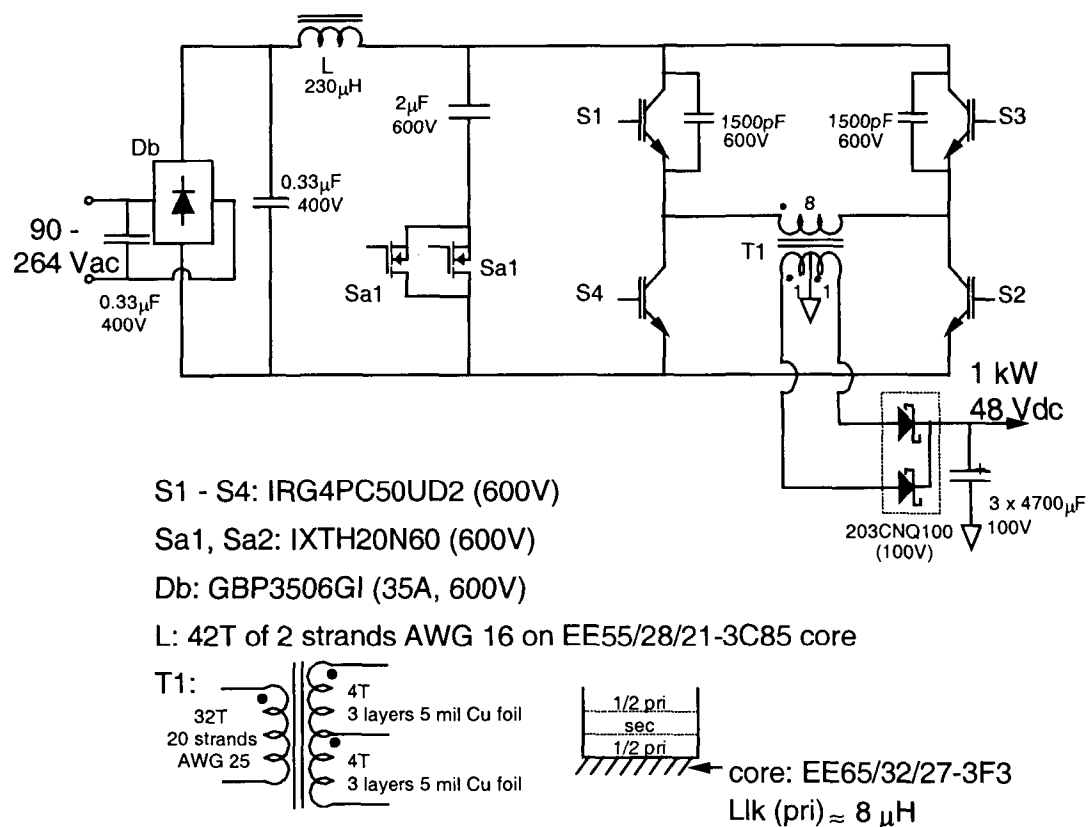


Fig. 4.7 Power stage component selection.

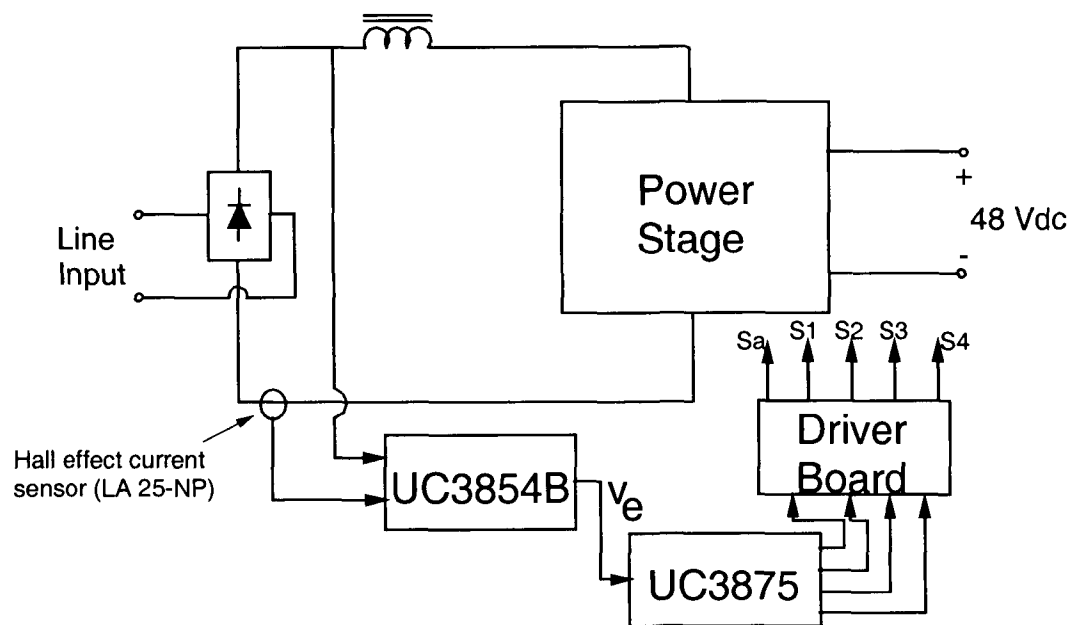


Fig. 4.8 Block diagram of FB control for PFC.

However, since there are no commercially available full-bridge ϕ -shift control integrated circuits for PFC, the method shown in Fig. 4.8 was adopted. A single-ended PFC controller IC, the UC3854B, was used to generate a control voltage, V_e , as part of the standard application of this IC [48]. This control voltage was then used as the error amplifier input for a ϕ -shift control IC, the UC3875. The UC3875 then generates the four ϕ -shifted drive signals for the bridge switches. The driver board is then responsible for developing the active-clamp switch drive based on the ϕ -shifting of the bridge switches.

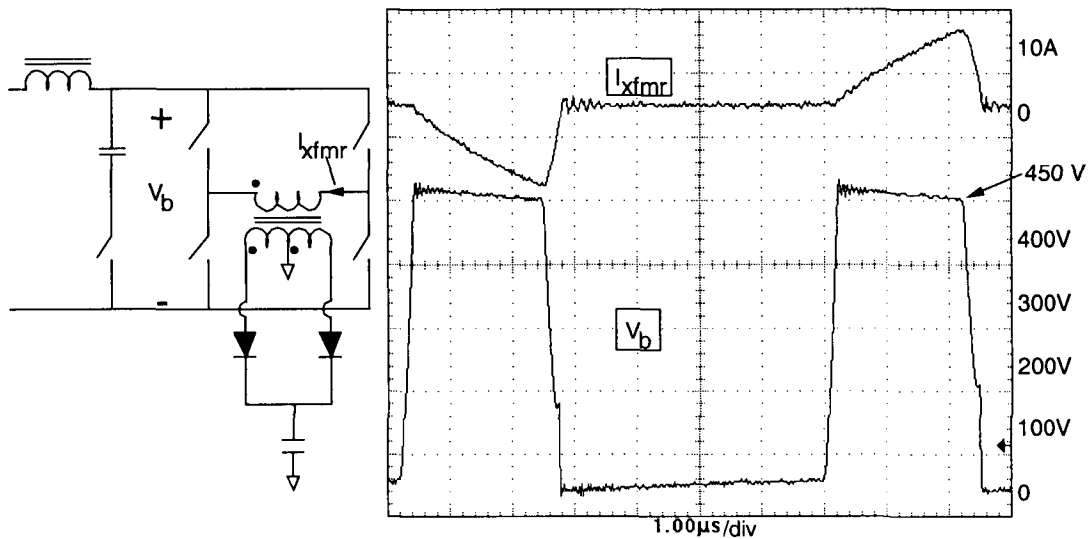
4.4.2 Experimental waveforms and efficiency measurements

Figures 4.9 through 4.11 summarize the fundamental experimental waveforms of the converter. Figures 4.9 and 4.10 were obtained with the converter operating from a DC input voltage in order to more clearly show these waveforms over a switching cycle. Figure 4.9 summarizes the operating waveforms associated with the active-clamp network. The fundamental characteristics of these waveforms is identical to those of the active-clamp flyback converter presented in Chapter 3. The waveforms illustrating the soft-switching characteristics are shown in Fig. 4.10. The line input waveforms are shown in Fig. 4.11 for $V_{in} = 120 \text{ Vac}$ and $P_o = 1 \text{ kW}$. As can be seen, the power factor is very high and has minimal harmonic distortion, both typical of average current mode control PFC and CCM operation of the boost inductor.

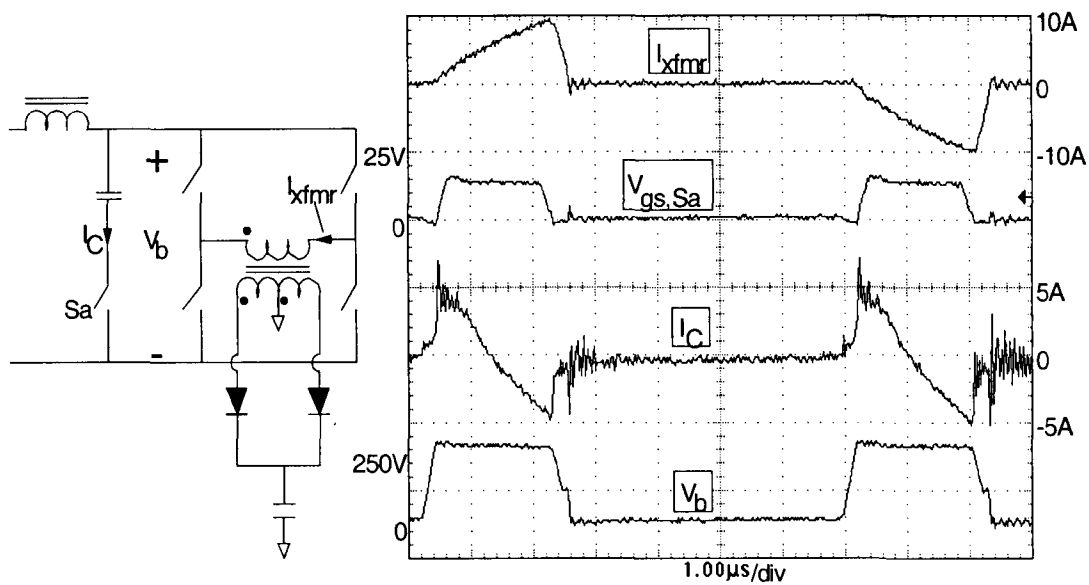
The experimental efficiencies, as a function of line and load are presented in Fig. 4.12. High-line/maximum load efficiencies are above 90 %.

4.4.3 Experimental comparison with the two-stage approach

In order to provide a relative efficiency comparison with the two-stage front-end approach, data was taken from the two-stage power train design shown in Fig. 4.13. This design cascades a ZVT PFC boost converter, switching at 140 kHz, with a standard ZVS ϕ -shifted full-bridge DC/DC converter switching at 100 kHz, using the components

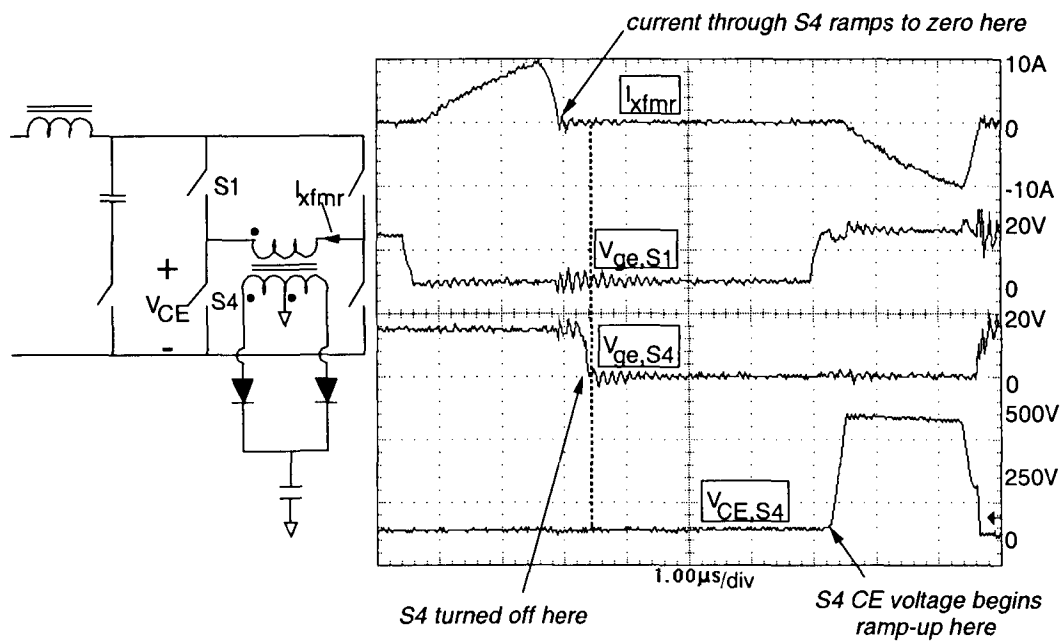


(a) Bridge-voltage clamping, $V_{in} = 150$ Vdc, $P_o = 960$ W.

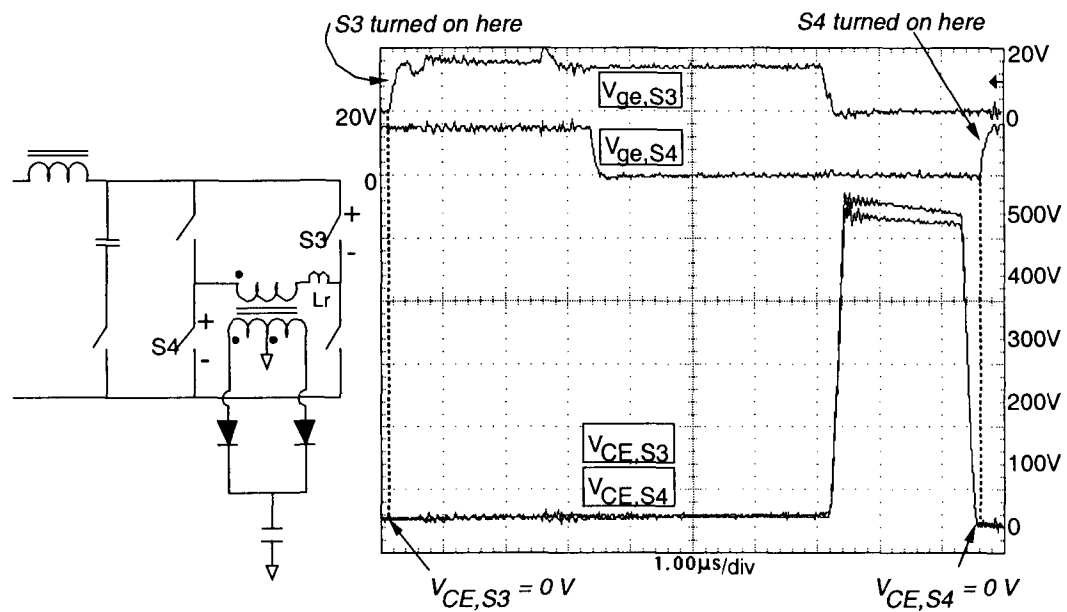


(b) Clamp circuit waveforms, $V_{in} = 140$ Vdc, $P_o = 700$ W.

Fig. 4.9 Experimental waveforms - clamp operation.



(a) ZCS for S4, $V_{in} = 140$ Vdc, $P_o = 700$ W.



(b) ZVS for S3/S4 ($L_r = 5 \mu H$), $V_{in} = 140$ Vdc, $P_o = 958$ W.

Fig. 4.10 Experimental waveforms - soft-switching.

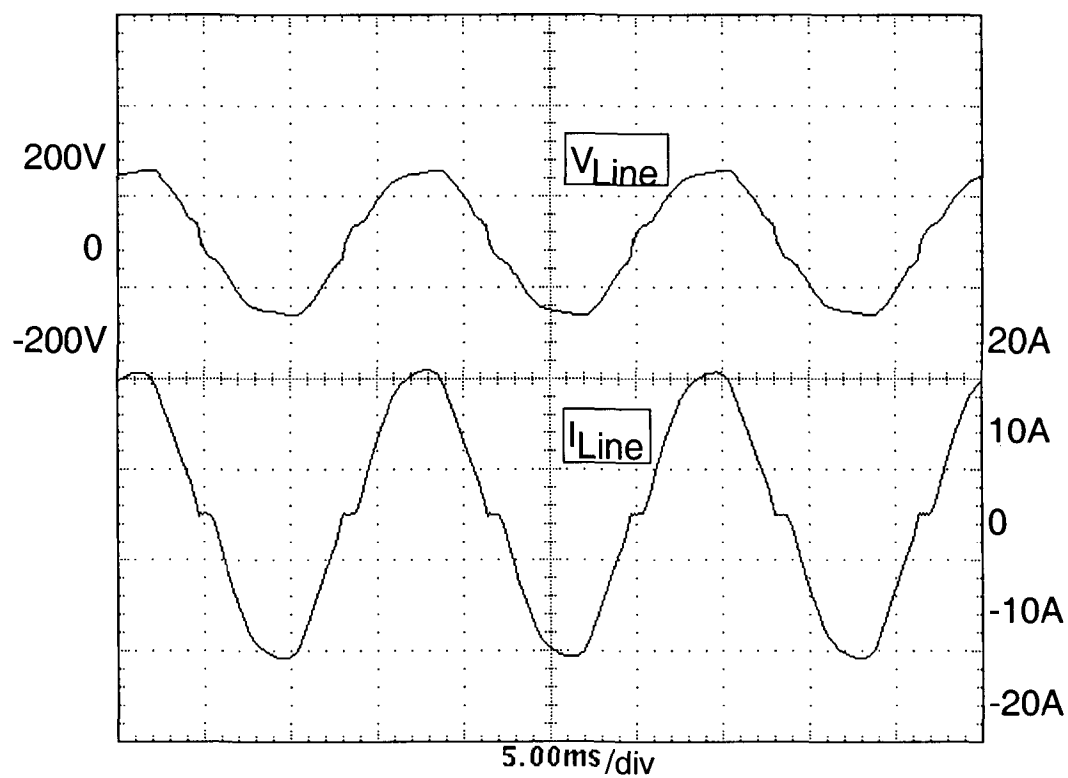


Fig. 4.11 Experimental waveforms - line voltage and current, $V_{in} = 120 \text{ Vac}$,
 $P_o = 1 \text{ kW}$.

Efficiency (%)

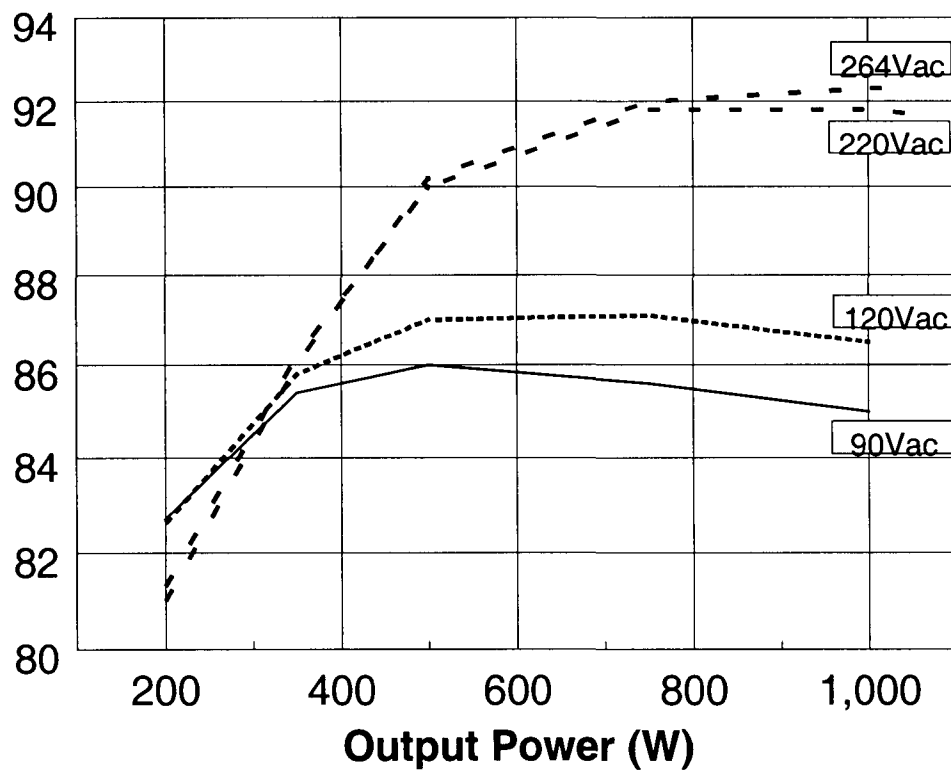


Fig. 4.12 Experimental efficiency as a function of line and load (includes control circuit losses).

indicated. The large value of bulk capacitance (close to 900 μF) used in this design is to provide for a significant hold-up function.

The efficiency data, as a function of line voltage with a constant output power of 1 kW, is given in Fig. 4.14. At low line voltages the efficiencies are very comparable, but as the line voltage is increased past about 150 Vac the two curves diverge. At $V_{in} = 220$ Vac the active-clamp boost is nearly 3 % more efficient. This makes sense as at high line the circulating energy losses introduced by the active-clamp network are reduced (boost inductor current is at a minimum) and maximum advantage can be taken of the fact that the power is processed in one conversion stage. From the efficiency data it would seem the active-clamp boost converter is best suited for applications requiring PFC and equipment operation from a single-phase 220 Vac input.

4.5 Summary

In this chapter, the phase-shift controlled, active-clamp full-bridge PFC boost converter has been shown to be an attractive solution for higher power front-end converters utilized in DC distributed power systems. The introduction of the active-clamp circuit introduces minimum complexity in the implementation of the overall converter and provides a mechanism to recover the transformer leakage energy and achieve zero-voltage switching. This approach reduces device voltage and switching stress at the expense of increasing circulating energy. It was also demonstrated how phase-shift operation of the bridge switches results in zero-current switching for two of the four bridge switches [52]. The control concept is straightforward, with constant frequency, PWM phase-shift control being utilized. Experimental results indicate a 3 % efficiency improvement (at high input line and with a 1 kW, 48 Vdc output) over a boost PFC/ZVS FB DC/DC two-stage design.

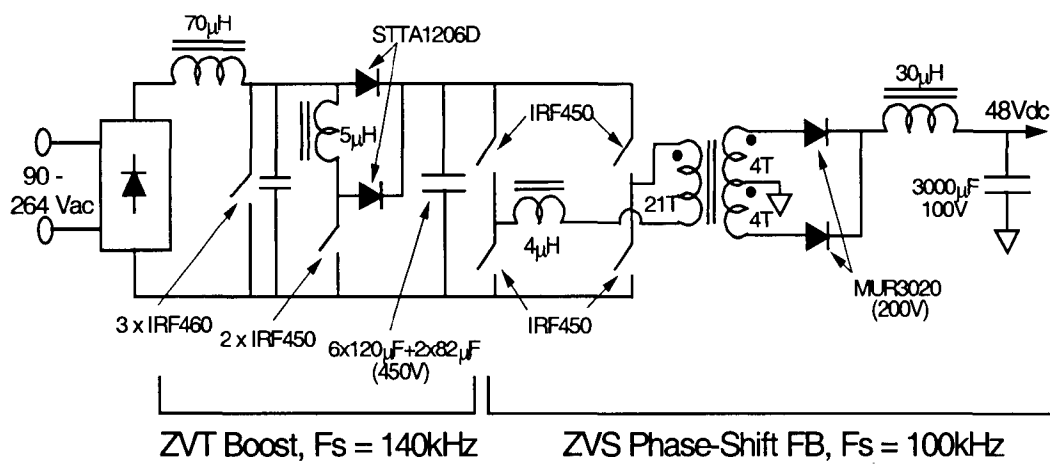


Fig. 4.13 Experimental two-stage power train design.

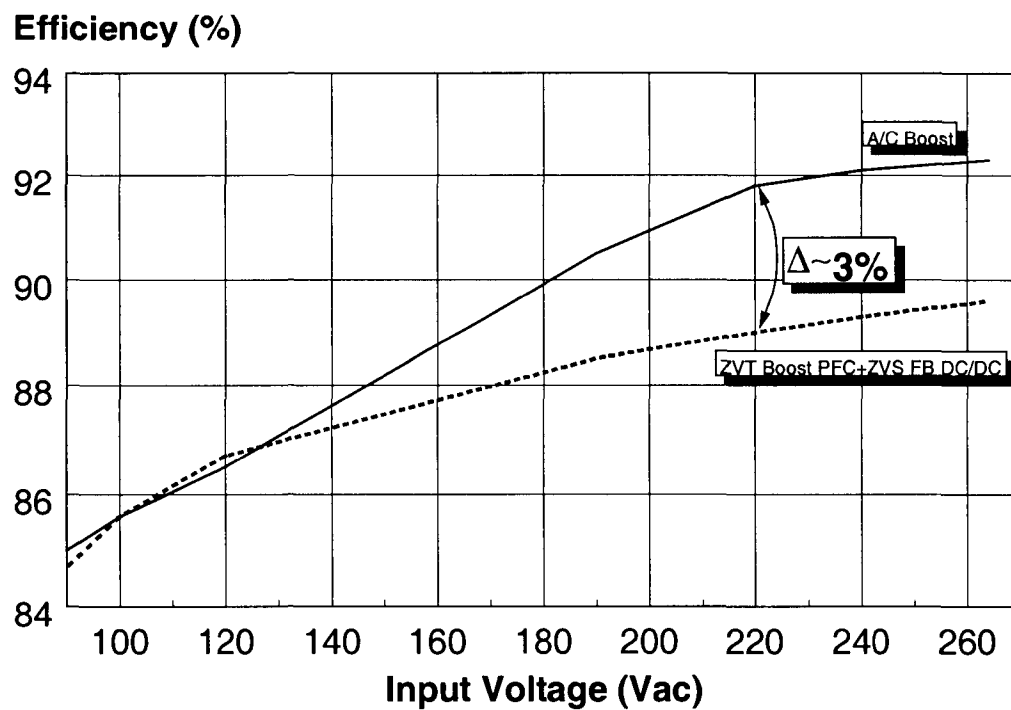


Fig. 4.14 Experimental efficiency comparison between 2 stage and single stage implementations. $P_o = 1$ kW.

Table 1: Comparison with CCMPA and general CMPA

	General CMPA	Linear CCMPA	Circular CCMPA
Return losses	-33.8 dB	-29 dB	-15.15 dB
-10 dB return loss bandwidths	55.5 MHz (3.5%)	71 MHz (4.5%)	140 MHz (8.9%)
Gains at 0°	7.1 dBd	6.2 dBd	2.8 dBd
-3 dB beamwidths	<i>E</i> -plane	72°	61°
	<i>H</i> -plane	76°	72°
Axial ratio			1.3 dB (1.575 GHz)
Patch diameter	99 mm	91 mm (8.08%)	92 mm (7.07%)

Conclusion: A corrugated antenna showing a reduction of 8.08% in patch diameter and 21.12% in patch area by maintenance of the fundamental mode for general CMPA as well as shortening of current length has been reported. Each linearly-polarised CCMPA and circularly-polarised CCMPA has the improved characteristic of 71 MHz (4.5%) and 144 MHz (8.9%) in comparison with the general CMPA in the aspect of -10 dB return loss bandwidth at 1.575 GHz. Therefore, the CCMPA proposed in this Letter has been confirmed to be appropriate for the miniaturisation of antennas.

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Generalised analysis of switched-capacitor step-down quasi-resonant converter

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Conventional switched-capacitor converter circuits have a high current stress because of the short circuit when charging and discharging the capacitors. A novel switched-capacitor quasi-resonant step-down converter family with a generalised analysis that can improve the current stress problem is presented. Measured efficiency is high and all the circuits can operate under zero-current switching.

Introduction: Switched-capacitor DC-DC converters [1] use capacitors only for energy storage, therefore the main advantage is that it is not necessary to include an inductor in the design and circuit size can be reduced. The fabrication of the converter in an integrated circuit is also possible. However, the main drawback of the circuit is the momentary short circuit when charging and discharging the capacitor; this generates switching loss and electromagnetic interference. A zero-current switching method for the switched-capacitor quasi-resonant (SCQR) converter was proposed in [2]. In this Letter we describe extending the principle of the switched-capacitor resonant converter into a generalised method. A switching-capacitor cell is proposed to the switched-capacitor converter and hence a series of circuits of various conversion ratios and zero-current switching of all switches can be obtained.

Concepts: The proposed SCQR converters are shown in Fig. 1 and 2 where the circuits give the output voltage equal to $1/3$ and $1/n$ of input voltage, respectively. The output voltages of the proposed family of converters are dependent on the number of switched-capacitor cells, which is formed by C_2 , D_a , D_b , D_c as shown in Fig. 1. For other fractional output, this cell can be extended to the $1/n$ version as shown in Fig. 2. Each capacitor C_i ($i > 1$) in the switched-capacitor cells is for sharing the input voltage together in series and releasing the energy in parallel with C_1 to the load where the diodes decide the direction of the current flow to charge the capacitors. For a converter with $1/n$ conversion ratio ($1/n$ mode converter), there should be $(n-2)$ switched-capacitor cells. L_r is to ensure zero-current switching, and the current of all the transistors must pass through L_r .

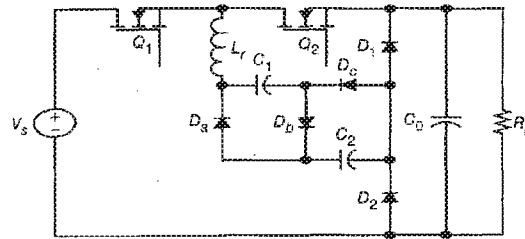


Fig. 1 $1/3$ mode SCQR converter

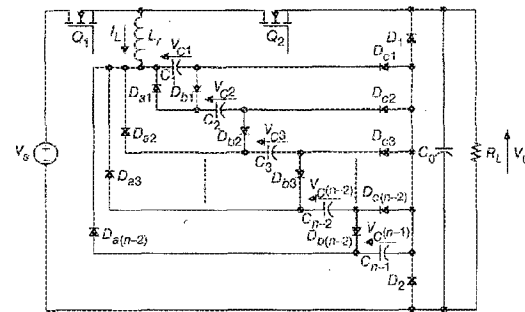


Fig. 2 $1/n$ mode zero-current switching SCQR converters

Principles of operations: The $1/3$ mode SCQR converter is analysed. There are a total of four states of operation in each switching period. Fig. 3 shows the computer simulation waveforms of the converter. Assuming the output capacitor C_o is very large to keep the output voltage constant, the capacitance of C_1 is equal to that of C_2 (called C), and there is no power loss in the circuit, the analysis of each state is as following.

(a) *States of operations:* (i) *State I* [$t_0 - t_1$]: At t_0 , Q_1 is switched on while Q_2 is switched off. C_1 and C_2 are charged in series with D_b , D_1 , L_r and the load. At this moment, C_1 and C_2 resonate with L_r in series. The charging current of the capacitors resonates from zero at t_0 to the peak and is back to the zero at t_1 in sinusoidal waveform so that Q_1 is switched on under zero-current switching condition. The equations of this state are:

$$V_s = L_r \frac{di_L}{dt} + (n-1)v_C + V_0 \quad (1)$$

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

where V_s is the input voltage, V_0 the output voltage, v_C the resonant capacitors (C_1 or C_2) voltage, i_L the resonant inductor (L_r) current, and n represents the $1/n$ version of the circuits.

(ii) *State II* [$t_1 - t_2$]: Q_1 keeps to be on and Q_2 is still off. At t_1 , the inductor current reaches zero and the diodes D_b and D_1 stop the resonance. Both the inductor and input currents are equal to zero. C_0 is discharged to the load. v_C stays constant and $i_L = 0$.

(iii) *State III* [$t_2 - t_3$]: Q_1 is switched off and Q_2 is switched on at t_2 . C_1 and C_2 are discharged in parallel to L_r and the load. D_a , D_c and D_2 are conducting. These two capacitors resonate with L_r in sinusoidal wave which is similar to State I. Since the current of Q_1 is zero at t_2 , Q_1 is switched off in zero-current switching condition while Q_2 is zero-current switched on. At t_3 , the inductor current is resonated to be zero.

The equations of this state are:

$$v_C + L_r \frac{di_L}{dt} = V_0 \quad (3)$$

$$i_L = (n-1)C \frac{dv_C}{dt} \quad (4)$$

(iv) *State IV* [$t_3 - t_4$]: Similar to State II, the diodes stop the resonance at t_3 . Inductor current is equal to zero and only C_0 is discharged to the load. v_C stays constant and $i_L = 0$.

(b) *General equations of 1/n mode converter*: The above equations can be solved by using input and output power balancing, and the continuity of the four states and the inductor current.

For State I,

$$V_S \int_0^{t_1} i_L dt = V_0 I_0 T_S \quad (5)$$

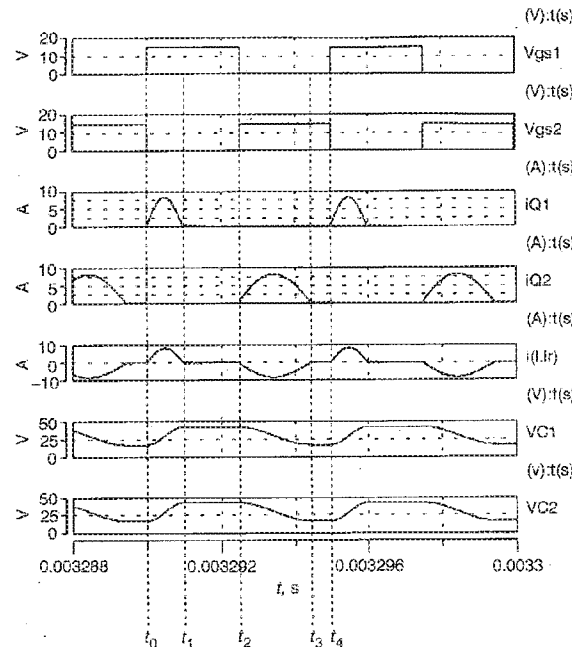


Fig. 3 Simulation waveforms of 1/3 mode SCQR converter

By solving (5), and for a 1/n mode converter,

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \cos \omega_0(t - t_0) \quad (6)$$

$$i_L = \frac{\pi I_0 T_S}{nT_0} \sin \omega_0(t - t_0) \quad (7)$$

where

$$\omega_0 = \sqrt{\frac{n-1}{L_r C}} \quad \text{and} \quad Z_0 = \sqrt{\frac{(n-1)L_r}{C}} \quad (8)$$

T_S is the switching frequency, T_0 is the resonant period in State I, and I_0 is the output current.

In State II,

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \quad (9)$$

In State III,

$$v_C = V_0 + \frac{(n-1)\pi I_0 T_S Z_1}{nT_1} \cos \omega_1(t - t_2) \quad (10)$$

$$i_L = -\frac{(n-1)\pi I_0 T_S}{nT_1} \sin \omega_1(t - t_2) \quad (11)$$

where

$$\omega_1 = \sqrt{\frac{1}{(n-1)L_r C}} \quad \text{and} \quad Z_1 = \sqrt{\frac{L_r}{(n-1)C}} \quad (12)$$

where T_1 is the resonant period of State III.

In State IV,

$$v_C = V_0 - \frac{(n-1)\pi I_0 T_S Z_1}{nT_1} \quad (13)$$

Experimental results: The 1/3 mode SCQR converter has been tested. The specification of the prototype is input voltage = 90 V, output voltage = 30 V, switching frequency = 200 kHz, $C_1 = C_2 = 0.22 \mu\text{F}$, $C_0 = 50 \mu\text{F}$ and $L_r = 1 \mu\text{H}$. The measured waveforms, the gate-source voltage, the drain current and drain-source voltage of Q_1 and Q_2 , of the converter in 80 W output power are shown in Fig. 4. From Figs. 4a and 4b it is clear that two transistors are operated under zero-current switching. The measured output voltage of the converter in 80 W operation is 25.8 V and is slightly lower than the theoretical output voltage of 30 V because of the loss. The measured efficiency varied between 80 and 93% when the load power varies between 5 and 80 W, respectively.

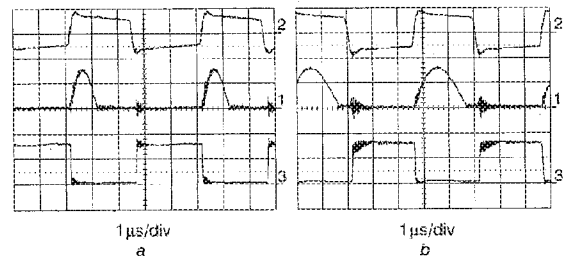


Fig. 4 Experimental waveforms of converters

a Upper trace: gate signal Q_1 , 20 V/div; middle trace: Q_1 current, 5 A/div; lower trace: $V_{ds} Q_1$, 40 V/div
b Upper trace: gate signal Q_1 , 20 V/div; middle trace: Q_1 current, 5 A/div; lower trace: $V_{ds} Q_1$, 40 V/div

Conclusion: A generalised analysis of switched-capacitor circuits is presented. They are operated under zero-current switching; the inherited disadvantage of high pulsational current has been eliminated, and the number of switches needed is low. The main contribution is that a number of fractional output voltage conversion zero-current switching converters and a generalised analysis has been produced. Experimental results confirmed that the operation agrees with the theoretical prediction. All the transistors can achieve zero-current switching, and efficiency is high.

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Zero-current switching switched-capacitor quasiresonant step-down converter

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Abstract: Current stress is usually high for conventional switched-capacitor converter circuits, because of short-circuiting when charging up the capacitors. In this paper, two zero-current switching switched-capacitor resonant step-down converter families are presented which can improve the current stress problem. They are able to provide $1/2$ to $1/n$, and -1 to $-1/n$ conversion ratios respectively by using only two switches. Principles of operation, computer simulation and experimental results of the proposed converters are presented. The measured efficiency is high and all parts of the circuits are under zero-current switching conditions.

1 Introduction

The switched-capacitor DC-DC converter [1–6] uses capacitors only for energy storage. Its novel feature is that only capacitors and switches are required in the power stage. The main advantage of this kind of converter is that no magnetic component is required for storing energy, making it possible to fabricate a smaller size and lighter weight converter in an integrated circuit.

The drawback of the switched-capacitor DC-DC converter is that the voltage conversion ratio is fixed. If the voltage conversion ratio of the circuit changes widely, more switches, capacitors and diodes are required. This means that more gate drive circuits will be required for the extra switches. Also, very high current spikes can be observed when the switched capacitors are being charged from the voltage source causing high EMI.

Another shortcoming of the switched-capacitor converter is voltage regulation. Because it is based on shorting switched-capacitors to a certain voltage, the voltage output depends on the number of switched-capacitor stages hence it is theoretically difficult to regulate. One way to overcome this is to use the equivalent series resistance (ESR) of the capacitor and the resistance of the transistor when it is on as an RC charging and discharging control [3]. The drawback of such a method is an increase in switching losses since the transistor does not operate in the saturation region [6]. These converters are therefore usually used only for low power and operating at lower efficiency (80% or less). The efficiency also deteriorates seriously as the switching frequency increases.

A family of switched-capacitor DC-DC converters, which can solve the high current spike problem has been proposed [2]. All the converters in this family use zero-current switching [7]. There is only a very small inductor in each circuit which can resonate with the switched capacitor, so the charging current is limited. The turn-off switching loss

can also be eliminated. However, this family of converters only includes the circuits which provide 0.5, -1 and 2 voltage conversion ratios.

In this paper, two switched-capacitor step-down DC-DC resonant converters are presented: noninverting mode and inverting mode. These two circuits are based on the inverting mode and the half mode converters [2], which are shown in Figs. 1 and 2. A simple switched-capacitor cell is proposed. This switched-capacitor cell consists of only one capacitor and three diodes. By adding this switched-capacitor cell to the converters, the voltage conversion ratios of the converters can decrease fractionally. No other

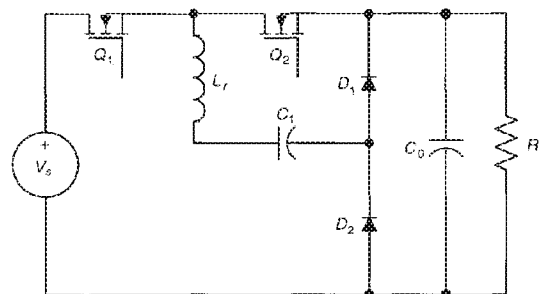


Fig. 1 Conventional half-mode zero-current switching switched-capacitor resonant converter

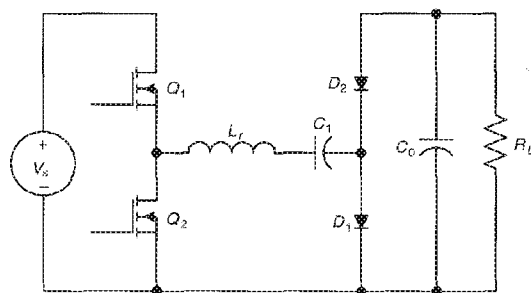


Fig. 2 Conventional inverting mode zero-current switching switched-capacitor resonant converter

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switch is required to change the voltage conversion ratio. Also, all the circuits of these two families are operated under zero-current switching conditions. They have no current spike problems and can operate at high frequency with high efficiency. Therefore, these converters have the potential to be used where DC isolation is unnecessary and fixed conversion is needed.

2 Concepts of proposed step-down switched-capacitor resonant converters

The proposed noninverting mode step-down switched-capacitor resonant converters are shown in Figs. 3–5 and the inverting mode step-down switched-capacitor resonant converters are shown in Figs. 6–8, where the circuits in Figs. 3–8 give an output voltage equal to $1/3$, $1/4$, $1/n$, $-1/2$, $-1/3$ and $-1/n$ input voltage, respectively.

The output voltages of both noninverting mode and inverting mode converters are dependent on the number of switched-capacitor cells, shown in Fig. 9, applied on the converter circuit. Referring to Figs. 3–5, in each cell for the proposed converters, ports a and b are connected across the capacitor of the former cell. Port c is connected to the node between D_1 and D_2 . Port d is connected to the anode of the D_b of the next cell. Port e is connected to the cathode of D_b of the former cell. Port f is connected to L_r .

Each capacitor in the switched-capacitor cells shares the input voltage in series and releases the energy in parallel with C_1 to the load where the diodes are used to decide the direction of the current flow which charges the capacitors.

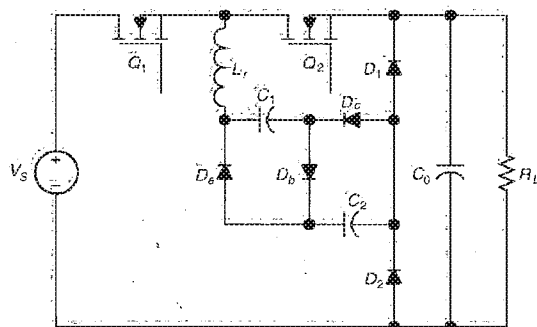


Fig. 3 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/3 mode converter

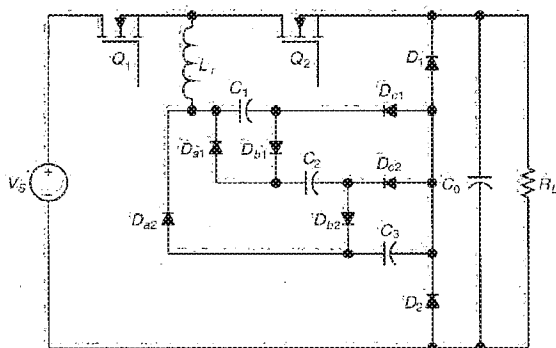


Fig. 4 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/4 mode converter

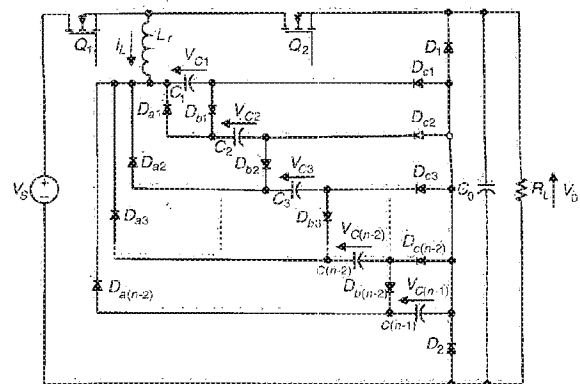


Fig. 5 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/n mode converter

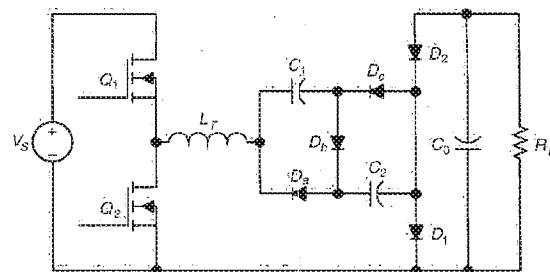


Fig. 6 Proposed inverting zero-current switching switched-capacitor resonant step-down converter
1/2 mode converter

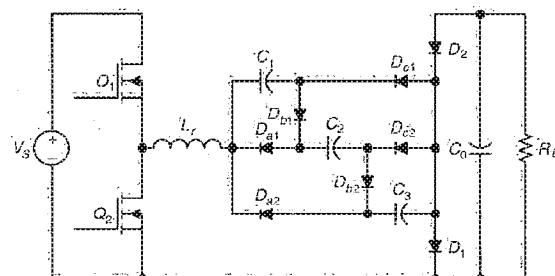


Fig. 7 Proposed inverting zero-current switching switched-capacitor resonant step-down converters
-1/3 mode converter

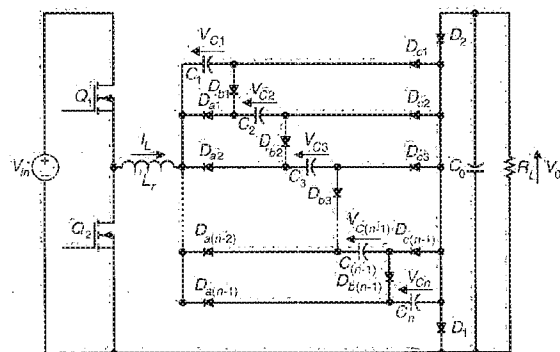


Fig. 8 Proposed inverting zero-current switching switched-capacitor resonant step-down converters
-1/n mode converter

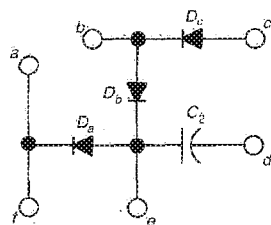


Fig. 9 Switched-capacitor cell

Using this method, the converter can step down the input voltage fractionally. For a converter with $1/n$ voltage conversion ratio ($1/n$ mode converter), there should be $(n-2)$ switched-capacitor cells.

3 Principles of operations

3.1 Noninverting mode step-down resonant converters

The noninverting $1/3$ mode switched-capacitor resonant step-down converter is analysed. There are four states of operation in each switching period. Figs. 10–13 shows the equivalent circuits of each state of operations of the noninverting $1/3$ mode converter and Fig. 14 shows the computer simulation waveforms of the converter. Assuming the output capacitor C_0 is very large to keep the output voltage constant, the capacitance of C_1 is equal to that of C_2 (called C), and there is no power loss in the circuit, the analysis of each state is given below.

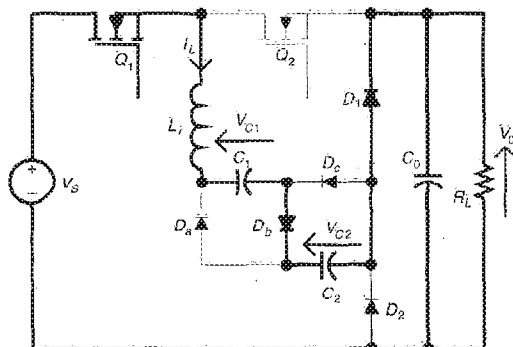


Fig. 10 State I of noninverting $1/3$ mode switched-capacitor resonant step-down converter

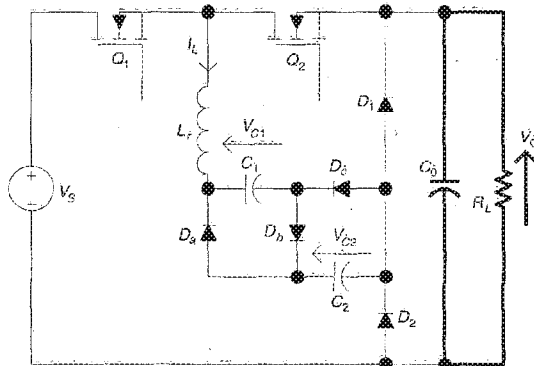


Fig. 11 State II of noninverting $1/3$ mode switched-capacitor resonant step-down converter

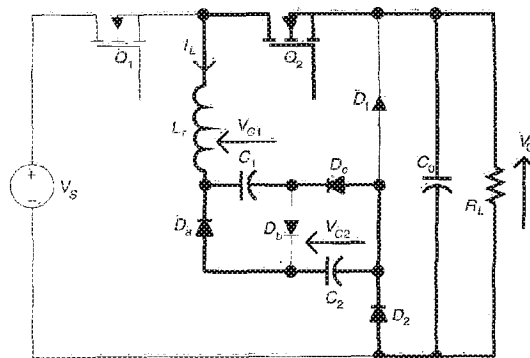


Fig. 12 State III of noninverting $1/3$ mode switched-capacitor resonant step-down converter

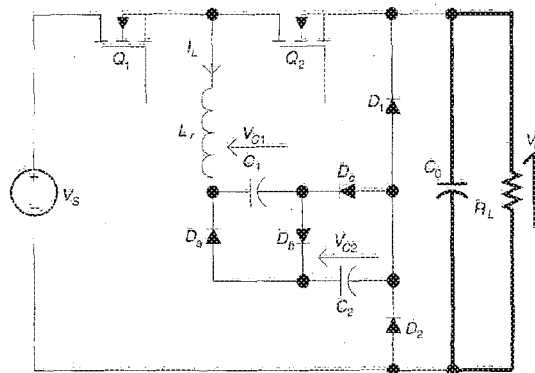


Fig. 13 State IV of noninverting $1/3$ mode switched-capacitor resonant step-down converter

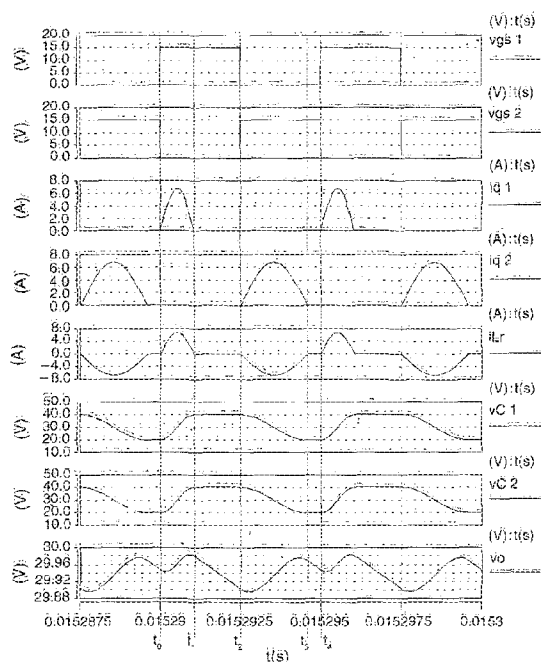


Fig. 14 Idealised simulation waveforms of the noninverting $1/3$ mode switched-capacitor resonant step-down converter

3.1.1 Operating states

(i) State I [t_0-t_1]

At t_0 , Q_1 is switched on while Q_2 is switched off. C_1 and C_2 are charged in series with D_b , D_1 , L_r and the load. At this moment, C_1 and C_2 resonate with L_r in series. The charging current of the capacitors resonates from zero at t_0 to the peak and the back to the zero at t_1 in sinusoidal waveform so that Q_1 is switched on under the zero-current switching condition. The equations of this state are

$$V_S = L_r \frac{di_L}{dt} + 2v_C + V_0 \quad (1)$$

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

where V_S is the input voltage, V_0 is the output voltage, V_C is the resonant capacitors (C_1 or C_2) voltage, and i_L is the resonant inductor (L_r) current.

Let V_{C0} be the value of V_C at t_0 , the solutions of (1) and (2) are

$$v_C = \frac{V_S - V_0}{2} - \left(\frac{V_S - V_0 - 2V_{C0}}{2} \right) \cos \omega_0(t - t_0) \quad (3)$$

$$i_L = \left(\frac{V_S - V_0 - 2V_{C0}}{Z_0} \right) \sin \omega_0(t - t_0) \quad (4)$$

where the resonant angular frequency of this state is

$$\omega_0 = \sqrt{\frac{2}{L_r C}} \quad (5)$$

The resonant impedance of this state is

$$Z_0 = \sqrt{\frac{2L_r}{C}} \quad (6)$$

(ii) State II [t_1-t_2]

Q_1 remains on and Q_2 remains off. At t_1 , the inductor current reaches zero and the diodes D_b and D_1 stop the resonance. Both the inductor current and input current are equal to zero. C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C1} \quad (7)$$

$$i_L = 0 \quad (8)$$

where V_{C1} is the voltage of C at t_1 .

(iii) State III [t_2-t_3]

Q_1 is switched off and Q_2 is switched on at t_2 . C_1 and C_2 are discharged in parallel to L_r and the load. D_a and D_2 and D_2 are conducting. These two capacitors resonate with L_r in sinusoidal manner, which is similar to state I. Since the current of Q_1 is zero at t_2 , Q_1 is switched off in the zero-current switching condition while Q_2 is zero-current switched on. At t_3 , the inductor current is resonated to zero. The equations of this state are

$$v_C + L_r \frac{di_L}{dt} = V_0 \quad (9)$$

$$i_L = 2C \frac{dv_C}{dt} \quad (10)$$

The solutions of (9) and (10) are

$$v_C = V_0 - (V_0 - V_{C2}) \cos \omega_1(t - t_2) \quad (11)$$

$$i_L = \frac{V_0 - V_{C2}}{Z_1} \sin \omega_1(t - t_2) \quad (12)$$

where V_{C2} is the voltage of C at t_2 .

The resonant angular frequency of this state is

$$\omega_1 = \sqrt{\frac{1}{2L_r C}} \quad (13)$$

The resonant impedance of this state is

$$Z_1 = \sqrt{\frac{L_r}{2C}} \quad (14)$$

(iv) State IV [t_3-t_4]

Similar to state II, the diodes stop the resonance at t_3 . The inductor current is equal to zero and only C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C3} \quad (15)$$

$$i_L = 0 \quad (16)$$

where V_{C3} is the voltage of C at t_3 .

3.1.2 Generalised equations of 1/3 mode converter

By using input and output power balancing

$$V_S I_m = V_0 I_0 \quad (17)$$

For state I

$$V_S \int_{t_0}^{t_1} i_L dt = V_0 I_0 T_S \quad (18)$$

By solving (18), and assuming the voltage conversion ratio is 1/3, then

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{6 T_0} \cos \omega_0(t - t_0) \quad (19)$$

$$i_L = \frac{\pi I_0 T_S}{3 T_0} \sin \omega_0(t - t_0) \quad (20)$$

where T_S is the switching frequency, T_0 is the resonant period in state I, and I_0 is the output current.

For state II

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{6 T_0} \quad (21)$$

$$i_L = 0 \quad (22)$$

For state III

$$V_S \int_{t_2}^{t_3} i_L dt = V_0 (I_0 - I_m) T_S \quad (23)$$

By solving (23)

$$v_C = V_0 + \frac{2\pi I_0 T_S Z_1}{3 T_1} \cos \omega_1(t - t_2) \quad (24)$$

$$i_L = -\frac{2\pi I_0 T_S}{3 T_1} \sin \omega_1(t - t_2) \quad (25)$$

where T_1 is the resonant period of state III.

For state IV,

$$v_C = V_0 - \frac{2\pi I_0 T_S Z_1}{3 T_1} \quad (26)$$

$$i_L = 0 \quad (27)$$

3.1.3 Generalised equations of 1/n mode converter

For a 1/n mode converter, in state I

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \cos \omega_0(t - t_0) \quad (28)$$

$$i_L = \frac{\pi I_0 T_S}{n T_0} \sin \omega_0 (t - t_0) \quad (29)$$

where

$$\omega_0 = \sqrt{\frac{n-1}{L_r C}} \quad (30)$$

$$Z_0 = \sqrt{\frac{(n-1)L_r}{C}} \quad (31)$$

In state II

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \quad (32)$$

$$i_L = 0 \quad (33)$$

In state III

$$v_C = V_0 + \frac{(n-1)\pi I_0 T_S Z_1}{n T_1} \cos \omega_1 (t - t_2) \quad (34)$$

$$i_L = -\frac{(n-1)\pi I_0 T_S}{n T_1} \sin \omega_1 (t - t_2) \quad (35)$$

where

$$\omega_1 = \sqrt{\frac{1}{(n-1)L_r C}} \quad (36)$$

$$Z_1 = \sqrt{\frac{L_r}{(n-1)C}} \quad (37)$$

In state IV

$$v_C = V_0 - \frac{(n-1)\pi I_0 T_S Z_1}{n T_1} \quad (38)$$

$$i_L = 0 \quad (39)$$

3.2 Inverting mode step-down resonant converters

Each state of the operation of the inverting $-1/2$ mode step-down converter is analysed. The equivalent circuits of each state of the converter are shown in Figs. 15–18. Fig. 19 shows the simulation waveforms of this converter. Assuming the converter is lossless and the output filtering capacitor, C_o , is very large to maintain a constant output voltage, the calculations of the converter are given below:

3.2.1 Operating states

(i) State I [$t_0 - t_1$]

Q_1 is switched on and Q_2 is switched off. The current passing through L_r , D_b and D_1 from the source is charging

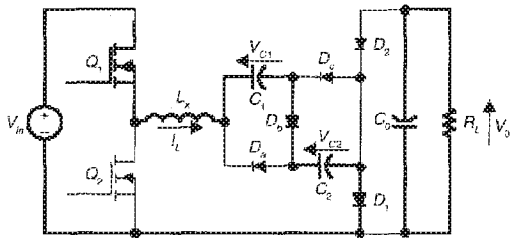


Fig. 15 State I of inverting $-1/2$ mode switched-capacitor resonant step-down converter

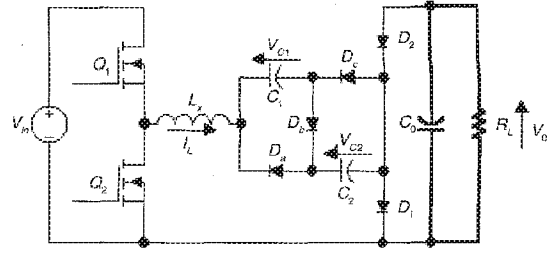


Fig. 16 State II of inverting $-1/2$ mode switched-capacitor resonant step-down converter

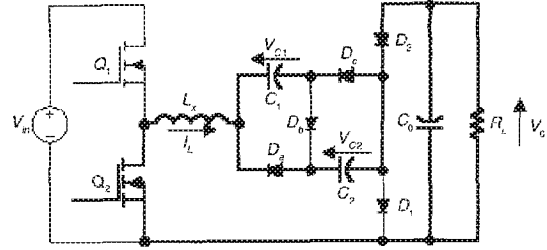


Fig. 17 State III of inverting $-1/2$ mode switched capacitor resonant step-down converter

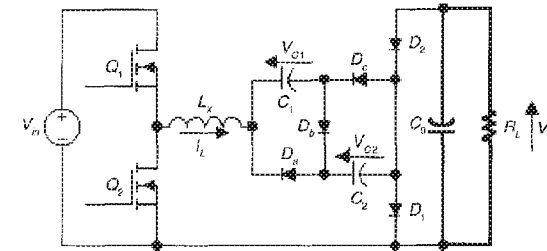


Fig. 18 State IV of inverting $-1/2$ mode switched capacitor resonant step-down converter

the capacitors C_1 and C_2 in series. The output capacitor C_o is discharged to the load. The inductor current i_L starts resonating with C_1 and C_2 in sinusoidal manner so that zero-current switching-on can be achieved. The resonant inductor current increases from zero at t_0 and then decreases from the peak. It reaches zero again at t_1 . Since Q_2 is switched off at t_0 while the current is zero, Q_2 is switched off under zero-current switching conditions. The equations of this state are

$$V_S = L_r \frac{di_L}{dt} + 2v_C \quad (40)$$

$$i_L = C \frac{dv_C}{dt} \quad (41)$$

By solving (40) and (41)

$$v_C = \frac{V_S}{2} - \frac{V_S - 2V_{C0}}{2} \cos \omega_0 (t - t_0) \quad (42)$$

$$i_L = \frac{V_S - 2V_{C0}}{Z_0} \sin \omega_0 (t - t_0) \quad (43)$$

where V_{C0} is the voltage of C at t_0 .

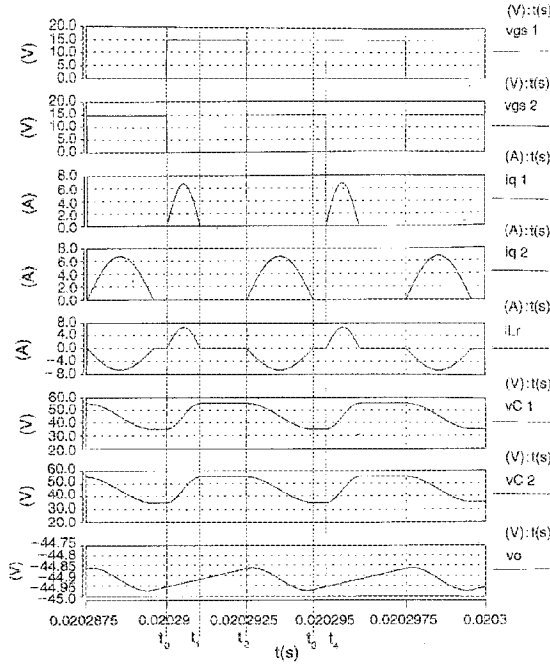


Fig. 19 Idealised simulation waveforms of the inverting $-1/2$ mode switched-capacitor resonant step-down converter

The resonant angular frequency of this state is

$$\omega_0 = \sqrt{\frac{2}{L_r C}} \quad (44)$$

The resonant impedance of this state is

$$Z_0 = \sqrt{\frac{2L_r}{C}} \quad (45)$$

(ii) State II [$t_1 - t_2$]

Q_1 is on and Q_2 is off. At t_1 , the inductor current reaches zero. The resonance is stopped by the reverse bias of the diodes. The current is kept to be zero. C_0 is still being discharged to the load in this state. The equations of this state are

$$v_C = V_{C1} \quad (46)$$

$$i_L = 0 \quad (47)$$

where V_{C1} is the voltage of C at t_1 .

(iii) State III [$t_2 - t_3$]

Q_1 is switched off and Q_2 is switched on. Since the inductor current is zero at t_2 when Q_1 is switched off, Q_1 is switched off under zero-current conditions. At t_2 , C_1 and C_2 are discharged to the load through L_r , D_a , D_b and D_2 . The current starts resonating at t_2 in sinusoidal manner which is similar to state I and hence, Q_2 is switched on under zero-current switching conditions. The inductor current i_L starts decreasing from zero at t_2 to the peak and then increases to zero at t_3 . The equations of this state are

$$v_C + L_r \frac{di_L}{dt} + V_0 = 0 \quad (48)$$

$$i_L = 2C \frac{dv_C}{dt} \quad (49)$$

The solution of (48) and (49) are

$$v_C = -V_0 + (V_0 + V_{C2}) \cos \omega_1(t - t_2) \quad (50)$$

$$i_L = -\frac{V_0 + V_{C2}}{Z_1} \sin \omega_1(t - t_2) \quad (51)$$

where V_{C2} is the voltage of C at t_2 .

The resonant angular frequency of this state is

$$\omega_1 = \sqrt{\frac{1}{2L_r C}} \quad (52)$$

and the resonant impedance is

$$Z_1 = \sqrt{\frac{L_r}{2C}} \quad (53)$$

(iv) State IV [$t_3 - t_4$]

Q_1 is off and Q_2 is on. Similarly to state II, the resonance of the inductor current is stopped by the reverse bias of the diodes. The inductor current is zero in this state. The output capacitor C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C3} \quad (54)$$

$$i_L = 0 \quad (55)$$

where V_{C3} is the voltage of C at t_3 .

3.2.2 General equations of $-1/2$ mode converter

By using input and output power balancing, for state I

$$V_S \int_{t_0}^{t_1} i_L dt = V_0 I_0 T_S \quad (56)$$

By solving (56), and assuming the voltage conversion ratio is $-1/2$, then

$$v_C = -V_0 + \frac{\pi I_0 T_S Z_0}{4T_0} \cos \omega_0(t - t_0) \quad (57)$$

$$i_L = -\frac{\pi I_0 T_S}{2T_0} \sin \omega_0(t - t_0) \quad (58)$$

where T_S is the switching frequency, T_0 is the resonant period in state I, and I_0 is the output current.

For state II

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_0}{4T_0} \quad (59)$$

$$i_L = 0 \quad (60)$$

For state III

$$\int_{t_2}^{t_3} i_L dt = I_0 T_S \quad (61)$$

By solving (61),

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \cos \omega_1(t - t_2) \quad (62)$$

$$i_L = \frac{\pi I_0 T_S}{T_1} \sin \omega_1(t - t_2) \quad (63)$$

where T_1 is the resonant period of state III.

For state IV

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \quad (64)$$

$$i_L = 0 \quad (65)$$

3.2.3 General equations of $-1/n$ mode converter

For a $1/n$ mode converter, in state I

$$v_C = -V_0 + \frac{\pi I_0 T_S Z_0}{n^2 T_0} \cos \omega_0 (t - t_0) \quad (66)$$

$$i_L = -\frac{\pi I_0 T_S}{n T_0} \sin \omega_0 (t - t_0) \quad (67)$$

where

$$\omega_0 = \sqrt{\frac{n}{L_r C}} \quad (68)$$

$$Z_0 = \sqrt{\frac{n L_r}{C}} \quad (69)$$

In state II

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_0}{n^2 T_0} \quad (70)$$

$$i_L = 0 \quad (71)$$

In state III

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \cos \omega_1 (t - t_2) \quad (72)$$

$$i_L = -\frac{\pi I_0 T_S}{T_1} \sin \omega_1 (t - t_2) \quad (73)$$

where

$$\omega_1 = \sqrt{\frac{1}{n L_r C}} \quad (74)$$

$$Z_1 = \sqrt{\frac{L_r}{n C}} \quad (75)$$

In state IV

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \quad (76)$$

$$i_L = 0 \quad (77)$$

$$C_1, C_2 = 0.22 \mu\text{F}$$

$$C_0 = 50 \mu\text{F}$$

$$L_r = 1 \mu\text{H}$$

From the simulation waveforms, the output voltage is about 30 V. It also shows that the inductor current and the voltage of C_1 and C_2 resonate in sinusoidal manner on the regions t_0 – t_1 and t_2 – t_3 . The voltages of C_1 and C_2 are the same all the time. For 80 W output power, the output current I_0 is equal to 2.67 A. According to (19) and (20), v_c at t_0 is 19.9 V and at t_1 is 40.1 V. The highest peak value of i_L is 6.7 A in the period t_0 – t_1 . According to (24) and (25), v_c at t_2 is 40.1 V and at t_3 is 19.9 V. The lowest peak value of i_L is -6.7 A. All the calculated results are equal to the simulation results. It is evident that the generalised equations of the noninverting converter can describe the waveforms of the converter correctly.

4.1.2 Inverting $-1/2$ mode switched capacitor converter

Fig. 19 shows the idealised computer simulation waveforms of inverting $-1/2$ mode switched capacitor converter. The simulation was also done by using a Saber Designer program. All switches and diodes in the simulation were ideal components. The specification of the simulated circuit is:

Input voltage	90 V
Output power	80 W
Switching frequency	200 kHz
C_1, C_2	0.22 μF
C_0	50 μF
L_r	1 μH

From the simulation waveforms, the output voltage is about -45 V. It also shows that the inductor current and the voltages of C_1 and C_2 resonate in sinusoidally between t_0 – t_1 and t_2 – t_3 . The voltages of C_1 and C_2 are equal all the time. For 80 W output power, the output current I_0 is equal to -1.78 A. According to (57) and (58), v_c at t_0 is 34.9 V and at t_1 is 55.1 V. The highest peak value of i_L is 6.7 A in the period t_0 – t_1 . According to (62) and (63), v_c at t_2 is 55.1 V and at t_3 is 34.9 V. The lowest peak value of i_L is -6.7 A. All the calculated results are equal to the simulation results. It is evident that the generalised equations of the noninverting converter can describe the waveforms of the converter correctly.

4 Simulation results

4.1 Comparison of generalized equations and idealized computer simulation results

4.1.1 Noninverting $1/3$ mode switched capacitor converter

Fig. 14 shows the idealised computer simulation waveforms of a noninverting $1/3$ mode switched capacitor converter. The simulation was done by using the Saber Designer program. All switches and diodes in the simulation were ideal components. The specification of the simulated circuit is:

Input voltage	90 V
Output power	80 W
Switching frequency	200 kHz

4.2 Comparison of idealised and practical computer simulation results

The specifications of the circuit for practical computer simulations are the same as those for idealised computer simulations in both converters except the former ones use MOSFETs IRF 630 and Schottky diodes MBR10100 for the switches and diodes, respectively. The practical simulation results of the noninverting $1/3$ mode converter and the inverting $-1/2$ mode converter are shown in Figs. 20 and 21, respectively.

Comparing Fig. 14 with Fig. 20 and Fig. 19 with Fig. 21, they show that when using practical switches (Figs. 20 and 21) instead of ideal switches (Figs. 14 and 19), the output voltages of the converters in the practical circuits are lower than those of the ideal circuits. This is because of the drain-to-source resistance (R_{ds}) of the MOSFET. For IRF 630, R_{ds} is about 0.4Ω . When the MOSFETs are turned on and conduct, there will be a voltage

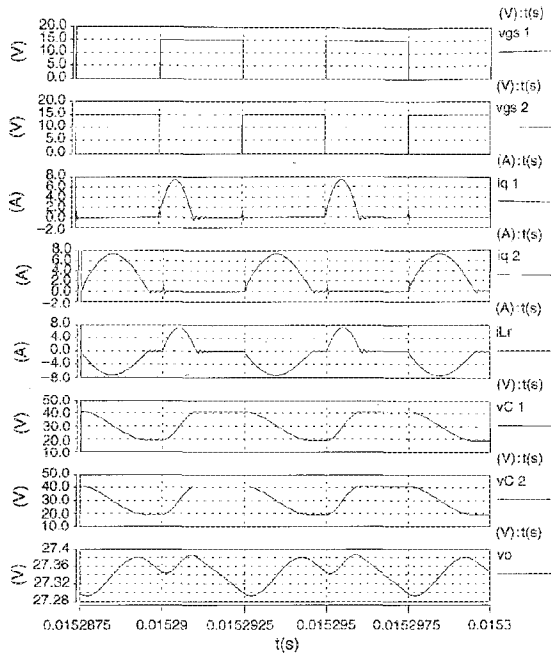


Fig. 20 Practical simulation waveforms of the noninverting 1/3 mode switched-capacitor resonant step-down converter

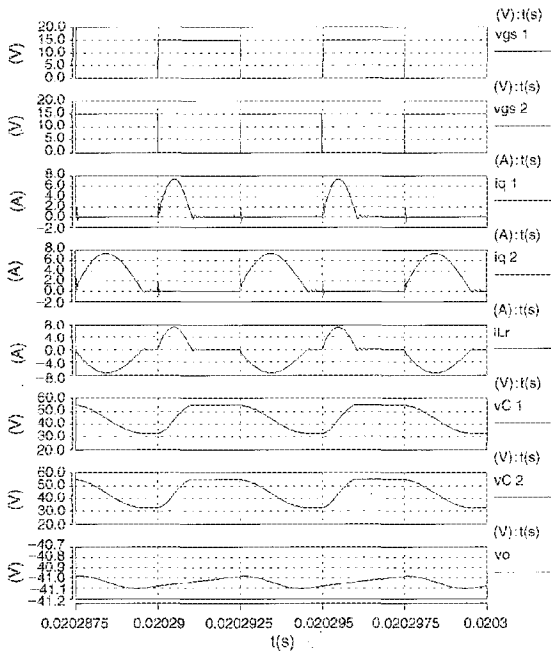


Fig. 21 Practical simulation waveforms of the inverting -1/2 mode switched-capacitor resonant step-down converter

across them. The practical diodes also have forward voltages (V_F). For MBR10100, V_F is about 0.8 V. These two components ensure that the output voltage will be lower than that in the ideal case. In fact, it is the main power loss in the circuit. It also makes the voltages of the switched capacitors V_{C1} and V_{C2} lower than those in the ideal case.

Figs. 9 and 10 show that when either transistor Q_1 or transistor Q_2 is switched off, the transistor currents I_{Q1} and I_{Q2} give short pulses at that moment. All practical transistors have junction capacitors across their legs. Since Q_1 and Q_2 are connected together, when one transistor is switched off, its junction capacitor will be charged through another transistor and hence, there will be a short pulse charging current. The junction capacitor will discharge internally when that transistor is switched on so that it gives power loss. This power loss is directly proportional to the switching frequency. Also, since both the transistors and the Schottky diodes have junction capacitors and the junction capacitors resonate with the resonant inductor in the practical case, when the main resonance stops, the currents still ring at a very small amplitude.

To maintain the output power at 80 W, the input power should be higher to compensate the power loss practically. Since the input voltage of each converter is constant, in practice the input current is higher than that in the ideal case. As a result, the amplitudes of all the transistor currents and inductor currents are higher than those in the ideal case.

5 Design

The design of a 1/3 mode step-down converter is shown in the following procedure as an example:

(i) Define the specification:

$V_s = 90$ V, $P_o = 10$ –80 W, switching frequency = 200 kHz

(ii) The resonant frequency is chosen to be higher than the switching frequency so that a zero-current switching can be naturally achieved. Because states I and III have two frequencies, the worst case, i.e. ω_1 , is used for the calculation. A 20% higher frequency is used to ensure zero current switching

$$\omega_1 = 1.2(2\pi f_s) \quad (78)$$

This gives a resonant frequency $f_o = 240$ kHz.

(iii) The ripple voltage on the resonant capacitor C_{1a} and C_{1b} must be small compared to their DC level. The peak-to-peak ripple voltage is derived from (21) and (26), or (32) and (38)

$$\Delta v_{C1} = 0.943 \frac{\pi I_o T_s}{T_1} \sqrt{\frac{L_r}{C}} \quad (79)$$

A maximum voltage ripple of 60% of the DC component is allowed on the resonant capacitor. Hence, it gives $Z = \sqrt{L_r/C} = 1.96 \Omega$

(iv) The resonant components can therefore be calculated by angular frequency and impedance from (13) or (36)

$$C = \frac{1}{\sqrt{2}\omega_1 Z} = 0.24 \mu\text{F} \quad (80)$$

$$L_r = \frac{Z}{\sqrt{2}\omega_1} = 0.92 \mu\text{H} \quad (81)$$

C and L_r are chosen to be $0.22 \mu\text{F}$ and $1 \mu\text{H}$ in the experiment.

(v) C_o is a larger capacitance and is to maintain constant voltage for the output stage. Its value can be estimated by the basic capacitor ripple voltage calculation

$$\Delta V_{Co} = \frac{I_o}{C_o} \left(\frac{\cos \theta}{f_s} - \frac{1}{f_o} \left(1 - 2 \frac{\theta}{\pi} \right) \right) \quad (82)$$

where $\theta = \sin^{-1} f_s / \pi f_0$ and $I_0 = P_0 / V_0$. For a 0.1 V ripple voltage, and 25% overload reservation, this gives $C_0 = 50 \mu\text{F}$.

(vi) The transistor rating voltage is at least equal to V_s . The diode rating voltage is also equal to V_s/n , but with a tolerance of ripple voltage on C_1 . Therefore a rating of at least 90 V is chosen.

6 Experimental results

The noninverting 1/3 mode step-down resonant converter as shown in Fig. 22 has been tested in the laboratory. The circuit is very similar to that shown in Fig. 7 but only a filtering capacitor C_m was added to ensure a good input voltage for experimental purposes. The specification for the prototype is

input voltage	90 V
output power	10 W 80 W
output voltage	30 V
switching frequency	200 kHz
Q_1 and Q_2	IRF630
D_1, D_2, D_m, D_b and D_c	MBR10100
C_1, C_2	0.22 μF
C_m	100 μF
C_0	50 μF
L_r	1 μH

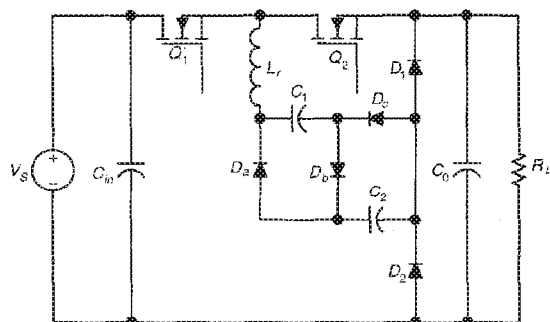


Fig. 22 Circuit of prototype for noninverting 1/3 mode switched-capacitor resonant step-down converter

The measured waveforms of the converter operating with 80 W output power are shown in Figs. 23–27. Fig. 23 shows the resonant inductor current while Figs. 25 and 26 show the currents and drain-to-source voltages of the transistors. When compared with Figs. 25 and 26, it can be observed that the resonant inductor current consists of the Q_1 and Q_2 currents.

The output voltage of the converter is shown in Fig. 24. The measured output voltage of the converter when operating at 80 W is 25.84 V. The measured output voltage is lower than the expected output voltage (30 V) and the practical simulated output voltage because of the drain-to-source resistance of the transistors and the voltage drop of the diodes in forward bias. The copper loss of the wires and the inductors, and also the equivalent series resistance (ESR) of all capacitors give output voltage drop and power loss in the circuit. To minimise the influence of the ESR of the capacitors, polypropylene capacitors, which have low ESR, are used for the switched capacitors.

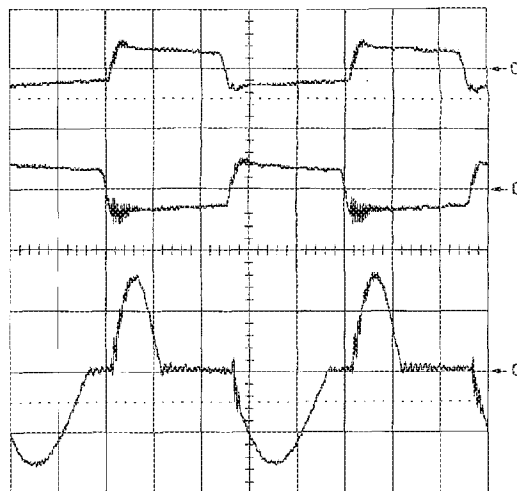


Fig. 23 Measured waveforms of converter output power 80 W
upper trace: gate signal of Q_1 , 40 V/div
middle trace: gate signal of Q_2 , 40 V/div
lower trace: i_L , 5 A/div

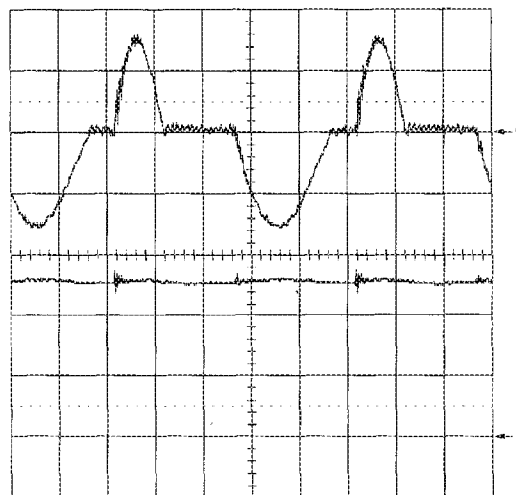


Fig. 24 Measured waveforms of converter
output power 80 W
upper trace: i_L , 5 A/div
lower trace: output voltage: 10 V/div

Figs. 25 and 26 show that the transistor currents resonate sinusoidally. The resonance stops when the current reaches zero while the drain-to-source voltage of the corresponding transistors is zero. The transistors are switched off when the currents are equal to zero so that a zero-current switching condition is obtained. This is very similar to the simulated waveforms. Because of the power loss of the circuit and to obtain 80 W output power, the input current and hence the transistors and inductor currents are higher than in the simulated waveforms.

Fig. 27 shows that the resonant inductor current and the voltages of the switched capacitors resonate together. The voltage waveform of C_1 is almost equal to that of C_2 . The amplitude is lower than those in simulated waveforms.

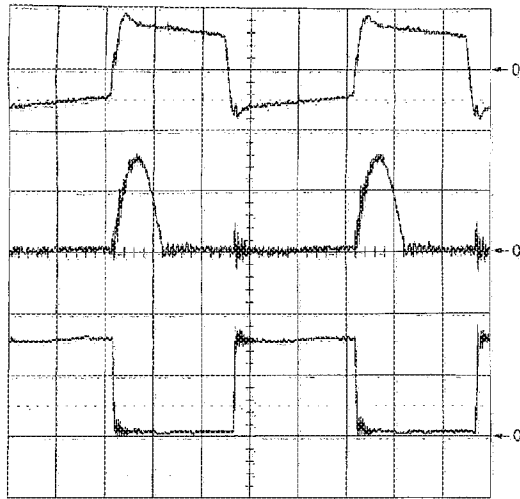


Fig. 25 Measured waveforms of converter
output power 80 W
upper trace: gate signal of Q_1 , 20 V/div
middle trace: Q_1 current, 5 A/div
lower trace: V_{ds} of Q_1 , 40 V/div

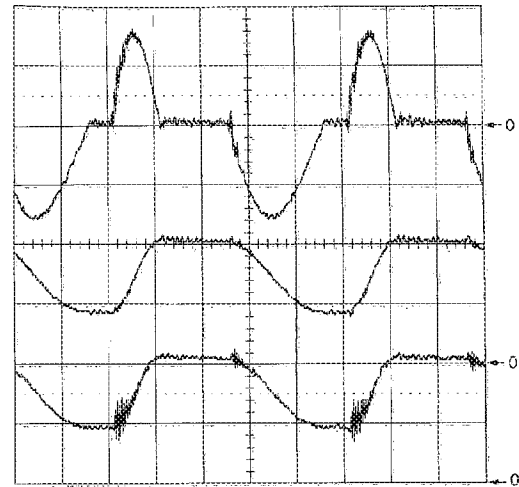


Fig. 27 Measured waveforms of converter
output power 80 W
upper trace: I_L , 5 A/div
middle trace: V_{C1} , 20 V/div
lower trace: V_{C2} , 20 V/div

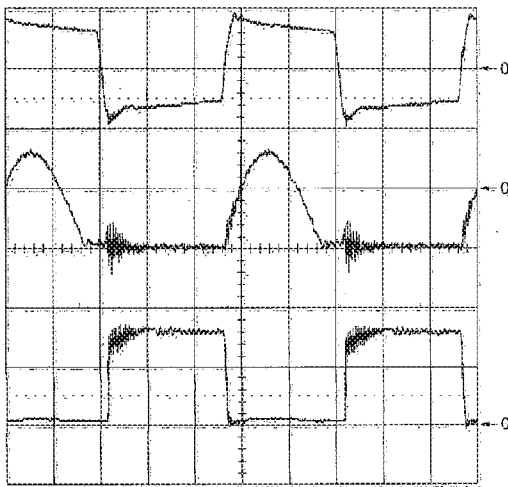


Fig. 26 Measured waveforms of converter
output power 80 W
upper trace: gate signal of Q_1 , 20 V/div
middle trace: Q_1 current, 5 A/div
lower trace: V_{ds} of Q_1 , 40 V/div

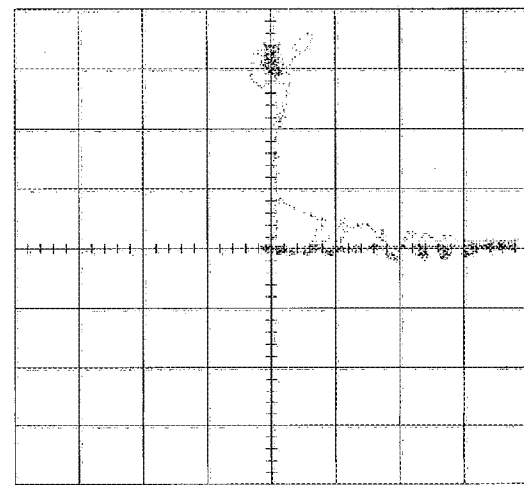


Fig. 28 Noninverting H3 mode switched-capacitor resonant step-down converter
 $v-i$ trajectory of Q_1
x-axis: Q_1 current, 2 A/div
y-axis: V_{ds} of Q_1 , 20 V/div

because of the voltage drop of the transistors, diodes and the wires.

The diagrams of the $v-i$ trajectory of each transistor are shown in Figs. 28 and 29. Figs. 23 and 24 again confirm that both transistors are in zero-current switching condition.

Fig. 30 plots the efficiency against the output power of the converter. By testing the circuit with the output power between 10–80 W, the efficiency varied between 85.5% at both 10 W and 80 W output power to 93.2% at 22 W output power. Fig. 31 plots the voltage conversion ratio against the output power of the converter. The voltage

conversion ratio varies between 0.325 at 10 W output power to 0.287 at 80 W output power. Figs. 23–27 show that the efficiency was highest when the output power was 22 W. At light load, power loss in the circuit is mainly due to the diodes, which have voltage drop in forward bias. At high load, the power loss is mainly due to the drain-to-source resistance of the transistors. On the other hand, Figs. 28 and 29 show that the voltage conversion ratio decreases when the output power increases. Other than the voltage drop on the diodes, when the current increases, the voltage drop on both transistors increases and hence, the voltage conversion ratio decreases.

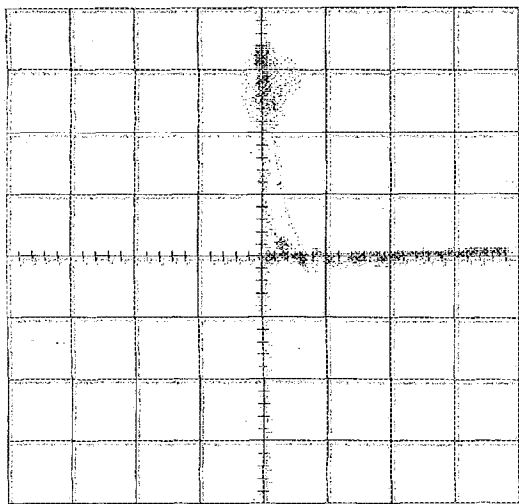


Fig. 29 Noninverting $1/3$ mode switched-capacitor resonant step-down converter
 v - i trajectory of Q_2
 x-axis: Q_2 current, 2 A/div.
 y-axis: V_{ds} of Q_2 , 20 V/div

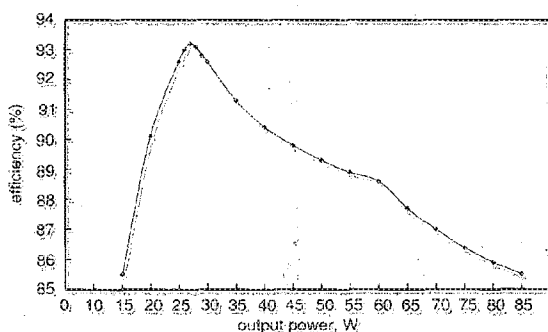


Fig. 30 Measured efficiency and output power of noninverting $1/3$ mode switched-capacitor resonant step-down converter

7 Conclusions

This paper has proposed two families of switched-capacitor circuits. One is the noninverting power converter and the other is the inverting converter. Both families operate under zero-current switching. The inherent disadvantage of a high

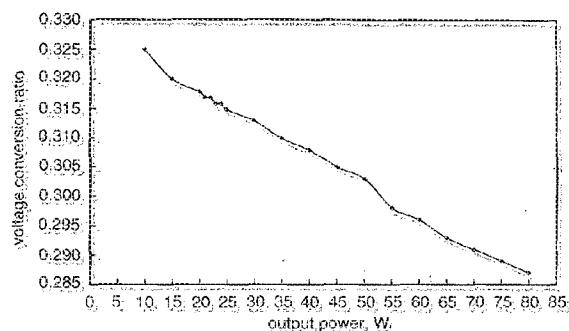


Fig. 31 Measured voltage conversion ratio and output power of noninverting $1/3$ mode switched-capacitor resonant step-down converter

pulsational current has been eliminated. The number of switches needed is low. The complete analysis of the converters has also been presented. Each family has a number of fractional output voltage conversion converters. Experimental results have confirmed that the operation agrees with the theoretical prediction in that all transistors can achieve zero-current switching. The efficiency is also very high.

8 Acknowledgment

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First Named Inventor/Applicant Name:	Patrizio Vinciarelli
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Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

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Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Prior to examination, please amend the application as indicated on the following pages.

Amendments to the Specification:

Please replace the paragraph beginning at paragraph [036] with the following amended paragraph:

[036] Preferably, the transformer and output cells of the multi-cell power converter may be formed on a multi-layer printed circuit board (“PCB”) 321 and encapsulated to form a package amenable to mount directly to the semiconductor load. For example, as shown in Fig. 3, the multi-cell converter 310 is shown in an encapsulated package having a generally rectangular outline or footprint. Figs. 5A and 5B show edges of the internal PCB 321 of the converter 310 sandwiched between two cured encapsulant layers 322 and 323. One of the large surfaces 320 of the converter 310 is shown in Fig. 5A including a plurality of electrical terminations which will be discussed in more detail below in connection with Fig. 3. The other large surface 319 of the converter 310 is shown in Fig. 5B having a conductive layer 324 covering a majority of the surface 319 and which preferably may be electrically connected to a common terminal, e.g. using one or more electrical and thermal connections 325, to provide EMI shielding and also enhance thermal performance. For example, the surface 319 may be mated with a cold plate or heat sink. Encapsulated packages of the type shown in Figs. 3, 5A, and 5B, amenable for housing the multi-cell power converter in a format suitable for in-package and on-package applications is described in Vinciarelli et al., *Panel Molded Electronic Assemblies with Multi-Surface Contacts*, U.S. Patent Appln. Ser. No. 14/731,287 filed June 4, 2015 (the “Leadless Disclosure”) and in Vinciarelli et al., *Panel Molded Electronic Assemblies with Integral terminals*, U.S. Patent Appln. Ser. No. ~~xxxxxxx~~ 16/218,395, filed December 12, 2018, ~~December xx 2018~~, (the “Trench Disclosure”) both of which are assigned to VLT, Inc. of Andover, MA., the entire disclosure of each is incorporated herein by reference.

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 3 of 3

Attorney Docket 00614-0183001 / 0269-US-1

REMARKS

The specification has been amended.

Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: September 11, 2019_____

/Rex I. Huang/_____

Rex I. Huang

Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Electronic Acknowledgement Receipt

EFS ID:	37126597
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	11-SEP-2019
Filing Date:	12-DEC-2018
Time Stamp:	09:07:26
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Prelim_Amen d.pdf	111448	yes	3
			80591273a2b1b03d92bc016856d9c06c71 ddad76		

Multipart Description/PDF files in .zip description			
Document Description	Start	End	
Preliminary Amendment	1	1	
Specification	2	2	
Applicant Arguments/Remarks Made in an Amendment	3	3	

Warnings:

Information:

Total Files Size (in bytes):	111448
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111
If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371
If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office
If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/218,418		Filing Date 12/12/2018		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)	(Column 2)						
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)		
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A		N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A				
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL				
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT	12/17/2018	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 47	Minus	** 47	= 0	x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 6	Minus	*** 6	= 0	x \$460 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/MYRTLE B LEIGH/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed before the receipt of a first action on the merits.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: February 27, 2020_____

/Rex I. Huang/

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary)		Applicant VLT, Inc.	
		Filing Date December 12, 2018	Group Art Unit
(37 CFR §1.98(b))			

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	3846741	11/1/1974	Kunkle et al.			
	2.	4223970	9/23/1980	Walter			
	3.	4230385	10/28/1980	Ammon et al.			
	4.	4394711	7/19/1983	Conley			
	5.	4526429	7/2/1985	Kirkman			
	6.	4742477	5/3/1988	Phillips et al.			
	7.	4917526	4/17/1990	Paterson			
	8.	5168432	3/6/2019	Murphy et al.			
	9.	5321351	6/14/1994	Swart et al.			
	10.	5408189	4/18/1995	Swart et al.			
	11.	6031253	2/29/2000	Kobayashi			
	12.	6110213	8/29/2000	Vinciarelli et al.			
	13.	6116916	9/12/2000	Kasai			
	14.	6230403	5/15/2001	Skoolicas et al.			
	15.	6646886	11/11/2003	Popovich et al.			
	16.	6847853	1/25/2005	Vinciarelli et al.			
	17.	7294007	11/13/2007	Lawlyes			
	18.	10231333	3/12/2019	Liu et al.			
	19.	10390437	8/20/2019	Kudou et al.			
	20.	10512182	12/17/2019	Suzuki			
	21.	20010018285	8/30/2001	Mizumura			
	22.	20010018286	8/30/2001	Mizumura			
	23.	20010023980	9/27/2001	Ohmori			
	24.	20010032388	10/25/2001	Morris			
	25.	20030162434	8/28/2003	Kamiya			
	26.	20030168499	9/11/2003	Tanabe et al.			
	27.	20040207089	10/21/2004	Masuda			

Examiner Signature	Date Considered
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	28.	20040251554	12/16/2004	Masuda			
	29.	20050048692	3/3/2005	Hanada et al.			
	30.	20060127652	6/15/2006	Kanaya et al.			
	31.	20060180346	8/17/2006	Knight et al.			
	32.	20060272150	12/7/2006	Eguchi et al.			
	33.	20130083495	4/4/2013	Moon et al.			
	34.	20160128215	5/5/2016	Kanzaki et al.			
	35.	20170115144	4/27/2017	Watanabe et al.			
	36.	20170336584	11/23/2017	Ariga et al.			
	37.	20190080931	3/14/2019	Ito et al.			
	38.	RE32212	7/22/1986	Walter et al.			
	39.	RE36442	12/14/1999	Kardos			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Electronic Acknowledgement Receipt

EFS ID:	38714712
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Mollie Kufner
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	27-FEB-2020
Filing Date:	12-DEC-2018
Time Stamp:	16:25:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0183001_Transmittal.pdf	72324 857c6f5da2aed4979282b03c8087fa9dee3593f5	no	1

Warnings:

Information:					
This is not an USPTO supplied IDS fillable form					
2	Information Disclosure Statement (IDS) Form (SB08)	00614-0183001_1449.pdf	131918	no	2
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Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
			Total Files Size (in bytes):	204242	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Prior to examination, please amend the application as indicated on the following pages.

Amendments to the Specification:

Please replace the paragraph beginning at paragraph [009] with the following amended paragraph:

[009] In general, in another aspect, a method of making power converters is provided. The method includes assembling an encapsulated panel including a printed circuit board (“PCB”) having electronic components mounted to the PCB and a plurality of conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the panel comprising one or more power conversion modules having boundaries defined by one or more predetermined cut lines; configuring each of the one or more power conversion modules in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output nodes ~~(it seems a conversion cell does not include a whole converter output node, but rather shares portions of several output nodes, see comment in claim 1)~~, wherein the output nodes are arranged in a first pattern having a first spacing between the output nodes, and wherein each output node is electrically connected to a respective conductive feature; selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; and cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module.

Please replace the paragraph beginning at paragraph [016] with the following amended paragraph:

[016] In another general aspect, an apparatus includes: a semiconductor device having power input terminals; a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals (~~it seems a conversion cell does not include a whole converter output terminal, but rather shares portions of several output terminals, see comment in claim 1~~), the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals. The converter output terminals of the power conversion module is mated with and electrically connected to respective x-input terminals of the interconnection module. The interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern. The power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 4 of 4

Attorney Docket 00614-0183001 / 0269-US-1

REMARKS

The specification has been amended.

Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: March 11, 2020_____

/Rex I. Huang/_____

Rex I. Huang

Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Electronic Acknowledgement Receipt

EFS ID:	38838430
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	11-MAR-2020
Filing Date:	12-DEC-2018
Time Stamp:	17:00:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Prelim_Amen d_031120.pdf	89920	yes	4
			d76072117bd2c300aaadec38cf7a26c1cfab d90d		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Specification	2	3
	Applicant Arguments/Remarks Made in an Amendment	4	4

Warnings:

Information:

Total Files Size (in bytes):	89920
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed before the receipt of a first action on the merits.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: April 30, 2020_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0183001	Application No. 16/218,418
	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Filing Date If Appropriate	Class	Subclass
	1.	4211603	7/8/1980	Reed			
	2.	20030087538	5/8/2003	Ueno			
	3.	20170013748	1/12/2017	Saji et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Translation	Class	Subclass

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Electronic Acknowledgement Receipt

EFS ID:	39313160
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Mollie Kufner
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	30-APR-2020
Filing Date:	12-DEC-2018
Time Stamp:	13:14:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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Warnings:

Information:					
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Information:					
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NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 06/30/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2849

DATE MAILED: 06/30/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

TITLE OF INVENTION: Delivering Power To Semiconductor Loads

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	09/30/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 06/30/2020
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

	(Typed or printed name)
	(Signature)
	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001	1092

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nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	09/30/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2849	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____